

MEDST R



CLINICAL GUIDE AND SYNOPSIS

SECOND EDITION

OBSTETRICS AND GYNECOLOGY

PREPARED BY

THE GRADUATING CLASS OF JIMMA UNIVERSITY

2015 E.C (2022/23)

FOR
REVISION
ONLY

MEDSTAR CLINICAL GUIDE AND SYNOPSIS

OBSTETRICS AND GYNECOLOGY

SECOND EDITION

DEC, 2022

Prepared by Graduating Class of Jimma University 2015 EC
(2022/23)

Preface

The second edition of MEDSATR CLINICAL GUIDE AND SYNOPSIS TO OBSTETRICS AND GYNECOLOGY (OBGYN) is one of the three similar works, which include similar sister documents for Internal medicine and Pediatrics. The second edition is prepared based on the positive feedbacks & the constructive criticism we received from the previous one. In this edition we have tried to include more topics and updates.

The guide is mainly prepared by combining our ward experience and the detailed science on textbooks for the major cases in obstetrics and gynecology. The approach we used is case oriented. The management plans for the most cases is mainly taken from the FMOH Obstetrics management protocol and Jimma University OBGYN Management Guideline.

For history and physical examination in OBGYN we strongly recommend you to read, Introduction to Obstetrics and Gynecology diagnosis, by *Dr. Asheber Gaym*. You should also check the simplified format for history taking and physical examination prepared by Jimma Medical Center OBGYN Department.

In conclusion, we would like to say, we have tried to make this guide as readable as possible and we hope it will come in handy for revision in a short period of time.

We have prepared an Email address and Telegram bot link so that we could get your feedback on the book and come up with a better second edition in upcoming years.

Email address: MEDSTARcomment@Gmail.com

The Contributors

Acknowledgment

We would like to express our sincere gratitude to the medical students who have sacrificed their time in relentlessly committing to this book.

Special thanks are due to Dr. Rajif Shawl (Medstar Coordinator and Medstar Internal Medicine organizer) for his initiative and encouragement to prepare this document.

Contributors

- Dr. Balkewkal Kebede (***MI***)
- Dr. Daniel Delessa (***MI***)
- Dr. Abubeker Nuredin (***MI***)
- Dr. Amanuel Tefera (***MI***)
- Dr. Bulcha Lemma (***MI***)
- Dr. Abdulhafiz Hussein (***MI***)
- Dr. Birhanu Olani (***MI***)
- Dr. Bethel Ayele (***MI***)
- Dr. Abraham Gebeyehu (***MI***)
- Dr. Bemnet G/Micheal (***MI***)

Coordinator: *Dr. Balkewkal Kebede (MI)*

Cover page by: Dr. Nahom Asnake (Nax, MI)



Medstar OBGYN Team

Contents

<i>Preface</i>	I
<i>Acknowledgment</i>	II
<i>Contributors</i>	III
SECTION ONE: OBSTETRICS	15
Chapter 1	1
ANTENATAL CARE (ANC)	1
Definition, Classification and Objectives of ANC	1
First ANC contact	4
Diagnosis of pregnancy	4
Gestational age determination	6
Detail Hx, P/E and Baseline IXs	8
Subsequent ANC contacts	10
Functions of Obstetric Ultrasound	11
First trimester U/S	11
Second and Third Trimester U/S	12
Anemia in pregnancy	13
HIV in pregnancy	16
Syphilis in pregnancy	16
Viral hepatitis in pregnancy	20
UTI in pregnancy	21
Acute urethritis/acute urethral syndrome	21
Asymptomatic Bacteriuria and Acute Cystitis	22
Acute Pyelonephritis	24
Rh alloimmunization	26
Chapter 2	27
Antenatal Assessment of Fetal Well Being	27
Introduction	27
Modalities of antepartum testing	30
Chapter 3	40
ASSESSMENT OF FETAL PULMONARY MATURATION	40
Tests of Fetal Pulmonary Maturity	41

Antenatal Corticosteroid Therapy (ACS).....	45
Chapter 4.....	50
HYPERTENSIVE DISORDERS OF PREGNANCY (HDP)	50
Definition and Classifications	50
Gestational HTN (GH).....	53
Preeclampsia (PE)	54
Eclampsia	67
Chronic HTN and Superimposed PE	70
Superimposed PE	71
Management of PE-E syndrome	71
Management of eclampsia	78
Subsequent risks of women who developed HDP	79
SAMPLE HISTORY.....	80
Chapter 5	85
ANTEPARTUM HEMORRHAGE (APH)	85
Causes of Third-Trimester bleeding.....	86
Placental Abruptio	87
Placenta Previa	93
Morbidly adherent placentas.....	97
Vasa Previa.....	100
SAMPLE HISTORY.....	102
Chapter 6	105
PROM & PRETERM LABOR	105
PRETERM LABOR/BIRTH	105
PREMATURE/PRELABOR RUPTURE OF MEMBRANE (PROM)	111
Sample history	118
Chapter 7	121
MALPRESENTATION	121
Introduction	121
Incidence and Etiologies/Risk factors	123
Breech presentation	124
Face presentation.....	135

Brow presentation	138
Compound presentation	140
Shoulder presentation (Transverse lie).....	142
Chapter 8.....	146
MULTIFETAL GESTATION	146
Twin pregnancy	147
Monozygotic twins.....	147
Dizygotic twins	151
Diagnosis of MG	153
Complications of MG.....	154
Management of MG pregnancies	160
Chapter 9.....	165
Red Cell Alloimmunization (Rh Alloimmunization).....	165
Introduction.....	165
Rh (CDE) Blood Group Incompatibility	168
Management of Pregnancy in Rh Negative Women.....	171
Management of ICT Negative Pregnancy – Prevention of Sensitization.....	172
Antepartum Prophylaxis for Pregnancy Complications with FMH	174
Postpartum prophylaxis.....	175
Management of ICT Positive Pregnancy - the Alloimmunized pregnancy.....	179
Outcomes	184
Hemolytic Disease of the Fetus and Newborn (HDFN)	185
Chapter 10.....	187
GESTATIONAL DIABETES MELLITUS (GDM)	Error! Bookmark not defined.
Types of DM.....	187
Pathophysiology of GDM	189
Screening and Diagnosis	190
Complications of diabetes during pregnancy.....	191
Congenital Malformations	192
Fetal Macrosomia	193
Neonatal Hypoglycemia	194
Respiratory Distress Syndrome.....	194

Management of DM during pregnancy	195
Management of the patient with type 1 or type 2 DM	195
Management of the woman with GDM.....	202
Long-Term Effects of Glucose Intolerance on Mother and Fetus.....	204
Pre-pregnancy counseling of women with preexisting diabetes mellitus	205
Chapter 11	206
Intrauterine Growth Restriction/Retardation	206
Introduction	206
Etiologies of IUGR	210
Diagnosis of FGR	212
Management	217
Chapter 12	220
Amniotic fluid disorders	220
Introduction	220
Assessment of AFV	221
Oligohydramnios	222
Polyhydramnios/hydramnios	225
Chapter 13	229
POSTPARTUM HEMORRHAGE (PPH)	229
Introduction	229
Classification	230
Etiology	230
Late PPH	246
Chapter 14	251
Intra-amniotic Infection (Chorioamnionitis)	251
Introduction	251
Pathogenesis	251
Epidemiology	252
Clinical presentation	253
Diagnosis	254
Management	255
Complications	258

Prevention	259
Chapter 15	260
Puerperal Fever	260
Definitions	260
Differential Diagnosis for Puerperal fever	261
Postpartum endomyometritis.....	261
SECTION TWO: GYNECOLOGY.....	272
Chapter 16.....	273
ABDOMINOPELVIC MASS (APM)	273
Approach	273
Differential diagnosis of APM.....	278
Chapter 17	287
Approach to Abnormal Uterine Bleeding	287
Definitions	287
Incidence and Etiology	290
Endometrial cancer	298
Patient Evaluation	301
Chapter 18.....	304
LEIOMYOMAS.....	304
Factors associated with leiomyoma	305
Pathophysiology.....	306
Classification.....	307
Secondary changes in myoma.....	308
Clinical features.....	310
Differential diagnosis	313
Complications.....	313
Myomas and Pregnancy.....	313
Management.....	314
Sample History	323
Chapter 19.....	327
OVARIAN TUMORS	327
Epidemiology and Risk factors	327

Prevention of Ovarian Cancer	330
Classification of Ovarian Tumors	333
Benign Ovarian Masses	334
Borderline Ovarian Masses	336
Malignant Ovarian Masses.....	337
Epithelial Ovarian Cancer	337
Differentiating between malignant and Benign ovarian masses	342
Malignant Ovarian Germ Cell Tumors	349
Ovarian Sex Cord-Stromal Tumors (SCST).....	352
Ovarian cancer during pregnancy	355
Sample history	357
Chapter 20	359
CERVICAL NEOPLASIA	359
Cervical Intraepithelial Neoplasm (CIN)	359
Pathogenesis and Risk factors.....	360
CIN and Human papilloma Virus (HPV)	362
Cervical Cancer Prevention and Screening	365
Cervical cancer.....	373
Sample history	381
Chapter 21.....	383
GESTATIONAL TROPHOBLASTIC DISEASE	383
Definitions	383
Classification.....	383
Epidemiology and Risk factors	384
Pathogenesis	385
Clinical features.....	387
Complications.....	391
Treatment of Benign GTD	392
GESTATIONAL TROPHOBLASTIC NEOPLASIA	394
SAMPLE HISTORY.....	404
Chapter 22	407
Pelvic Pain.....	407

Classification.....	407
Acute pelvic pain	408
DDx for Acute pelvic pain.....	408
Cyclic pelvic pain.....	416
Chronic pelvic pain.....	417
DDx Chronic Pelvic Pain	418
Endometriosis	419
Adhesion	423
Approach to Pelvic/lower abdominal pain.....	424
Chapter 23.....	429
ABORTION.....	429
Definition and Terminology	429
Spontaneous first trimester abortion.....	430
Etiologies and Risk Factors.....	431
Clinical classification of spontaneous abortion.....	433
Spontaneous midtrimester abortion	436
Induced abortion	437
Complication of abortion.....	438
Approach	438
Management.....	440
Spontaneous abortion	440
Induced abortion.....	441
Components of Post abortion care (PAC)	442
Chapter 24	443
Ectopic pregnancy.....	443
Definition and terminology	443
Risk factors	444
Pathophysiology.....	445
Clinical manifestation	446
Diagnosis.....	447
Management.....	449
Chapter 25.....	453

PELVIC INFLAMMATORY DISEASE	453
Etiology.....	453
Pathogenesis	454
Risk factors for PID	456
Clinical manifestation	457
DIAGNOSIS	458
Complication.....	461
Treatment	462
Chapter 26	466
HYPEREMESIS GRAVIDARUM (HEG)	466
Definition and Incidence	466
Pathogenesis	466
Risk factors	468
Approach	469
Management.....	473
Chapter 27	477
PELVIC ORGAN PROLAPSE	477
Definition and Epidemiology.....	477
Pathophysiology.....	477
Description and classification	481
POP Quantification system	482
Baden-Walker Halfway system	484
Clinical manifestations of POP	485
Management principle	487
SAMPLE HISTORY.....	491
Chapter 28	494
INFERTILITY.....	494
Definitions and Classification	494
Etiologies.....	495
Approach to Infertility.....	495
Evaluation for Infertility	500
Evaluation of Female Infertility.....	500

Evaluation of male infertility.....	510
Treatment	512
Chapter 29	514
AMENORRHEA	514
Primary amenorrhea	514
Etiologies.....	515
Evaluation of primary amenorrhea.....	521
Management.....	528
Secondary amenorrhea.....	530
Evaluation of secondary amenorrhea.....	536
Management.....	539
Chapter 30	541
GENITOURINARY FISTULA (GUF).....	541
Physiology of Micturation	541
Urinary Incontinence.....	542
Genitourinary fistula.....	542
Incidence	542
Classification	543
Etiology and Risk factor.....	545
Pathophysiology.....	547
Clinical manifestation.....	548
Diagnosis	548
Complications.....	549
Management.....	550
Prevention.....	551
Sample history	552
References.....	553

This page is intentionally left blank



SECTION ONE: OBSTETRICS

Chapter 1

ANTENATAL CARE (ANC)

Prepared by Dr. Balkewkal Kebede (MI)

Definition, Classification and Objectives of ANC

Antenatal care (ANC) is defined as the complex of interventions that a pregnant woman and adolescent girl receives from skilled health care professionals in order to ensure the best health conditions for both mother and baby during pregnancy.

- Timely and appropriate ANC: -
 - reduces complications from pregnancy and childbirth
 - reduces stillbirth and perinatal death

Historically the practice of comprehensive and organized delivery of ANC passed through three phases

- Traditional ANC model
- Focused ANC model
- ANC for a positive pregnancy experience, (New WHO 2016 ANC model)
 - The new model uses the term **contact** between a pregnant woman and a health care provider.
 - Objectives (what it means to have positive pregnancy experience)
 - A health pregnancy for mother and baby (including preventing or treating risks, illness and death)
 - Physical and sociocultural normality during pregnancy
 - Effective transition to positive labour and birth
 - Positive motherhood (including maternal self-esteem, competence and autonomy)

ANC contacts

Generally prenatal care can be classified in to: -

- Basic Care and
- Specialized Care

WHO (2016) recommends at least **eight ANC contacts** as a basic prenatal care for all pregnant women.

Specialized care is given to pregnant ladies with risk factors associated with adverse pregnancy outcomes. The specific risk factors are listed in the FMOH integrated pregnancy, labor, delivery, newborn and postnatal care card.

Basic prenatal care schedule

- 1st contact in the 1st TM, up to 12wks of gestation
- 2nd and 3rd contacts in the 2nd TM, @20th and 26th wks of gestation
- 4th - 8th contacts in the 3rd TM, @30th, 34th, 36th, 38th & 40th wks of gestation

What are the general aims/objectives/components of ANC?

- Early detection and management of preexisting disease
 - ✓ DM, HTN, Cardiac, thyroid
- Early detection and management of pregnancy complications

- | | |
|---|---|
| <ul style="list-style-type: none"> ▪ Early <ul style="list-style-type: none"> ○ Ectopic pregnancy ○ Molar pregnancy ○ Abortion ○ HEG... | <ul style="list-style-type: none"> ▪ Late <ul style="list-style-type: none"> ○ HDP ○ APH ○ GDM ○ PTL ○ PPRM/PROM ○ Malpresentation ○ Anemia... |
|---|---|

- Health promotion and disease prevention
 - What do you promote?
 - Good Nutrition and rest
 - Abstinence from Harmful substance (tobacco, alcohol, traditional remedies)
 - Hygiene and infection prevention practice
 - Breast feeding
 - Family planning
 - What do you prevent?
 - **Anemia**=>
 - Fe and folate supplementation, with 30 mg to 60 mg of elemental iron and 400µg (0.4 mg) of folic acid
 - What is the elemental iron in different preparations of iron?
 - Fe supplementation is also recommended to prevent puerperal sepsis, LBWt and preterm labor (Ayder's)

- *Intermittent Fe (120 mg elemental iron) and folic acid(1280µg) weekly is recommended if daily iron is not acceptable due to side effects (Ayder's)*
- Deworming=>using single-dose albendazole (400mg) or mebendazole (500 mg) is recommended after the first trimester in endemic areas (areas with greater than 20% prevalence of infection with any soil-transmitted helminths)
- **Neonatal tetanus**=> Td-Immunization,
 - A total of five doses needs to be administered, 1st dose at 1st visit/contact, 2nd dose 1 month from 1st dose, 3rd dose 6month from 2nd dose, 4th dose after 1yr and last dose after 1yr from 4th.
 - The level and duration of protection increases with each dose.

TT	Level of protection	Duration of protection
TT ₁	Nil	None
TT ₂	80%	3yrs
TT ₃	95%	5yrs
TT ₄	99%	10yrs
TT ₅	99%	30yrs

- NB. Most mothers take only the 1st two, and won't return for the remaining doses after delivery
- **HIV(PMTCT)**=> start/continue ART
 - Option "B plus" = start treatment right after dx, irrespective of the CD₄
- **Malaria**=> ITNs
- **HDFN** (HDN/erythroblastosis fetalis) => Anti-D Ig in non-sensitized Rh-negative women at 28th and 34th wks of gestation
- **Viral hepatitis** (fulminant & chronic hepatitis in the neonate) => see below
- **Calcium deficiency**=>1.5-2.0 g oral elemental calcium starting from 14 weeks of gestation
- **Preeclampsia**=>
 - low dose Aspirin(50-150mg) is given for women, who is considered to have high risk for PE, like those with
 - ✓ Hx of PE with adverse outcome
 - ✓ Chronic HTN
 - ✓ MG
 - ✓ Comorbidities like DM (type 1 or 2), renal disease
 - ✓ Autoimmune disease (SLE, Antiphospholipid antibody syndrome)

- Should be initiated between 12 and 28wks of gestation (optimally before 16wks), and continuing till delivery
- Birth preparedness and complication readiness
 - Advise on the need to prepare place of delivery and a skilled attendant, money, transportation, social/family support...
 - Readiness for any complication during or after delivery, especially for high-risk pregnancies

First ANC contact

Objectives:

- Confirmation of pregnancy
- Gestational Age (GA) determination
- Detail Hx, P/E and Baseline IXs

Diagnosis of pregnancy

- Not all amenorrhea and/or protuberant abdomen is pregnancy/a baby. You need to widen your approach when evaluating a lady presenting with amenorrhea and/protuberant abdomen
- The manifestations of pregnancy can be classified into **three groups: -Presumptive, Probable and Positive**

Presumptive manifestations

- Amenorrhea => the abrupt cessation of menses in a healthy reproductive-aged female with predictable cycles is highly suggestive of pregnancy
- Nausea and vomiting (morning sickness)
 - Occur in 50% of pregnancies), starting as early as 2wks of gestation and usually resolves at 13th-16thwks of gestation
 - HEG- is a severe form of nausea and vomiting characterized by dehydration, weight loss (up to 5%) and ketonuria. (See chapter)
 - Advise the women to have frequent small meals, dry diet, and provide emotional support
- Breast changes
 - Mastodynia
 - Breast engorgement
 - Colostrum secretion=> can occur as early as 16thwks of G
 - Development of secondary breast tissue along the nipple line
- Elevated basal body temperature=>progesterone produces a 0.5°F increase in the basal body temperature, which persists after the missed menses.

- Quickening => initial fetal movement perception by the mother, occurs at around 16th-18thwks of gestation in multiparous and 18th-20thwks of gestation in primiparous women
- Skin changes
 - Chloasma/mask of pregnancy=> darkening of the forehead, bridge of the nose and cheeks
 - ✓ Occur after 16thwks of gestation and is linked to genetic predisposition
 - ✓ Exacerbated by sunlight
 - Linea nigra & nipple darkening=> MSH associated
 - Stria marks, on breast, abdomen
 - ✓ Appear late in pregnancy and is associated with collagen separation
 - Spider telangiectasia and palmar erythema=> due to elevated estrogen
- Bladder irritability, frequency, nocturia, UTI

Probable Manifestations

- Pelvic organ changes
 - **Chadwick's sign**=> bluish discoloration of the vagina and cervix
 - Due to congestion of the pelvic vasculature
 - **Hegar's sign**=> widening and softening of the body or isthmus of the uterus, occur @6th-8thwks of gestation
 - **Leukorrhea**
 - Relaxation of pelvic ligaments
- Abdominal enlargement
- Uterine contraction (Braxton Hick's contraction,)

Positive signs of pregnancy

- Fetal heart tone=>are detectable by handheld Doppler (after 10 weeks' gestation) or by fetoscope (after 18-20 weeks' gestation).
- **Uterine Size/Fetal Palpation**=>the fetus can be palpated through the maternal abdominal wall (after 22 weeks), and the position can be determined by Leopold's maneuvers.
- **Imaging Studies**
 - Sonography (ultrasound) is one of the most useful technical aids in diagnosing and monitoring pregnancy.
 - Cardiac activity is discernible at 5-6 weeks via transvaginal sonogram, limb buds at 7-8 weeks, and finger and limb movements at 9-10 weeks.
 - At the end of the embryonic period (10 weeks by LNMP), the embryo has a human appearance.

- The GA can be determined by the crown rump length (CRL) between 6- and 13-weeks' gestation, with a margin of error of approximately 8% or 3–5 days.
- *See more about function of obstetric ultrasound below.*

➤ **Pregnancy Tests**

- Sensitive, early pregnancy tests measure changes in the level of human chorionic gonadotropin (hCG)
- The β subunit of hCG is produced by the syncytiotrophoblast **8 days after fertilization** and may be detected in the maternal serum 8–11 days after conception or as early as **21–22 days(3wks) after the LNMP**.
- β -hCG levels peak at 10–12 weeks' gestation and decrease afterward
- Generally, serum and urine levels return to normal ($<5\text{mIU/mL}$), 21–24 days after delivery or after a fetal loss.
- **Home pregnancy test**— hCG is a qualitative test that is performed on the first voided morning urine sample. A positive test is usually indicated by a color change and it should always be repeated in the office.
- **Urine pregnancy test**— an antibody assay recognizing the β -hCG, is performed in the office to diagnose pregnancy.
 - The test is reliable, rapid (1–5 minutes), and inexpensive, with a positive test threshold between 5 and 50mIU/mL, characterized by a color change.
 - This is the most common method to confirm pregnancy.
- **Serum pregnancy test**— β -hCG can be detected within 7 days after conception or at a menstrual age of 21 days' gestation. The threshold for a positive test can be as low as 2–4mIU/mL
- Serial quantitative tests of β -hCG are used to evaluate threatened abortion, ectopic pregnancy, or a molar pregnancy

Gestational age determination

Early milestones to calculate GA

- **Reliable LNMP**, when do we say reliable and how do we use it? (See JUMC dept, of OBGYN Hx taking and P/E format).
 - NB. Recall bias is an important factor affecting this parameter.
- **Early ultrasound**, when do we say “early”? -up to 24wks of G(23⁺⁶wks)
 - Reliable LNMP and Early U/S are the most reliable way of determining GA. But which one do you prefer if there is discrepancy between the two?

- ✓ *Answer*; It depends on the days/weeks of the discrepancy; use U/S if discrepancy is **greater** than the margin of error for U/S in that specific GA group.
- ✓ *Explanation*; U/S is most reliable (*with 95% confidence intervals*) in determining the GA within a certain margin of error in different GA ranges (the time the U/S is done),

GA (wks.)	Margin of error of U/S(days)	U/S measurement method
<8 ^{+6/7}	5(3-5)	CRL
9 ^{0/7} -15 ^{6/7}	7	CRL
16 ^{0/7} -21 ^{6/7}	10	BPD, HC, AC, FL
22 ^{0/7} -27 ^{6/7}	14	BPD, HC, AC, FL
>28 ^{0/7}	21	BPD, HC, AC, FL

- So, if you find a discrepancy between GA from LNMP and U/S **greater than** the specified margin of error, use U/S. for example the GA from LNMP is 20wks, and from early U/S it is 18wks from CRL done at 12th wk, which one do u take?
- Fetal growth rate and factors affecting it are important determinant of U/S dating of pregnancy
- *NB. In GA assessment with U/S, it is important to consider multiple anthropometric measurements (CRL, BPD, HC, AC, and FL) and computing a composite GA is better than a single variable measurement.*
 - ✓ *The CRL becomes less accurate after the first trimester and should not be used after it reaches 84 mm (14 weeks of gestation).*
- **Urine HCG** (exact date of 1st pregnancy test done),
 - NB. This isn't used to calculate the exact GA; it just helps you *to say term or preterm*.
 - If 35wks have lapsed from the date of the 1st positive pregnancy test (making the GA at least 38wks), you can say the pregnancy is term.
- **Fetal heart beat**=> if u hear the fetal heart beat with fetoscope, the GA must be at least...?
- **Serial fundal height**=> is generally not recommended
 - Between **18 & 34-weeks'** gestation, the uterus size or fundal height measured in **centimeters** from the pubic symphysis to the upper edge of the uterine corpus, **correlates well** with the GA in weeks
- **Quickening**, (never seen it used, personally)

NB. Only Early U/s and Reliable LNMP is routinely used for GA determination in our setup,

Detail Hx, P/E and Baseline IXs

So, you have confirmed it is a pregnancy and determined the GA with the most accurate available method. Next step on the 1st ANC contact is to: -

- Take detail current & past obstetric and general medical hx,
 - which will help you to classify which type of care you are going to provide to the women, Let's see some important points to focus on the hx
 - **Age**=> age extremes are common risk factors for many obstetric complications
 - Young age (adolescent pregnancy)
 - Preeclampsia-eclampsia
 - IUGR, low birthweight (LBWt)
 - Maternal malnutrition
 - Labor abnormalities, like OL/CPD
 - Older age
 - Abortion
 - MG, GDM, Chromosomal abnormalities
 - APH(PA&PP), PPH, Uterine rupture
 - Malpresentation
 - PTL, PROM
 - **Mode of conception** (spontaneous Vs ART)
 - ✓ Conception using ART is a risk factor for MG, PTL/preterm delivery, perinatal mortality, congenital anomalies, LBWt.
 - Any exposure to teratogens, harmful substance
- **Danger signs**, the one thing your ANC Hx (any Hx of a pregnant lady) shouldn't miss is assessment of danger signs. Like
 - Vaginal bleeding,
 - HA, ABM, Blurring of vision
 - Leakage of liquor (may be confused with vaginal discharge syndrome, therefore do sterile speculum examination,
 - abdominal pain (lower as well as epigastric)
 - decreased fetal movement,
 - vertigo, tinnitus, palpitation
 - high grade fever

○ Past obstetrics Hx,

- Almost all pregnancy associated complications in a previous px are risk factors for subsequent pregnancy, with increasing percentage nonetheless.
- When do we say bad obstetric hx (BOH)?
 - Ans. When a woman has hx of
 - recurrent abortion (2 abortions for age <35 and 3 for age >35,)
 - stillbirth,
 - early neonatal death (within 1wk postpartum)
- For a woman with hx of previous infant with genetic disorder or/& congenital anomalies, cytogenetic testing is recommended prior to conception
- Rh-alloimmunization (maternal BG)
- Interpregnancy interval (interval >7yrs is considered too long and is a risk factor for similar complications associated with young age, and interval <18month is considered too soon)

▪ Do complete physical examination, with special consideration to;

- V/S (BP, HR & Temp),
- BMI
 - What is the recommended weight gain during pregnancy?
 - Ans. It's based on pre-pregnancy BMI

BMI (kg/m ²)	Recommended weight gain (kg)
<18.5(underweight)	12.5-18
18.5 – 24.9(normal)	11.5-16
25-29.9(overweight)	7-11
≥30(obese)	5-9

- Signs of anemia, pale conjunctiva, palmar pallor
- LGS (thyroid and breast abnormalities),
- CVS, gallop, signs of high output HF
- Abdomen (including obstetric palpation and FHB auscultation)
- GUS=> vulvar ulcer, discharge, speculum examination: evaluation of the vagina & cervix. Digital pelvic examination (when indicated); pelvic mass
- MSS, edema

➤ Do baseline investigations

- *When you mention an Ix, you should know what you are looking for in that Ix and an approach if u get a positive result, see below)*

- **Blood tests:**

- CBC/Hemoglobin, BG & Rh,
- PICT, VDRL, HBsAg
- Indirect Coomb 's test for RH negative women

- **Urine tests:**

- Urine strip test and microscopy (Albumin, Sugar, ketone, WBC etc.)
- Urine (midstream) gram stain or **culture** to diagnose asymptomatic bacteriuria (ASB).
 - ASB is defined as true bacteriuria in the absence of specific symptoms of acute UTI. It is common in pregnancy.

- **Ultrasound scan**

➤ Well, you've evaluated your client, it's time for intervention to accomplish your aims

- Provide vaccines, medications and nutrient supplementations
- Advice on nutrition, harmful substance and practice, hygiene, birth preparedness and family planning
- Advice on danger signs
- Appoint for subsequent visit and tell her to come at any time before the appointed date if she experiences any of the danger symptoms.

Subsequent ANC contacts

Content of subsequent antenatal care visits are: -

- Ask about general well-being, fetal movements, danger symptoms and any problems.
- Check BP, PR, weight, and color of the mucous membranes and conjunctivae.
- Measure the SFH and compare with previous measurements.
- Palpate the presenting part after 34 weeks of gestation.
- One step screening using 75 gm oral glucose tolerance test at 24 to 28wks, if the woman is high risk for GDM (obese, previous hx of GDM, previous hx of macrosomia)
- Repeat investigations
 - Repeat HIV counseling and testing in the 3rd TM preferably between 28 to 36 weeks for those tested negative at initial testing.

- Repeat blood tests: Hgb at 32 and 38 weeks
- Repeat urine test at 26 and 34 weeks.
- Repeat information for danger signs of pregnancy, and review delivery and transport plans as well as feeding and contraception choices.
- Document the detail contact schedule, risk identification and list of interventions at each contact.

Functions of Obstetric Ultrasound

First trimester U/S

- **Transvaginal ultrasound is almost always superior to transabdominal ultrasound in evaluation of very early pregnancy.**
 - Because the distance between the probe and pregnancy structures is often just a few centimeters, attenuation of sound waves is minimal, and high-frequency probes may be used
- **Main purposes of 1st TM U/S are: -**
 - Diagnosis of pregnancy (including intrauterine, ectopic or molar pregnancy)
 - Determination of fetal viability and DX of failed pregnancy
 - Criteria for deciding that a pregnancy of uncertain viability is in fact a failed pregnancy should be set to virtually eliminate the possibility of a false positive diagnosis. These include: -
 - ✓ the presence of a fetus with a CRL of more than 7 mm and no heartbeat,
 - ✓ the absence of an embryo when the mean sac diameter is >25 mm,
 - ✓ the absence of an embryo with a heartbeat more than 2 weeks after a scan showed a gestational sac without a yolk sac, and
 - ✓ the absence of an embryo with a heartbeat more than 11 days after a gestational sac with a yolk sac was seen
 - If there are borderline findings and uterine evacuation is being considered, it is prudent to repeat the ultrasound in 7 to 10 days to be absolutely certain that a viable pregnancy is not interrupted.
 - GA determination
 - Identification of chromosomal abnormalities
 - First-trimester ultrasound findings predictive of a chromosome abnormality include:
 - ✓ *A thick nuchal translucency,*
 - ✓ *absent nasal bone,*

- ✓ *abnormally fast or slow fetal heart rate, and*
- ✓ *some structural malformations*

Second and Third Trimester U/S

Components/functions of 2nd and 3rd TM U/S

- Fetal cardiac activity (present or absent, normal or abnormal) and fetal wellbeing assessment (BPP, in 3rd TM)
- Fetal growth assessment, estimation of fetal weight (Dx of IUGR)
 - **Repeated ultrasound exams for growth should be performed at least 2 to 4 weeks apart**
- Number of fetuses (Dx of MG) (if multiples, document chorionicity, amnionicity, comparison of fetal sizes, estimation of amniotic fluid normality in each sac, and fetal genitalia)
- Qualitative or semiquantitative estimate of amniotic fluid volume (Dx of oligohydramnios or polyhydramnios)
- Placental location, especially its relationship to the internal Os, and placental cord insertion site (Dx of placenta previa and vasa previa)
 - The diagnosis of vasa previa is critical because the recognition of this finding at the time of a screening ultrasound greatly affects the chance of fetal survival; the fetal mortality rate is high when vasa previa is not diagnosed before labor.
- Evaluation of the uterus that includes fibroids (Dx of tumor previa), adnexal structures, and the cervix (short cervix, as measured with transvaginal ultrasound, is associated with an increased risk of preterm birth)
- Anatomic survey to include:
 - **Head and neck**
 - Dx of NTD and associated cranial abnormalities
 - Dx of cleft lip and cleft palate
 - **approximately two thirds of those with cleft lip also have cleft palate,**
 - Nuchal skin fold may be helpful for aneuploidy risk
 - **Chest**
 - Four-chamber view of the heart & Outflow tracts (if possible)
 - **Abdomen**
 - Stomach (presence, size, and situs, dx of duodenal atresia)
 - Kidneys & Bladder
 - Umbilical cord insertion into the abdomen
 - Number of umbilical cord vessels

- **Spine**
- **Extremities** (presence or absence of legs and arms)
- Gender determination
- Presentation

Anemia in pregnancy

Most clinicians diagnose anemia when the hemoglobin concentration is less than 11 g/dL or the hematocrit is less than 32%.

- Using these criteria, 50% of pregnant women are anemic.

BOX 44-3 CAUSES OF ANEMIA DURING PREGNANCY

Common Causes: 85% of Anemia

Physiologic anemia

Iron deficiency

Uncommon Causes

Folic acid deficiency

Vitamin B₁₂ deficiency

Hemoglobinopathies

- Sickle cell disease

- Hemoglobin SC

- β -Thalassemia minor

Bariatric surgery

Gastrointestinal bleeding

Rare Causes

Hemoglobinopathies

- β -Thalassemia major

- α -Thalassemia

Syndromes of chronic hemolysis

- Hereditary spherocytosis

- Paroxysmal nocturnal hemoglobinuria

Hematologic malignancy

Physiologic anemia of pregnancy

- Occurs because of disproportionate expansion of the plasma volume and the RBC mass
- During a singleton pregnancy, maternal plasma volume gradually expands by approximately 50% (1000 mL). The total RBC mass also increases but only by approximately 300 mg (25%), and this starts later in pregnancy
- By **6th weeks postpartum**, in the absence of excessive blood loss during the puerperium, hemoglobin and hematocrit levels have returned to normal if the mother has adequate iron stores

Iron deficiency anemia (IDA)

- Approximately 75% of anemia that occurs during pregnancy is secondary to iron deficiency
- There is ~2.3g of total body iron in women (~65% of this is in RBCs) An additional 1g(6-7mg/day) iron is needed to support the RBC mass increment(450mg), fetus & placenta(360mg) and the loss during delivery(190mg)
- Predisposing factors for IDA
 - Poor dietary intake
 - Short interpregnancy interval (<2yrs)
 - Delivery complicated by hemorrhage
 - Teenage pregnancy
 - Obesity
- Development of IDA follows three stages,
 - Iron depletion=>Iron deficient erythropoiesis=>IDA (defined by the identification of microcytic hypochromic RBCs)
 - The first pathologic change to occur in iron deficiency anemia is the depletion of bone marrow, liver, and spleen iron stores. This may take a few weeks to a few months depending on the level of the woman's iron stores
 - Over a period of a few weeks, the serum iron level falls, as does the percentage saturation of transferrin. The total iron-binding capacity (TIBC) rises simultaneously with the fall of iron, because this is a reflection of unbound transferrin.
 - A falling hemoglobin and hematocrit follow within 2 weeks. Microcytic hypochromic RBCs are released into the circulation.
 - NB. If iron deficiency is combined with folate/Vit B12 deficiency, normocytic (normal MCV) normochromic RBCs are observed in peripheral morphology
- Lab parameters in IDA
 - Low MCV, MCH, MCHC, and increased RDW on CBC
 - Serum iron <60µg/dl, Transferrin saturation <16%, TIBC >400µg/dl and serum ferritin <15µg/l on iron studies
 - a single normal serum iron concentration does not rule out iron deficiency
 - the **serum ferritin level** indicates the total status of iron store and the best parameter to judge degree of iron deficiency
 - Bone marrow aspiration isn't necessary to make dx
 - It is reserved for persistent anemia with confusing hematologic parameters, and can be safely performed in pregnancy.

▪ Management

- A patient who has a very low hemoglobin as a result of iron deficiency immediately postpartum should return to normal by her 6-week postpartum visit
- Iron supplementation
 - One 325mg ferrous sulfate tab daily provides adequate prophylaxis,
 - In iron deficient lady, one-tab 3x/d has to be taken
 - What are the elemental iron contents of different oral preparations of iron supplements and what %of supplemental iron is absorbed?
 - NB. Iron needs acidic environment for absorption, thus antacids and chronic use of PPIs & H₂blockers diminish its absorption, while vit-c (from citric fruits like lemon and orange) increases absorption.
 - Iron should be taken 30minutes before meal, but this may lead to dyspepsia & nausea, so u may suggest taking with food.
 - For those patients who are noncompliant or unable to take oral iron and are severely anemic (**not anemic enough to require transfusion**) IM/IV iron can be given.
 - Iron dextran can result in **anaphylaxis** and overdose will lead to **hemosiderosis**
- Response to iron therapy
 - Improvement in general wellbeing, malaise, and gain of appetite are initial responses
 - Reticulocytosis will occur within 3-4 days of initiating therapy, and the
 - Hemoglobin/Hct will start to increase in subsequent 2weeks
 - It takes more than 3months to completely replete iron stores

Effects of anemia on pregnancy

- Anemia is associated with: -
 - Preterm delivery & LBWt
 - Fetal loss/spontaneous abortion
 - PPH
 - Postpartum depression (ACOG)

HIV in pregnancy

This topic is well written in the Obstetric Management Protocol for Hospitals by MOH, 2021.

While reading focus on

- Factors that increase risk of transmission
- The four prongs to PMTCT and what to do at each prong
- ART prophylactic regimen for the newborn and ways of Dx in the infant

Syphilis in pregnancy

Syphilis is a chronic systemic infectious process secondary to infection with the spirochete *Treponema pallidum*

Stages/phases of infection

Primary syphilis

- Is characterized by **the Chancre**
 - The chancre arises at the point of entry by the spirochetes and is a broad-based, typically non-tender ulcerated lesion that gives a characteristic “woody” or “rubbery” feel on palpation.
 - It is seldom secondarily infected, and it resolves without medical treatment in 3 to 6 weeks.

Secondary syphilis

- is characterized by
 - a generalized **maculopapular eruption**,
 - constitutional symptoms,
 - major organ involvement (CNS involvement in 40% of cases), and
 - lymphadenopathy
- resolves over 2 to 6 weeks as the patient enters the latent phase of the disease

Latent phase

- Is divided into *early latent* (<1 year) and *late latent* (>1 year) disease, during which **no symptoms or signs** of clinical disease are noted. If left untreated, progression to tertiary disease
- In immunocompromised women the progression to tertiary disease, especially of the CNS, is known to occur early after the onset of disease.

Tertiary syphilis

- Tertiary disease mainly involves the CVS, CNS (neurosyphilis), and MSS
- The organism has an affinity for the arterioles, and the inflammatory response that follows results in obliterative endarteritis & subsequent end-organ damage.

Diagnosis

It is difficult to dx syphilis because; -

- the clinical manifestations to mimic a variety of other diseases
- the microorganism can't be cultured on an artificial medium

Diagnostic modalities

- Serologic tests
 - Two types of serologic tests are currently available, nontreponemal & treponemal tests.
 - The *nontreponemal tests* include the rapid plasma reagin (RPR) and the Venereal Disease Research Laboratories (VDRL) tests.
 - False-positive reactions occur with nontreponemal tests secondary to viral infections or autoimmune disease
 - If a nontreponemal test is positive, a *treponemal test* is subsequently **performed to confirm the diagnosis** of syphilis.
 - The treponemal tests include the fluorescent treponemal antibody absorbed (FTA-ABS) tests, the *T. pallidum* passive particle agglutination assay (TP-PA), as well as other EIAs and chemiluminescence immunoassays.
 - Once appropriately treated, most individuals lose reactivity to these nontreponemal serologic tests.
 - For most individuals, the treponemal tests remain positive lifelong
- Darkfield microscopy=> examination of the serous exudates from the chancre can **definitively diagnose** the condition by identification of the flexuous-bodied (spiral) organism
 - darkfield microscopy lacks sensitivity, owing in part to specimen collection technique
- PCR has increased the sensitivity for detection in samples obtained from genital lesions, cerebrospinal fluid (CSF), amniotic fluid, serum, and paraffin-embedded tissue

Screening syphilis infection in pregnancy

TABLE 52-4 SCREENING SYPHILIS INFECTIONS IN PREGNANCY

When to screen	• All women should be screened at their first prenatal visit. • Repeat screening should be performed in all pregnancies early in the third trimester. • Patients should be screened at delivery if not screened previously or if at high risk.
How to screen*	• Treponemal and nontreponemal test
Diagnostic criteria	• Positive treponemal <i>and</i> nontreponemal test

Congenital syphilis

- Congenital syphilis infection results from trans-placental migration of the organism to the fetus.
- It can occur at any stage of maternal infection and at any gestational age.
 - It is now believed that ***trans-placental infection must be close to 100% during the early stages of maternal disease*** because of the known hematogenous spread, with rates of transmission falling to 10% as bacteremia abates with the subsequent mounting of the maternal immunologic response during late latent disease
- The risk for adverse pregnancy outcome in pregnant women with syphilis that was untreated is ***approximately 52%***. The specific risks include: -
 - Higher risk for miscarriage or stillbirth (21%)
 - Neonatal death (9.3%)
 - Premature birth or low birthweight (5.8%)
 - Clinical evidence of congenital infection (15%)
- markers for congenital syphilis
 - nonimmune fetal hydrops
 - intrauterine fetal demise
 - hepatomegaly
 - polyhydramnios
 - placentomegaly

Treatment

- Penicillin has been effective in the treatment of disease and in the prevention of disease progression in both nonpregnant and pregnant women, and it has also been used in the prevention and treatment of congenital syphilis

- For penicillin allergic patient alternative treatment could be considered, **except** for pregnant ladies with syphilis, those with congenital syphilis or neurosyphilis.
 - In the later cases penicillin is the only effective treatment, and if penicillin allergy is reported, do skin testing with both major and minor determinants.
 - If positive, patient should undergo **penicillin desensitization** with subsequent administration of penicillin therapy
- **Almost 90% of stillbirths** attributed to congenital syphilis occur among women who are **never treated or who are treated inappropriately**.
- Factors associated with treatment failure
 - preterm delivery (<36 weeks)
 - a short interval between treatment and delivery
 - a higher VDRL titer at the time of treatment and delivery
 - early stage of maternal disease

BOX 52-3 RECOMMENDED TREATMENT REGIMENS* FOR *TREPONEMA PALLIDUM* INFECTIONS

Primary, Secondary, Early Latent Disease

Benzathine penicillin G, 2.4 million U IM as a single dose

Late Latent and Latent Disease of Unknown Duration

Benzathine penicillin G, 2.4 miU IM weekly for three doses (7.2 miU total)

Tertiary Disease

NEUROSYPHILIS

Aqueous crystalline penicillin G, 18 to 24 miU/day IV (3 to 4 miU every 4 hours or continuous infusion) for 10 to 14 days *or*

Procaine penicillin G, 2.4 miU IM once daily with probenecid 500 mg PO four times a day for 10 to 14 days if compliance can be ensured

WITHOUT NEUROSYPHILIS

Benzathine penicillin G, 2.4 miU IM weekly for 3 weeks (7.2 miU total)

Penicillin Allergy (Documented)

IN PREGNANCY

Desensitization and penicillin therapy as above

Modified from Workowski KA, Bolan GA; Centers for Disease Control and Prevention (CDC): Sexually transmitted diseases treatment guidelines, 2015. *MMWR Morb Mortal Wkly Rep*. 2015;64(RR-03):1-137.

*The CDC recommends penicillin as the treatment of choice in individuals with syphilis.

IM, intramuscularly; IV, intravenously; miU, million units; PO, per os.

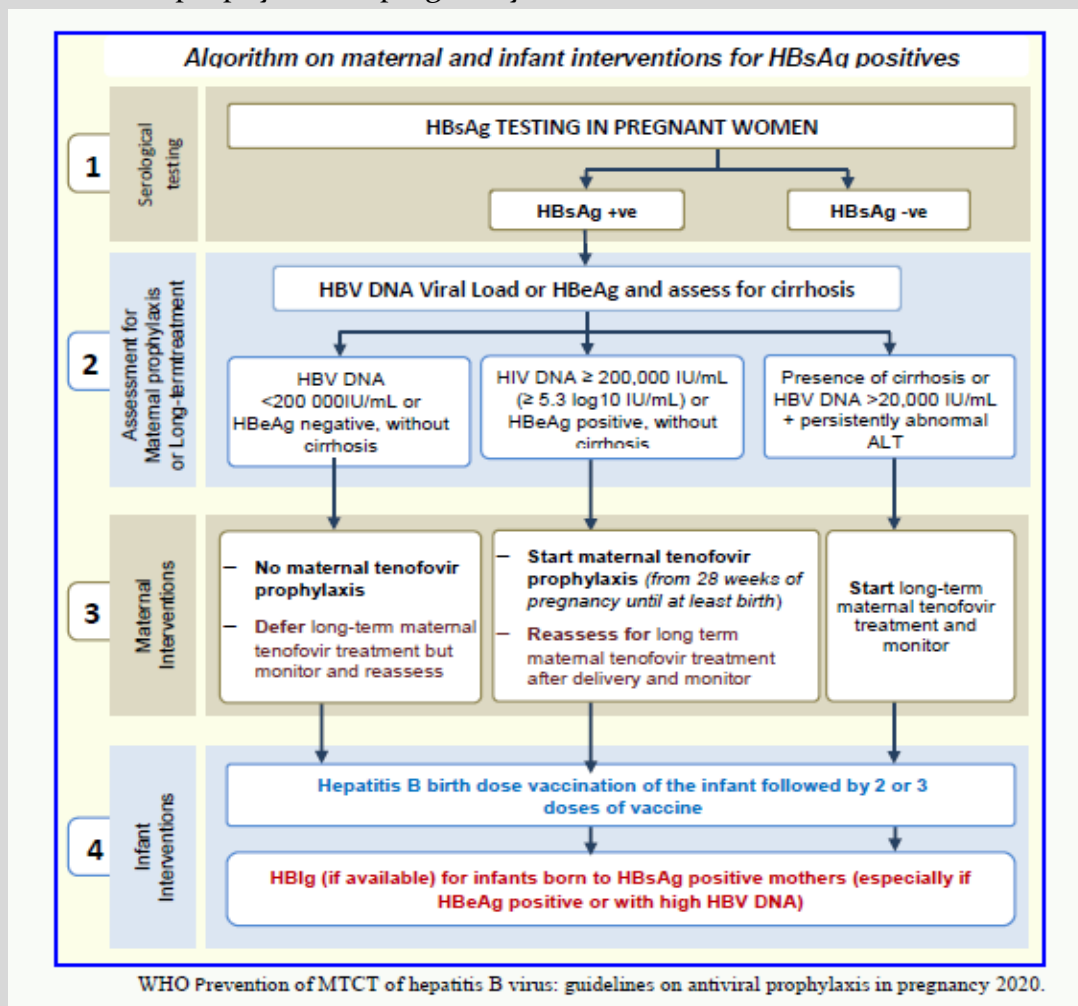
Viral hepatitis in pregnancy

Why do we need to dx and treat HBV infection during pregnancy?

- Because the probability of chronic infection is very high in newborns (85-95%), with a 25-30% life time risk of serious or fatal liver disease.

How do you approach if you find HBsAg positive?

- You need to take measures to minimize the risk of transmission and disease propagation in the newborn. This includes: -
 - Prophylactic antiviral to the mother, depending on the infectivity level (determined by viral load and presence of the envelop antigen (HBeAg))
 - Provision of Hep B birth dose vaccine
 - Giving IgG (HBIG) at birth
- See the algorithm from WHO Prevention of MTCT of hepatitis B virus: guidelines on antiviral prophylaxis in pregnancy 2020, below



UTI in pregnancy

- Acute Urethritis
- Asymptomatic Bacteriuria and Acute Cystitis
- Acute Pyelonephritis

Acute urethritis/acute urethral syndrome

Etiologies

- Coliforms (principally *Escherichia coli*),
- *Neisseria gonorrhoeae*, and
- *Chlamydia trachomatis*.
 - ✓ Coliform organisms are part of the normal vaginal and perineal flora and may be introduced into the urethra during intercourse or when wiping after defecation.
 - ✓ *N. gonorrhoeae* and *C. trachomatis* are sexually transmitted pathogens.

Diagnosis

- Hx => frequency, urgency, & dysuria are usually present. They may also have hesitancy, dribbling, and a mucopurulent urethral discharge
- Microscopic examination=> the urine usually has WBCs, but bacteria may not be consistently present.
- Urine cultures may have low colony counts of coliform organisms, and cultures of the urethral discharge may be positive for gonorrhea and chlamydia.
- A rapid diagnostic test, such as a NAAT, is now the preferred method for identification of gonorrhea and chlamydia.

Treatment

- Most patients with acute urethritis warrant empiric treatment before the results of laboratory tests are available.
- Infections caused by coliforms usually respond to the antibiotics described later for treatment of asymptomatic bacteriuria and cystitis.
- If gonococcal infection is suspected give IM ceftriaxone (250 mg in a single dose) plus 1000 mg oral azithromycin.
 - ✓ If the patient is allergic to β -lactam antibiotics, an effective alternative is azithromycin 2000 mg orally in a single dose. This high dose of azithromycin is more likely to be associated with gastrointestinal side effects.
 - ✓ An alternative choice in the penicillin-allergic patient is ciprofloxacin 500 mg orally in a single dose.

- If chlamydial infection is suspected or confirmed, the patient should be treated with azithromycin 1000 mg in a single dose.

Asymptomatic Bacteriuria and Acute Cystitis

The prevalence of asymptomatic bacteriuria in pregnancy is **5% to 10%** & the frequency of acute cystitis in pregnancy is **1% to 3%**.

Some cases of cystitis arise de novo, whereas others develop as a result of failure to identify and treat asymptomatic bacteriuria.

Etiology

- *E. coli* is responsible for at least 80% of cases of initial infections and about 70% of recurrent cases.
- *Klebsiella pneumoniae* and *Proteus* species also are important pathogens, particularly in patients who have a history of recurrent infection.
- Up to 10% of infections are caused by gram-positive organisms such as GBS, enterococci, and staphylococci.

All pregnant women should have a urine culture at their first prenatal appointment to detect preexisting asymptomatic bacteriuria.

- If the culture is negative, the likelihood of the patient subsequently developing an asymptomatic infection is less than 5%.
- If the culture is **positive** (defined as **>10⁵ colonies/mL urine** from a midstream, clean catch specimen), prompt treatment is necessary to prevent ascending infection.
- In the **absence of effective treatment**, about **one third** of pregnant women with asymptomatic bacteriuria will **develop acute pyelonephritis**.

Diagnosis

Hx

- Patients with acute cystitis usually have symptoms of frequency, dysuria, urgency, suprapubic pain, hesitancy, and dribbling.
- Gross hematuria may be present,
- High fever and systemic symptoms are uncommon.

Lab

- Urine analysis positive leukocyte esterase and nitrate tests
- Urine culture from a catheterized sample is preferred because it minimizes the probability that urine will be contaminated by vaginal flora.

- ✓ With a catheterized specimen, a colony count greater than $10^2/\text{mL}$ is considered indicative of infection.

Treatment

Asymptomatic bacteriuria and acute cystitis characteristically respond well to short courses of oral antibiotics.

Single dose therapy is not as effective in pregnant women as in nonpregnant patients. However, a 3days course of treatment appears to be comparable to a 7 to 10days regimen for an initial infection.

Longer courses of therapy are more appropriate for patients with recurrent infections.

TABLE 54-4 ANTIBIOTICS FOR TREATMENT OF ASYMPTOMATIC BACTERIURIA AND ACUTE CYSTITIS

DRUG (TRADE NAME)	STRENGTH OF ACTIVITY	ORAL DOSE	COST
Amoxicillin	Some <i>Escherichia coli</i> , most <i>Proteus</i> species, GBS, enterococci, some staphylococci	500 mg TID or 875 mg BID	Low
Amoxicillin-clavulanic acid (Augmentin)	Most gram-negative aerobic bacilli and gram-positive cocci	875 mg BID	High
Ampicillin	Some <i>E. coli</i> , most <i>Proteus</i> species, GBS, enterococci, some staphylococci	250-500 mg 4 times daily	Low
Cephalexin (Keflex)	Most <i>E. coli</i> , most <i>Klebsiella</i> and <i>Proteus</i> species, GBS, staphylococci	250-500 mg 4 times daily	Low
Nitrofurantoin monohydrate macrocrystals—sustained-release preparation (Macrobid)	Most uropathogens except enterococci and <i>Proteus</i> species	100 mg BID	Moderate
Double-strength trimethoprim-sulfamethoxazole (Bactrim DS, or Septra DS)	Most uropathogens except some strains of <i>E. coli</i>	800 mg/160 mg BID	Low

BID, twice a day; DS, double strength; GBS, group B streptococci; TID, three times a day.

In recent years, **20% to 30%** of strains of *E. coli* and **more than half the strains of *Klebsiella*** have **developed resistance to ampicillin**.

When choosing among the drugs listed in the table above you should consider several factors.

- **First**, the sensitivity patterns of ampicillin, amoxicillin, and cephalexin are the most variable.
- **Second**, these drugs—along with amoxicillin-clavulanic acid—also have the most pronounced effect on normal bowel and vaginal flora and thus are the most likely to cause diarrhea or monilial vulvovaginitis.

- In contrast, **nitrofurantoin** monohydrate has only **minimal effect on vaginal and bowel flora**. Moreover, it is more uniformly effective against the common uropathogens, except for *Proteus* species, than trimethoprim-sulfamethoxazole (TMP-SMX).
- **Third**, amoxicillin-clavulanic acid and TMP-SMX usually are the best empiric agents for treatment of patients with **suspected drug resistant pathogens**.

However, **sulfonamides** should be avoided in the first trimester of pregnancy because of possible **teratogenicity**, and they should be avoided immediately prior to deliver because of concern about displacement of bilirubin from protein binding sites, with resultant **neonatal jaundice**

Follow up

- Cultures during or immediately after treatment are indicated for patients who have a poor response to therapy or who have a history of recurrent infection.
- During subsequent clinic appointments, the patient's urine should be screened for nitrites and leukocyte esterase. If either of these tests is positive, repeat urine culture and retreatment are indicated.

Acute Pyelonephritis

- **The incidence of pyelonephritis in pregnancy is 1% to 2%.**
- Most cases develop as a consequence of **undiagnosed or inadequately treated** lower urinary tract infection. What predispose pregnant ladies to ascending infection of the urinary tract?
 - ✓ First, the high concentration of progesterone secreted by the placenta has an inhibitory effect on ureteral peristalsis.
 - ✓ Second, the enlarging gravid uterus often compresses the ureters, **particularly the right**, at the pelvic brim, thus creating additional stasis. Stasis, in turn, facilitates migration of bacteria from the bladder into the ureters and renal parenchyma
- About **75% to 80% of cases of pyelonephritis occur on the right side**, 10% to 15% are left sided, and slightly smaller percentages are bilateral.

Etiologies

- **E. coli** is again the principal pathogen.
- *K. pneumoniae* and *Proteus* species also are important causes of infection, particularly in women with **recurrent** episodes of pyelonephritis.
- Highly virulent gram-negative bacilli such as *Pseudomonas*, *Enterobacter*, and *Serratia* are unusual isolates except in **immunocompromised patients**.
- Gram-positive cocci do not frequently cause upper tract infection. Anaerobes also are unlikely pathogens unless the patient is **chronically obstructed or instrumented**.

Diagnosis

- Hx=> most patients will have fever, chills, flank pain and tenderness, urinary frequency or urgency, hematuria, and dysuria.
 - Patients also may have signs of preterm labor, septic shock, and acute respiratory distress syndrome (ARDS).
- Urinalysis=> is usually positive for white blood cell casts, red blood cells, and bacteria.
- Urine culture=> Urine colony counts greater than 10^2 colonies/mL in samples collected by catheterization confirms the diagnosis of infection.

Treatment

Outpatient treatment

- If their **disease manifestations are mild**, they are **hemodynamically stable**, and they have **no evidence of preterm labor**.
- The patient should be treated with agents that have a high level of activity against the common uropathogens.
 - Acceptable oral agents include Augmentin 875 mg twice daily or double-strength TMP-SMX (Cotrimoxazole) twice daily for 7 to 10 days.
 - Although an excellent drug for lower tract infections, nitrofurantoin monohydrate does not consistently achieve the serum and renal parenchymal concentrations necessary for successful treatment of more serious infections.

Inpatient treatment

- If patient appear to be moderately to severely ill or who show any signs of preterm labor should be hospitalized

- **Start IV antibiotic therapy and supportive treatment**
 - They should be monitored closely for complications such as sepsis, ARDS, and preterm labor.
 - One reasonable choice for empiric IV antibiotic therapy is ceftriaxone 2 g every 24 hours.
 - ✓ If the patient is critically ill or is at high risk for a resistant organism, a second antibiotic, such as gentamicin (7 mg/kg/ideal body weight every 24 hours) or aztreonam (500 mg to 1 g every 8 to 12 hours) should be administered, along with ceftriaxone, until the results of susceptibility tests are available.
 - Once antibiotic therapy is initiated, about 75% of patients' defervesce within 48 hours. By the end of 72 hours, almost 95% of patients are afebrile and asymptomatic.
- The two most likely causes of treatment failure are a resistant microorganism and obstruction
 - obstruction is best diagnosed with CT scan or renal U/S
- Once the patient has begun to defervesce and her clinical condition has improved, she may be discharged from the hospital. Oral antibiotics should be prescribed to complete a total of 7 to 10 days of therapy.

Follow up

- A repeat urine culture should be obtained after therapy is completed to ensure that the infection has been adequately treated.
- About **20% to 30%** of pregnant patients with acute pyelonephritis develop a **recurrent urinary tract infection** later in pregnancy.
 - The most cost effective way to reduce the frequency of recurrence is to administer a daily prophylactic dose of an antibiotic such as nitrofurantoin monohydrate 100 mg.
 - Patients receiving prophylaxis should have their urine screened for bacteria at each subsequent clinic appointment. They also should be questioned about recurrence of symptoms.
 - If symptoms recur, or the dipstick test for nitrite or leukocyte esterase is positive, a urine culture should be obtained to determine whether retreatment is necessary.

Rh alloimmunization

- *See chapter 9*

Chapter 2

Antenatal Assessment of Fetal Well Being

Prepared by Dr. Amanuel Tefera (MI)

Introduction

The goal of evidence-based antepartum fetal evaluation is to decrease:

- perinatal mortality
- permanent neurologic injury through judicious use of reliable and valid methods of fetal assessment

In animals and humans, FHR pattern, level of activity, and degree of muscular tone are ***sensitive to hypoxemia and acidemia***.

Redistribution of fetal blood flow in response to hypoxemia may result in diminished renal perfusion and oligohydramnios.

Surveillance techniques such as cardiotocography, real-time ultrasonography, and maternal perception of fetal movement can ***identify*** the fetus that may be undergoing some degree of ***uteroplacental compromise***.

- Identification of suspected fetal compromise provides the opportunity to intervene before progressive metabolic acidosis results in fetal death.

Causes of fetal death (Stillbirth)

Antepartum fetal death is attributed to: -

- **Asphyxia** (IUGR, prolonged gestation) => in about 30%
- **Maternal complications** (placental abruption, hypertension, preeclampsia, and diabetes mellitus) => in about 30%
- Congenital malformations and chromosome abnormalities => in about 15%
- Infections => 5%
- No obvious fetal, placental, maternal, or obstetric etiology => in about 20%
 - this percentage increases with advancing gestational age

Fetal States

To be able to diagnose suspected fetal compromise using tests of fetal biophysical state, blood flow, and heart rate, we must be able to appreciate how these parameters appear under normal conditions and in response to suboptimal conditions.

Four normal fetal states have been identified.

I. State 1F – Quite sleep state

- ✓ Fetus spends 25% of its time
- ✓ Narrow oscillation of FHR
- ✓ Slow FHR, reduced variability
- ✓ Can last 20Min.

III. State 3F

- ✓ Continues eye movement in the absence of FM
- ✓ No acceleration of FHR with movement
- ✓ *Existence is questioned*

II. State 2F- Active sleep state

- ✓ Fetus spends 60-70% of its time
- ✓ Frequent gross movement.
- ✓ wider oscillation of FHR,
- ✓ Continuous eye movement,
- ✓ Increase variability, and Increase acceleration with FM
- ✓ Can last 40min

IV. State IV- awake state

- ✓ Vigorous body movement
- ✓ Continues eye movement
- ✓ FHR acceleration and inc variability

- State 1F and 2F can be affected by activity, drugs & nutritional status
- Near term, periods of quiet sleep may last 20 minutes, and those of active sleep, about 40 minutes.

The mechanisms that control these periods of rest and activity in the fetus are not well established.

External factors such as the mother's activity, her ingestion of drugs, and her nutrition may play a role. Specific factors that may decrease fetal movement in the third trimester include:

- fetal anomalies, particularly central nervous system (CNS) anomalies;
- maternal exposures, including corticosteroids, sedatives, smoking, and anxiety;
- low amniotic fluid volume; and
- decreased placental blood flow due to placental insufficiency

When evaluating fetal condition using the NST or the biophysical profile (BPP), the clinician must consider whether a fetus who is not making breathing movements or showing accelerations of its baseline heart rate is in a ***quiet sleep state or is neurologically compromised***.

- In such circumstances, prolonging the period of evaluation usually allows a change in fetal state, and more normal parameters of fetal well-being will appear
 - VAS (Vibro-Acoustic Stimulation)

Fetal movement is a more ***indirect*** indicator of fetal oxygen status and CNS function, and decreased fetal movement is noted in response to hypoxemia.

- However, gestational development of fetal movement must be considered when evaluating fetal well-being as marked by fetal activity.
 - Periods of absent fetal movement become more prolonged as gestation advances,
 - Longer than 40 minutes of fetal inactivity at 40 weeks may be a normal finding, compared with less than 10 minutes at 20 weeks and less than 20 minutes at 32 weeks.

Indications for testing

Pregnancy-related conditions	Maternal conditions
▪ GH / PE ▪ Decreased fetal movement ▪ Gestational diabetes mellitus (poorly controlled or medically treated) ▪ Oligohydramnios ▪ IUGR ▪ Late term or postterm pregnancy ▪ Isoimmunization ▪ Previous fetal demise (unexplained or recurrent risk) ▪ Monochorionic MG (with significant growth discrepancy)	▪ Pregestational DM ▪ Hypertension ▪ SLE ▪ Chronic renal disease ▪ Antiphospholipid syndrome ▪ Hyperthyroidism (poorly controlled) ▪ Hemoglobinopathies (sickle cell, sickle cell– hemoglobin C, or sickle cell–thalassemia disease) ▪ Cyanotic heart disease

Recommended frequency of testing

If the maternal medical condition is stable and test results are reassuring, tests of fetal well-being (NST, BPP, modified BPP, or CST) are typically repeated at ***weekly intervals***;

- However, in the presence of certain ***high-risk conditions***, some investigators have performed ***more frequent testing***, although the optimal regimen has not been established.

Modalities of antepartum testing

- Fetal movement count
- Non stress testing
- Contraction stress testing
- Biophysical profile
- Modified biophysical profile
- Doppler Velocimetry

Fetal movement count

During the third trimester, the human fetus spends 10% of its time making gross fetal body movements and that 30 such movements are made each hour.

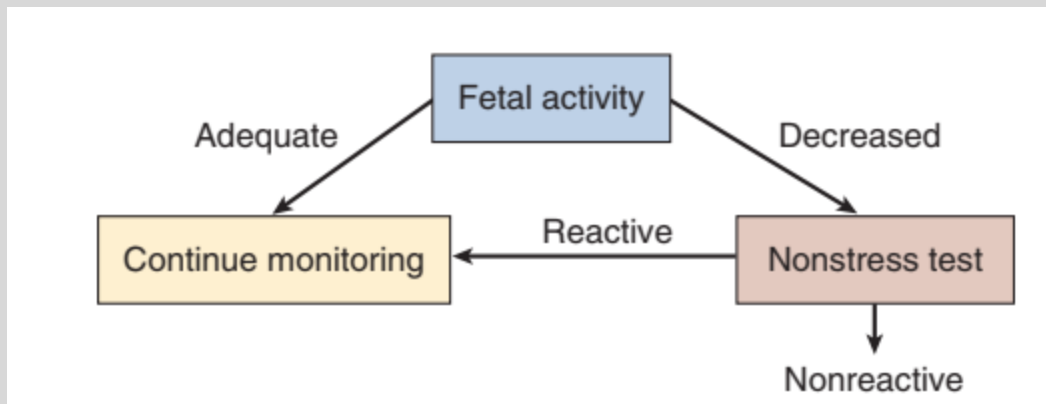
- Periods of active fetal body movement last about 40 minutes, whereas quiet periods last about 20 minutes.
- Fetal movement appears to peak between 9:00 PM and 1:00 AM, a time when maternal glucose levels are falling.
- The longest period without fetal movements in a normal fetus was about 75 minutes.
- The mother is able to perceive about 70% to 80% of gross fetal movements.

Maternal factors that may influence the evaluation of fetal movement include: maternal activity, parity, obesity, medications, and psychological factors, including anxiety.

The decrease in fetal movement with hypoxemia makes maternal assessment of fetal activity a potentially simple and widely applicable method of monitoring fetal well-being.

Methods of FM counting

- | | |
|---|---|
| <ul style="list-style-type: none">▪ The Cardiff “count-to-10” methods<ul style="list-style-type: none">○ mother count FM once/day (6am to 6pm)○ <10 movement over 12 hours is alarming | <ul style="list-style-type: none">▪ Sadovsky method<ul style="list-style-type: none">○ mother count FM 2-3x daily○ <3 movement/one hour is alarming▪ Rayburn method<ul style="list-style-type: none">○ count once per day for 60 minutes○ <3 movement/1hr for two consecutive days is alarming |
|---|---|
- The Cardiff “count-to-10” method has received wide scrutiny and is used most frequently in clinical practice.



Non stress Test

- The NST is the most widely applied technique for antepartum fetal evaluation
- At term, fetal accelerations are associated with fetal movement more than 85% of the time, and more than 90% of gross movements are accompanied by accelerations.
 - FHR Acceleration is increment in FHR by 15bpm from the baseline and lasts for 15seconds
- The NST is based on the premise that the heart rate of a fetus that is not acidotic or neurologically depressed will temporarily accelerate with fetal movement.
 - Heart rate reactivity is thought to be a good indicator of normal fetal autonomic function.
- Fetal heart rate accelerations may be absent during periods of quiet fetal sleep.
 - Maximum period for absence of FHR acceleration still be normal is ~80min.

The NST is usually performed in an outpatient setting.

- In most cases, only 10 to 15 minutes are required to complete the test.
- The patient may be seated in a reclining chair, with care taken to ensure that she is tilted to the left to avoid the supine hypotensive syndrome.

- The patient's blood pressure should be recorded before the test is begun and then repeated at 5- to 10-minute intervals.
- The fetal heart rate is monitored using the Doppler ultrasound transducer, and the tocodynamometer is applied to detect uterine contractions or fetal movement.

Finding of NST

- **Reactive NST- 85% of cases**
 - At least two accelerations of FHR by **15bpm** above the baseline which lasts for **15sec in 20min** of observation in those above 32wks of gestation.
 - At least two accelerations of FHR by **10bpm** above the baseline which lasts for **10sec in 20mins** of observation in those less than 32 weeks of gestation
- **Non-reactive NST- 15% of cases**
 - If less than two accelerations in 20 minutes
 - **Factors causing Non reactivity**
 - ✓ Quiet fetal sleep state,
 - ✓ CNS depressants such as narcotics and phenobarbital, as well as the
 - ✓ β -blockers like, propranolol
 - ✓ Chronic smoking
 - What to do in case of non-reactivity?
 - ✓ Prolong duration of test to 40min
 - ✓ Do Vibroacoustic stimulation (VAS)
 - Use of sound wave to stimulate FM
 - It shortens observation time
 - Interpretation is the same
 - ✓ Do CST or BPP

Contraction stress tests (CST)

- The contraction stress test (CST), also known as the oxytocin challenge test (OCT), was the first biophysical technique widely applied for antepartum fetal surveillance.
- The CST is based on the response of the FHR to uterine contractions.
 - It relies on the premise that fetal oxygenation will be transiently worsened by uterine contractions, leading to the FHR pattern of **late decelerations**
 - A **late deceleration** is defined as a visually apparent and usually symmetrical gradual decrease and return to baseline of FHR in association with uterine contractions, with the time from onset of the deceleration to its FHR nadir as 30 seconds or longer
- An adequate CST requires uterine contractions of moderate intensity that last about 40 to 60 sec with a frequency of three in 10 minutes.
 - These criteria were selected to approximate the stress experienced by the fetus during the first stage of labor.
 - If uterine activity is absent or inadequate,
 - ✓ intravenous oxytocin is begun to initiate contractions, and it is increased until adequate uterine contractions have been achieved
 - ✓ nipple stimulation has been used to induce adequate uterine activity, and the success rate at achieving adequate contractions and test results is comparable to that of oxytocin infusion

Contraindications to CST

- Patient at high risk for premature labor, such as in patients with PROM, multiple gestations, and cervical incompetence,
- The CST should also be avoided in conditions in which uterine contractions may be dangerous, such as placenta previa and a previous classic cesarean delivery or uterine surgery.

Interpretation of contraction stress test

Findings	Description	Incidence (%)
Negative	No late decelerations appear anywhere on the tracing with adequate uterine contractions (three in 10 min).	80
Positive	Late decelerations that are consistent and persistent present with the majority (>50%) of contractions without excessive uterine activity; if persistent late decelerations are seen before the frequency of contractions is adequate, the test is interpreted as positive.	3-5
Suspicious	Decelerations are late and inconsistent.	5
Hyperstimulation	Uterine contractions <i>closer than every 2 min or lasting more than 90 sec or five uterine contractions in 10 min</i> ; if no late decelerations are seen, the test is interpreted as negative.	5
Unsatisfactory	The quality of the tracing is inadequate for interpretation or adequate uterine activity cannot be achieved.	5

- Overall likelihood of perinatal death after a positive CST ranging from 7% to 15%.
- The high incidence of false positive (35% -65%) CSTs is one of the greatest limitations of this test because such results could lead to unnecessary premature intervention.

Factor causing positive CST

- Misinterpretation of tracing
- Supine hypotension
- Uterine hyperstimulation
- Fetal jeopardy

Management of abnormal result

- For suspicious CST, repeat the CST
- For positive result do BPP

Biophysical Profile

The BPP consists of an NST combined with four observations made by real time ultrasonography.

Biophysical profile scoring

Biophysical variable	Normal (score = 2)	Abnormal (score = 0)
Fetal breathing movements	At least one episode of ≥ 30 sec duration in 30-min observation	Absent or no episode of ≥ 30 sec duration in 30 min
Gross body/limb movement	At least three discrete body/limb movements in 30 min (episodes of active continuous movement considered a single movement)	Up to two episodes of movements in 30 min
Fetal tone	At least one episode of active extension with return-flexion of fetal limb or trunk, with opening and closing of the hand considered to reflect normal tone	Either slow extension with return-partial flexion, movement of limb in full extension, or absent fetal movement
Non stress test	At least two episodes of acceleration of ≥ 15 beats/min and 15 sec duration associated with fetal movement in 20 min	Fewer than two accelerations or acceleration < 15 beats/min in 20 min
Amniotic fluid volume	At least one pocket of amniotic fluid measuring ≥ 2 cm in two perpendicular planes	Either no amniotic fluid pockets or a pocket < 2 cm in two perpendicular planes

Management based on biophysical profile

SCORE	INTERPRETATION	MANAGEMENT
8 – 10	Normal; low risk for chronic asphyxia	Repeat testing at weekly to twice-weekly intervals.
6	Suspect chronic asphyxia (Possible fetal Asphyxia)	- If ≥ 36-37 wks gestation or < 36 wk with positive testing for fetal pulmonary maturity, consider delivery; - if < 36 wk and/or fetal pulmonary maturity testing is negative, repeat biophysical profile in 4-6 hrs; - Deliver if oligohydramnios is present.
4	probable fetal asphyxia	If ≥ 36 wk gestation, deliver; if < 32 wk gestation, repeat score.
0 – 2	Strongly suspect chronic asphyxia (Almost certain fetal asphyxia)	Extend testing time-120 min; if persistent score is 4 or less, deliver regardless of gestational age.

Modified Biophysical Profile

In an attempt to ***simplify and reduce the time*** necessary to complete testing, a variety of modifications of the full BPP have been evaluated by focusing on the components of the BPP that are most predictive of perinatal outcome.

The modified BPP combines:

- NST, as a short-term indicator of fetal acid–base status, with
- AFV assessment, as an indicator of long-term placental function

Thus, the results of the modified BPP are considered:

- **Normal** if the NST is reactive and the amniotic fluid volume is greater than 2 cm in the deepest vertical pocket and
- **Abnormal** if either the NST is nonreactive or amniotic fluid volume in the deepest vertical pocket is 2 cm or less (i.e., oligohydramnios is present).

If the NST is nonreactive despite VAS or extended monitoring, or if the AFV is abnormal, either a full BPP or CST is performed.

Doppler Velocimetry

Noninvasive method to assess blood flow; based on the observation that flow velocity waveforms in the umbilical artery of normally growing fetuses differ from those of growth-restricted fetuses.

Specifically, the umbilical flow velocity waveform of normally growing fetuses is characterized by high-velocity diastolic flow, whereas in growth-restricted fetuses, there is decreased umbilical artery diastolic flow

- In some cases of severe fetal growth restriction, diastolic flow is absent or even reversed.
 - The perinatal mortality rate in such pregnancies is significantly increased

Abnormal result when

- S/D ratio is above 95th percentile
- Absent diastolic flow
- Reversed diastolic flow

Umbilical artery systolic/diastolic ratio is commonly used

- Usually employed in case of IUGR: not recommended for other obstetric problems other than IUGR
- Perinatal mortality rate for absent end-diastolic flow is 10%

Comparison of selected antenatal tests

Test	False-negative rate (%)	False-positive rate (%)
Contraction stress test	0.04	35-65
Non stress test	0.2-0.8	55-90
Biophysical profile	0.07-0.08	40-50
Modified biophysical profile	0.08	60

Chapter 3

ASSESSMENT OF FETAL PULMONARY MATURATION

Prepared by Dr. Daniel Delessa (MI)

Introduction

RDS in the newborn is caused by a deficiency of pulmonary surfactant, *an anti-atelectasis factor that is able to maintain a low stable surface tension at the air-water interface within alveoli.*

Surfactant is packaged in lamellar bodies, discharged into the alveoli, and carried into the amniotic cavity with pulmonary fluid.

- It contains 70-80% phospholipids, 8-10% protein, and 10% neutral lipids, primarily cholesterol.
- *Dipalmitoyl phosphatidylcholine (DPPC), or lecithin, is functionally the principal phospholipid.*
- Phosphatidylglycerol makes up 4-15% of the phospholipids

When is fetal lung maturity testing performed?

- A test for fetal lung maturity may be performed before *semi-elective* but medically indicated births **<39 weeks** when this information significantly impacts assessment of the balance between the maternal-fetal risks of continuing the pregnancy versus the maternal-fetal risks of preterm birth

When should testing be avoided?

- Tests for fetal lung maturity are generally **not** performed
 - Before 32 weeks of gestation, given the high prevalence of fetal pulmonary immaturity and the lower predictive value of a mature test result at this GA.
 - In pregnancies with a reliable gestational age ≥ 39 weeks. At this GA, fetal lung maturity can be inferred
 - *test results are not more predictive of respiratory outcome than gestational age alone*

Tests of Fetal Pulmonary Maturity

- With the exception of amniotic fluid specimens obtained from the vaginal pool, the evaluation of fetal pulmonary maturation requires that a sample of amniotic fluid be obtained by amniocentesis.
- *A test that is positive for fetal pulmonary maturity will much more accurately predict the absence of RDS than a negative test will predict the presence of RDS*

1. Lecithin/sphingomyelin (L/S) ratio

- The amniotic fluid concentration of lecithin increases markedly at about 35 weeks' gestation, whereas sphingomyelin levels remain stable or decrease.
- Amniotic fluid sphingomyelin exceeds lecithin until 31 to 32 weeks, when the L/S ratio reaches 1.
- Lecithin then rises rapidly, and an ***L/S ratio of 2 is observed at about 35 wks.***
 - a ratio of 2 or greater has repeatedly been associated with pulmonary maturity
- The presence of blood or meconium in the amniotic fluid sample can cause erroneous results

2. Presence of PG

- PG, which does not appear until 35 weeks' gestation and increases rapidly between 37 and 40 weeks,
 - It is a marker of completed pulmonary maturation
- An advantage of testing for PG is that blood or meconium usually does not affect test results
- A disadvantage is that the absence of PG, especially before 36 weeks of gestation, is less predictive of the occurrence of respiratory distress than immature results from other tests

3. Surfactant/albumin(S/A) ratio

- Currently, there are no commercially available tests for measuring this ratio.
 - *Values greater than 55 mg of surfactant per gram albumin-mature*
 - *Values of 40-54-indeterminate*
 - *Values less than 40-immature*

4. Visual inspection of fluid

- During the first and second trimesters, amniotic fluid is yellow and clear.
- It becomes colorless in the third trimester.
- By 33 to 34 weeks' gestation, cloudiness and flocculation are noted, and as term approaches, vernix appears.
 - Amniotic fluid with obvious vernix or fluid so turbid it does not permit the reading of newsprint through it will usually have a mature L/S ratio.

5. Lamellar body count

- Lamellar bodies are the storage form of surfactant released by fetal type 2 pneumocytes into the amniotic fluid.
- A standard hematology analyzer can be used for quantification because of the similar size of lamellar bodies and platelets???
 - *Values less than 15,000 per microliter are almost always associated with lung immaturity*
 - *values $\geq 50,000$ per microliter strongly suggest lung maturity*
 - However, there is no consensus on the optimal threshold for predicting lung maturity; values greater than 30,000 to 40,000 per microliter, as well as higher levels, have been suggested
- The LBC has been used as an initial screening test, followed by the lecithin/sphingomyelin ratio where available when the LBC is neither clearly mature ($\geq 50,000$ per microliter) nor immature ($< 30,000$ per microliter).
- Blood contamination can lead to false elevation of the LBC because platelets are counted as lamellar bodies; the effect of meconium is minimal

6. Optical density at 650 nm

- An indirect measurement of lamellar bodies can be obtained by measuring the optical density of amniotic fluid at a wavelength of 650 nanometers.
- It is based upon the concept that increasing opalescence is due to increasing numbers of lamellar bodies.
- *An optical density reading ≥ 0.15 is used as the indicator of lung maturity*

7. Foam stability index

- FSI assesses total surfactant activity.
- It is a rapid predictor of fetal lung maturity based upon the ability of surfactant to generate stable foam in the presence of ethanol
 - Ethanol is added to a sample of amniotic fluid to eliminate the effects of non-surfactant factors on foam formation.
 - The mixture is then shaken and will generate a stable ring of foam if surfactant is present.
 - The FSI is calculated by utilizing serial dilutions of ethanol to quantitate the amount of surfactant present
- *The discriminating value indicative of lung maturity is usually set at ≥ 47 .*
- A positive result virtually excludes the risk of respiratory distress; however, a negative test often occurs with mature lungs.
- The presence of blood or meconium interferes with results of the FSI.

Factors that may affect interpretation

- **Gestational age**
 - For each test, the ability to predict respiratory distress is influenced by the prevalence of respiratory distress in the population tested; thus, the predictive value varies with gestational age
- **Blood, meconium**
 - *PG determination generally is not affected by blood, meconium, or other contaminants; its ability to predict lung maturity is the same whether or not contamination is present*
 - The S/A ratio is usually reliable if mature since contaminants tend to cause falsely immature findings,
 - the presence of blood or meconium can interfere with interpretation of the L/S ratio
 - Bloody samples give false information because blood contains sphingomyelin and decreases extraction of lecithin by cold acetone techniques (falsely low "L" and falsely high "S")
- **Oligohydramnios and polyhydramnios**
 - Theoretically, tests that are expressed as a ratio or proportion of two solutes released into the amniotic fluid should remain accurate independent of amniotic fluid volume, while tests that reflect the concentration of a substance in the amniotic fluid (e.g., LBC, PG) may be affected by amniotic fluid volume
- **Vaginal pool samples**
 - In women with intact membranes, amniotic fluid is obtained by amniocentesis.
 - In women with ruptured membranes, vaginal pool specimens can be collected from AF pooled in the posterior vaginal fornix by aspirating with syringe or by a sterile sponge
 - the L/S ratio can be higher in the vaginal pool than in fluid obtained by amniocentesis and false-positive PG have been reported in vaginal pool samples due to bacterial contamination

- **Antenatal corticosteroids**
 - Standard tests for predicting fetal lung maturity may be less reliable for predicting absence of fetal lung maturity in women who have received antenatal corticosteroids to enhance fetal lung maturation
- **Maternal diabetes mellitus**
 - RDS may occur in the IDM with a mature L/S ratio but absent PG.
 - onset of PG production is delayed in GDM
- **Twin pregnancy**
 - if amniocentesis to determine fetal lung maturity before 37wk is performed, amniocentesis of only one twin is adequate if gestation is ≥ 36 wk, but others test both twins in all cases because pulmonary maturation can be asynchronous
 - when only one sac is sampled, it would be reasonable to sample the sac of the fetus less likely to be mature

Antenatal Corticosteroid Therapy (ACS)

Trials have confirmed that a course of ACS administered to women at risk for preterm delivery:

- Reduces the incidence and severity of RDS and mortality in offspring
- Improves circulatory stability in preterm neonates, resulting in lower rates of IVH and necrotizing enterocolitis compared with unexposed preterm neonates.

Mechanism of action

- acceleration of development of type 1 and type 2 pneumocytes, leading to structural and biochemical changes that improve both lung mechanics and gas exchange (eg, surfactant production)
- induction of pulmonary beta-receptors, which play a role in surfactant release and absorption of alveolar fluid when stimulated

- induction of fetal lung antioxidant enzymes
- upregulation of genes for mediators of pulmonary epithelial sodium and liquid absorption, which are important for postnatal absorption of lung fluid

For these changes to occur, however, the lungs need to have reached a stage of development that is biologically responsive to corticosteroids

Timing before delivery

- Ideally, ACS is timed so that maximum efficacy is achieved before delivery,
 - *this window is **two to seven days** after administration of the first dose*
- Efficacy is incomplete <24 hours from administration of the first dose and appears to decline after seven days
- Neonatal benefits begin to accrue within a **few hours(as early as 6 hours)** of ACS administration
- Infants who received one dose of betamethasone in utero, but delivered before the second dose was given, had better outcomes than infants who did not receive any ACS
 - However, predicting when a patient is two to seven days before delivery is often highly imprecise.

Choice of drug and dosing

- **Betamethasone or Dexamethasone?**
 - Both drugs are effective for accelerating fetal maturity in randomized trials
 - *They are preferred over other steroids because they are less extensively metabolized by the placental enzyme 11 beta-hydroxysteroid dehydrogenase type 2, so they have maximum fetal impact.*
 - When both drugs are available, some prefer betamethasone because, in randomized trials where each drug was compared with placebo, **betamethasone showed a clear reduction in IVH**

- **Betamethasone**

- Two doses of 12 mg IM 24 hours apart
- The biologic half-life is 35 to 54 hours
- The onset and duration of action are affected by the vascularity at the injection site
- Drug concentrations in cord blood are approximately 20 percent of maternal levels one hour following maternal injection

- **Dexamethasone**

- Four doses of 6 mg IM 12 hours apart
- It has a more rapid onset and shorter duration of action than betamethasone; therefore, the dosing interval is shorter and more doses are required

- **Alternatives — Hydrocortisone**

- It is extensively metabolized by placental enzymes, so relatively little active drug crosses into the fetal compartment
 - therefore, beneficial fetal effects may not occur

However, if both betamethasone and dexamethasone are unavailable due to drug shortages, hydrocortisone 500 mg intravenously every 12 hours for four doses-last resort

Side effects of ACS use

Maternal side effects

- Case reports have described **pulmonary edema**, primarily associated with combination treatment with tocolytics, especially in the setting of chorioamnionitis, fluid overload, or multiple gestation
- **Transient hyperglycemia** occurs in many women; the steroid effect begins approximately 12 hours after the first dose and may last for five days.
 - *Screening for gestational diabetes, if indicated, should be performed either before ACS administration or at least five days after the first dose.*

- *In women with diabetes, hyperglycemia can be severe if not closely monitored and treated*
- The ***total leukocyte count increases by approximately 30 percent*** within 24 hours after ACS injection, and the lymphocyte count significantly decreases
 - *These changes **return to baseline within three days** but may complicate the diagnosis of infection*
- **Uterine activity may increase** slightly after betamethasone administration, particularly in multiple gestations and especially with increasing duration of pregnancy

Fetal side effects

- **Fetal heart rate (FHR) and biophysical parameters**
 - ACS may be associated with transient FHR and behavioral changes that typically return to baseline by four to seven days after treatment
 - The most consistent FHR finding is a ***decrease in variability on days 2 and 3*** after administration, which alone is not an indication for delivery
 - ***Reduced fetal breathing and body movements*** can result in a lower biophysical profile score or nonreactive NST; however, decreased fetal movement is not a consistent finding

Use of rescue (salvage, booster) ACS

- ACOG recommends considering a single repeat course of ACS in patients with **all** of the following characteristics :
 - <34+0 weeks of gestation, **plus**
 - At high risk of preterm delivery within the next seven days, **plus**
 - A prior course of ACS administered more than 14 days previously.
- ✓ However, rescue ACS can be provided as early as seven days from the prior dose, if indicated by the clinical scenario.

- A single repeat dose of betamethasone 12 mg rather than the standard two doses of 12 mg 24 hours apart is reasonable
 - A single dose may confer all the benefits of rescue treatment while minimizing potential risks
 - However, a two-dose betamethasone or four-dose dexamethasone regimen is also reasonable and commonly used worldwide
- **Multiple courses of steroids (>1 rescue course)**
 - Individual trials suggest increasing exposure to ACS is associated with increasing risk of adverse effect

Special populations

- Multiple gestations — The same dosing schedule is recommended for singleton and multiple gestations
- Diabetes — ACS should not be withheld from women with diabetes when indicated;
 - *however, secondary hyperglycemia must be closely monitored*

Chapter 4

HYPERTENSIVE DISORDERS OF PREGNANCY (HDP)

Prepared by Dr. Balkewkal Kebede (MI)

Definition and Classifications

HDP are one of the most common obstetric complications, affecting **5-10% of all pregnancy**.

- *Along with hemorrhage and infection it forms the **deadly triad** during pregnancy as a cause of maternal mortality.*

Hypertension (HTN) is diagnosed when appropriately# taken blood pressure exceeds 140mmHg systolic or 90mmHg of diastolic, observed at least on two occasions 4hrs apart but not more than 7days apart.

#Appropriate BP measurement include: -

- Avoiding smoking, caffeine and exercise for 30min prior to measurement
- Patient should sit quietly for 5min (10min, old JUSH guideline) in a chair with feet on the floor.
- Measurement is taken from right arm at the level of the heart
- Using appropriate BP cuff size
 - Width of the inflatable bladder of the cuff should be about 40% of the upper arm circumference (about 12- 14cm in the average adult)
 - Length 1.5x upper arm circumference or a cuff with bladder that encircle 80% or more of the arm
 - The standard cuff is 12X23cm, appropriate for arm circumference up to 28cm
- The 5th Korotkoff sound is used for diagnosis

Generally, avoid factors that will falsely or temporarily increase the BP, like: -

- smaller cuff size
- a loose cuff or a bladder that balloons outside the cuff
- taking stimulants within the 30min prior to measurement
- stressful environment
- sitting with legs crossed

On the contrary larger cuff size and taking antihypertensive medications will decrease BP measurement.

Read more on the steps and techniques of BP measurement on Bates Guide to Physical Examination 12ed.

NB. We don't necessarily need both the systolic and diastolic BP to be elevated to diagnose HTN; elevation of one of them is enough for diagnosis

Severe HTN is when sustained elevation of SBP of $\geq 160\text{mmHg}$ or DBP of $\geq 110\text{mmHg}$ is observed on two occasions for at least 4hrs apart but not more than 7days

- severe HTN can be confirmed within a short interval (minutes), to facilitate timely antihypertensive therapy

Proteinuria: Can be diagnosed before or during pregnancy, and it is defined as: -

- $\geq 0.3\text{g}(300\text{mg})$ protein in 24hr urine collection, or
- Protein: creatinine(P/C) ratio of ≥ 0.3 , or
- A dipstick measurement of $\geq +2$ on two occasions
 - Should be used if only the quantitative methods are not available, because it has high false positive and false negative test results
 - A test result of 1+ proteinuria is **false positive in 71% of cases**, and 3+ in 7% of cases compared with 300mg cutoff in 24hr urine collection
 - Therefore, if dipstick is the only available method the overall accuracy is better, when using 2+ as a determinant value (ACOG 2019)

The concentration of urinary protein is influenced by contamination with vaginal secretions, blood, bacteria, or amniotic fluid. It also varies with urine specific gravity and pH, exercise, and posture.

- It usually appears after HTN, but it may also appear before HTN too

Edema: Is defined as an excessive weight gain of >1.8kg in 1wk in the second or third trimester. It may be the first sign of the potential development of PE

Classification

According to ACOG (2013) there are four types of hypertensive disorders during pregnancy

1. Gestational HTN (GH)
2. Preeclampsia eclampsia syndrome (PE-E syndrome)
3. Chronic essential HTN
4. Preeclampsia-superimposed on chronic HTN

Delta HTN

- Normally during the first and second trimester of pregnancy the blood pressure declines to a nadir at 24-26wks of gestations and then rises afterward (mainly due to the decreased systemic vascular resistance, which is one of the physiologic changes during pregnancy).
- Some pregnant women may have more than usual increment in BP especially after 32wks of gestation who by term has substantially high BP, but still under 140/90mmHg. *The term delta HTN is used to describe this acute rise in BP*
- Some of these women will go on to have obvious PE, eclampsia or HELLP syndrome.

Gestational HTN (GH)

GH is a **new onset** elevated BP **without** proteinuria or signs/symptoms of preeclampsia related end organ dysfunction that is diagnosed for the first time **after 20 weeks** of pregnancy and resolves by 12 weeks postpartum in previously normotensive lady.

- It is the most frequent cause of HTN during pregnancy
- The incidence ranges from **6-29% in nulliparous** women and from **2-4% in multiparous**, the incidence increases in multiple gestation.
- Almost half of these women develop PE syndrome.
 - The likelihood of progression to PE depends on gestational age (GA) at the time of diagnosis, with higher rates if the onset of HTN is before 35wks (32wks, ACOG 2019) of gestation. Fortunately, most cases of GH develop at or beyond 37wks of gestation.

GH is divided in to two: Mild GH and Severe GH

Criteria for the diagnosis of mild GH

- Systolic blood pressure >140 mm Hg but <160 mm Hg and diastolic blood pressure >90 mm Hg but <110 mm Hg
- Proteinuria of <300 mg per 24hr collection
- Platelet count of >100,000/mm³
- Normal liver enzymes
- Absent maternal symptoms
- Absent intrauterine growth restriction and oligohydramnios by ultrasound

Criteria for the diagnosis of severe GH (see gabbe 7th edition, but there is some confusion about whether sever GH is separate thing or it is just PE with severity feature)

- *Women with GH who present with severe range BP should be managed in the same approach as for women with severe PE. GH and PE may also be undistinguishable in long term cardiovascular risks, like chronic HTN. (ACOG 2019)*

The overall pregnancy outcome of GH depends on GA at time of diagnosis and the type of GH.

- Most cases of mild GH which develop after 37wks of gestation usually have similar outcome as normotensive pregnancy, other than higher rate of induction of labor.
- Women with severe GH has increased risk of maternal and perinatal morbidities, (greater than those with mild PE), including placental abruption, preterm delivery, and delivery of small for gestational age baby

Some reclassify GH as transient hypertension, if evidence of PE doesn't develop and the BP returns to normal by 12 weeks postpartum.

Preeclampsia (PE)

Preeclampsia is a multi-system progressive disorder characterized by new onset of **hypertension and proteinuria** or hypertension and significant end-organ dysfunction with or without proteinuria after 20 weeks of gestation in a previously normotensive woman.

- ***PE = GH + proteinuria or severe features***

Incidence:

- ranges from 2-7% in otherwise healthy nulliparous women which is generally mild, with onset ***near term or during labor (in 75% of cases)***
- NB, on Williams Obstetrics 25th Ed, the incidence is slightly different,
 - ***3-10% for nulliparous and 1.4 -4% for multiparous women***
- PE complicates **2-8%** of pregnancies globally. (ACOG 2019)

Classification

- PE is categorized as, PE with severity features and PE without- severity features

Criteria for Preeclampsia with Severity Features

Preeclampsia, with severe features, is defined when the disorder is present in association with **any one** of the following abnormalities:

- **SBP ≥ 160 mm Hg or DBP ≥ 110 mmHg** on two occasions at least 4 hours apart while the patient is on bed rest or once if the patient has received prior antihypertensive therapy.
- **New-onset persistent cerebral symptoms:** headaches or visual disturbances
- **Impaired liver function** as indicated by abnormally elevated liver enzymes (at least twice the upper limit of normal [ULN]); severe, persistent right upper quadrant or epigastric pain that is unresponsive to medications and not accounted for by an alternative diagnosis; or both
- **Pulmonary edema** (often develop in postpartum)
- **Thrombocytopenia** (platelet count $< 100,000/\mu\text{l}$)
- **Progressive renal insufficiency** (serum creatinine > 1.1 mg/dl); some guidelines also include doubling in the patient's base line creatinine in absence of other renal diseases.

NB, the amount of proteinuria ($\geq 5\text{gm}$ in a 24hr urine specimen or $\geq 3+$ on dipstick), oliguria, and presence of intrauterine growth restriction (IUGR) or fetal growth restriction (FGR) by ultrasound **have been removed** as criteria for the diagnosis of severe disease.

Even though HTN is considered to be the hallmark of PE, some patient with PE may present with signs of PE without HTN, - *and it is referred to as **atypical PE***.

- The criteria for atypical preeclampsia include gestational proteinuria or FGR plus one or more of the following:

- Severity features of preeclampsia without hypertension: hemolysis, thrombocytopenia, elevated liver enzymes
- Early signs and symptoms of PE-E syndrome earlier than 20 weeks, (seen in patient with molar pregnancy, hydropic degeneration of chorionic villi and antiphospholipid antibody (APA) syndrome
- Late postpartum PE-E syndrome (>48 hours postpartum but less than six weeks after delivery).

Risk factors PE

- Nulliparity, (limited sperm exposure)
- Age <20yrs or >40yrs (35yrs ACOG)
- Race, (higher in the Black population because of genetic predisposition)
- Multifetal gestation (**increase risk to 13%**)
- Obesity/gestational diabetes mellitus (**BMI $\geq 35\text{kg/m}^2$ increase risk to 13.3%**)
- PE in previous pregnancy
 - In women with severe PE in previous pregnancy, **around 21%** will have severe PE in the subsequent pregnancy.
 - The earlier PE is diagnosed in the index pregnancy, the greater the likelihood of recurrence.
 - The risk of recurrence varies depending on the severity and GA at w/c PE developed
 - PE with severity feature in 2nd TM has 25-65% risk
 - PE w/o severity feature 5-7% risk
- Preexisting medical-genetic conditions
 - Chronic hypertension
 - Renal disease
 - Type 1 DM
 - Antiphospholipid antibody syndrome
 - Thrombophilia, Factor V Leiden mutation
- Pregnancy with the aid of artificial reproductive technology
 - Use ART increase risk because:
 - ✓ Increase chance that the women are >40yrs, nulliparous or obese with PCOS
 - ✓ Increase possibility of multiple gestation,
 - ✓ Its pregnancy with donated gametes or embryos
- Interpregnancy interval >7 years

- Family history of preeclampsia
- Woman born small for gestational age
- Poor outcome in previous pregnancy
 - Fetal growth restriction (FGR), placental abruption (PA), fetal death
- Paternal factors (men who fathered one preeclamptic pregnancy were nearly twice as likely to father a preeclamptic pregnancy with a different woman)
- Obstructive sleep apnea

Factors that reduce risk of PE

- **Smoking**, unlike its effect in many obstetric and gynecologic abnormalities, it carries a *reduced risk* for hypertension during pregnancy, because it up-regulates placental adrenomedullin expression which regulates volume homeostasis
- Long term sperm exposure with the same partner
- Placenta previa?

Etiopathogenesis

- The etiology of PE remains unknown, but d/t theories have tried to explain it.

Any satisfactory theory concerning the origins of preeclampsia must account for the observation that gestational hypertensive disorders are more likely to develop in women with the following characteristics: -

- Are exposed to chorionic villi for the first time
- Are exposed to a *superabundance of chorionic villi*, as with twins or hydatidiform mole
- Have preexisting conditions associated with endothelial cell activation or inflammation, such as diabetes, obesity, cardiovascular or renal disease, immunological disorders, or hereditary influences
- Are genetically predisposed to hypertension developing during pregnancy.

A fetus is not a requisite for preeclampsia to develop. And, although chorionic villi are essential, they don't need to be intrauterine.

The mechanisms that are currently considered important in explaining the pathogenesis of PE include:

- Placental implantation with abnormal trophoblastic invasion of uterine vessel
- Immunological maladaptive tolerance between maternal, paternal (placental), and fetal tissues
- Maternal maladaptation to cardiovascular or inflammatory changes of normal pregnancy
- Genetic factors including inherited predisposing genes and epigenetic influences.

○ *There are around 5 more theories, check them out*

Abnormal trophoblastic invasion

During normal implantation the uteroplacental (UP) arteries are formed by the action of migratory interstitial and endovascular trophoblasts in to the walls of the spiral arterioles, this transforms the UP arteries in to low resistance, low pressure and high flow system.

These trophoblasts induced changes extend all the way from intervillous space to the origin of spiral arteries (to inner 3rd of myometrium). These changes occur in two stages:

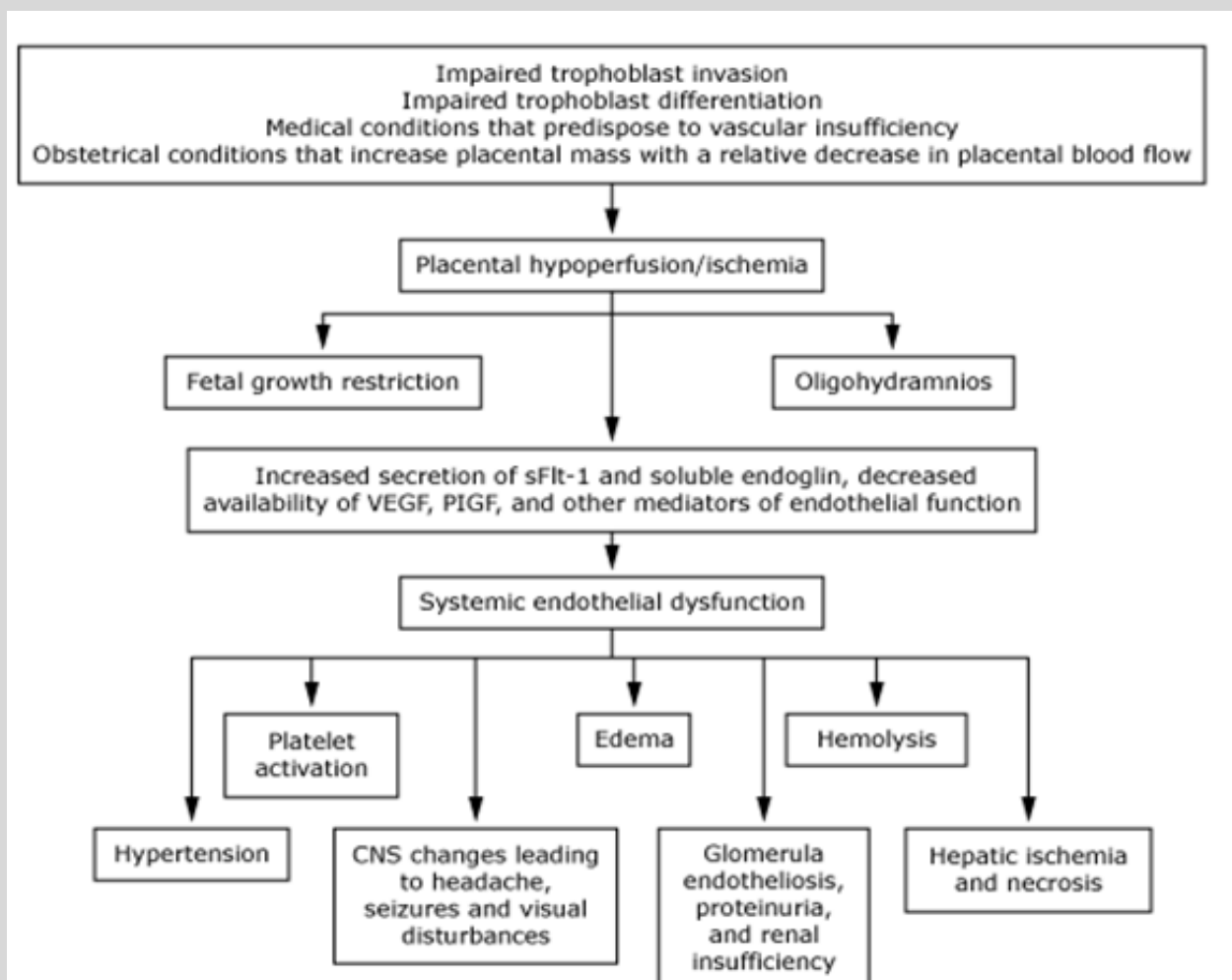
- 1st wave involves decidual segment of the arteries and occur in 1st trimester
- 2nd wave involving the myometrium segment and occur in 2nd trimester (after 16th week)

In pregnancies complicated with PE these changes are *limited to the decidual segment*, leaving the myometrial segments as high pressure and low flow system, resulting in limited UP blood flow (→ Placental ischemia and FGR).

The pathogenesis of PE is divided into two artificial stages: -

- Asymptomatic Placental Stage
 - Caused by faulty endovascular trophoblastic remodeling
- Symptomatic Maternal Stage
 - Causes by the changes in stage one
 - Is susceptible to modification by preexisting maternal conditions that are manifested by endothelial cell activation or inflammation, such as cardiovascular or renal disease, DM, immunologic disorders... hence they are risk factors!

The following diagram shows a model for Pathogenesis of PE



Pathophysiologic changes in PE

Cardiovascular changes

- There is increased afterload and reduced cardiac preload
- Ventricular remodeling
- Capillary leak and decreased colloid oncotic pressure (from? proteinuria)
 - Due to this changes, aggressive fluid therapy increases risk of pulmonary edema
- Hemoconcentration

Renal changes

- Proteinuria, may develop late, for example, approximately 10- 15% of HELLP syndrome and 17% of patient with eclampsia don't have proteinuria at the time of presentation.
 - Proteinuria in PE is nonselective, as a result of increased tubular permeability
- Increased urine sodium level
- Reduced renal perfusion and GFR (up to 25% below normal pregnancy)
- Oliguria,
 - Is a consequence of intrarenal vasospasm with approximately 25% reduction in GFR
 - NB, Intensive IV fluid therapy is not indicated as a treatment for preeclamptic women with oliguria, unless urine output is diminished from hemorrhage or fluid loss from vomiting or fever
- Increased serum creatinine
 - PE is the most common cause of AKI in pregnancy
 - AKI will develop soon in patients with comorbid hemorrhage with hypovolemia and hypotension (for example, in patient with severe placenta abruption)
- Increased uric acid level, due to increased production and decreased excretion

- Urinary calcium also decreases because of increased tubular reabsorption

Hematological changes

- Thrombocytopenia
 - Results from increased platelet activation, aggregation and consumption
 - Plt count $<150,000/\mu\text{l}$ occurs in 20% of PE (7% in those w/o severity feature and 50% in those with severity features (SF))
 - NB, once its developed, thrombocytopenia usually continues to worsen, so delivery is recommended
- Hemolysis, (Microangiopathic Hemolysis)
 - Microangiopathic hemolysis caused by endothelial disruption with platelet adherence and fibrine deposition
- Increase in LDH, can be from hemolysis or hepatic necrosis or ischemia

Hepatic changes

- Hepatic function may be significantly altered in women with PE with SF (indicated by elevated AST and ALP)
- AST is the dominant transaminase released in liver dysfunction due to PE, and is related to periportal necrosis. (AST>ALT)
 - This may help in distinguishing PE from other potential causes of parenchymal liver disease, in which ALT is usually higher than AST
- Increased LDH
- Alteration in hepatic synthetic function, as reflected by abnormalities of PT, aPTT, and fibrinogen, usually develop in advanced PE
 - Evaluation of these coagulation parameters is useful when there is:
 - ✓ Thrombocytopenia
 - ✓ Significant liver dysfunction
 - ✓ Suspected placental abruption
- RUQ or epigastric pain is thought to be from periportal or focal parenchymal necrosis, hepatic cell edema, or Glisson capsule distension or combination of these

HELLP syndrome

- Is one of the **more severe forms of PE**, because it is associated with increased rates of maternal morbidity and mortality
- Is mostly a 3rd TM condition, but in 30% of cases it is 1st expressed or progress postpartum (i.e., 70% antepartum)
- RUQ pain and generalized malaise are the main presenting symptoms in 90% of cases. Nausea and vomiting in 50% of cases
- Occurs in 1-2/1000 pregnancies overall
 - 10% of severe PE and 50% eclampsia

Criteria to establish the diagnosis of HELLP syndrome

- Hemolysis (at least two of the following signs of hemolysis)
 - Peripheral blood smear showing schistocytes, or burr cells
 - Serum Bilirubin ≥ 1.2 mg/dl
 - ✓ Late manifestation
 - Low serum haptoglobin
 - Severe anemia (Microangiopathic Hemolytic Anemia) unrelated to blood loss
 - LDH >600 U/L (>2 x the ULN)
 - ✓ >3000 U/L is associated with increased risk of mortality
- Elevated liver enzymes
 - ALT/AST at least twice the ULN (upper limits of the normal)
 - AST level >2000 U/L is associated with increased risk of mortality
- Low platelets ($\leq 100,000/\mu\text{l}$)

Classification of HELLP syndrome

- Based on the level of Plt count, LDH and AST/ALT

Mississippi Triple-class HELLP syndrome classification

Class	Lab values	Days from delivery-recovery	Incidence of bleeding (<i>Medscape</i>)
I	Plt count $\leq 50,000/\mu\text{l}$ AST/ALT $\geq 70\text{IU/L}$ LDH $\geq 600\text{IU/L}$	14days	13%
II	Plt count 50,000-100,000/ μl AST/ALT $\geq 70\text{IU/L}$ LDH $\geq 600\text{IU/L}$	7-10days	8%
III	Plt count 100,000-150,000/ μl AST/ALT $\geq 40\text{IU/L}$ LDH $\geq 600\text{IU/L}$	3-5days	No risk

Tennessee classification system

According to this classification, HELLP syndrome is classified in to: -

- Complete
 - When **all three** parameters are affected.
- Partial
 - When **only one or two of the three** parameters are affected.

Neurologic changes

- Both sudden elevation in BP (hyperperfusion=> disruption of end capillaries, and vasogenic edema) and hypoperfusion (=> Ischemia, cytotoxic edema and tissue infarction) are suggested in the development of neurologic manifestation.
- **Headache**, (in 75% of severe PE)
 - HA usually start from occipital area, because of the cerebrovascular hyperperfusion that has a predilection for the occipital lobes (because of limited sympathetic innervation in these areas).
 - Frequently improve after MgSO_4

- **Visual changes and & temporary blindness** (lasting a few hours to a week)
 - Is rare in PE, but complicates up to 15% of eclamptic cases
 - Blindness may arise from 3 major areas
 - ✓ Visual cortex of occipital lobe, occipital blindness is also called ***amaurosis fugax*** (*a transient loss of vision in one or both eyes*) (from occipital lobe vasogenic edema)
 - ✓ Lateral geniculate nuclei
 - ✓ Retina, from retinal ischemia, infarction or detachment, it's called ***Purtscher Retinopathy***
 - *Is a hemorrhagic & vasoocclusive vasculopathy, characterized by multiple white retinal patches and retinal hemorrhage, that are associated with severe vision loss*
- **Generalized cerebral edema**
 - Usually manifest by mental status changes that vary from confusion to coma.
 - Is particularly dangerous because fatal transtentorial herniation can result.
- **Cognitive decline**
 - In women with eclamptic pregnancies

Clinical manifestations of PE

- The clinical findings of PE can manifest as either a maternal syndrome or a fetal syndrome

Maternal syndrome

- The maternal syndrome of PE represents a clinical spectrum with major differences between near term PE without demonstrable fetal effects and PE that is associated with low birth weight and preterm delivery. Maternal manifestations include:

- **Symptoms**

- Headache (severe, global type, which is unresponsive to NSAIDs)
- Blurring of vision, or less commonly double vision and blindness (from retinal detachment)
- Right upper quadrant or Epigastric pain
- Swelling of the face or vulva
- Vaginal bleeding
- Nausea/vomiting
- Difficulty of breathing

- **Signs**

- Elevated BP, (*NB, BP could be normal and patient still can have PE*)
- Facial/generalized edema
- Signs of pleural effusion and ascites
- RUQ tenderness
- Mental status change (confusion to coma, result of cerebral edema)
- Hyperreflexia

- **Laboratory findings**

- Proteinuria
- MAHA, Thrombocytopenia
- Elevated serum AST/ALT (2x ULN), LDH, creatinine and bilirubin level
- Findings of DIC

Fetal manifestations include: - Oligohydramnios, FGR, Abruption, Vascular stillbirth and Preterm delivery

Prediction and Prevention of PE

- ACOG recommends the use of only risk factors for identifying women considered to be at risk for PE.
- There is insufficient data supporting preventive role of nutritional supplements like Ca, vitamin C, D, and E, garlic...

Currently **low dose aspirin** is the only recommended preventive measure in women considered to have high risk for PE, (and also for those with two or more of moderate level risk factors)

- Preventive role of aspirin is due to *its preferential inhibition of TXA₂* at low dose, correcting/preventing the imbalance of prostacyclin and TXA₂ in the pathogenesis of PE.
 - The rationale for recommending LDA prophylaxis is the theory that the *vasospasm and coagulation abnormalities* in preeclampsia are caused partly by an imbalance in the TXA₂/prostacyclin ratio
- It should be started preferably before 16wks of gestation (12-28), and continued till delivery.
- It decreases rate of severe PE and FGR significantly
 - reduced the risk of **PE** by an average of **24%**
 - reduced the average risk of **preterm birth** by **14%**
 - reduced the risk of **FGR** by **20%**

Table 1. Clinical Risk Factors and Aspirin Use*

Level of Risk	Risk Factors	Recommendation
High [†]	• History of preeclampsia, especially when accompanied by an adverse outcome • Multifetal gestation • Chronic hypertension • Type 1 or 2 diabetes • Renal disease • Autoimmune disease (ie, systemic lupus erythematosus, the antiphospholipid syndrome)	Recommend low-dose aspirin if the patient has one or more of these high-risk factors
Moderate [‡]	• Nulliparity • Obesity (body mass index greater than 30) • Family history of preeclampsia (mother or sister) • Sociodemographic characteristics (African American race, low socioeconomic status) • Age 35 years or older • Personal history factors (eg, low birth weight or small for gestational age, previous adverse pregnancy outcome, more than 10-year pregnancy interval)	Consider low-dose aspirin if the patient has more than one of these moderate-risk factors [§]
Low	• Previous uncomplicated full-term delivery	Do not recommend low-dose aspirin

Complications of PE

Maternal	Fetal/Neonatal
▪ Seizure/Eclampsia ▪ DIC ▪ Pulmonary edema ▪ Renal failure ▪ Liver failure ▪ Stroke ▪ Temporary blindness ▪ APH ▪ Death ▪ Chronic cardiovascular complications postpartum	▪ Uteroplacental insufficiency ○ FGR ○ Oligohydramnios ○ Perinatal death ▪ Consequences of prematurity

Eclampsia

Eclampsia is defined as the development of convulsions (generalized tonic-clonic) or unexplained coma during pregnancy or postpartum in patients with signs and symptoms of PE.

ACOG defines eclampsia as a new onset tonic-clonic, focal or multifocal seizure in the absence of other causative conditions, such as, Epilepsy, Cerebral arterial ischemia or infarction, ICH and Drug use.

- Alternative dx must be considered if seizure occurs after 48-72hrs of postpartum or while administration of MgSO₄.

In the western world, the reported incidence of eclampsia ranges from 1 in 2000 to 1 in 3448 pregnancies.

Eclampsia can develop antepartum, intrapartum (most of the time) or postpartum (in around 10% of cases, mostly within 48hr of delivery)

- **Late postpartum eclampsia** is defined as eclampsia that occurs after 48 hours but before 4 weeks of delivery.

Eclampsia that occurs before the 20th week of gestation is generally associated with molar or hydropic degeneration of the placenta with or without a coexistent fetus

Diagnosis

The diagnosis of eclampsia is secure in the presence of generalized edema, hypertension, proteinuria, and convulsions.

- *20-38% of women don't manifest classic signs of PE (HTN and proteinuria) before seizure episode*

Eclampsia often (in 78-83%, ACOG 2019) is preceded by premonitory signs of cerebral irritation, (**signs of impending eclampsia**) such as: -

- Persistent occipital or frontal headache
- Blurring of vision (blindness is seen in up to 10-15% of eclamptic patients, and it is mainly due to occipital lobe edema)
- Photophobia
- Altered mental status

Seizure may lead to severe maternal hypoxia, trauma and aspiration pneumonia

CT scan or MRI of the brain is not necessary for the diagnosis and management of eclampsia. However, it is indicated for patients with: -

- Focal neurologic deficit or prolonged coma
- Atypical presentation
 - Onset <20wk of gestation or >48hr postpartum
 - Eclampsia refractory to adequate therapy

Differential diagnosis for eclampsia (see Gabby 7th Ed, p692)

Severity of Eclampsia

Eclampsia is considered severe if one or more of the following are present (Eden's criteria)

- Coma of 6 or more hours
- Temp $>39^{\circ}\text{C}$
- PR $>120\text{bpm}$
- SBP $>200\text{mmHg}$
- RR >40
- >10 convulsions

Maternal and fetal outcomes of eclampsia

Maternal (with incidence): - PA (7-10%), DIC (7-10%), Pulmonary edema (3-5%), AKI (5-9%) Aspiration pneumonia (2-3%) and Cardiopulmonary arrest (2-5%)

Feta or perinatal: - Prematurity, PA, Severe FGR, Preterm delivery, is about 50% and about 25% occur before 32wks of gestation

Prevention of eclampsia

Can be: -

- **Primary:** by preventing the development and /or progression of PE,
- **Secondary:** by using pharmacologic agents that prevent convulsions in women with established PE.
- **Tertiary:** by preventing subsequent convulsions in women with established eclampsia

Some of the recommended preventive therapies include:

- close monitoring (in-hospital or outpatient),
- use of antihypertensive therapy to keep maternal BP below a certain level (< severe range or to normal values),
- timely delivery, and prophylactic use of magnesium sulfate during labor and immediately postpartum in those considered to have PE
- Prophylactic MgSO₄ is recommended only for women who are hospitalized with the established diagnosis of PE. MgSO₄ is associated with a significantly lower rate of recurrent seizures and a lower rate of maternal death than that observed with other agents, like diazepam.

NB, about 31-87% of eclampsia is considered unpreventable

Chronic HTN and Superimposed PE

Chronic HTN: Is defined as an elevated BP that is present prior to conception or is diagnosed before 20th week of gestation, or HTN that persist for more than 12wks postpartum.

- The frequency of chronic hypertension in pregnancy is estimated at 1% to 5%.
- Women with chronic hypertension are at increased risk for superimposed PE.

Maternal and Perinatal Risks

Maternal:

- Increased risk for the development of superimposed PE,
- PA (0.7% -2.7% in those with mild HTN and 5% -10% in those with severe or high-risk HTN), and
- FGR

Perinatal:

- perinatal mortality (3-4x greater),
- premature delivery and
- a growth restricted infant

Superimposed PE

The overall rate of superimposed PE is 25% (14% to 28% in mild HTN and 50% to 79% in severe HTN), ***the rate is not affected*** by maternal age, race, or presence of proteinuria early in pregnancy.

Criteria to diagnose preeclampsia in women with preexisting medical conditions

Conditions	Criteria
Hypertension only	Proteinuria of ≥ 300 mg per 24hr or thrombocytopenia.
Hypertension plus proteinuria (renal disease or class F diabetes)	Worsening severe hypertension plus proteinuria and either ▪ new onset of symptoms ▪ thrombocytopenia, or ▪ elevated liver enzymes

Management of PE-E syndrome

Management of PE depends on the severity of PE and the GA. It can be an expectant and definitive management.

NB, termination of pregnancy is the only definitive management for PE

Antepartum management

Mild PE (PE w/o severity feature) and GH

- Can be managed as outpatient, but if follow up is difficult, admit.
- Expectant management

Follow up

- Twice weekly ANC
- Daily fetal kick count
- Serial ultrasonography
- Weekly antepartum testing
- Closely monitoring BP, twice weekly
- Steroids, if <37wks
- Twice weekly testing for PE (in those with GH)
 - U/A, CBC, RFT, liver enzymes
 - Following initial documentation of proteinuria, and dx of PE, additional quantifications of proteinuria are not necessary
 - ✓ Although the amount of proteinuria is expected to increase with time during expectant management, it isn't predictive of perinatal outcome and shouldn't affect expectant management.
- The patient should be reminded to report any sign of severity if developed during the expectant management period.
- *NB. In women with GH w/o SF, progression to PE with SF can occur **within 1-3wks**, and in those with PE w/o SF, progression to PE with SF can occur **within days***

- Indication for delivery (definitive mgt)
 - GA $\geq 37^{+0/7}$ **wks.** (ACOG 2019) or ≤ 28 **wks**
 - If cervical status is favorable, induction is initiated. If the cervical status is unfavorable, pre- induction cervical ripening agents are used as needed
 - Abnormal antepartum testing
 - Fetal death
 - Fetal condition incompatible with life
 - If evidence of worsening preeclampsia is seen (development of any SF)
 - Eclampsia
 - Suspected PA or Vaginal bleeding in the absence of PP

Prophylactic intrapartum magnesium sulfate is indicated to prevent convulsions

Severe preeclampsia (PE with SF)

- Severe preeclampsia mandates hospitalization.

Expectant management

- Is given until development of maternal or fetal indications for delivery or until achievement of fetal lung maturity or 34 weeks' gestation.
- Acute BP control and maintenance of BP below severe level
- IV MgSO₄ is begun to prevent convulsions for the first 24-48 hours, anticonvulsant should be discontinued if no indications for delivery develop
- Corticosteroids to accelerate fetal lung maturity (Dexamethasone 6mg IM BID four doses) and delivery after 48 hours
- Follow up with preeclampsia chart
- Fetal kick chart daily
- Laboratory evaluation includes
 - CBC every other day
 - liver enzymes, and serum creatinine twice a weekly
- Fetal evaluation includes fetal heart monitoring, a BPP, and fetal growth assessment twice weekly

Antihypertensive therapy

The aim of this therapy is to: -

- keep the SBP between 140 & 160mmHg and DBP between 90 & 110mmHg, and
- prevent CHF, AKI, and Stroke

Acute blood pressure control

- **Hydralazine** (peripheral vasodilator) 5–10 mg IV. The onset of action is 10–20 minutes, and the dose can be repeated in 20–30 minutes if necessary (maximum of 60 mg IV or 300 mg PO in 24 hours)
- **Labetalol** (Beta blocker) 5–20 mg by slow IV push. The dose can be repeated in 10–20 minutes
- **Nifedipine** (calcium channel blocker) 5–10 mg orally. The dose can be repeated in 20–30 minutes, as needed

Table 3. Antihypertensive Agents Used for Urgent Blood Pressure Control in Pregnancy

Drug	Dose	Comments	Onset of Action
Labetalol	10–20 mg IV, then 20–80 mg every 10–30 minutes to a maximum cumulative dosage of 300 mg; or constant infusion 1–2 mg/min IV	Tachycardia is less common and fewer adverse effects. Avoid in women with asthma, preexisting myocardial disease, decompensated cardiac function, and heart block and bradycardia.	1–2 minutes
Hydralazine	5 mg IV or IM, then 5–10 mg IV every 20–40 minutes to a maximum cumulative dosage of 20 mg; or constant infusion of 0.5–10 mg/hr	Higher or frequent dosage associated with maternal hypotension, headaches, and abnormal fetal heart rate tracings; may be more common than other agents.	10–20 minutes
Nifedipine (immediate release)	10–20 mg orally, repeat in 20 minutes if needed; then 10–20 mg every 2–6 hours; maximum daily dose is 180 mg	May observe reflex tachycardia and headaches	5–10 minutes

Maintenance antihypertensive therapy

- Methyldopa up to 4gm/day PO in divided doses depending on the response
- Nifedipine 10-20mg PO every 6hrs

NB. Beside acute BP control, antihypertensive are indicated in: -

- ✓ Those already on medication for chronic HTN
- ✓ Those with end organ damage

Neuroprophylaxis—Prevention of Seizures

▪ Magnesium sulfate (MgSO_4)

- Is the most preferred and effective anticonvulsant prophylaxis
- Mechanism of action
 - ✓ reduced presynaptic release of the neurotransmitter glutamate
 - ✓ blockade of glutamatergic N-methyl D-aspartate (NMDA) receptors
 - ✓ potentiation of adenosine action
 - ✓ improved calcium buffering by mitochondria, and
 - ✓ blockage of calcium entry via voltage-gated channels
- Dosage
 - ✓ IV loading dose of 4gm over 20–60 minutes and 10gm IM half on each buttock, followed by a maintenance dose of 3gm/hour IV or 5gm IM in alternate buttocks every 4 hours. If convulsion occurs after MgSO_4 is started, give 2gm MgSO_4 IV.
- Toxicity
 - ✓ MgSO_4 has half-life of 5hrs, and it is eliminated predominantly through kidney
 - ✓ Is dependent on the plasma magnesium level
 - Loss of deep tendon reflexes occurs at 8 to 10 mEq/L
 - Respiratory paralysis at 10 to 15 mEq/L
 - Cardiac arrest at 20 to 25 mEq/L

Table 2. Serum Magnesium Concentration and Toxicities

Serum Magnesium Concentration			
mmol/L	mEq/L	mg/dL	Effect
2–3.5	4–7	5–9	Therapeutic range
>3.5	>7	>9	Loss of patellar reflexes
>5	>10	>12	Respiratory paralysis
>12.5	>25	>30	Cardiac arrest

- ✓ In the event of these adverse effects MgSO_4 should be discontinued and **Calcium gluconate (10 mL of 10% solution over 3min)** should be given.
- ✓ Urine output is monitored, and the magnesium dose is adjusted accordingly to prevent hypermagnesemia (mild AKI (Scr 1.0–1.5mg/dl) or UOP <30 ml/hour → **reduce MgSO_4 dose to half**).
- **Contraindications**
 - ✓ Severe renal impairment
 - ✓ Hypermagnesemia
 - ✓ Hypocalcemia
 - ✓ Skeletal muscle disorders, progressive muscle weakness with carcinoma
 - ✓ Myasthenia gravis and Allergies
 - ✓ Heart block, myocarditis
- Other than preventing seizure, MgSO_4 also has,
 - ✓ inhibitory effect in uterine contraction (**at higher plasma concentration**)
 - ✓ protective effect against the development of cerebral palsy in very-low-birth weight newborns
 - ✓ relieving effect on headache and blindness
- In absence of magnesium sulfate, one may use diazepam 30mg/1000ml D5W 30 drops per minute and 10mg IV bolus if convulsion occurs.

Indications for definitive management (delivery) in Severe PE

- The gestational age is ≥ 34 wks
 - If delivery is indicated before this, CS should be given, but maternal or fetal deterioration may preclude completion of course CS therapy.
- Failure to control hypertension with two antihypertensive drugs with a maximum dose in 48 hours. **Resistant HTN**
- Persistent maternal severity symptoms (severe headache, visual changes and abdominal and/or epigastric pain with elevated liver enzymes).
- HELLP Syndrome
- Eclampsia
- Pulmonary edema or left ventricular failure
- IUFD
- DIC
- Severe renal dysfunction

Intrapartum management

- The two main goals of management of women with PE during labor and delivery are
 - Prevention of seizure
 - ✓ Prophylactic anticonvulsant
 - Control of HTN
- If there are no contraindications to labor, vaginal delivery is the preferred approach.
- Cervical ripening agents and oxytocin are used as needed.
- To avoid pulmonary edema, total IV fluids should not exceed 100 mL/hour.
 - Even during augmentation use half the amount of fluid and the drop/min of the standard augmentation
- Follow FHB every 15min in AFSOL and every 5min in SSOL
- Shorten SSOL with instrumental delivery when necessary
- Pain control is achieved with regional anesthesia or with intramuscular or IV narcotic analgesics.

- The use of either epidural, spinal, or combined techniques of regional anesthesia is the method of choice during cesarean delivery.
 - But general anesthesia generally carries more risk, because of
 - ✓ Risk of aspiration
 - ✓ Failed intubation due to pharyngolaryngeal edema
 - ✓ Stroke secondary to increased systemic and intracranial pressure during intubation and extubation

Postpartum Management

- Close monitoring of BP, of symptoms consistent with severe disease, and accurate measurements of fluid intake and urinary output.
- Magnesium sulfate should be continued for at least 24 hours.
- Antihypertensive medications are discontinued if the BP remains below the hypertensive levels for at least 48 hours.
- 5 days of oral furosemide therapy (20 mg/day) enhances recovery and reduces the need for antihypertensive therapy in women with severe preeclampsia.
- If neurologic symptoms exist, brain imaging (if available) is undertaken to rule out the presence of cerebral pathology.
- When will LFT, RFT and platelet count normalize after delivery?
 - LFT= 4days
 - RFT= 10days
 - Platelet count= 3-5days
- Long term follow up is necessary because of the chronic cardiovascular complication of PE

Management of eclampsia

- ABCD of life, and basic supportive care
 - Prevent aspiration
 - Positioning in lateral decubitus position
 - Give oxygen and monitor saturation

- Control convulsion
 - Anticonvulsants –
 - MgSO₄ (same dosage as in severe PE)
 - NB. MgSO₄ isn't necessary to arrest the current seizure, rather to prevent recurrence
 - If refractory to MgSO₄, (still seizing after 20min or >2 recurrence)
 - Use sodium amobarbital (250mg IV in 3min), thiopental or phenytoin (1250mg IV at a rate of 50mg/min)
 - Alternative supplementary anticonvulsant medication include: -
 - Intravenous barbiturate (given slowly).
 - Midazolam or lorazepam may also be given in a small single dose, but prolonged use is avoided because it is associated with a higher mortality rate from aspiration pneumonia
 - Sedation and ICU may be necessary, if all medical interventions fail
- Acute BP control
- Supportive management to patients with loss of consciousness
- Deliver the baby after the patient is stabilized
 - Although seizure is associated with fetal heart beat abnormalities like; recurrent deceleration, reduced variability, tachycardia or even bradycardia, one shouldn't rush to delivery before stabilization of the mother
 - Maternal resuscitation is usually followed by normalization of fetal tracing

N.B. mere presence of eclampsia is not an indication for CD.

Subsequent risks of women who developed HDP

Women who developed PE have an increased risk of: -

- cardiovascular disease (HTN, MI, and CHF)
- cerebrovascular disease (stroke)
- peripheral arterial disease
- cardiovascular mortality later in life

SAMPLE HISTORY

Chief Complaint: Headache of 2days duration

HPP:

This is a 19yrs old primigravida lady whose LNMP was on 19-4- 12EC (reliable) making the EDD on 24-1-13EC and gestational age of 33+2wks (on 12-12-12EC, date of clerking). Menses was regular, coming every 28-30days and lasting for 3-4days, she usually used 2-3 tampons per day, and flow wasn't associated with clot or any abdominal discomfort. She used to take injectable contraception for two years, but stopped 6month before she saw her last period.

She suspected she was pregnant after her period was delayed for two weeks and when she started to have nausea & vomiting, which was usually in the mornings, then she went to the local health center, where urine pregnancy test was done and she was told that she was pregnant, (NB. if she remembers, write the specific date the test was done, and if ultrasound was done write the gestational age too), and She was told to return after 3months to start her ANC follow up. She has 4 ANC visits at () local health center.

Her first ANC visit was when she was around 4month pregnant, (NB. for all visits write specific date if she has a card or if she remembers correctly). On this visit history was taken, weight, height and BP were measured, and abdominal examination was done. Blood and urine sample were taken. She doesn't remember any of the specific results, but she was told that everything was fine. She was given an injection on her arm and a red tablet to take 1 tablet per day. (Write "iron supplement" if she knows what it is) and was told to return after 1month.

Her 2nd, 3rd and 4th visits were on 5th, 6th, and 7th months of pregnancy respectively (some seniors may expect you to write each visit separately even though it is going to be pretty much a repetition).

Similar examination and lab investigation (only urine) were done, and all were uneventful. 2nd injection was given on the 2nd visit and red tablet was given in all the visits.

She started to feel fetal movements at around 4-5month of pregnancy (if she remembers ask exact date of fetal quickening) and it has been increasing even since and it is a rolling and kicking type. The pregnancy was wanted, planned and supported. She hasn't made any significant change in her diet after she became pregnant. Her staple diet was injera made of teff with shiro wot 3 times per day, meat on holidays and fruit and vegetables at least once per week.

Currently she is presented with a persistent global type of headache of 2days duration, which hasn't improved even if she started taking medications for headache that she found in the house. Associated with this she also has swelling on her face, both hands and both legs, the swelling started from the leg and progress upward. After admission she was told that she has an elevated blood pressure and she was given medications. Now the headache and the swelling over her face and her hands have disappeared, otherwise: -

Danger signs

- ✓ She has no history of blurring of vision, ABM or LOC
- ✓ She has no history of epigastric pain, or right upper quadrant pain. She has no history of vaginal bleeding, or vaginal discharge, high grade fever
- ✓ She has no history of excessive vomiting
- ✓ She has no history of cough, or shortness of breath
- ✓ She has no history of urinary frequency, decreased urine output or pain during urination
- ✓ She has no history of passage of liquor or pushing down pain

Risk factors

- ✓ She has no previous history of hypertension, diabetes mellitus.

- ✓ She has no history of use of artificial reproductive technology She has no history of bleeding disorders
- ✓ She has no family history of similar complaints

DDx (for the headache)

- ✓ She has no history of neck stiffness or fever (meningitis)
- ✓ She has no history of eye pain or redness of the eye (glaucoma)
- ✓ She has no history of sleep disturbance, or worsening of headache at awakening or during bending, coughing or straining (brain tumor)
- ✓ She has no history of head trauma, vomiting or abnormal body movement (Subarachnoid hemorrhage, which is usually referred by the patient as “the worst headache ever”, with sudden onset, thunderclap headache)
- ✓ She has no history of fever, pain in the jaw or tongue during chewing or ringing sensation in the ears (Temporal Arteritis)

Physical examination

Vital signs: BP=> 165/105mmHg, Pulse, RR, temp, Weight, height and BMI... normal

Abdominal examination

- Inspection
 - The abdomen is protuberant and moves with respiration (sometimes the abdomen might appear asymmetrical in pregnant ladies, more on the right side (because of the sigmoid colon on the left side, which occupy the space)
 - The umbilicus is flat
 - There is midline hyperpigmentation (linea nigra) and purplish marks in the lower abdomen (striae gravida)
 - There is no scar over the abdomen (any previous C/S scar?)
- Palpation
 - Superficial palpation: There is no tenderness, rigidity, or superficial mass
 - Deep palpation: No organomegaly or deep mass

- Obstetric palpation
 - ✓ Fundal palpation (Leopold I)
 - The fundus is 6 fingers above the level of the umbilicus: - 32 weeks
 - The fundus is occupied by a soft bulky, irregular, non-ballotable mass: -breech
 - ✓ Lateral palpation (Leopold II)
 - The longitudinal axis of the fetus is in line with the mother's longitudinal axis, -longitudinal lie
 - There is a regular or smooth surfaced, hard mass on the right side of the abdomen, -the back
 - ✓ Pelvic palpation (Leopold III)
 - The occiput (hard ballotable mass) is palpated in the lower uterine pole: - cephalic presentation
 - The anterior shoulder is 5 fingers above the symphysis pubis: - floating
 - The cephalic prominence is opposite to the side of the back: - flexed attitude.
- Percussion: No sign of fluid collection
- Auscultation: FHB is 140bpm heard on the right side

Musculoskeletal system: Grade II bilateral pitting edema

Investigation

- CBC (anemia, or thrombocytopenia)
- BG (blood group, Rh)
- Urinalysis (proteinuria, bacteriuria, UTI?)
- LFT
- RFT
- VDRL (Venereal Disease Research Laboratory)

- PICT (Provider-Initiated Counseling and Testing)
- Serum LDH
- Abdominal Ultrasound (fetal growth and BPP)
- Coagulation profile (if thrombocytopenia or elevated liver enzymes)

Functions of obstetric ultrasound include: -

- Diagnosis of pregnancy
- Gestational age determination
- Placental localization
- Amniotic fluid volume determination
- Fetal wellbeing assessment
- Fetal sex identification
- Confirmation of fetal viability
- Fetal weight estimation
- Diagnosis of: -
 - fetal growth restriction
 - congenital anomaly
 - multiple gestation
 - molar pregnancy
 - uterine and adnexal pathology

See details on obstetric ultrasound and other investigation in obstetrics on Introduction to Obstetric and Gynecological diagnosis, Asheber Gaym (M.D.)

Management plan

See antepartum management of severe PE above.

Chapter 5

ANTEPARTUM HEMORRHAGE (APH)

Prepared by Dr. Bemnet G/Micheal (MI)

Introduction

APH or late pregnancy bleeding is vaginal bleeding during pregnancy after 28th wks of G, and before the delivery of the fetus (the last fetus in case of multiple pregnancies).

Incidence: 2 – 3 % of all pregnancies

Classes of hemorrhage and the physiologic response

Normal physiologic adaptations and changes during hemorrhage

- In normal physiology when 10% of circulatory blood volume is lost, vasoconstriction occurs in both arterial and venous compartments in order to maintain blood pressure and preserve blood flow to essential organs.
- As blood loss reaches 20% or more of the total blood volume, increase in systemic vascular resistance can no longer compensate for the lost intravascular volume and blood pressure decreases => cardiac output falls => poor end organ perfusion and finally cardiogenic shock will ensue.

Hemorrhage classification and physiologic response

Class	Acute blood loss (ml)	% Lost	Response
I	1000	15	Dizziness, palpitation, minimal BP change, narrow pulse pressure
II	1500	20-25	Tachycardia, tachypnea, sweating, weakness
III	2000	30-35	Significant tachycardia and tachypnea, restlessness, pallor and cool extremities
IV	≥2500	40	Cardiogenic shock, air hunger, Oliguria or anuria

NB, it is important that you know these classes with the response, because it will help you to estimate the blood loss in initial assessment of the patient

Causes of Third-Trimester bleeding

Differential diagnosis for third trimester vaginal bleeding

Placental causes	Non-placental causes
○ Abruptio placentae ○ Placenta previa ○ Vasa previa ○ Placenta membranacea ○ Circumvallate placenta	○ Bloody show ○ Uterine rupture ○ Bleeding diathesis ○ Local causes ~5% ▪ Cervicitis ▪ Cervical ca. ▪ Cervical polyp ▪ Leech infestation ▪ Vulvo-vaginal abnormalities ▪ Unknown/Undetermined* causes ~25%

**Undetermined causes: is to indicate that causes of APH other than local causes are ruled out but the local causes are not ruled out yet, because of the need to wait 48hr after last vaginal bleeding episode to perform speculum examination.*

**Unknown causes: no identified causes after performing all the necessary physical examination, including speculum examination and investigations.*

Most common once are:

- Bloody show associated with cervical change
 - This cervical bleeding is the consequence of effacement and dilation of the cervix, with tearing of small vessels.
- Abruptio placentae (AP)
- Placenta previa
- Vasa previa

Placental Abruption

Placental abruption (PA), or abruptio placentae (AP), refers to the premature separation of a normally implanted placenta from the uterus after 20 weeks of gestation but prior to delivery of the last fetus.

Mechanism of bleeding

- AP is commonly due to rupture of defective maternal vessels in decidua basalis causing separation of the placenta.
 - This cause bleeding which in turn results in a decidual hematoma which promotes placental separation, destruction of placental tissue and a loss of maternal-fetal surface area for nutrient and gas exchange.
- In rare cases it can be due to disruption of fetal-placental vessels.

Incidence

- The overall incidence of placental abruption is approximately 1 in 100 births.
- About 1/3rd of all APH can be attributed to PA. And 40-60% of abruptions occur prior to 37 weeks of gestation.

Risk factors of AP

- Increasing parity and maternal age
 - incidence in primigravida ~1%, multigravida ~2.5%
- Prior abruption
 - Risk of recurrence 5-17%, after two consecutive abruptions risk of recurrence rises to ~25%
- Maternal diseases
 - Hypertension (5-fold increased risk), Hypothyroidism, Asthma
- Preterm premature rupture of membranes; in 2-5%, abruption occurs

- Rapid uterine decompression associated with multiple gestation (3-fold higher risk) and polyhydramnios
- Uterine and placental factors
 - Anomalies, Synechiae, Fibroids, Cesarean scar, Abnormal placental formation and Chronic ischemia
- Maternal substance use
 - Cigarette smoking (40% increased risk), Cocaine abuse (10% higher risk)
- Trauma
 - Blunt or penetrating trauma to the gravid abdomen has been associated with placental abruption.
 - After a minor trauma, the risk for placental abruption is between 7% and 9%, whereas the risk may be as high as 13% after severe injury.
 - The two most common causes of maternal trauma are motor vehicle crashes and domestic abuse.
 - With motor vehicle crashes, uterine stretch, direct penetration, and placental shearing from acceleration-deceleration forces are the primary etiologies of trauma-related placental abruption
- Hyperhomocysteinemia

Clinical manifestation

Several factors determine the clinical manifestations of placental abruption. These factors include:

1. The temporal nature of the abruption (acute vs. chronic)
2. Clinical presentation; Overt (80%) vs. concealed (20%), and
3. Severity

An acute, overt abruption typically presents with **vaginal bleeding, abdominal pain, and uterine contractions**

- As placental separation worsens, uterine tenderness, tachysystole, fetal heart rate (FHR) patterns consistent with hypoxia, and fetal death may occur.

NB. The amount of vaginal bleeding correlates poorly with the extent of placental separation and its potential for fetal compromise. In fact, concealed abruption occurs in 10% to 20% of cases

Chronic abruption may be insidious in its presentation and is often associated with ischemic placental disease.

- Typically, these cases present with intermittent, light vaginal bleeding and evidence of chronic placental inflammation and dysfunction, such as
 - **oligohydramnios**,
 - **fetal growth restriction**,
 - preterm labor,
 - premature preterm rupture of membranes (PPROM), and
 - PE

*With **severe** abruptions **more than 50%** of the placental surface area separates. In these scenarios, maternal compromise in the form of **consumptive coagulopathy** may result from the triggering of the clotting cascade by hemorrhage and extensive thrombin deposition. i.e., **DIC** (~10% of cases)*

Causes of DIC in pregnancy

- Severe abruption
- IUFD
- Acute fatty liver of pregnancy
- HELLP syndrome
- Sepsis

Diagnosis

Placental abruption is primarily a clinical diagnosis that is supported by radiographic, laboratory, and pathologic studies.

Clinically: Any findings of vaginal bleeding, uterine contractions, abdominal and/or back pain, or trauma should prompt an investigation for potential placental abruption. Port wine discoloration of liquor and non-reassuring fetal status can also be indicative.

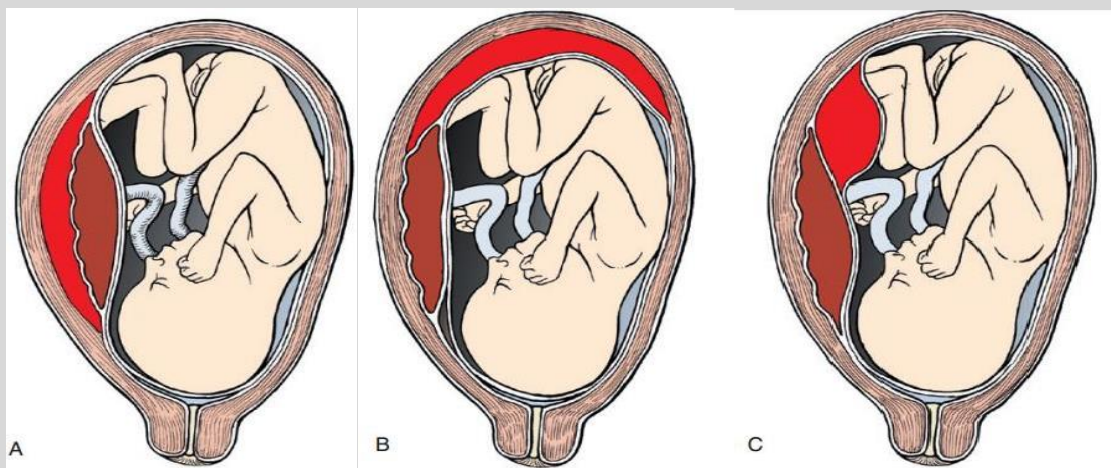
Radiological (Ultrasound): for the diagnosis of placental abruption identified less than 2% of cases.

- Early hemorrhage is typically hyperechoic or isoechoic,
- Resolving hematomas are hypoechoic within 1 week and sonolucent within 2 weeks of the abruption.

Acute hemorrhage may be misinterpreted as a homogeneous thickened placenta or fibroid.

There are three predominant locations for placental abruption: -

- A. Retroplacental (between the placenta and the myometrium)
- B. Subchorionic (between the placenta and the membranes)
- C. Preplacental (between the placenta and the amniotic fluid)



NB, Retroplacental hematomas are associated with a worse prognosis for fetal survival than subchorionic hemorrhage. And the size of the hemorrhage is predictive of fetal survival; the greater the size, the greater the fatality.

Laboratory findings: Hypofibrinogenemia and evidence of consumptive coagulopathy may accompany severe abruption.

Pathologic studies: Macroscopic inspection of the placenta may demonstrate adherent clot, depression (due to missing cotyledons) of the placental surface and bleeding into myometrium causing Couvelaire uterus.

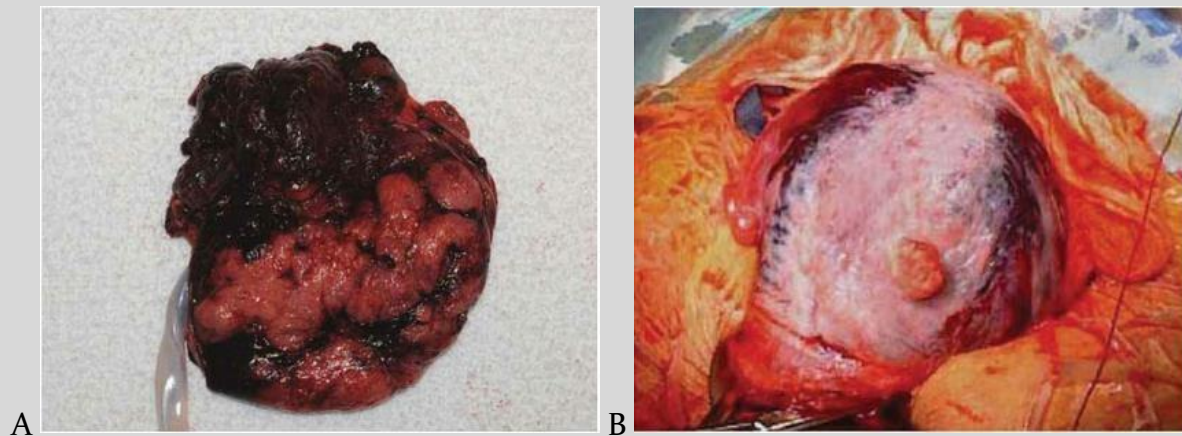


Figure 2, A. Partial placental abruption with a dark adherent clot. B. Couvelaire uterus from total placental abruption after CD

Complications of PA

▪ Maternal complications

- Blood loss
- Consumptive coagulopathy
- Need for transfusion
- End organ damage
- Cesarean delivery
- Death

▪ Fetal complications

- IUGR
- Oligohydramnios
- Prematurity
- Hypoxemia => NRFHRP
- Still birth

Grading of abruptio placentae

- Grade 0: Retrospective diagnosis of abruption
- Grade 1: Vaginal bleeding
- Grade 2: Vaginal bleeding, concealed hemorrhage, uterine tenderness, NRFHR
- Grade 3: Vaginal bleeding, shock, extensive concealed hemorrhage, uterine tenderness, IUFD
 - Grade 3 A: With no coagulopathy
 - Grade 3 B: With coagulopathy

Management of PA

All cases should be admitted.

Asses hemodynamic status, then

- Establish IV line
- Assess and monitor Fetal wellbeing strictly
- Determine HCT, BGP & Rh, platelets, fibrinogen, PT, aPTT

Grade 1:

- Conservative management
 - Steroid in < 34 weeks of gestation
 - Keep in ward till bleeding subsides
 - Tocolysis in < 33 weeks of gestation
 - Follow maternal V/S, bleeding, uterine contraction & tenderness, FHB, kick count, BPP, fetal growth
- Indications for delivery: Term, IUFD, malformed fetus, NRFHR, advanced labor, excessive bleeding

Grade 2 – 3:

- Asses maternal hemodynamic status, then stabilize the mother, then
- Maintain U.O.P > 30 ml/hr & HCT > 30%
- Platelets (6U) transfusion if thrombocytopenia
- FFP if fibrinogen < 100mg/dl
- Delivery
 - Vaginal: Amniotomy & induction with oxytocin
 - C/d for uncontrolled hemorrhage & other obstetric indications
- Hysterectomy; if uncontrolled hemorrhage occurs
- Couvelaire uterus---Uterotonic agents, hysterectomy if unresponsive

Placenta Previa

Placenta previa is defined as the presence of placental tissue over or adjacent to the cervical Os.

Recent revised classification of placenta previa consists of two variations:

1. True placenta previa; in which the internal cervical os is covered by placental tissue, and
2. Low-lying placenta; in which the placenta lies within 2 cm of the cervical os but does not cover it.

Incidence

The overall reported incidence of placenta previa at delivery is 1 in 200 births. In the second trimester, placenta previa may occur in up to 6% of pregnancies.

Placental migration has been used to explain this “resolution” of placenta previa that is noted near term. Three theories have been suggested to account for this phenomenon:

- **1st theory:** as the pregnancy advances, the stationary lower placental edge relocates away from the cervical Os with the development of the lower uterine segment.
 - The lower uterine segment has been noted to increase from 0.5 cm at 20 weeks to more than 5 cm at term.
- **2nd theory:** the placenta-free uterine wall has been proposed to grow at a faster rate than the uterine wall covered by the placenta.
- **3rd theory:** (trophotropism); the growth of trophoblastic tissue away from the cervical Os toward the fundus, results in resolution of the placenta previa.

Etiologies of PP

- Exact cause is unknown
- Theories include:
 - Dropping down theory
 - Persistence of chorionic activity
 - Defective decidua
 - Big surface area of placenta

Risk factors

- Intrinsic maternal factors
 - Increasing parity (Grand multipara 5% risk, Nulliparous women~ 0.2% risk)
 - Advanced maternal age (If age >35, 4-fold increase risk, if age>40, 9-fold increase risk)
 - Maternal race (higher rates in Asian women)
- Extrinsic maternal factors
 - Cigarette smoking (as high as 3-fold increased risk)
 - Cocaine use
 - Residence at higher elevation (increased placental surface area due to decreased uteroplacental oxygenation)
 - Infertility treatments

- Fetal factors
 - Multiple gestations
 - Male fetus (due to larger placental size and delayed implantation of the blastocyst)
- Prior placenta previa (8-fold increased risk)
- Prior uterine surgery and cesarean delivery.
 - PP incidence increases with the numbers of cesarian deliveries (CD).
 - ✓ It occurs in 0.9% of women with one prior CD, in 1.7% of women with two prior CDs, and in 3% of those with three or more CDs.
 - ✓ In patients with four or more cesarean deliveries, the risk for placenta previa has been reported to be as high as 10%.

Clinical manifestation

Placenta previa typically presents as painless vaginal bleeding in the second or third trimester.

- Between 70% and 80% of patients with placenta previa will have at least one bleeding episode, and from those with bleeding,
 - One third of women will present before 30 weeks of gestation,
 - One third between 30 and 36 weeks, and
 - One third after 36 weeks.
- About 10% to 20% of patients present with uterine contractions before bleeding, and
- Fewer than 10% remain asymptomatic until term.

Time of bleeding in placenta previa is during:

- Lower uterine segment formation
- Fetal head engagement
- Manipulation upon physical examination (e.g. PV)
- Labor

Diagnosis

Whenever there is uterine bleeding after mid-pregnancy, placenta previa or abruption are always considered.

NB, PP should not be excluded until sonographic evaluation has clearly proved its absence.

Radiology (ultrasound):

- Transabdominal (diagnostic accuracy of at least 95%) and
- Transvaginal (diagnostic accuracy that approaches 100%) ultrasound provide the best means for diagnosing placenta previa.

If a true placenta previa or low-lying placenta is diagnosed in the second trimester, repeat sonography should be obtained in the early third trimester at 32 weeks.

More than 90% of placenta previa cases diagnosed in the second trimester resolve by term.

Management

- Admit all ladies with APH secondary to placenta previa at time of diagnosis.
- Resuscitation based on clinical condition.
- Vaginal & rectal examinations are absolutely contraindicated.
- Monitor closely maternal & fetal conditions.
- HCT, BG & Rh, cross-match at least two units of blood
- Decide on conservative management versus immediate delivery
 - Indications for immediate delivery:
 - ✓ Term pregnancy, IUFD, fetal growth restriction, NRFS, excessive bleeding, gross fetal congenital malformation which may not be compatible with life, lady in labor.
 - Route of delivery depends on the type of placenta previa and presence of other obstetric indications:
 - ✓ Low lying & anterior marginal → vaginal delivery
 - ✓ Partialis, totalis, & posterior marginal → c/d

- ✓ Excessive bleeding, NRFS, other indications → c/d
- In absence of any of the above indications to deliver the fetus, conservative management is instituted till an indication comes to picture.
 - Conservative management in preterm lady includes:
 - ✓ Bed rest (in hospital) -----No place for outpatient management
 - ✓ Dexamethasone 6mg IM Q 12 hours for a total of 4 doses
 - ✓ Follow: Maternal V/S, vaginal bleeding, uterine contractions, fundal height, FHB, kick count, BPP, serial HCT.
 - Deliver at 37 completed weeks of gestation after maturity is confirmed
- Double setup examination:
 - Is used in areas where U/S is not widely available/not done by experienced people.
 - It is important to decide on mode of delivery
 - The procedure include: -
 - ✓ Prep and take client to operation room
 - ✓ Everything must be ready for delivery
 - ✓ Speculum examination is done gently to rule out local causes & see the cervical status → then gentle digital vaginal examination is done to check for presence of placenta between fornices & presenting part, cervical dilatation & effacement, presence of placenta through open cervix.

Morbidly adherent placentas

Morbidly adherent placenta (a.k.a **accrete syndromes**) describes aberrant placentation characterized by abnormally implanted, invasive, or adhered placenta.

Classified by the depth of trophoblastic growth, as: -

1. Placenta accrete; indicates that villi are attached to the myometrium
2. Placenta increta; villi actually invade the myometrium, and
3. Placenta percreta; defines villi that penetrate through the myometrium and to or through the serosa.

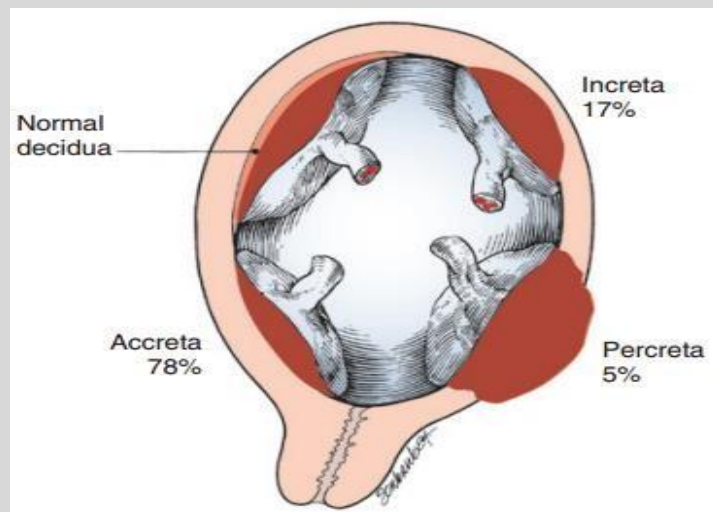


Fig 3. Uteroplacental relationships found with invasive placentation

Incidence

- The prevalence of accrete syndromes was 3.7 per 1000 births—1 per 270

Risk factors

- Placenta previa
- Prior cesarean delivery
 - A classical hysterotomy incision has a higher risk for a subsequent accrete placenta

- Risk of placenta accreta with placenta previa and prior cesarean delivery

Number of prior CD	Placenta accrete risk (%)
0	3
1	11
2	40
3	61
≥4	67

- Increasing parity and maternal age
- Submucosal uterine fibroids
- Curettage or endometrial ablation

Clinical manifestation

- In cases of first- and second-trimester accrete syndromes, there is usually hemorrhage that is the consequence of coexisting placenta previa. Such bleeding will typically prompt evaluation and management.
- In some women who do not have an associated previa, accreta may not be identified until third-stage labor when an adhered placenta is encountered. Unfortunately, imaging modalities are less than perfect to identify all of these placentas early.
- Hematuria can be a feature of placenta percreta with bladder invasion

Diagnosis

Radiographic techniques: Ultrasound is the preferred radiographic modality for the diagnosis of placenta accreta. Findings suggestive of placenta accreta include:

- loss of the normal hypoechoic retroplacental myometrial zone,
- thinning and disruption of the uterine serosa–bladder wall interface,
- focal exophytic masses within the placenta, and numerous intraplacental vascular lacunae

Laboratory findings: Placenta accreta has been associated with unexplained elevations in maternal serum α -fetoprotein (MSAFP).

Pathologic studies: Placenta accreta is confirmed by the pathologic examination of a hysterectomy specimen.

Management

- Because of its associated risk for massive postpartum hemorrhage, placenta accreta accounts for a large percentage of peripartum hysterectomies.
- When performing the surgery, it is recommended that the uterus be incised above the placental attachment site and that the placenta be left in situ after clamping the cord because disruption of the implantation site may result in rapid blood loss.
- Timing of delivery depends upon clinical circumstances; however, most authorities favor delivery at 34⁰/7 to 35⁶/7 weeks with or without antenatal corticosteroid administration.
- Adequate IV access with two large-bore catheters and ample blood product availability are mandatory.
- In specific circumstances, uterine conservation may be attempted. These situations include focal accreta, desired future fertility, and fundal or posterior placentae accrete.

Vasa Previa

- Vasa previa is defined as the presence of fetal vessels over the cervical os.
- These fetal vessels lack protection from Wharton jelly (velamentous cord insertion) and are prone to rupture and compression. When the vessels rupture, the fetus is at high risk for exsanguination.

Incidence

- The overall incidence of vasa previa is 1 in 2500 deliveries; however, data have shown a range from 1 in 2000 to 1 in 5000 deliveries.

Risk factors

- Bilobed and succenturiate-lobed placentas
- Pregnancies that result from assisted reproductive technology (ART)
- Multiple gestations
- History of second-trimester placenta previa or low-lying placenta.

Clinical manifestations

- Most cases of vasa previa presented after rupture of membranes with acute onset of vaginal bleeding from a lacerated fetal vessel. If immediate intervention was not provided, fetal bradycardia and subsequent fetal death occurred.
- Recently, many cases of vasa previa are diagnosed antenatally by ultrasound. In rare cases, pulsating fetal vessels may be palpable in the membranes that overlie the cervical Os.

Diagnosis

- Vasa previa is often diagnosed antenatally by ultrasound with color and pulsed Doppler mapping. Transabdominal and transvaginal approaches are most often used.

Management

- When diagnosed antenatally, vasa previa should be managed similarly to placenta previa.
- If an intrapartum diagnosis of vasa previa is made, expeditious delivery is needed. Immediate neonatal blood transfusion is often required in these circumstances.

SAMPLE HISTORY

Chief complaint: Vaginal bleeding of 1 hour duration

HPP

This is a 30years old G₄P₃ (all alive) mother whose LNMP was on 22/05/2011EC (reliable) making the EDD on 27/02/2012EC and gestational age of 36+1 weeks (on 30/01/2012) EC, date of clerking). Menses was regular coming every 26-28 days, lasting for 3-4 days. She used 2-3 sanitary pads per day, menses has no clots associated and she has no abdominal discomfort. She has no history of contraceptive use. She doesn't remember the exact date she felt fetal movement but claimed that she still feels the kicks and it's increasing through time.

She suspected she was pregnant when she missed her menses and went to a nearby health center where the diagnosis of her pregnancy was confirmed by a urine pregnancy test, (write the exact date of urine pregnancy test or ultrasound if done and if she can recall).

She had total of four ANC visits at a local health center. First visit was at 3month of pregnancy (write exact date if available), where history was taken, weight, height and blood pressure were measured, and abdominal examination was done. Blood and urine test was done, she doesn't remember any results, she was told everything was fine. She was given an injection on her left upper arm and a red round tablet to take once daily. She was appointed after a month.

Same physical examinations and investigation (only urine examination) were done on her 2nd, 3rd and 4th visits and they were all uneventful. She had another injection on her left upper arm on the 2nd visit and had a red round tablet refill in all the visits.

The pregnancy was not planned but is wanted and supported by her family. She hasn't made any significant changes in her diet after she got pregnant.

Her staple diet is Injera made of teff with shiro wot 3 times a day, fruits and vegetables occasionally and meat on rare occasions.

Currently she presented with a sudden onset of vaginal bleeding which was bright red, non-clotting and odorless. The blood soaked her underwear and dress. It was not associated with abdominal pain, gush of fluid or decreased fetal movement.

She has no history of easy fatigability, blurring of vision, ringing in the ears or palpitation (symptoms of anemia).

Otherwise:

Danger signs

- She has no history of cough, or shortness of breath
- She has no history of vision disturbances, ABM or LOC
- She has no history of epigastric pain, or right upper quadrant pain.
- She has no history of headache
- She has no history of excessive vomiting
- She has no history of urinary frequency, decreased urine output or pain during urination
- She has no history of passage of liquor or pushing down pain

Risk factors

- She has no history of trauma to the abdomen
- She has no history of cigarette smoking or alcohol drinking
- She has no history of infertility treatment
- She has no history of twinning
- She has no history of gynecologic operations
- She has no similar previous history

DDx

- She has no history of post-coital bleeding (Cervical ca.)
- She has no history of river water contact (Leech infestation)
- She has no history of bleeding from other sites (Autoimmune disorders like ITP)
- She has no history of vaginal discharge, genital burning sensation (cervicitis)

PHYSICAL EXAMINATION

V/S: BP, pulse, RR, TO, Weight, Height, BMI.....normal

HEENT: Pale conjunctiva

Abdominal examination: thorough examination,

(See the Preeclampsia history in chapter 1 to know how to write findings in obstetric palpation)

Pelvic examination (avoid Per Vaginal examinations unless placenta previa has been ruled out)

Assessment: Late preterm pregnancy + APH secondary to R\O Placenta previa

Chapter 6

PROM & PRETERM LABOR

Prepared by Dr. Bethel Ayele (MI), edited

Definitions

Preterm rupture of membrane: is spontaneous rupture of membrane occurring before 37th week of gestation.

Premature rupture of membrane (PROM): is spontaneous rupture of membrane occurring before onset of labor.

- Currently the term **Prelabor ROM** is also used.

Preterm premature rupture of membrane (PPROM): is if both occur together i.e. if spontaneous rupture of fetal membrane occurs before 37 completed weeks and before the onset of labor.

Prolonged rupture of membrane: is any ROM for greater than 12 hours (8hrs, JUMC).

- It becomes a risk for **chorioamnionitis** when it is longer than 18 hours.

Latency Period: is the period between rupture of membrane and onset of labor.

PRETERM LABOR/BIRTH

Classification of the time of labor onset or delivery

	Classification	GA (Wks.)
Preterm	Very early preterm	28-31 ⁺⁶
	Early preterm	32-33 ⁺⁶
	Late preterm	34-36 ⁺⁶
Term	Early term	37-38 ⁺⁶
	Full term	39-40 ⁺⁶
	Late term	41-41 ⁺⁶
Post term		≥42 completed

Slightly different classification from the 2021 Ethiopian MOH Obstetric Management Protocol

- Early preterm: 28–32 completed weeks.
- Moderate preterm: 32 plus 1 day to 33 weeks plus 6 days.
- Late preterm: 34 completed weeks –36 weeks plus 6 days.

Preterm birth:

Describes neonates who are born early, i.e. born before term (before 37 full weeks), but after fetal viability (after 28 full weeks).

- It is with respect to gestational age, but not with respect to size, neonates can be born before term and be small for gestational age, appropriate for gestational age or large for gestational age.

Etiopathogenesis and risk factors

The **four** primary processes in pathogenesis of preterm labor are:-

1. Premature activation of the maternal or fetal hypothalamic- pituitary-adrenal axis
2. Inflammation or infection
3. Decidual hemorrhage
4. Pathological uterine distention

Most risk factors of PTL will result in one or more of the above mechanisms,

Risk factors of PTL

- **Prior preterm birth:** is the most important risk factor
 - It increases risk by **three fold** greater when compared with women whose first neonate was born at term.
 - The risk of recurrent preterm birth is influenced by three factors:
 - ✓ frequency of prior preterm deliveries,
 - ✓ severity as measure by gestational age, and
 - ✓ the order in which the prior preterm delivery occurred (consecutively? Or every other pregnancy)
- **Infection:**
 - chorioamnionitis is the most common associated condition in our setup
- **PPROM**
- **Multifetal pregnancy**
- **Hydramnios,**

- **APH/bleeding during px,**
 - Threatened abortion in early pregnancy is associated with later adverse outcomes.
 - Both light and heavy bleeding are associated with subsequent preterm labor, placental abruption and pregnancy loss before 24 week
- **Interval between pregnancies:**
 - Intervals <18 months and >59 months were associated with risks for both preterm and small-for-gestational age newborns.
 - Short interpregnancy interval (<6month) is associated with higher risk
- **Uterine condition: Like: -**
 - Mullerian malformation
 - Cervical insufficiency
- **Obstetric conditions:**
 - Preeclampsia, eclampsia, fetal growth restriction, fetal macrosomia, ECV, amniocentesis
- **Other medical conditions:**
 - Chronic hypertension, DM, renal disease, malaria, UTI, HIV, Syphilis
- **Birth defects may also predispose to preterm birth.**
- **Lifestyle factors:**
 - Cigarette smoking, illicit drug use
 - Inadequate maternal weight gain and underweight, young or advanced maternal age, poverty, short stature, vitamin C deficiency.
 - obesity
- **Psychological factors, like depression,**
- **Genetic factors:**
 - Recurrent, familial and racial nature of preterm birth suggests that genetics may play a causal role.

Diagnosis

History

- **Patient may complain:**
 - vaginal bleeding or bloody show
 - pelvic pressure, menstrual-like cramps & lower back pain
 - increased vaginal discharge
 - Gush of fluid =>rupture of membrane

Physical examination

- Four uterine contractions per 20 minutes or eight contractions per 60 minutes which are accompanied by one of the following:
 - Rupture of membrane
 - Cervical dilation greater than 2 cm
 - Effacement exceeding 80%
 - Change in cervical dilation or effacement detected by serial examinationssterile speculum examination is essential to assess fetal and cervical status.
- Early differentiation between true and false labor is difficult especially before demonstrable cervical effacement and dilation.
 - Uterine activity alone can be misleading because of **Braxton Hicks Contractions** (irregular, non-rhythmical contraction).
 - Therefore, preterm labor is defined as regular contractions before 37 weeks that are associated with cervical change.

Investigation

- If there is vaginal bleeding present, an ultrasound must be performed to rule out placenta Previa before digital examination is performed.
- Urine specimen is sent for culture and fetal heart rate monitoring is done.

Management of PTL

- Treatment depends on gestational age, estimated fetal weight, presence or absence of contraindications for tocolytics.
- Preterm labor should be managed in a setup where there is best possible neonatal care of the preterm newborn. Hence, in-utero transfer should be considered whenever possible.

Management of preterm labour includes:

- **Corticosteroids**
 - Dexamethasone 6 mg IM BID for 48 hours or betamethasone 12 mg every 24 hours for 48 hours.
 - A single repeat course of antenatal corticosteroid is recommended if preterm birth does not occur within 7 days after the initial dose, and a subsequent clinical assessment demonstrates a high risk of preterm birth in the next 7 days.
 - This recommendation should only be applied if the gestational age is less than 34 weeks of gestation.

▪ Tocolytics:

- Provide window for administration of antenatal corticosteroids and/or in-utero fetal transfer to an appropriate neonatal health care setting;
- Tocolytic therapy is considered when cervical dilatation is less than 4cm; uterine contraction is fewer than 4-5 within an hour with no cervical change.
- Nifedipine is the preferred drug for tocolysis. Do not give a combination of tocolytic agents as there is no additional benefit. See box below for dosing of Nifedipine.
- *Tocolytic therapy is considered when:*
 - ✓ Contractions are resulting in a demonstrated cervical change and cervical dilatation is less than 4 cm.
 - ✓ Uterine contraction is more than 4-5 within an hour with no cervical change.
- *Contraindications for tocolytics include: -*
 - ✓ Preterm prelabor rupture of membranes (PPROM),
 - ✓ Cervical dilatation >4 cm and effacement >80%.
 - ✓ Chorioamnionitis,
 - ✓ Antepartum hemorrhage,
 - ✓ Cardiac disease,
 - ✓ Fetal death,
 - ✓ Fetal congenital abnormality not compatible with life

Nifedipine dose

- ✓ Loading oral dose of 20 mg followed by 10– 20 mg every 4–8 hours for up to 48 hours.
- ✓ Inform the woman to be aware of side effects of Nifedipine such as headache, flushing, dizziness, tiredness, palpitations and itching.
- ✓ Monitor maternal and fetal condition: pulse, blood pressure, signs of respiratory distress, uterine contractions, loss of amniotic fluid or blood, fetal heart rate, fluid balance

▪ Neuroprotection:

- Administer MgSO_4 up to 32 weeks of gestation to prevent preterm birth-related neurologic complications (neuroprotection). See box below for dosing of MgSO_4

▪ Antibiotics:

- Antibiotics should be administered for spontaneous preterm labour with unknown GBS status.

- Provide Ampicillin 2 gm IV as initial loading dose then 1gm IV every four hours until delivery for GBS prophylaxis
- **Labor and delivery**
 - Avoid vacuum-assisted birth for pregnancies less than 34 weeks of gestation
 - Prepare for management of preterm or low birth weight baby and anticipate the need for resuscitation.

Prevention

Can be described according to the public health model as

- **Primary** (prevention and reduction of risk in the population).
 - reduce the prevalence of smoking, increase the use of condoms to prevent sexually transmitted infections
 - weight control
 - regulate artificial reproductive ways, to reduce high order pregnancies
- **Secondary** (identification and treatment for individuals with increased risk)
- **Tertiary** (treatment initiated **after** the parturitional process has begun to limit perinatal morbidity and mortality)

Secondary preventions

Cerclage:

- Women who have the following conditions can benefit from application of cerclage:
- Pregnant women who have history of recurrent mid trimester pregnancy losses and who are diagnosed with cervical insufficiency (*history indicated cerclage*).
- Pregnant women who are diagnosed to have short cervical length (< 25 mm) and have history of preterm birth (*ultrasound indicated cerclage*)
- Pregnant women who are accidentally found to have dilated cervix prior to 24 weeks of gestation can benefit from emergency/ rescue cerclage.

Progesterone compounds:

- The criteria for progesterone are a history of prior preterm birth or no prior preterm birth but a sonographically identified short cervix.
- Progesterone vaginal tablet 100 mg to 200 mg vaginally each night or progesterone caproate 250 mg IM weekly starting from 16 weeks of gestation until 36 weeks of gestation can be considered

PREMATURE/PRELABOR RUPTURE OF MEMBRANE (PROM)

PPROM

- Affects about 1% of pregnancies overall
- *The relative contribution of PROM to prematurity appears to vary greatly among patient populations, affecting about 10% of preterm births in national databases but over 20% in certain high-risk populations*
- The risk factors for PPRM are similar to those for preterm labor. And the pathogenesis is similar with PROM the only difference is that it occurs before term (preterm).

PROM

- Affects approximately 8-10% of all pregnancies
- About 94% of cases occur at term and 5% preterm.
- It is associated with considerable maternal infection and perinatal morbidity and mortality due to sepsis and prematurity.

Etiopathogenesis and Risk factors

As the pregnancy progresses, changes in ***collagen content and type, intercellular matrix, and cellular apoptosis*** result in structural weakening of the fetal membranes.

- Membrane remodeling is more evident near the internal cervical Os and can be stimulated by
 - Thrombin-mediated increases in matrix metalloproteinases (e.g., MMP-1, MMP-2, MMP-9) and
 - Decreased levels of tissue inhibitors of matrix metalloproteinases (e.g., TIMP-1, TIMP-3) within the membranes, as well as
 - Increased poly (ADP-ribose) polymerase (PARP) cleavage.

Contractions subject the amniochorionic membranes to additional physical strain that can lead to membrane rupture.

Preterm PROM likely results from a variety of factors that ultimately ***lead to accelerated membrane weakening*** through: -

- an increase in local cytokines and an imbalance in the interaction between MMPs and TIMPs,
- increased collagenase and protease activity, or
- other factors that cause increased intrauterine pressure

Most PROM cases occur due to subclinical chorioamnionitis resulting from ascending infection from lower genital tract.

- Genital infections produce phospholipases which stimulate production of prostaglandins which further stimulate uterine contraction and subsequent ROM.

Risk factors for PROM

Maternal	Fetal
▪ Subclinical or Overt chorioamnionitis ▪ Past history of PROM ▪ Prior history of spontaneous preterm delivery ▪ Vaginal bleeding (2nd and 3rd trimester bleeding) ▪ Low body mass index and nutritional deficiency ▪ Collagen synthesis defect ▪ Cervical incompetence ▪ Cigarette smoking ▪ Emergency cerclage ▪ Low socioeconomic status ▪ Pulmonary disease (chronic cough) ▪ Straining (constipation) ▪ Heavy physical exercise	▪ Polyhydramnios (uterine over distention) ▪ Multiple pregnancy ▪ Malpresentation Iatrogenic ▪ Amniocentesis ▪ External cephalic version ▪ Cervical conization ▪ Percutaneous umbilical blood sampling ▪ Abdominal trauma

Diagnosis and Investigations

- PROM can be diagnosed from historical claim of gush of fluid through vagina by the pregnant lady

The diagnosis of membrane rupture is confirmed by the presence of the following findings:

- Visualization of amniotic fluid passing from the cervical canal or
- Vaginal sidewall or posterior fornix pH of more than 6.0 to 6.5 and
- Microscopic arborized crystals ("Ferning"), owing to the interaction of amniotic fluid proteins and salts, from dried vaginal secretions obtained by swabbing the posterior fornix with a sterile swab.

The diagnosis of membrane rupture can be made **unequivocally** by **ultrasound-guided dye amnioinfusion** (1 mL indigo carmine plus 9 mL sterile saline), followed by observation for passage of dye onto a perineal pad.

A sterile speculum examination

- is performed to evaluate the fetal membrane status and to inspect the cervix
- this examination is a key to differentiating PROM from hydrorrhea gravidarum, vaginitis, increased vaginal secretions and urinary incontinence
- the cervix should also be visually inspected to determine the degree of dilatation and effacement and the **presence of cord prolapse**
- Findings suggestive of ROM
 - Pooling:
 - ✓ Is the collection of amniotic fluid in the posterior fornix
 - Positive Nitrazine test:
 - ✓ This test is based on the pH difference between the vagina and AF (normally vagina pH is 4.5 to 6.0 and the pH of amniotic fluid is 7.1 to 7.3).
 - ✓ Nitrazine test is a sterile cotton tipped swab that is used to collect fluid from the posterior fornix and apply it to nitrazine (phenolphthazine) paper; **a change from yellow to blue suggests presence of AF.**
 - ✓ **False positive** results can be due to the presence of alkaline fluids in the vagina, such as blood, seminal fluid, or soap. In addition, the pH of urine can be elevated to near 8.0 if infected with Proteus species
 - Ferning test:
 - ✓ Fluid from the posterior fornix is placed on a slide and allowed to air-dry.
 - ✓ Amniotic fluid will form a fern-like pattern of crystallization.
 - ✓ This test is not affected by meconium, vaginal PH & blood.
 - ✓ Cervical mucus can cause ferning, but it is usually patchy and less extensive than that with AF.
 - Observing loss of fluid from the cervical Os when patient coughs or performs a Valsalva maneuver during speculum examination
 - ✓ If no free fluid is found, a dry pad should be placed under the patient's perineum and observed for leakage

Digital examination

- ***should be avoided*** because it
 - may decrease the latency period (i.e. time from rupture of membranes to delivery) and
 - increase the risk of chorioamnionitis

Ultrasound examination

- Oligohydramnios

Noninvasive cervicovaginal markers

- Such as fFN, AFP, prolactin, human chorionic gonadotropin (hCG), placental α -microglobulin 1 (PAMG-1), and insulin-like growth factor-binding protein 1 (IGFBP-1) have been studied for their ability to confirm or exclude membrane rupture, but most are unavailable for clinical use

Lab evaluation for chorioamnionitis

- CBC, CRP, (see chapter ... for chorioamnionitis)

Differential Diagnosis

- Leucorrhea gravidarum
- Vaginal discharge (excessive)
- Urinary incontinence
- Perspiration
- Ruptured cyst (like in the vagina)

Complications

Maternal

- ***Chorioamnionitis*** is the most common maternal complication after PPRM
 - This risk increases as the duration of membrane rupture becomes more prolonged and decreases with advancing GA
 - The risk varies in different patient populations (13% to 60% for chorioamnionitis and 2% to 13%, for endometritis).
- ***APH***
 - Abruptio placentae can cause PROM or can occur subsequent to membrane rupture, and ***it affects 4% to 12% of these pregnancies***

- **Preterm labor/birth**
- Uncommon but serious complications of PROM managed conservatively near the limit of viability include
 - Retained placenta and hemorrhage, requiring dilation and curettage (12%)
 - Maternal sepsis (0.8%); and
 - Maternal death (0.14%).
- Operative deliveries: due to failed induction
- Prolonged hospitalization and financial implication

Fetal and Neonatal Risks

- **Fetal distress** due to umbilical cord compression or placental abruption
 - Umbilical cord compression due to oligohydramnios is not uncommon after PROM.
 - Frank or occult umbilical cord prolapse can also occur, particularly with fetal malpresentation.
- **Fetal death** complicates 1% to 2% of cases of conservatively managed PROM
- **Prematurity** and its complication
 - RDS, IVH, NEC,
 - ✓ **Pulmonary hypoplasia** is a severe complication of **oligohydramnios** in the second trimester ROM.
 - results from a lack of terminal bronchiole and alveolar development during the canalicular phase of pulmonary development
- Fetal and **neonatal sepsis** (EONS)
 - PROM that is prolonged for >18hrs is a RF for EONS
 - PPROM increases the risk of neonatal sepsis **two fold**

Management

Management of PROM Management depends on:

- **Gestational age**, duration of rupture of fetal membranes, fetal condition (dead, fetal distress, congenital malformations), presence of infection, labor and previous cesarean scar, and other obstetric indications.

Indications for expedite delivery:

- onset of labor,
- gestation age ≥ 37 wks,
- evidence for nonreassuring fetal status,
- evidence for chorioamnionitis,
- lethal congenital anomalies,
- intrauterine fetal death,
- if there is high risk of cord prolapse (e.g., transverse lie) and
- abruptio placenta
 - Note that if the gestational is below 34 weeks and both the fetal and maternal conditions are stable, expectant management can be considered for abruptio placenta in a setting where close follow up is possible

Expectant management

- Admit to the ward (Transfer patients with early preterm PROM to a higher health facility with newborn intensive care, if possible).
- Avoid digital cervical (pelvic) examination.
- Advise bed-rest, to potentially *enhance amniotic fluid re-accumulation & possibly delay onset of labor.*
- Provide corticosteroids and antibiotics

Corticosteroids

- Administer antenatal corticosteroids (betamethasone 12 mg intramuscularly 24 hours apart for two doses or *dexamethasone 6 mg IM 12 hours apart for four doses*) for lung maturity.
- Note that if preterm birth is considered imminent, treatment for short duration still improves fetal lung maturity and chances of neonatal survival. Therefore, the first dose of corticosteroids should be administered even if the ability to give the second dose is thought to be unlikely.
- Antenatal corticosteroid therapy should not be administered in women with chorioamnionitis.

Antibiotics

- Ampicillin 2gm IV QID and Erythromycin 250 mg P.O QID for 48 hours followed by Amoxicillin 500 mg P.O TID & Erythromycin 250 mg. P.O QID for 5 days.
- Azithromycin may be substituted for Erythromycin with regimen of 500mg PO on day 1 followed by 250mg PO daily for 6 days.

- In our setup ceftriaxone 1gm IV BID is given for 48hrs together with Erythromycin 500 mg P.O TID, then ceftriaxone is changed to cephalexin 500mg P.O TID for 5days
- If there is onset of labor and in the absence of signs of uterine infection, discontinue antibiotics after delivery.

Neuroprotection

- If gestational age is less than 32 weeks and preterm birth is likely within the next 24 hours, consider magnesium sulfate for neuroprotection

Monitoring and follow up

- Monitor the following clinical features during expectant management of PROM:
- Maternal pulse & temperature - every 4-6 hours
- FHR - every 4-6hrs (& if possible CTG 2x daily)
- Uterine tenderness or irritability (or pain) – daily
- WBC count & differential - changes, every 2-3 days
- Amniotic fluid appearance & odor – daily
- If possible, examine for presence of subclinical intraamniotic infection with amniocentesis.

If there is a need for termination of expectant management, deliver the baby. If there is no contraindication for priming, induction or vaginal delivery, ripen the cervix and induce labor.

Labor and delivery for term PROM without infection:

- If cervix is favorable, labor is induced, unless there are contraindications to labor or vaginal delivery, in which case cesarean delivery is performed.
- If cervix is unfavorable, ripen the cervix (preferably with PO misoprostol).
- Institute antibiotic (Ampicillin 2gm IV QID) when the duration of ROM >12hrs.
- Follow for features of chorioamnionitis (maternal fever, tachycardia, leukocytosis, uterine tenderness, offensive vaginal discharge and fetal tachycardia).
- The antibiotic should continue throughout labor and for at least one dose after delivery.

Management of near-term PROM (34-37 weeks)

- Induction or expectant management is acceptable management options depending on local resources.

Sample history

C/c: Gush of fluid of 4hour duration

HPP

This is a 25years old G2 P1 (alive) woman who had her LNMP on 10/05/11 EC with an EDD of 15/2/12 EC and GA of 37+1 (on 30/01/12 EC). Menses was regular and it came every 28-30 days, lasting for 4 days and there was no associated clot and discomfort. She never used contraception.

The pregnancy was first confirmed when she went to a local health center for missing her menses for two weeks and got suspicious of being pregnant. And after urine sample was taken she was told she was pregnant. She had 6 ANC follow ups at a local hospital. Her 1st ANC visit was on her second month of pregnancy. Complete history was taken and physical examination was done. Urine sample, blood sample was taken and U/S (ultrasound) was done. She remembers her blood type is B+ (we ask if she knows her blood type because if she is RH negative we have to make sure she took/takes anti-D), but she does not remember her other results. She was given an injection on her left arm (we can ask if she knows what the injection was for- TT (tetanus toxoid) vaccine) and she was given red tablet to take one tablet per day (iron supplementation).

And she was told everything was on good status. Her 2nd ANC follow up was at her third month of pregnancy. Complete physical examination was done and she was given another injection on her arm (second dose of TT vaccine). Her iron supplement was continued and she was told everything was good and that there was no problem. Her 3rd, 4th, 5th and 6th were at her 4th, 5th, 6th and 7th months of pregnancy. Routine history taking and complete physical examination was done and it was uneventful. On her 5th visit she was advised on birth. And U/S was done on her 6th visit.

She doesn't remember when she started to feel fetal movement, but fetal movement has not decreased since first felt. She had an increase in appetite during the pregnancy. She increased her meals from 3 times per day to 5 times per day.

Her staple food is injera made of 'tef' and 'shiro wot' and 'denech wot'. She eats vegetables and fruits. She occasionally eats meat. She claims she gained around 5kg since she changed her diet.

The pregnancy was wanted, planned and supported.

4 hours prior to admission she felt a sudden onset of leakage of gush of fluid while she was washing her hands and it soaked her clothing. Its consistency is water like.

The fluid was clear watery and it was odorless, and there was no associated pushing down pain. Associated to this she had back pain. 1 hour prior to admission while she was on the way to the hospital she started to have pushing down pain.

Otherwise: -

Risk factors

She has no history of fever, chills

She has no history of foul smelling vaginal discharge (besides showing sign of infection, large amount of vaginal discharge can be mistaken for leakage of liquor)

She has no prior history if PROM

She has no self or family history of spontaneous preterm delivery

She has no history of vaginal bleeding during pregnancy

She has no history of smoking (or alcohol consumption) She has no history of trauma to the abdomen

She has no history of chronic cough or straining during pregnancy She denies and serious stress (stress incontinence)

Danger signs

She has no history of blurring of vision

She has no history of epigastric pain or right upper quadrant pain

She has no history of abnormal body movement or loss of consciousness

She has no history of headache

She has no history of nausea or excessive vomiting

She has no history of swelling of the face and hand (swelling of independent areas of the body)

She has no history of flank pain

She has no history of cough or shortness of breath Differential diagnosis

She has no frequency, urgency (urinary incontinence) Others

She has no history of blood transfusion

She has no personal or family history of HTN or DM

Her sero-status for RVI is negative

(Then state her economic status)

Physical Examination

Vital sign: All, patient may be febrile (?)

HEENT: Conjunctiva

Abdomen: Check for uterine tenderness, uterine size (if fundal height less than that of expected for gestational age, it indicate oligohydramnios)

Pelvic examination:

- External genital examination: may show ongoing AF flow
- Speculum examination: the most important examination in PROM (see the diagnosis section above)
- *Once membrane rupture has been confirmed, **avoid digital vaginal examination until labor or induction of labor** (avoid in PPROM given the risk of infection).*

Investigation

- Investigations (Tests) that confirm diagnosis
 - See above in the diagnosis section
- Other investigation
- CBC with differential
- Blood type and RH
- Urinalysis, culture and sensitivity testing done by collecting urine by catheterization
- Amniocentesis
 - If between 32 and 34 weeks, amniotic fluid specimen sent for fetal lung maturity test
 - In all women with PPROM before 34 weeks to test for evidence of intra-amniotic infection
- High vaginal swab for culture
- Ultrasound
 - Fetal size and amniotic fluid index
 - Biophysical profile

Chapter 7

MALPRESENTATION

Prepared by Dr. Bemnet G/Micheal (MI), edited

Introduction

The ability of the fetus to successfully negotiate the pelvis during labor and delivery is dependent on the complex interaction of three variables

- **Passenger**
 - Especially *fetal presentation/position* and Weight
- **Passage**
 - Maternal pelvis, both bony pelvis and Soft tissue structures
- **Power**
 - Uterine contraction

Shapes of Human Pelvis

1. Gynecoid pelvis: Classic female pelvis (50% of women)
2. Android Pelvis: Typical male pelvis (25% of women)
3. Anthropoid pelvis; predisposes to **occiput posterior**
4. Platypelloid pelvis (*rare*); predisposes to **transverse lie** hence shoulder presentation

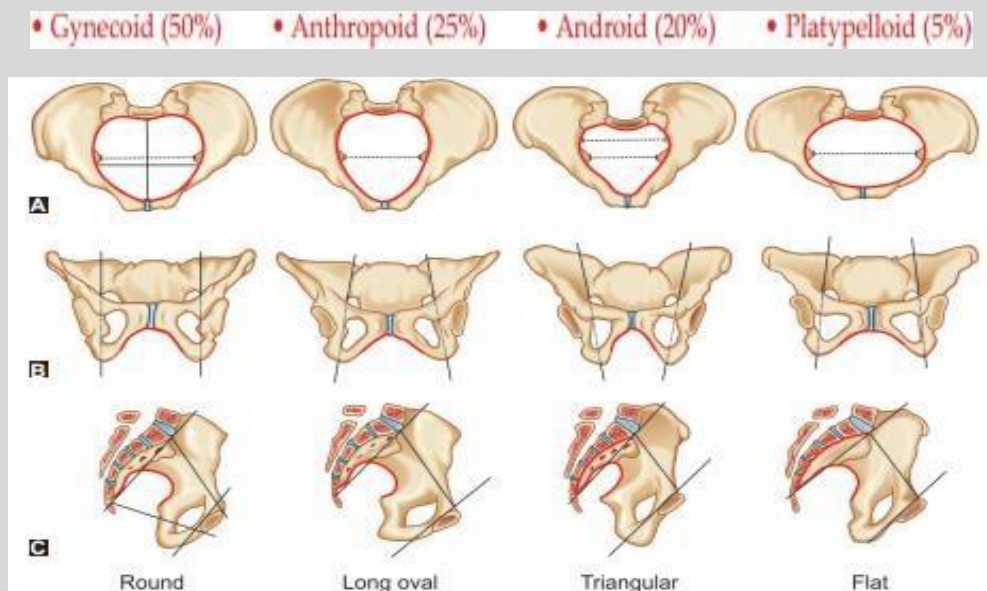


Fig. Anatomical features of parent pelvic types: (A) Inlet; (B) Cavity; (C) Outlet

Definition of Terms

▪ Fetal presentation

- Part of fetus overlying the pelvic inlet, or part of the fetus that occupy the lower uterine segment
- The normal fetal presentation is **vertex/cephalic** (**vertex**, NB. Vertex presentation is a PV examination finding)
 - ✓ NB. Vertex and cephalic aren't the same

▪ Fetal lie

- The fetal lie indicates the orientation of the fetal spine relative to the spine of the mother.
- The normal fetal lie is **longitudinal** and by itself does not indicate whether the presentation is cephalic or breech

▪ Fetal position

- Is the relationship of a definite fetal part to maternal bony pelvis.
- It's different in early and late labor.
 - ✓ In early labor; Left occipitotransverse
 - ✓ At delivery; **Occipitoanterior** are normal findings

▪ Fetal attitude

- It's the degree of extension and flexion of fetal head. Can be classified as:
 - ✓ **Flexed**: Vertex; most common, chin is pressed against the chest.
 - ✓ Military; head is straight
 - ✓ Extended; Brow (increased extension) and Face (complete extension)

▪ Fetal station

- Is the degree of descent of the presenting part through the birth canal
- It's expressed in centimeter above or below the maternal **ischial spine**.
 - ✓ If at the level of the spines it is at "0 (zero)" station, if it passed it by 2cm it is at "+2" station

▪ Presenting part

- Is defined as part of the fetus that occupies the lower uterine segment
 - ✓ The **vertex** is the normal and most common presenting part
 - ✓ The alternatives include: - face, brow, breech, shoulder, or compound

Malpresentation

- is **any presentation other vertex** presentation, with the flexed fetal vertex presenting to the pelvis

Incidence and Etiologies/Risk factors

- Malpresentation is seen in up to **3% to 5% of singleton gestations at term.**

Factors associated with malpresentation include:

1. Diminished vertical polarity of the uterine cavity
2. Increased or decreased fetal mobility,
3. Obstructed pelvic inlet,
4. Fetal malformation, and
5. Prematurity.

The Risk factors mentioned below will result in one or more of these (the above 5)

Maternal	Fetal	Placental and membrane
▪ High parity ▪ Pelvic shape ▪ Pelvic contracture ▪ Pelvic tumors ▪ Uterine malformations	▪ Polyhydramnios ▪ Oligohydramnios ▪ Multiple pregnancy ▪ Fetal macrosomia ▪ Fetal anomalies ▪ Fetal attitude	▪ Cornual placenta ▪ Placenta previa ▪ PROM

Malpresentations Includes: - Breech, Face, Brow, Compound and Shoulder presentations

Diagnosis of fetal presentation

- **Abdominal examination**
 - Experienced clinicians can identify malpresentation with 88% sensitivity and 94% specificity.
 - Can be difficult if:
 - ✓ The patient is Obese
 - ✓ There is polyhydramnios
 - ✓ The placenta is implanted anteriorly
- Digital vaginal examination and
- Ultrasound examination
 - Especially in Obese women and those with rigid abdomen

Breech presentation

- The most common malpresentation and the infant presenting occupy a longitudinal axis with the cephalic pole in the uterine fundus
- It occurs in 3% - 4% of labors overall, although it is found in 7% of pregnancies at 32 weeks and in 25% of pregnancies of less than 28 weeks' duration.

Classification

Based on their varying relations between the lower extremities and buttocks, breech presentation can be classified as

- Frank,
- Complete and
- Incomplete breech

Types of breech presentation

	Frank Breech	Complete Breech	Incomplete (Footling) Breech
Features	Lower extremities are flexed at the hips and extended at the knees	Both hips are flexed, and one or both knees are also flexed	Both hips & knees are partially flexed or knees may even be extended
Presenting part	two buttocks and external genitalia only	two buttocks, external genitalia and two feet	one or both feet (single/double footling breech)
Incidence	Common 60-70% (Commonly present in primigravida*; about 70%)	Less common (10%) occurs mainly in multigravida → 60%	is about 25%
Risk of cord prolapse	0.5-1%	5%	15%(15-20%)
Association with prematurity	38%	12%	50%
Risk of labor abnormality	<complete breech	<Incomplete/footling	High

**The increased prevalence in primigravida is due to a tight abdominal wall, good uterine tone and early engagement of breech*



Fig. A. Frank Breech B. Complete Breech C. Incomplete (Footling) breech

Etiology and risk factors

The following are the known factors responsible for breech presentation. In a significant number of cases, the cause remains obscure.

- **Prematurity:** It is the commonest cause of breech presentation.
- **Factors preventing spontaneous version:**
 - Breech with extended legs,
 - Twins,
 - Oligohydramnios,
 - Congenital malformation of the uterus such as septate or bicornuate uterus,
 - Short cord, relative or absolute,
 - Intrauterine death of the fetus
- **Favorable adaptation:**
 - Hydrocephalus—big head can be well accommodated in the wide fundus,
 - Placenta previa,
 - Contracted pelvis
 - Cornufundal attachment of the placenta—minimizes the space of the fundus where the smaller head can be placed comfortably.
- **Undue mobility of the fetus:**
 - Hydramnios
 - Multipara with lax abdominal wall.
- **Fetal abnormality:**
 - Trisomies 13, 18, 21, anencephaly and myotonic dystrophy due to alteration of fetal muscular tone and mobility.

Diagnosis

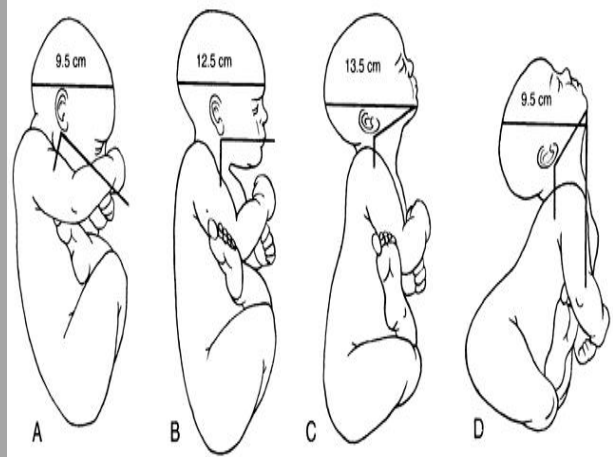
The diagnosis of breech presentation may be made by abdominal obstetric palpation or vaginal examination and confirmed by ultrasound.

P/E:

- Leopold's maneuver

- First maneuver, the hard, ballotable, round fetal head occupies the fundus.
- Second maneuver, identifies the smooth surface (back) to be on one side of the abdomen and the small irregular parts (extremities) on the other.
- Third maneuver, if not engaged, the softer breech is movable above the pelvic inlet.
- After engagement, the fourth maneuver shows the breech to be beneath the symphysis.

- Fetus in breech presentation can have different attitudes
- The position in picture “d”, with **hyperextended head** is called **star-gazing position**
 - Similar hyperextension of head in transverse lie is called **flying fetus**



The accuracy of this palpation varies. Thus, with suspected breech presentation or any presentation other than cephalic—sonographic evaluation is indicated.

- Vaginal examination

- With a frank breech, no feet are appreciated, but the fetal ischial tuberosities, sacrum, and anus are usually palpable. After further fetal descent, the external genitalia may also be distinguished.

When labor is prolonged, the fetal buttocks may become markedly swollen, rendering digital differentiation of a face and breech difficult. In some cases, the anus may be mistaken for the mouth and the ischial tuberosities for the malar eminences.

With careful examination, however, the finger encounters muscular resistance with the anus, whereas the hard, less yielding jaws are felt through the mouth. The finger, upon removal from the anus, may be stained with meconium. The mouth and malar eminences form a triangular shape, whereas the ischial tuberosities and anus lie in a straight line

- With a complete breech, the feet may be felt alongside the buttocks. In footling presentations, one or both feet are inferior to the buttocks.

Although physical examination is very important Ultrasonography is most informative in the assessment of breech presentation.

- It confirms the clinical diagnosis— especially in primigravida with engaged frank breech or with tense abdominal wall and irritable uterus.
- It can detect fetal congenital abnormality and also congenital anomalies of the uterus.
- Type of breech (complete or incomplete)
- It measures biparietal diameter, gestational age and approximate weight of the fetus.
- It also localizes the placenta.
- Assessment of liquor volume (important for ECV).
- Attitude of the head—flexion or hyperextension (important for decision making at the time of delivery).

The point of designation in a breech labor is the fetal sacrum.

Complications

Maternal risk/complications

- Operative delivery
- Prolonged labor

Fetal risk/complications

- High PMR* 2-4 fold increased
- Intracranial hemorrhage
- Asphyxia
- Injuries

*Perinatal mortality rate

Management

Antenatal management

- Identification of complicating factors associated with breech and
- External Cephalic Version (ECV) can be done if not contraindicated. And formulation of the line of management should be done, if version fails or contraindicated.

External Cephalic Version (ECV)

- ECV should be carried out in an area that has ready access to a facility equipped to perform emergency cesarean delivery (ACOG, 2016a).
- Because of the risk for surgical intervention, intravenous access is obtained, and patients abstain from eating for 6 or more hours.
- Sonographic examination is performed to confirm nonvertex presentation, document amniotic fluid volume adequacy, exclude obvious fetal anomalies if not done previously, and identify placental location and fetal spine orientation.

The success rate of version is about 60%. Successful version reduces the risk of cesarean delivery significantly.

ECV is considered from 36 weeks onwards. While version in the early weeks is easy but chance of reversion is more. Late version may be difficult because of increasing size of the fetus and diminishing volume of liquor amnii. However, the use of uterine relaxant (tocolysis) has made the version at later weeks less difficult

Contraindications of ECV

- Antepartum hemorrhage (placenta previa or abruption); risk of placental separation
- Fetal causes—hyperextension of the head, large fetus (>3.5 kg), congenital abnormalities (major), dead fetus, fetal compromise (IUGR)
- Multiple pregnancy
- Ruptured membranes—with drainage of liquor
- Known congenital malformation of the uterus
- Abnormal cardiotocography
- Contracted pelvis
- Previous cesarean delivery—risk of scar rupture
- Obstetric complications: Severe pre-eclampsia, obesity, elderly primigravida, bad obstetric history (BOH)
- Rhesus isoimmunization

Success and dangers of ECV

Successful version is likely in cases of:

- Complete breech
- Non-engaged breech
- Sacroanterior position
 - Fetal back anteriorly
- Adequate liquor
- Non-obese patient

Dangers of version:

- Premature onset of labor
- PROM
- PA & bleeding
- Entanglement of the cord round the fetal part or formation of a true knot leading to impaired
- circulation
- Increased chance of FMH
- Amniotic fluid embolism

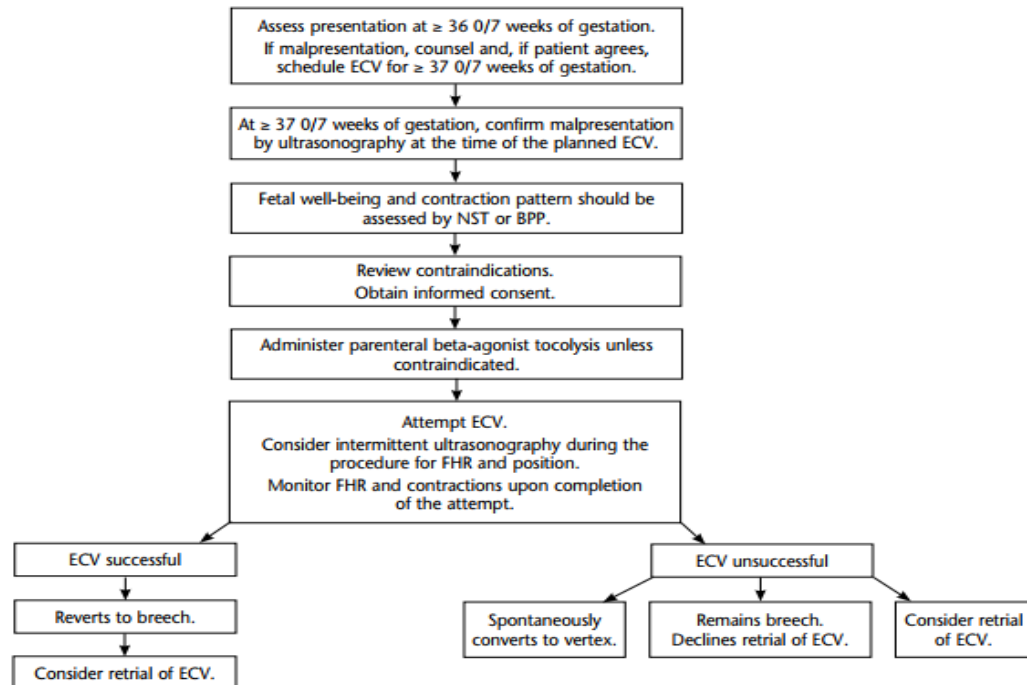


Fig. 1. An algorithm for patient management for external cephalic version. Note: All Rh-negative women who undergo an ECV attempt, whether successful or not, should receive Rh immune globulin unless they are known to have an Rh-negative fetus, are already sensitized, or will give birth in less than 72 hours and can have an assessment for risk of sensitization. Abbreviations: BPP, biophysical profile; ECV, external cephalic version; FHR, fetal heart rate; NST, nonstress test. ⇐

Management at Term/Labor

First stage

The management protocol is similar to that mentioned in normal labor. The following are the important considerations.

- Spontaneous onset labor increases the chance of successful vaginal delivery.
- Vaginal examination is indicated
 - at the onset of labor for pelvic assessment,
 - Soon after rupture of the membranes to exclude cord prolapse.
- An intravenous line is sited with Ringer's solution, oral intake is avoided, blood is sent for group and cross matching (considering the chance of CS).
- Adequate analgesia is given, epidural is preferred.
- Fetal status and progress of labor are monitored.
- Oxytocin infusion may be used for augmentation of labor.

Indications of cesarean delivery

- Weight : >3500 or <1800 & >1000 gm.
- Hyperextended fetal head and neck
- Contracted pelvis
- Footling breech
- Abnormal fetal wellbeing tests
- Cord presentation/prolapse
- Expertise in breech delivery is absent
- Arrest of progress has occurred
- Presence of any risk factors – ☛ Post term; previous CS; APH; DM;
- **Zatuchni- Andros score is <4**
 - *This scoring system is developed to select candidates for trial of labor in breech presentation*
 - *It has six parameters, each given a value of 0, 1, or 2*

TABLE 17-3 ZATUCHNI-ANDROS SYSTEM

FACTOR	0	1	2
Parity	Nullipara	Multipara	Multipara
Gestational age	39	38	37
Estimated fetal weight	8 lb	7-8 lb	7 lb
Previous breech	No	One	Two
Dilation (cm)	2	3	≥4
Station	-3 or greater	-2	-1 or less

From Zatuchni GI, Andros GJ. Prognostic index for vaginal delivery in breech presentation at term. *Am J Obstet Gynecol.* 1965;93:237.

Second stage:

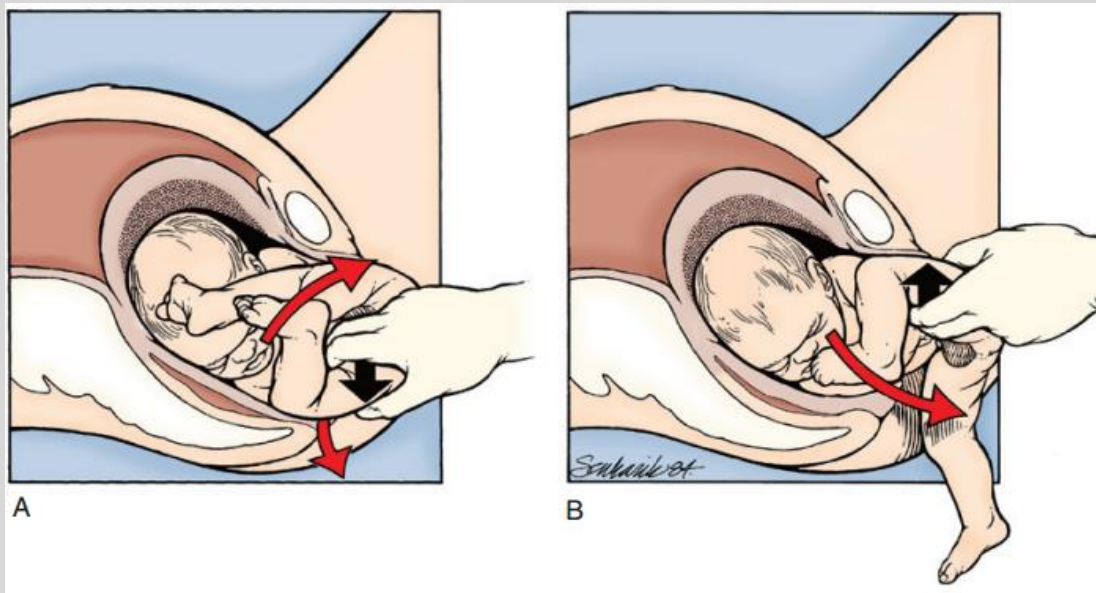
There are three methods of vaginal breech delivery:

- **Spontaneous** (10%): Expulsion of the fetus occurs with very little assistance. This is not preferred.
- **Assisted breech:** The delivery of the fetus is by assistance from the beginning to the end. This method should be employed in all cases (see below).

- **Breech extraction** (partial or total): When part or the entire body of the fetus is extracted by the obstetrician. It is rarely done these days as it produces trauma to the fetus and the mother. Indications are:
 - Delivery of the second twin after IPV (Internal Podalic Version),
 - Cord prolapse,
 - Extended legs arrested at the cavity or at the outlet.

Different maneuvers are used to assist vaginal breech delivery

- **Pinnard maneuver**
 - to facilitate **delivery of the legs** in a frank breech presentation
 - pressure is applied to the medial aspect of the knee, which causes flexion and subsequent delivery of the lower leg

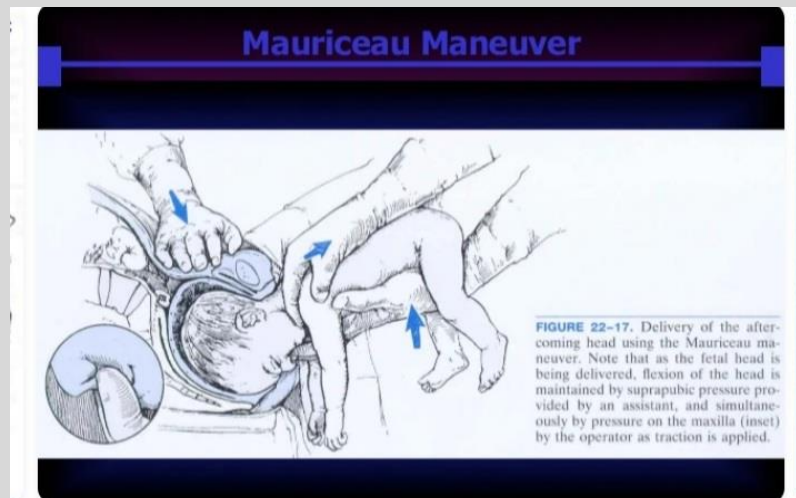


- **Louvset's maneuver**
 - Can be used to **deliver the shoulders** when one or both arms may be fully stretched along the side of the head or lie behind the neck (**nuchal displacement**) resulting in arrest with delivery of the trunk up to the costal margins.

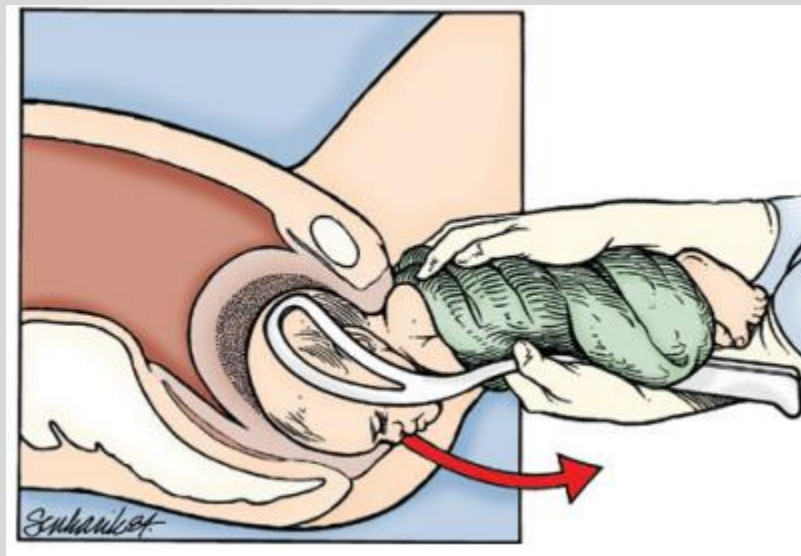
- **Delivery of the after coming head**

- Maneuvers

- ✓ Mauriceau – Smellie - Veit (**MSV**) Maneuver
 - ✓ Modified Prague maneuver
 - ✓ Burn Marshal Method



- Forceps for after-coming head

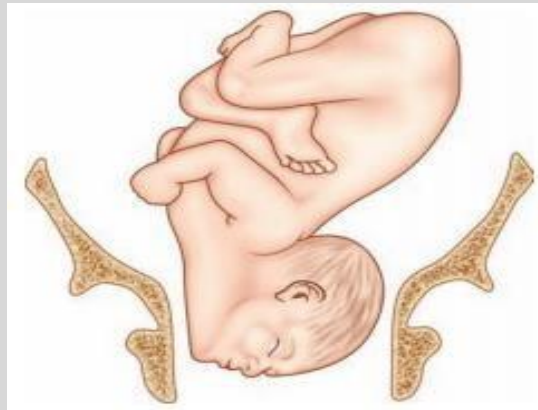


Face presentation

Face is a rare variety of cephalic presentation where the presenting part is the face.

The attitude of the fetus shows complete flexion of the limbs with extension of the spine.

There is ***complete extension of the head*** so that the occiput is in contact with the back.



face presentation

Incidence

- The reported incidence of face presentation ranges from 0.14% to 0.54% and averages about 0.2%, or 1 in 500 live births overall.

Etiology of face presentation

The cause of extreme extension of the head is not clear in all the cases. The following are the factors which are often associated.

Maternal:

- Multiparity with pendulous abdomen,
- Lateral obliquity of the uterus especially, if it is directed to the side towards which the occiput lies,
- Contracted pelvis is associated in about 40% cases. Flat pelvis favors face presentation,
- Pelvic tumors.

Fetal:

- **Congenital malformations (15%)**
 - The commonest one is anencephaly. The almost non-existent neck with absence of the cranium makes it easy to feel the facial structure even with semi-extended head,
 - Congenital goiter—prevalent in endemic areas,
 - Dolichocephalic head with long anteroposterior diameter,
 - Congenital branchocele.
- Twist of the cord several turns round the neck
- Increased tone of the extensor group of neck muscles

Diagnosis

Antenatal diagnosis is rarely made. Diagnosis is made only during labor but in about half, the detection is made at the time of delivery. It is more often discovered by vaginal examination.

P/E

- **Second Leopold's Maneuver**, if mentoanterior:
 - Fetal limbs are felt anteriorly.
 - Back is on the flank and is difficult to palpate.
 - The chest is thrown anteriorly against the uterine wall and is often mistaken for back
- **Third Leopold's Maneuver**, if mentoanterior:
 - Head seems big and is not engaged.
 - Cephalic prominence is to the side towards which back lies
 - Groove between the head and back is not so prominent

FHB is distinctly audible anteriorly through the chest wall of the fetus towards the side of limbs.

Vaginal examination

- The diagnostic features are palpating the mouth with hard alveolar margins, nose, malar eminences, supraorbital ridges and the mentum.

Sonography/Radiography:

- This should be done to confirm the diagnosis, to exclude bony congenital malformation of the fetus and to note the size of the baby.

Management

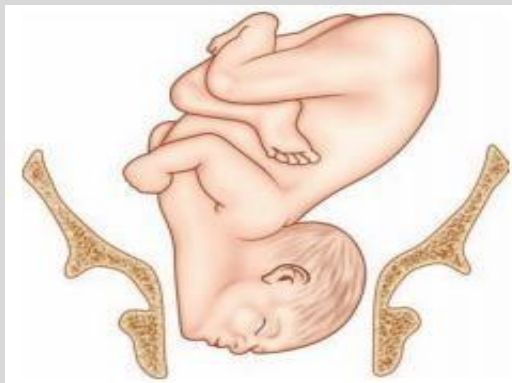
Vaginal delivery

- **Mentoanterior**
 - **First stage:**
 - ✓ In uncomplicated cases, a wait and watch policy is adopted.
 - ✓ Labor is conducted in the usual procedure and the special instructions, as laid down in occipitoposterior positions, are to be followed.
 - **Second stage:**
 - ✓ One should wait for spontaneous delivery to occur.
 - ✓ Perineum should be protected with liberal mediolateral episiotomy.
In case of delay, forceps delivery is done.
- **Mentoposterior**
 - **First stage:**
 - ✓ In uncomplicated cases, vaginal delivery is allowed with strict vigilance hoping for spontaneous anterior rotation of the chin.
 - **Second stage:**
 - ✓ If anterior rotation of the chin occurs, spontaneous or forceps delivery with episiotomy is all that is needed.

- ✓ In incomplete or malrotation: Early decision for the method of delivery is to be taken soon after full dilatation of the cervix. The following methods may be employed to expedite the delivery.
 - Cesarean section is the preferred method and is commonly done these days.
 - Manual rotation of the chin anteriorly followed by immediate forceps extraction is rarely done these days. The principles and the methods are similar to those employed in unrotated occipitoposterior position.

Brow presentation

Brow is the *rarest variety of cephalic presentation* where the presenting part is the brow and the attitude of the head is short of that degree of extension necessary to produce face presentation, i.e. the head lies in between full flexion and full extension.



Brow presentation

Incidence

- The reported incidence of brow presentation varies widely, from 1 in 670 to 1 in 3433, averaging about 1 in 1500 deliveries.

Brow presentation is detected more often in early labor before flexion occurs to a normal attitude. Less frequently, further extension results in a face presentation.

Etiology

The causes of persistent brow are more or less the same as those of face presentation.

The position is commonly unstable and converts to either vertex or face presentation.

Diagnosis

- Antenatal diagnosis is rarely made. The findings are more or less like those of face presentation.

P/E

- **Leopold's maneuver:**
 - The cephalic prominence and the groove between it and the back are less prominent. The head feels very big and is nonengaged.
- **Vaginal examination**
 - The position is to be confirmed on vaginal examination by palpating supraorbital ridges and anterior fontanelle.
 - If the anterior fontanelle is on mother's left, with the sagittal suture in transverse pelvic diameter, it is left frontum transverse position. In late labor, the landmarks may be obscured by caput formation.

Sonography:

- Is confirmatory and also helps in excluding bony congenital malformation of the fetus.

Management

Antenatal

- If the presentation is diagnosed during pregnancy and there are no other contraindications for vaginal delivery, nothing is to be done.
- Contracted pelvis and congenital malformation of the fetus are to be excluded.
- Spontaneous correction into face is likely to occur.

Elective cesarean section:

- Cases with persistent brow presentation are delivered by elective cesarean section.

During labor:

- In uncomplicated cases, if spontaneous correction to either vertex or face fails to occur early in labor, cesarean section is the best method of treatment.
- Manual Correction to face with full dilatation of cervix is seldom practiced nowadays.
- Craniotomy—if the labor becomes obstructed and the baby is dead, craniotomy is done. Rupture of the uterus should be excluded.

Compound presentation

Whenever an extremity, most commonly an upper extremity, is found prolapsed beside the main presenting fetal part, the situation is referred to as a compound presentation.



Compound presentation

Incidence

The reported incidence ranges from 1 in 377 to 1 in 1213 deliveries. The combination of an upper extremity and the vertex is the most common.

Etiology

Conditions preventing engagement of the head can result in slipping of either upper or lower limbs by the side of the head.

- Prematurity (commonest), contracted pelvis, pelvic tumors, multiple pregnancy, macerated fetus, high head with premature or early rupture of the membranes and hydramnios are the known etiological factors.

Diagnosis

- The diagnosis is not difficult when the cervical Os is sufficiently dilated to feel the limb by the side of the presenting part, especially after rupture of the membranes.
- Premature or early rupture of the membranes occurs in about one-third of the cases.
- Cord prolapse is to be excluded because of its frequent association; **10–15%**.

Management

Factors to be considered are:-

1. stage of labor,
2. maturity of the fetus,
3. singleton or twins,
4. pelvic adequacy and
5. associated cord prolapse

The fetal risks in compound presentation are birth trauma and cord prolapse.

Indication of cesarean section:

- Mature singleton fetus associated with contracted pelvis or cord prolapse with the fetus alive should be safely delivered by caesarean section.

Expectant Management:

- In otherwise uncomplicated cases (vertex/hand), an attitude of wait and watch policy is preferable.
- The labor process needs to be monitored very carefully (preferably by electronic fetal monitoring).
- Elevation of the prolapsed limb with descent of the presenting part usually takes place spontaneously. Slight elevation of the prolapsed limb during uterine contraction is a favorable sign. Temptation to replace the limbs early is not only unnecessary but carries increased maternal and fetal risks.

Shoulder presentation (Transverse lie)

When the long axis of the fetus lies perpendicularly to the maternal spine or centralized uterine axis, it is called ***transverse lie***.

- But more commonly, the fetal axis is placed oblique to the maternal spine and is then called ***oblique lie***.

In either of the conditions, the shoulder usually presents over the cervical opening during labor and as such both are collectively called shoulder presentations.

Incidence

- The incidence is about 1 in 200 births.
- It is common in premature and macerated fetuses, 5 times more common in multipara than primigravida.
- Transverse lie in twin pregnancy is found in 40% of cases.

Etiology

The causes are:

- Multiparity — lax and pendulous abdomen, imperfect uterine tone and extreme uterine obliquity are the responsible factors.
- Prematurity — center of the gravity lies almost in the middle of the body.
- Twins — it is more common for the second baby than the first one to be in transverse position.
- Hydramnios
- Contracted pelvis
- Placenta praevia
- Pelvic tumors
- Congenital malformation of the uterus—arcuate or subseptate and
- Intrauterine death

Diagnosis

Abdominal examination

- Leopold's Maneuver
 - The fundal height is less than the period of amenorrhea.
 - Fundal grip—fetal pole (breech or head) is not palpable.
 - Lateral grip—
 - Soft, broad and irregular breech is felt to one side of the midline and smooth, hard and globular head is felt on the other side. The head is usually placed at a lower level on one iliac fossa.
 - The back is felt anteriorly across the long axis in dorso- anterior or the irregular small parts are felt anteriorly in dorso-posterior.
 - Pelvic grip— the lower pole of the uterus is found empty. This, however, is evident only during pregnancy but during labor, it may be occupied by the shoulder.

Vaginal examination

- During pregnancy; the presenting part is so high that it cannot be identified properly but one can feel some soft parts.
- During labor; Elongated bag of the membranes can be felt if it does not rupture prematurely. The shoulder is identified by palpating the following parts—acromion process, the scapula, the clavicle and axilla.

Management

Antepartum

External cephalic version should be done in all cases beyond 35 weeks provided there is no contraindication as mentioned in breech presentation.

- If the lie fails to stabilize even at 36th week, the case is to be managed as outlined in unstable lie.

If version fails or is contraindicated:

- The patient is to be admitted at 37th week, because risk of early rupture of the membranes and cord prolapse is very much there.
- Elective cesarean section is the preferred method of delivery.
 - Vaginal delivery may be allowed in a dead or congenitally malformed (small size) fetus. The labor may be allowed to continue under supervision till full dilatation of the cervix, when the baby can be delivered by internal version.

During labor:

The principles in management are as outlined below:

- **Early labor:**
 - External cephalic version: provided there is good amount of liquor amnii and there is no contraindication.
 - ✓ External cephalic version should be tried in all cases.

- Cesarean delivery is the preferred method of delivery if version fails or is contraindicated.
 - ✓ Difficulties are faced during cesarean section as the lower uterine segment is poorly developed. However, the fetus can be manipulated to a polar (cephalic/breech) presentation before entering the uterus and a low transverse incision may be made. Otherwise lower segment vertical incision may be made.
- **Late labor:**
 - **Baby alive** — there is hardly any scope of external version in late labor because of invariable rupture of the membranes and drainage of liquor.
 - ✓ If the baby is mature and the fetal condition is good, it is preferable to do cesarean section in all cases.
 - ✓ Internal version— in a singleton fetus, the risk of internal version is high. Not only it might inflict injury to the uterus (rupture uterus) but also the fetal mortality is increased to the extent of about 50%.
 - In modern obstetric practice, internal version is not recommended except in the case of second twin.
 - **Baby dead** — cesarean section even in such cases is much safer in the hands of those who are not conversant with destructive operations
 - ✓ If the obstetrician is conversant with destructive operation, decapitation or evisceration is to be done. Following destructive operation, the uterine cavity is to be explored to exclude rupture of the uterus. Internal version should not be done.

Chapter 8

MULTIFETAL GESTATION

Prepared by Dr. Abubeker Nuredin (MI)

Definition and Incidence

Multifetal gestation (MG): is when more than one fetus simultaneously develops within the uterus. It could be:

- Two- twins, **commonest**
- Three- Triplet
- Four-Quadruplets
- Five- Quintuplet, etc...

MG result from:-

- two or more fertilization events, or
- a single fertilization followed by a splitting of the zygote, or
- a combination of both

Incidence

- The overall multifetal birth rate was 33.1 per 1000, with twins representing nearly 97 percent of these births in western countries
- The frequency of monozygotic twin births is relatively constant worldwide, and approximately one set per 250 births (4-5% of 1000 pregnancies).
 - It is largely **independent** of race, heredity, age, and parity.
- Incidence of multiple pregnancies is approximated using:
 - Helen's Formula $\Rightarrow 1: 80^{n-1}$
 - ✓ Twins $1:80$
 - ✓ Triplets $1: 80^2 = 1:6400$
 - ✓ Quadruplets (etc.) $1:80^3 = 1:512,000$

Twin pregnancy

- It could be monozygotic (identical/uniovular) or dizygotic (fraternal/ biovular)
- Dizygotic is **common**, (2/3rd) and monozygotic-1/3rd

Monozygotic twins

- Result from division of a single fertilized ovum
- Normally they are always of same sex
- They share:-
 - Same physical characteristics; color of skin, hair and eye
 - Same body fluid
 - Same genetic features; blood type and histocompatibility genes
- They are often mirror images of one another
 - Sometimes identical twins can be paradoxically antithesis of identical.
 - Earliest split may be associated with simultaneous chromosomal error, resulting in heterokaryotypic monozygotes, for example, one with down syndrome (Trisomy 21) the other normal
- Although they are similar in a lot of ways, they have ***different fingerprints***.

Genesis of Monozygotic twins

- Genesis of monozygotic twin is poorly understood, but it is assumed to be from:-
 1. Delay in the timing of normal developmental events.
 2. Delayed ovum transport (pregnancies conceived in close temporal proximity, progestational agents & COC pills delay tubal motility)
 3. Minor trauma to the blastocyst during assisted reproductive technology (ART)
- The division of one fertilized zygote into two does not necessarily result in equal sharing of protoplasmic material.
 - Because of this the process of ***monozygotic twinning is in a sense a teratogenic event***, and monozygotic twins have an increased incidence of often discordant malformations

- About a third as often, twins arise from a single fertilized ovum that subsequently divides into two similar structures, each with the potential for developing into a separate individual.

Zygosity and Chorionicity in MZ twins

Zygosity: is genetic make-up of twin pregnancy.

- Normally monozygotic twins have same genetic makeup.

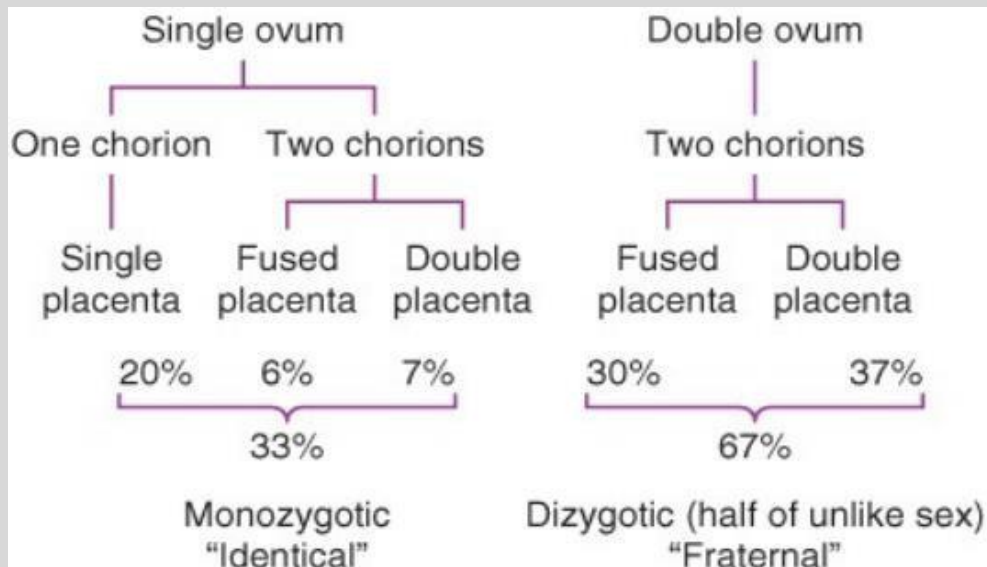
Chorionicty: is the membrane composition, it is determined by mechanism of fertilization and in monozygotic twins by timing of embryo division

Early determination of chorionicity is vital because it is a major factor in determining obstetric risks, management, and outcomes.

The table below shows the variation in fetal membranes and placentation in time correlated genesis of MZ twins

TABLE 32-1 DETERMINATION OF MONOZYGOTIC TWIN PLACENTATION		
TIMING OF CLEAVAGE OF FERTILIZED OVUM	RESULTING PLACENTATION	PERCENTAGE OF MONOZYGOTIC TWINS
<72 hours	Diamniotic dichorionic	25-30
Days 4-7	Diamniotic monochorionic	70-75
Days 8-12	Monoamniotic monochorionic	1-2
≥Day 13	Conjoined	Very rare

The percentage distribution of placental variation is shown below.



Assessment of chorionicity

- **Ultrasonography** is an effective prenatal tool for determining amnionicity and chorionicity.
 - The optimal time for performing the u/s examination is in the first trimester after 7 weeks (sensitivity $\geq 98\%$) with lower but acceptable accuracy in the early second trimester (sensitivity ≥ 90 percent).
- Identification of **two separate placentas is a highly reliable indicator of dichorionic twins**. This indicator is generally only useful in early pregnancy since separate placentas often appear fused later in gestation
- The presence/absence of the inter-twin membrane should be determined and its characteristics assessed early in pregnancy.
 - **Monochorionic/monoamniotic;(MCMA)**
 - ✓ The intertwin membrane is absent
 - **Dichorionic/diamniotic;(DCDA)**
 - ✓ An intertwin membrane with the "**twin peak**" or "**lambda**" sign indicates dichorionic twins.

- ✓ The twin peak or lambda sign refers to a triangular projection of tissue that extends between layers of the intertwin membrane from a fused dichorionic placenta.
- ✓ It is best seen at 10 to 14 weeks, becomes less prominent after 20 weeks of gestation, and may disappear.
- ✓ An additional clue that twins are dichorionic is that the intertwin membrane is thicker with dichorionic than monochorionic twins since **the DCDA membrane consists of four layers** (i.e. two layers of both amnion and chorion) whereas the intertwin membrane in a MCDA pregnancy only consists of two layers of amnion.
- **Monochorionic/diamniotic;(MCDA)**
 - ✓ An intertwin membrane with the "T" sign indicates a monochorionic/diamniotic placenta.
 - This sign refers to the appearance of the thin intertwin membrane composed of two amnions as it takes off from the placenta at a 90 degree angle.
- After the first trimester, identification of fetuses of different sex is a highly reliable means of confirming a dichorionic pregnancy.

TABLE 32-2**DETERMINATION OF CHORIONICITY AND AMNIONICITY IN FIRST-TRIMESTER PREGNANCIES**

PLACENTATION	GESTATIONAL SACS	YOLK SACS	AMNIOTIC CAVITIES
Dichorionic diamniotic	2	2	2 (thick dividing membrane)
Monochorionic diamniotic	1	2	2 (thin dividing membrane)
Monochorionic monoamniotic	1	1*	1

*Although this is nearly always true, there have been case reports of two yolk sacs in early pregnancy in twins later confirmed to be monoamniotic.

Dizygotic twins

- Result from separately fertilized ova
- They:-
 - bear resemblance of a sibling
 - may or may not have same blood type
 - can have different sex, but in about 75% of cases they have same sex

Risk factors for DZ twins

- **Use of fertility stimulating drugs or artificial reproductive technology (ART)**
 - Dizygotic twins are more common in pregnancies conceived with in vitro fertilization (IVF) than in naturally conceived pregnancies (≥ 95 percent versus 70 percent) since double embryo transfer is commonly performed as part of IVF.
 - Clomiphene citrate is associated with 5-10% increase risk of DZ twin
- **Maternal age**
 - The rate of natural twinning peaks at age 37 years, when maximal FSH stimulation increases the rate of multiple follicles developing.
 - The fall in incidence thereafter probably reflects physiological follicle depletion.
- **Race/geographic area**
 - Significant ethnic/racial variations in the prevalence of naturally conceived dizygotic twins occur worldwide.
 - women of African descent have higher rates of DZ twinning than white women, who in turn have higher rates than women of Asian descent
 - For instance,
 - ✓ in Japan, 1 in 250 newborns
 - ✓ in Nigeria, 1 in 11 babies is a product of a twin gestation.
- **Parity** – Increasing parity correlates with an increased likelihood of dizygotic twin birth.

- Pregnancy soon after cessation of long term oral contraception
 - Due to high rebound gonadotropin secretion
- **Family history**
 - Dizygotic twinning appears to have a genetic component that is expressed in women but can be inherited from either parent.
- **Maternal weight and height**
 - Obese (body mass index [BMI] ≥ 30 kg/m²) and tall women (≥ 164 cm) are at greater risk for dizygotic twin birth than underweight (BMI < 20 kg/m²) and short women (< 155 cm).
- **Diet** - some studies have reported that folic acid supplementation increased the rate of twinning.
- Increases in DZ twins have been reported in summer months and in locations with more daylight hours.

Zygosity and Chorionicity in DZ twins

- Dizygotic or "fraternal" twins occur from ovulation and fertilization of two oocyte, which results in dichorionic/diamniotic placentation and two separate placentas

Sex ratios in MG

- In humans, as the number of fetuses per pregnancy rises, **the percentage of male conceptuses declines**. Percentage of Male fetus in
 - singleton pregnancy- 51.6%
 - twins, it was 50.9%
 - triplets, 49.5% and
 - quadruplets, 46.5%
- This female predominance was explained by two explanations.
 - First, beginning in utero and extending throughout the life cycle mortality rates are lower in females.

- Second, female zygotes have a greater tendency to divide.

Physiological changes in twin

- Increased weight gain-For normal BMI (increased by 16.8-24.5 Kg)
- Increased cardiac output
- Plasma volume increased by 500ml from singleton
- Increased GFR, Tidal volume & fetoprotein
- Increase BP at least 15 mm Hg in 95 percent of women
- The uterus and its non-fetal contents may achieve a volume of 10 L or more and weigh in excess of 20 pounds!

Diagnosis of MG

History

- Pregnancy heavier(bigger abdominal girth) than previous pregnancies
- Personal or family history of twins
- Early onset preeclampsia
- Pregnancy following assisted reproductive technology
- Excessive weight gain
- Exaggerated minor ailments,
 - Earlier and more severe pressure in the pelvis results in, nausea, back pain, constipation, abdominal distention, varicosities hemorrhoids
 - The hormonal change associated with pregnancy which is more severe in MG is also responsible for the ailments

Physical examination

- Barrel- shaped “large” abdomen
- Excessive weight gain not associated with edema or obesity
- Obstetric palpation revealing:-
 - Outline or ballottement of more than one fetus
 - Big for date uterus
 - More than two poles
 - Multiple small parts

- FHR heard by two people at different areas with rate difference of at least 10 bpm (with different rate to the maternal pulse).
- Palpation of 1 or more fetus after delivery of one neonate

Investigations

- Ultrasound:
 - Diagnosis of MG,
 - Determination of:-
 - ✓ Chorionicity
 - ✓ Gestational age
 - ✓ Viability,
 - ✓ Fetal anomalies
 - ✓ Presentation, etc...
- Biochemical:
 - Serum hcG, MSAFP & unconjugated estriol are almost doubled.
- Radiography: rarely done these days

Complications of MG

- MG is associated with ***higher rates of almost every potential complication of pregnancy***, with the exceptions of post-term pregnancy and macrosomia.
- The most serious risk is ***spontaneous preterm delivery***, which plays a major role in the increased perinatal mortality and short-term and long-term morbidity observed in these infants.
- The complications are classified in to maternal and fetal

Maternal complication	Fetal complication
▪ Hyperemesis ▪ Abortion ▪ Maternal anemia ▪ PIH(PE) ▪ APH ▪ Dysfunctional labor ▪ Malpresentation ▪ PPH ▪ Postpartum Depression, etc...	▪ Fetal-fetal hemorrhage ▪ Perinatal mortality(4X,12X,26X–twin, triplet, quadruplet) ▪ Low birth weight ▪ Malformations ▪ Discordant ▪ Cord accident ▪ Polyhydramnios
	Unique complications of MZ twins ▪ Cord entanglement ▪ Conjoined Twins ▪ Vascular Anastomosis ▪ Discordant twin ▪ Locked twins

1. Cord entanglement

- Cord entanglement occurs in most, if not all, monoamniotic twin pregnancies.
- Loose cord entanglements probably begin in the first trimester, and have the potential to tighten at any time.
- Intermittent occlusion of umbilical blood vessels may be associated with neurologic morbidity; significant prolonged occlusion can be lethal.

2. Conjoined twins

Incidence

- Conjoined twins are a rare type of monoamniotic twins, estimated to occur in 1.5 per 100,000 births worldwide.
- Females are affected more often than males.

Types

When twins are conjoined, the fusion occurs between same body parts. Conjoined twins are classified as:

Types of conjoined twins	Fused part
▪ Cephalopagus	Head
▪ Thoracopagus	Chest
▪ Omphalopagus	Abdomen
▪ Ischiopagus	Caudal
▪ Pygopagus	Sacrum

Prenatal diagnosis

- The diagnosis should be suspected in 1st trimester MA twin pregnancies when the embryonic/fetal poles ***are closely associated and do not change in position with respect to each other.***
- Their findings, which are **not all specific to conjoined twins**, include:-
 - fetal hyperextension,
 - increased nuchal translucency or cystic hygroma,
 - no sign of separate movement of the twins,
 - juxtaposed embryos with a single midline cardiac motion,
 - fewer limbs than expected,
 - a single umbilical cord with more than three vessels,
 - both heads or breeches consistently at the same level to each other
- Color Doppler, fetal echocardiography, and three- dimensional ultrasound examination can confirm the diagnosis and clarify anatomy, which is critical for assessing prognosis and pre- and postnatal decision- making.

3. Vascular anastomosis between fetuses

- Nearly 100 % of monochorionic twin placentas have vascular anastomosis
- Most of these vascular communications are hemodynamically balanced and of little fetal consequence.

- The two patterns of hemodynamically significant anastomotic circulations include acardiac twinning and the twin-to-twin transfusion syndrome

Acardiac twin

Pathogenesis

- Large artery-to-artery placental shunts in the embryo, often also accompanied by a vein-to-vein shunt, result in the TRAP sequence.
- The perfusion pressure of the donor twin overpowers that in the recipient twin, who thus receives reverse blood flow from its twin sibling.
- The "used" arterial blood reaching the recipient twin preferentially goes to the iliac vessels and thus perfumes only the lower part of the body, leading to disruption or deterioration of growth and development of the upper body
- In the TRAP sequence, there is usually a normally formed donor twin who has features of heart failure as well as a recipient twin who lacks a heart (acardius) and various other structures like:-
 - Failure of Head development
 - ***acardius acephalus***
 - Partially developed head with identifiable limbs
 - ***acardius myelacephalus***
 - Failure of any recognizable structure to form
 - ***acardius amorphous***
- ***Without treatment, the donor or "pump" twin has been reported to die in 50 to 75 percent of cases.***

Twin-to-twin transfusion syndrome (TTTS)

- TTTS results from imbalances in fetoplacental blood flow in the shared placental circulation

- It is a potentially lethal disorder that develops in 10 to 15 percent of monochorionic twins.
- Is characterized by oligohydramnios (including anhydramnios) in one amniotic sac and polyhydramnios in the other sac.
- Cerebral palsy, microcephaly, porencephaly, and multicystic encephalomalacia are serious complications associated with vascular anastomoses in twin gestations.
- Outcomes of TTTS

Donor twin	Recipient twin
▪ small, pale, ▪ dehydrated ▪ IUGR ▪ oligohydramnios (due to oliguria) ▪ die from anemic heart failure	▪ plethoric ▪ edematous, ▪ hypertensive, ascites, ▪ kernicterus (need amniocentesis for bilirubin), ▪ hepatomegaly, ▪ polyhydramnios (due to polyuria) ▪ die from congestive heart failure, and jaundice

- **Diagnosis**

- **In utero**

- ✓ Two ultrasound evidences:

- 1st – MCDA pregnancy is identified
 - 2nd - Hydramnios (SDP>8cm) in one sac and oligohydramnios defined by SDP <2 cm in the other twin is found

- ✓ Although growth discordance or growth restriction may be found with TTTS, these per se are not considered diagnostic criteria.

- **Postnatal**
 - ✓ is made based on weight discordancy between twins of 15 or 20 percent and a hemoglobin level difference of 5 g/dL or greater, with the smaller twin being anemic
- **Staging**
 - TTTS is typically classified by the Quintero (1999) staging system:-
 - ✓ **Stage I**—discordant amniotic fluid volumes as described in the earlier paragraph, but urine is still visible sonographically within the bladder of the donor twin
 - ✓ **Stage II**—criteria of stage I, but urine is not visible within the donor bladder
 - ✓ **Stage III**—criteria of stage II and abnormal Doppler studies of the umbilical artery, ductus venosus, or umbilical vein
 - ✓ **Stage IV**—ascites or frank hydrops in either twin
 - ✓ **Stage V**—demise of either fetus
- **Management**
 - Several therapies are currently used for twin-to-twin transfusion, including
 - ✓ amnioreduction,
 - ✓ septostomy,
 - ✓ laser ablation of vascular anastomoses, and
 - ✓ selective feticide

4. Discordant twin

- Restricted growth of one twin fetus usually develops late in the second and early third trimester and is often asymmetrical.
- The cause of birth weight inequality in twin fetuses is often unclear
- In MC twins, discordancy is usually attributed to placental vascular anastomosis
- Dizygotic fetuses may have different genetic growth potential

5. Locked twin

- Occurred only once in 817 twin gestations.

- For twins to lock, the first fetus breech & second cephalic. With descent of the breech, the chin of first fetus locks between the neck and chin of second, cephalic fetus.
- Cesarean delivery is recommended when the potential for locking is identified

Management of MG pregnancies

Antepartum management

General Mgt:

- Supplementation of Iron & folic acid: Iron 60 to 120 mg per day, Folic acid 1mg/day
- Nutrition - increased caloric intake by 300kcal/day, equivalent to an extra snack, above that of singleton pregnancy.
- Rest - Limited physical activities; early work leave.
- Frequent ANC follow up: For uncomplicated twin pregnancy-
 - Weeks 4 to 24: a prenatal visit once a month.
 - Weeks 24 to 32: a prenatal visit every two weeks.
 - Weeks 32 to 38: a prenatal visit every week.
- For higher order multiples and complicated multiple pregnancies, more frequent follow up is required.
- At each visit perform thorough evaluation for signs and symptoms suggestive of complications (like pregnancy induced hypertension (PIH), preterm labor, premature rupture of membranes (PROM))
- Give advice on danger symptoms/signs of preeclampsia, abruptio placentae, preterm labor and PROM
- Birth preparedness and on the need for hospital delivery with Cesarean Section facility and in the presence of skilled birth attendant and neonatal resuscitation.
- Referral to tertiary centers is recommended for MC and/or MA twins.

Test of fetal well being

- Serial growth monitoring (ultrasound every 3-4 wks.) especially in monochorionic twins
- Antepartum fetal surveillance starting from 28wks for monochorionic, undetermined chorionicity and other complicated twins, from 32 weeks of gestation for uncomplicated twins, every week is indicated.
- Non-stress testing (If there is CTG in the facility)
- Biophysical profile/modified biophysical profile
- Assessing amniotic fluid (deepest vertical pool)

Timing of delivery

- MA twins should be delivered at 32-34wks of GA after a course of steroids by cesarean section
- Monochorionic twins should be delivered between 36-38 wks of gestation
- Dichorionic twins should be delivered by 40 weeks of gestation
- Elective cesarean sections for twin pregnancy should be done at 38wks of gestation
- Note that Induction & augmentation of labor are contraindicated in twins.
- For term single intrauterine death (co-twin death), deliver without delay but if it is preterm, prolonging the pregnancy for the benefit of increased maturity of the surviving twin is recommended.
- In triplet pregnancies, delivery is recommended if pulmonary maturity is assured or at gestational age of not more than 36 wks.

Mode of delivery

- The route of delivery of a multiple gestation depends on
 - the number of fetuses,
 - presentation of the fetuses and
 - Whether spontaneous onset of labor is present.

- For all twins with 1st vertex, allow delivery by vaginal route and manage the delivery of the second as for singleton - depending on the presentation.
- For all twins with 1st non-vertex, irrespective of the presentation of the 2nd, delivery should be by cesarean section.
- The route of delivery for higher order gestations should be cesarean section.

Intra-partum management

General

- Follow labor using partograph.
- Secure IV line and anticipate PPH.
- Make all preparations for delivery.
- Prepare delivery room and equipment for the birth of multiple babies.
- Arrange for a helper to assist you with the births and care of the babies.

First stage of labor

- Admit in early labor
- Open IV line; hydrate as needed
- Ascertain fetal number, presentations, estimated fetal weight and placental location.
- Close monitoring of fetal heart rate (FHR) in both fetuses.
- Augmentation is contraindicated before delivery of the 1st twin

Second stage of labor

- Deliver the first baby following the usual procedure. Label - Twin-A
- Ask helper to attend to the first baby.
- Leave the other clamp on the maternal end of the umbilical cord and do not attempt to deliver the placenta until the last baby is delivered.
- Do not give the mother bolus IM/ IV oxytocin until after the birth of all the babies

- Immediately after the first baby is delivered, palpate the abdomen to determine the lie and presentation of the second twin;
- Check fetal heart rate
- Perform a vaginal examination to determine the presentation of the second twin, presence or absence of prolapsed cord and whether membranes are intact or ruptured. Bedside ultrasound can be used to confirm the presentation and lie of the fetus.
- After spontaneous or artificial rupture of the membranes, perform vaginal examination to check for prolapsed cord. If the cord has prolapsed, manage as a case of cord prolapse.
- If the membranes are intact and if there is no contraindication (e.g., high station), artificial rupture of membranes (ARM) of the second sac facilitates the labor.
- Birth interval of about 30 minutes is considered a reasonable time, after which delivery should be expedited. This requires:
 - If contractions are inadequate after birth of first baby, augment labor with oxytocin in vertex presentation using rapid escalation to produce good contractions.
 - Operative vaginal or abdominal delivery depending on the case.
- Stay with the woman and continue monitoring her and the fetal heart rate closely.
- When strong contractions restart, ask the mother to bear down when she feels ready.
- If spontaneous delivery does not occur within 2 hours of good contractions, do operative vaginal delivery or CS depending on the case.
- After delivery of the second baby, resuscitate if necessary. Label - Twin B.
- Make sure there is no more baby and proceed to 3rd stage management

Third stage of labor

- Manage third stage of labor actively after the delivery of the last fetus following the steps in active management of the third stage of labor (AMTSL).

- Before and after delivery of the placenta and membranes, observe closely for vaginal bleeding because this woman is at greater risk of postpartum hemorrhage.
- Examine the placenta for completeness, vascular anomalies, and communications, and zygosity (mono or Dizygotic twin)

Immediate postpartum care

- Monitor closely as risk of postpartum bleeding is increased.
- Provide immediate postpartum care.
- Plan to measure hemoglobin postpartum if possible
- Special support and follow up for feeding and care of preterm/low birth weight babies

Chapter 9

Red Cell Alloimmunization (Rh Alloimmunization)

Prepared by Dr. Abdulhafiz Hussein (MI)

Introduction

Alloimmunization refers to an immune response (sensitization) to foreign antigens from another human, most commonly occurring after pregnancy or blood transfusions.

- The risk of developing severe reaction is proportional to the antigen load, i.e. the more amount of antigen exposed, the greater the risk for alloimmunization reactions to occur.

Rh Alloimmunization is therefore, formation of anti-red cell antibodies by a Rh negative (Rh-) pregnant women who is exposed to Rh positive (Rh+) fetus's RBC due to fetomaternal hemorrhage (FMH).

Epidemiology

- The prevalence of red cell alloimmunization in pregnancy approximates **1 %**.
- Most cases of severe fetal anemia requiring antenatal transfusion are attributable to ***anti-D, anti-Kell, anti-c, or anti-E alloimmunization***.
- The prevalence of D alloimmunization complicating pregnancy ranges from 0.5 to 0.9 percent.
- Alloimmunization is uncommon for the following reasons:
 - low prevalence of incompatible erythrocyte antigens
 - insufficient trans-placental passage of fetal antigens and maternal antibodies
 - ***maternal-fetal ABO incompatibility***, which leads to rapid clearance of fetal erythrocytes before they elicit an immune response
 - variable antigenicity
 - variable maternal immune response to the antigen

Blood Group Systems

Currently, 36 different blood group systems and 360 erythrocyte antigens are recognized by the International Society of Blood Transfusion. (43 and 343 respectively on up-to-date 2022)

Clinically significant Blood group systems are: ABO, Rh (CDE), Kell, Duffy, Kidd, etc...

At first Antenatal care, Blood grouping should be offered for every pregnant woman even if the woman is aware of her status

ABO Blood Group System

- Responsible for 4 major RBC phenotypes: A, B, O, and AB
- It has 3 antigens A, B and H and 2 antibodies; anti-A and anti-B
- Antibodies to ABO antigens generally appear in the blood by 4 to 6 months of age following exposure to bacterial antigens that are similar in structure to the A and B antigens as the gut becomes colonized in early infancy.
- ABO incompatibility can cause
 - acute hemolytic transfusion reactions (AHTR),
 - hemolytic disease of the fetus and newborn (HDFN), and
 - solid organ transplant rejection

Rh (CDE) Blood Group System

- More than 45 serologically defined antigens have been recognized within the Rh system.
 - The most common and clinically significant include D, C, c, E, and e (there is no "d" antigen)
 - The D antigen is detectable on embryonic RBCs by 38 days after conception (i.e. 7⁺ weeks of gestation)

Phenotypes under Rh blood group system

- **Rh-positive** - Individuals whose RBCs express the RhD antigen.
 - Approximately 40 % of RhD +ve individuals are homozygous for the D antigen (DD); the remainders are heterozygous (Dd).
 - Since the Rh allele transmission is in an autosomal recessive fashion, if the father is heterozygous the baby has a 50% chance of being Rh -ve. But if father is homozygous the baby would always be Rh +ve.
- **Rh-negative** – Individuals whose RBCs do not express the RhD antigen.
 - This phenotype is the result of either an absence of the RHD gene or alterations in the RHD gene resulting in gene inactivation.
 - The prevalence of the D antigen varies among populations:
 - ✓ Africans – 4 to 6 %,
 - ✓ Indians – 5 %
 - ✓ Black or African Americans – 8 %,
 - ✓ White North Americans or Europeans – 15 %,
 - ✓ Chinese – 0.3%
- **Partial or weak D** – Several D variants exist, referred to as partial D, weak D, Rh mod, D (u), and D (el).
 - In some cases, these cells may type as RhD-positive by all available serologic reagents, and in some cases they may not.
- **Rh-null** – In contrast to Rh-negative, which refers to lack of the RhD antigen, Rh-null refers to ***absence of any Rh antigens*** (absence of RhD, RhC, and RhE).

Genetics of Rh system

- Genes encoding for Rh surface antigen are found on short arm of chromosome 1.
- Approximately 40% of Rh +ve individuals are homozygous (DD) and the remainders are heterozygous (Dd).

- The allele of Rh is transmitted in an autosomal recessive manner.
 - Therefore, if the father is heterozygous the baby has a chance of 50% to be Rh -ve.

Rh (CDE) Blood Group Incompatibility

- The CDE system includes five erythrocyte antigens: C, c, D, E, and e.
 - The C, c, E, and e antigens have lower immunogenicity than the D antigen but can cause hemolytic disease.
- The prevalence of D alloimmunization complicating pregnancy ranges from 0.5 to 0.9 percent.

Unless the pregnant woman is sensitized before pregnancy, the 1st pregnancy remains unaffected but the fetal affection increases with subsequent pregnancies i.e the 3rd pregnancy is more affected than 2nd pregnancy.

- This is because of the repeated re-exposure to Rh+ RBC resulted in greater immune response- **anamnestic reaction**

Does every Rh -ve women who delivered Rh +ve baby develop Rh alloimmunization if not given anti-D Ig prophylaxis? ... No.

- Without anti-D Ig prophylaxis, an Rh-ve woman delivered of an Rh +ve, ABO-compatible newborn has only a **16%** (13% on gabby 7th edition) likelihood of developing alloimmunization.
 - 2% will become sensitized by the time of delivery,
 - 7 % by 6 months postpartum,
 - 7 % will be “sensitized”—producing detectable antibodies only in a subsequent pregnancy
- If there is **ABO incompatibility**, the D alloimmunization risk decreases to **2 %** because erythrocyte destruction of incompatible cells limits sensitization
- With immunoprophylaxis, at 28th week and postpartum, however, the alloimmunization risk may be reduced to **<0.2 % (0.1 - 0.3%)**

The advent of the routine administration of antenatal and postpartum Anti-D Ig has resulted in a marked reduction in cases of red cell alloimmunization secondary to the RhD antigen and a shift to other red cell antibodies associated with HDFN has occurred as a result of the decreasing incidence of RhD alloimmunization

- Anti-E alloimmunization is the most common, but the need for fetal or neonatal transfusions is greater with anti-c alloimmunization than with anti-E or anti-C.
- Anti-c antibody should be considered equivalent to anti-D regarding its potential to cause HDFN.

Grandmother effect theory,

In virtually all pregnancies, small amounts of maternal blood enter the fetal circulation. An Rh-negative female fetus exposed to maternal Rh-positive red cells might develop sensitization, and later might produce anti-D antibodies before or during pregnancy. This mechanism is called the grandmother effect because the fetus in the current pregnancy is jeopardized by maternal antibodies that were initially provoked by his or her grandmother's erythrocyte

Alloimmunization to Minor Antigens

Kell Alloimmunization may develop more rapidly and may be more severe than with sensitization to D and other blood group antigens.

- Unlike the case of other hemolytic antibodies, fetal anemia due to Kell (anti-K₁) sensitization is thought to be secondary to ***not only hemolytic but also to suppression of fetal erythropoiesis***. This is because Kell antibodies attach to erythrocyte precursors in the fetal bone marrow and thereby impair the normal hemopoietic response to anemia.
- With fewer erythrocytes produced, there is less hemolysis, and thus severe anemia may not be predicted by the maternal Kell antibody titer.

Given the potential for severe anemia, the ACOG (2019a) has recommended

- ultrasound surveillance regardless of titer
- antibody titers not be used to monitor Kell-sensitized pregnancies

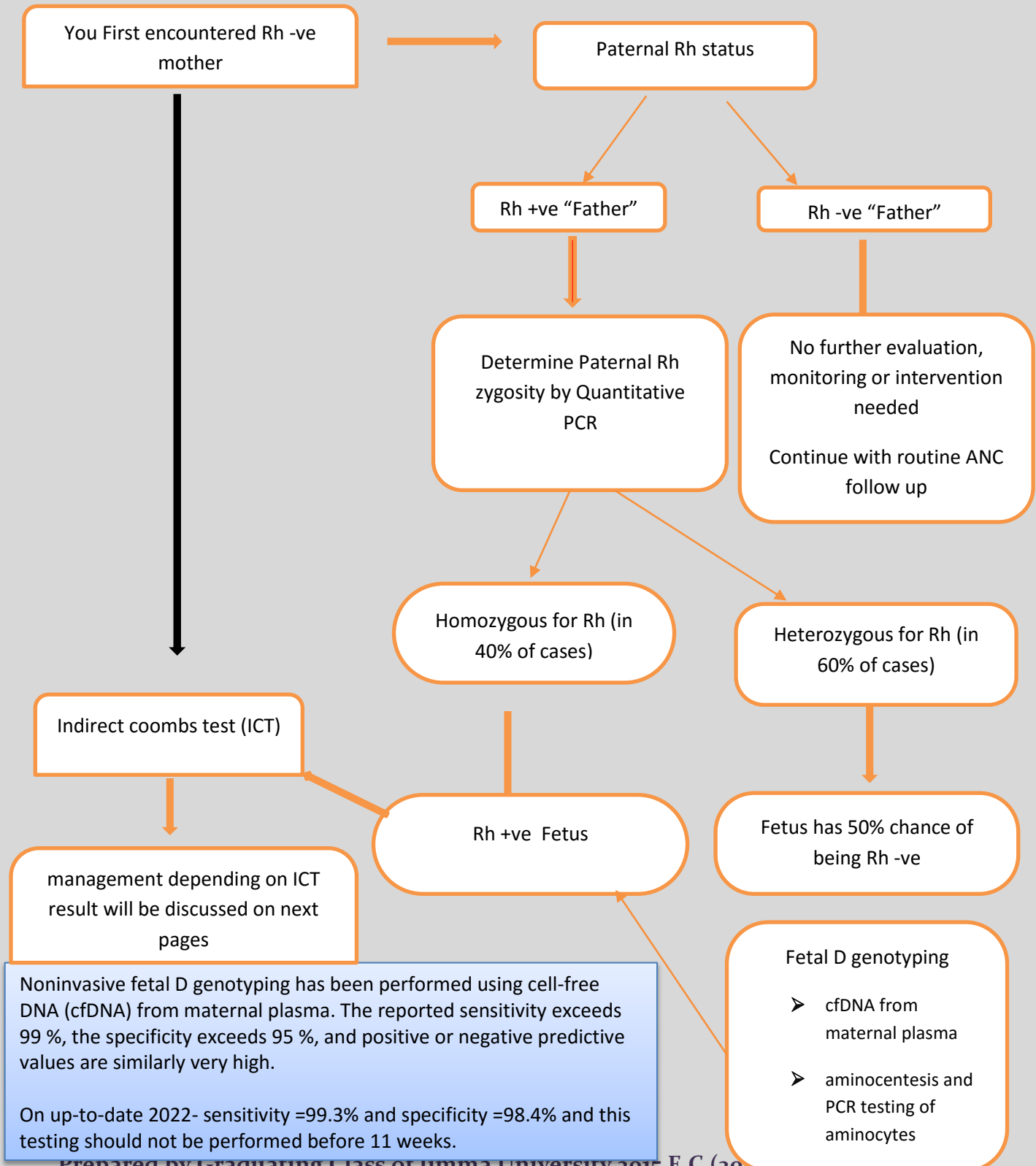
Transfusion history is important, as nearly 90 % of Kell sensitization cases result from transfusion with Kell-positive blood usually as a result of postpartum hemorrhage in a previous pregnancy.

Because 92% of individuals are Kell negative, the initial management of the K₁-sensitized pregnancy should entail paternal red cell typing and genotype testing

At present, prophylactic Ig preparations to prevent other forms of red cell alloimmunization such as anti-K₁ do not exist.

Management of Pregnancy in Rh Negative Women

Fetal Risk Assessment



Management of ICT Negative Pregnancy – Prevention of Sensitization

- Rh negative women with negative anti-D Antibody Screen (ICT) are candidates for Anti-D Ig at:
 - 28 weeks of gestation – RAADP.
 - After Antepartum events associated with an increased FMH
 - After delivery of Rh positive neonates – Postpartum prophylaxis

Routine Antenatal Anti-D prophylaxis (RAADP)

- After initial ICT screening become negative, ideally repeat ICT every 4 weeks until 28th week of Gestation and give Anti-D if ICT remains negative.
 - The issue of repeating an antibody screen after initial negative result is controversial and ACOG leaves the decision to repeat the antibody screen up to the obstetric provider.
- Practically we repeat ICT at 28th week of gestation and provide IM injection of 300 mcg (1500 IU) anti-D IgG prophylaxis if the ICT result is negative
 - **Why at 28 weeks?** Risk of FMH is more in the 3rd Trimester and exogenously administered anti-D Ig protective effect stay for 12 weeks.

Pharmacology of anti-D IgG

Mechanism of Action - unproven, but proposed mechanism includes,

- **Masking of Rh positive fetal RBC's** before they sensitize maternal immune system and rapid macrophage-mediated clearance of those RBCs and/or
- Down-regulation of antigen specific B cells before immune response occurs.

Mean half time - is **16-24 days irrespective of route of administration** i.e. whether IV or IM. However, peak serum levels are reached faster with IV injection.

Efficacy- rate of alloimmunization fall **from 16% to 1-2% with single dose** of routine postpartum administration of anti-D and **further reduced to 0.1 to 0.3% with addition of RAADP**

Preferred site for injection

- is *deltoid muscle*
- If gluteal region is used absorption may be delayed and efficacy reduced by inadvertent subcutaneous administration. (NHS guideline - UK).

The dosage of anti-D Ig after 20 weeks of gestation is calculated from the estimated volume of the FMH, as described below.

- One 300-µg dose is given for each 15 mL of fetal red cells or 30 mL of fetal whole blood to be neutralized.

Anti-D Administration Regimens in RAADP

- **Single dose regimen** - a single dose of 300mcg of anti-D Ig at 28 weeks.
 - It is practiced in US and also in our setup and *more convenient for the patient* than two dose regimen.
- **Two dose regimens** – 100 to 120mcg of anti-D Ig are administered *at 28 and 34 weeks of gestation*.
 - It is usually practiced in UK, Canada. Some meta-analysis found that two dose regimen was more effective than a single dose regimen.

D Alloimmunization has not been completely eliminated because of:

- Failure to administer anti-D Ig in accordance with medical society guidelines and
- Unsuspected FMH bleeding early in gestation, before administration of RAADP.

“Whenever there is doubt whether to give anti-D immune globulin, it should be given. If not needed, it will not cause harm, but failure to provide it when needed may have severe consequences”

“The rule of thumb should be to administer Rh immune globulin when in doubt, rather than to withhold it.” Vincent Freda,

Antepartum Prophylaxis for Pregnancy Complications with FMH

Fetomaternal hemorrhage (FMH) is defined by the transfer of fetal blood into normally separated maternal circulation during pregnancy or delivery. In most cases of Rh alloimmunization, FMH occurs in the antenatal period or, **more commonly at the time of delivery (90%)**.

- Sensitization may occur after a single exposure to as little as **0.1 mL** of fetal RBC

The immune response of a Rh-negative individual to RhD-positive red cells has been characterized into one of three groups:

1. **Responders** (60% to 70%) of individuals are responders who develop an antibody to relatively small volumes of red cells; in these individuals, the probability of immunization increases with escalating volumes of cells.
 - A small percentage of responders can be called **hyper responders** in that they will be immunized by very small quantities of red cells.
2. **Hypo-responders** (10% to 20%), can be immunized only by exposure to very large volumes of cells.
3. **Non-responders** (10% to 20%) of individuals who remain appear to be non-responders

Causes of FMH -Sensitizing events, if occur anytime, need to be separately covered with anti-D.

These include:

- **Pregnancy loss**
 - Abortion, spontaneous or induced
 - Threatened Abortion
 - Ectopic pregnancy
 - IUFD (any trimester)
- **Procedures**
 - Amniocentesis
 - Chorionic villus sampling
 - Fetal blood sampling
 - Molar pregnancy evacuation

- **Others**

- APH: **AP > PP**
- Delivery, Vaginal or cesarean
- ECV
- Abdominal trauma during pregnancy

Management of pregnancy loss before 20 weeks in Rh negative women

- Without intervention risk of alloimmunization from *spontaneous* abortion is **1.5 -2.0%** and **4-5%** from *induced* one (up-to-date 2022)
- In 2022 WHO recommend against administer Anti-D Ig for pregnancy terminations < 12 weeks (medical or surgical) but there is no consensus among experts and organizations.
- When Anti-D is to be given for <12 weeks, 50mcg (can protect against FMH of 2.5ml) is enough as mean fetal RBC volume at 12 weeks is 1.5ml; however, giving standard 300mcg do not cause harm
- FMH quantification is unnecessary to determine whether more than 1 vial (300mcg) of anti-D is needed before 20 weeks, but it should generally be performed after 20 weeks.
- Some guideline does not recommend provision of anti-D prophylaxis for threatened abortion before 10 weeks of gestation unless there is significant clinical bleeding, not just spotting.

Postpartum prophylaxis

An appropriate dose of anti-D Ig should be administered to ICT negative, Rh negative women within 72 hrs. of delivery of Rh positive neonates.

- Recognizing that 40 % of neonates born to Rh-negative women are also Rh-negative, administration of Ig is recommended only after the newborn is confirmed to be Rh positive.

- Cord blood typing is recommended even if prenatal cfDNA testing suggests Rh negativity of the fetus.

If Ig is inadvertently not administered following delivery, it should be given as soon as the omission is recognized, because there may be ***some protection up to 28 days postpartum.***

If delivery occurs less than 3 weeks from the administration of anti-D Ig used for antenatal indications such as ECV, a repeat dose is unnecessary unless a large FMH is detected at the time of delivery

It is estimated that in 2 to 3 per 1000 (~0.3%) pregnancies, the volume of FMH exceeds 30 mL of whole blood.

- A single dose of anti-D Ig would be insufficient in such situations. For this reason, all Rh-negative women should have estimation of FMH for possible additional doses after standard dose postpartum anti-D Ig is provided.

FMH Estimation

ACOG recommend that, all D-negative women should be screened at delivery, typically with a qualitative test, followed by quantitative testing if indicated.

Qualitative test

- **The rosette test**
 - Identifies whether fetal D positive cells are present in the circulation of a D-negative woman.
 - A sample of maternal blood is mixed with anti-D antibodies that coat any D positive fetal cells present in the sample. Indicator red cells bearing the D antigen are then added, and rosettes form around the fetal cells as the indicator cells attach to them by the antibodies. Thus, if rosettes are visualized, there are fetal D-positive cells in that sample.

- The test is designed to give **a negative** result when the amount of **FMH is <2 ml** and thus will not necessitate additional doses of anti-D Ig.
- When **rosette test is positive**, a quantitative test with kleihauer-Betke test or flow cytometry is recommended to determine the percentage of fetal RBCs in maternal blood.

Quantitative test

- **Kleihauer-Betke (KB) or Acid elation (AE) test.**
 - Is the most commonly used quantitative test
 - Fetal erythrocytes contain HbF, which is more resistant to acid elution than HbA. Following exposure to acid (the acid reagent used in KB test is **citrate**), the maternal hemoglobin dissolves away and only fetal hemoglobin remains. Therefore, after staining, the fetal erythrocytes appear **red (rose-pink)** and adult erythrocytes appear **as “ghosts”**. The fetal cells are then counted and expressed as a percentage of adult cells.
 - There are two scenarios in which the KB may be inaccurate:
 - ✓ Maternal hemoglobinopathies in which the fetal hemoglobin level is elevated, such as β -thalassemia, and
 - ✓ At or near term, because the fetus may already be producing hemoglobin A.
- **Flow cytometry,**
 - Uses monoclonal antibodies to hemoglobin F or to the D antigen and then measures the degree of fluorescence
 - Flow cytometry's advantages over KB test:
 - ✓ It is an automated test that can analyze a greater number of cells than the KB test.
 - ✓ It is also unaffected by maternal levels of fetal hemoglobin and by fetal levels of hemoglobin A.
 - ✓ more sensitive and accurate than the KB test,

- Its disadvantage is that, it uses specialized technology not routinely available in many hospitals.

The fetal blood volume involved in the FMH may be calculated using the following formulas:

$$\text{Fetal blood volume} = \frac{\text{MBV} \times \text{maternal Hct} \times \% \text{ fetal cells in KB test}}{\text{newborn Hct}}$$

As an example, for a pregnant woman of normal size who is normotensive and has reached full-term, the **maternal blood volume (MBV)** may be estimated as 5000 ml. thus, in a woman with a hematocrit of 35 % and whose fetus has a hematocrit of 50 %, the calculation for a KB test demonstrating staining of 1.7 percent of sample cells is:

$$\text{Fetal blood volume} = \frac{5000 \times 0.35 \times 0.017}{0.5} = 60 \text{ mL}$$

- A loss of this magnitude should be well tolerated hemodynamically but would require **two 300-μg doses** of anti-D immune globulin to prevent alloimmunization.

Another formula recommended by American Association of Blood Banks (AABB)

- Is that the percentage of fetal blood cells by KB or flow cytometry be multiplied by a factor of 50 (to account for an estimated maternal blood volume of 5000 mL) to calculate the volume of the FMH. This volume is divided by 30 to determine the number of vials of anti-D to be administered. A decimal point is rounded up to next higher whole number for values ≥ 0.5 . Because this calculation includes an inaccurate estimation of the maternal blood volume, one additional vial of anti-D is added to the calculation.

- **As an example**, a 2.8% % KB stain is calculated to indicate a 140 ml FMH. Dividing this number by 30 yields 4.6 vials of anti-D which will be rounded up to 5 vials, with one additional vial added; therefore, the blood bank would prescribe 6 vials of anti-D (a total of 1800 µg) for this patient.

If using an IM preparation of anti- D Ig, no more than five doses may be given in a 24-hour period.

- Should a large dose of RhIG be necessary, an alternative method would be, to give the calculated dose using one of the IV preparations of RhIG, two ampules—totaling 600 µg— may be given every 8hrs.

Twenty four hours after total estimated dose has been administered, performing an ICT is recommended and ***a positive ICT test or the presence of circulating fetal cells on a rosette*** test demonstrates that the dose was sufficient. If none detected, additional anti-D Ig should be administered and the assay repeated until positive result obtained.

Management of ICT Positive Pregnancy - the Alloimmunized pregnancy

Management of the alloimmunized pregnancy typically consists of: -

- Maternal antibody titer surveillance followed by ultrasound monitoring of the fetal MCA peak systolic velocity if a critical antibody titer is reached.
- Fetal blood sampling is generally performed if the MCA peak systolic velocity exceeds the threshold for severe anemia, with plan for concurrent intrauterine transfusion as needed.

Of fetuses from D-alloimmunized pregnancies,

- 25 to 30 percent will have mild to moderate hemolytic anemia, and
- **25 percent** have anemia severe enough to cause hydrops

If alloimmunization is detected and the titer is below the critical value, the titer is generally repeated every 4 weeks for the duration of the pregnancy (ACOG 2019a).

Maternal titers are screening tests, not diagnostic of severe anemia, and should be discontinued once critical value is reached; there is no benefit to repeating the titer.

- Similarly, if a prior pregnancy was complicated by alloimmunization, the current pregnancy is considered at risk and monitoring with weekly MCV-PSV is indicated starting from 16 to 18 weeks of gestation regardless of titer.

Critical titer is the anti-red cell titer associated with a risk for development of severe anemia and hydrops fetalis.

- It's laboratory dependent, but typically somewhere between 1:8 and 1:32

Middle Cerebral Artery Doppler Velocimetry

The anemic fetus shunts blood preferentially to the brain to maintain adequate oxygenation. The velocity rises because of increased cardiac output and decreased blood viscosity.

Serial measurement of the peak systolic velocity (PSV) of the fetal middle cerebral artery (MCA) is the recommended test to detect fetal anemia once critical titer is reached or if there is history of affected prior pregnancy.

Serial MCA-PSV monitoring starts:

- At 16-18 weeks of gestation if there is previous affected pregnancy and
- At 24 weeks of gestation in first affected pregnancy with critical titer reached or exceeded
- These are then repeated every 1 to 2 weeks

The threshold value of 1.5 multiples of the median (MoM) for gestational age, correctly identified all fetuses with moderate or severe anemia.

- This provided a sensitivity of 100 % (**88% on Gabbe 7th edition**), with a false positive rate of 12%.

If the velocity is between 1.0 and 1.5 MoM and the slope is rising—such that the value is approaching 1.5 MoM—MCA Doppler surveillance is generally performed at least weekly

If the MCA peak systolic velocity exceeds 1.5 MoM or if hydrops develops and anemia is the leading etiology, fetal blood sampling and intrauterine transfusion should be considered.

MCA -PSV is not measured beyond 35 weeks, but ultrasound evaluation for hydrops is performed as needed.

- The **false-positive rate** of MCA-PSV increases significantly beyond 34 weeks' gestation and stems from the normal rise in cardiac output that develops at this GA.

Lilley Curve: Sir William Lilley proposed the use of **amniotic fluid bilirubin** assessment as an indirect measure of the degree of fetal hemolytic. However; this Spectrophotometric analysis of amniotic fluid bilirubin, also known as the ΔOD_{450} test, is *no longer recommended and has now been replaced by noninvasive MCA-PSV*.



Fetal Blood Transfusion

Transfusion is generally recommended only if:

- the fetal hematocrit is **<30% and**
- GA **<35 weeks**

Later in gestation (≥ 35), the benefits of transfusion may be outweighed by the risks of delaying delivery and procedure related complications.

Transfusion is **most commonly intravascular**.

- However, the umbilical vein may be too narrow in the early 2nd™ to readily permit needle entry, and severe hemolysis may necessitate **intrapertoneal fetal transfusion**.

If **hydrops** has developed, the hematocrit is usually $\leq 15\%$.

- In the setting of hydrops, peritoneal absorption may be impaired, and some prefer to transfuse into both the fetal peritoneal cavity and umbilical vein.

The red cells transfused are:

- freshly donated, type O negative,
- cross matched and compatible to maternal blood sample,
- cytomegalovirus-negative,
- packed to a **hematocrit of ~ 80 %** to prevent volume overload,
- irradiated to prevent fetal graft-versus-host reaction, and
- leukocyte-poor

Before transfusion, a paralytic agent such as vecuronium with or without short-acting narcotic such as fentanyl may be given to the fetus to minimize movement.

In a non-hydrotic fetus, the **target hematocrit is 40 to 50 %**.

The volume transfused may be estimated by **multiplying the estimated fetal weight in grams by 0.02 for each 10-percent rise in hematocrit needed**.

- As an example, if EFW is 1000g, fetal Hct is 20% and our target Hct is 40%, the volume of transfusion will be $1000 \times 0.02 \times 2 = 40\text{ml}$.

In the **severely anemic fetus** at 18 to 24 weeks' gestation do not tolerate the acute correction of their hematocrit to normal values; a smaller volume is transfused initially, and another transfusion may be planned for approximately 2 days later. Subsequent transfusions take place every 2 to 4 weeks, depending on the hematocrit.

The MCA-PSV threshold for severe anemia is higher following an initial transfusion—1.70 MoM rather than 1.50 MoM. After the second procedure, the MCA Doppler loses its validity in predicting fetal anemia.

Outcomes

The overall survival rate approximates 95 % (91% on Gabbe 7th).

- For hydrotic fetuses, the neonatal survival rate is **about 80%**.

Complications include

- fetal death in 2 - 3.8 %
 - The stillbirth rate exceeds 15 % if transfusion is required before 20 weeks
- need for emergent cesarean delivery in 1 %
- infection 0.3% and
- preterm rupture of membranes in 0.3 %

Hemolytic Disease of the Fetus and Newborn (HDFN)

Red cell alloimmunization results from transplacental passage of **maternal antibodies that destroy fetal red cells**.

- Transportation of these antibodies across the placenta during pregnancy results in **fetal and/or neonatal anemia, neonatal hyperbilirubinemia**, and ultimately **hydrops fetalis**.

Fetal Anemia;

- Anti-D IgG is a non-agglutinating antibody that does not bind complement. This results in:
 - a lack of intravascular hemolysis
 - sequestration and subsequent destruction of antibody-coated red cells in the fetal liver and spleen are the mechanism of fetal anemia
- Alloimmunization leads to **overproduction** of immature fetal and neonatal red cells—**erythroblastosis fetalis**

Hydrops fetalis

- Is defined by the accumulation of extracellular fluid in at least two body compartments
- Is a **late finding** in cases of fetal anemia –
 - When hydrops is present, fetal hemoglobin deficits of 7 to 10 g/dL from the mean hemoglobin value for the corresponding gestational age can be expected.
- Its exact pathophysiology is unknown but is likely multifactorial.

Hydrops is diagnosed by identifying two or more fetal effusions—pleural, pericardial, or ascites—or one effusion plus anasarca.

- Sonographically measured *skin thickness of >5 mm constitutes edema or anasarca.*
- As hydrops progresses in severity, anasarca is an invariable feature and is usually accompanied by *placentomegaly and hydramnios.*
 - **Placentomegaly** is defined as placental thickness
 - ✓ ≥ 4 cm in the second trimester or
 - ✓ ≥ 6 cm in the third trimester
- If hydrops is found in association with red cell alloimmunization, hydrops is termed immune (10%). Otherwise it is nonimmune (most common type – 90%).

Chapter 10

DIABETES MELLITUS AND PREGNANCY

Prepared by Dr. Daniel Delessa (MI)

Definition and Incidence

- GDM is defined as glucose intolerance with onset or first recognition during pregnancy
- Preexisting diabetes affects 1% of all pregnancies
- GDM occurs in 7% of pregnant women

Types of DM

- **Pregestational diabetes** or diabetes that has been diagnosed prior to the onset of pregnancy includes
 - Type 1 DM
 - Type 2 DM
 - Other specific types of diabetes related to a variety of genetic-, drug-, or chemical-induced diabetes
- **Gestational DM**
 - The hallmark of GDM is **insulin resistance**, and as such, it is etiologically similar to type 2 diabetes
 - GDM can be considered to be type 2 disease that is unmasked by the metabolic changes of pregnancy
 - Risk factors for both conditions are similar and include obesity, family history, minority ethnicity, and older age

During pregnancy, classification of women with diabetes has often relied on the White classification

Table 32-3. Modified White classification of diabetes complicating pregnancy.

Class	Onset (age in years)	Duration (years)	Type of Vascular Disease
A1	Gestational – diet-controlled	–	None
A2	Gestational – treated with medication/insulin	–	None
B	20	< 10	None
C	10–19 or	10–19	None
D	< 10 or	20	Benign retinopathy
F	Any	Any	Nephropathy
R	Any	Any	Proliferative retinopathy
T	Any	Any	Prior renal transplant
H	Any	Any	Coronary artery disease

- ACOG no longer recommends the White classification. The current focus is whether diabetes *antedates* pregnancy or is first diagnosed *during* pregnancy.
- Many now recommend the system proposed by the American Diabetes Association

TABLE 60-3. Proposed Classification System for Diabetes in Pregnancy

Gestational diabetes: diabetes diagnosed during pregnancy that is not clearly overt (type 1 or type 2) diabetes

Type 1 Diabetes:

Diabetes resulting from β -cell destruction, usually leading to absolute insulin deficiency

- Without vascular complications
- With vascular complications (specify which)

Type 2 Diabetes:

Diabetes from inadequate insulin secretion in the face of increased insulin resistance

- Without vascular complications
- With vascular complications (specify which)

Other types of diabetes: genetic in origin, associated with pancreatic disease, drug-induced, or chemically induced

Data from American Diabetes Association, 2017a; Powers, 2018.

Pathophysiology of GDM

- Insulin sensitivity decreases in normal women as gestation advances.
- Pathogenesis of GDM is mediated by different hormones and pregnancy related changes, these includes: -
 1. **human placental lactogen (hPL) (aka human chorionic somatomammotropin)**
 - it increases up to 30-fold during pregnancy,
 - is thought to be the hormone mainly responsible for insulin resistance and lipolysis.
 - It is similar in structure to growth hormone and acts by reducing the insulin affinity to insulin receptors.
 - It also decreases the hunger sensation and diverts maternal carbohydrate metabolism to fat metabolism in the third trimester.
 - *The net effect is to favor placental transfer of glucose to the fetus and to reduce the maternal use of glucose.*
 2. **Maternal cortisol**
 - which likewise rise during pregnancy, may also contribute to insulin resistance by stimulating endogenous glucose production and glycogen storage and decreasing glucose utilization
 3. **Adipokines such as tumor necrosis factor- α (TNF- α), adiponectin, and leptin**
 - the change in *TNF- α* level has been found to be a significant predictor of insulin resistance during pregnancy.
 - *TNF- α* is produced by the placenta as well as by adipose tissue and may act in a paracrine fashion to impair insulin signaling mechanisms, resulting in decreased insulin sensitivity

Screening and Diagnosis

- Because in most cases, patients with GDM have normal fasting glucose levels, some challenge of glucose tolerance must be undertaken

There are two approaches to screening and diagnosis of GDM

- Two step approach
- One step approach

Two-step approach

- Step 1 consists of a 1-hour 50-g oral glucose challenge test (GCT).
 - It can be performed at any time of day and without regard to time of prior meal
 - A plasma value between 130 and 140 mg/dL is commonly used as a threshold
- If this screening test is positive, it is followed by the diagnostic test, a 3-hour 100-g OGTT.
 - Done after an overnight fast
 - A diagnosis of GDM is made when ≥ 2 thresholds are met or exceeded

One-step approach

- Using a 2-hour 75-g oral glucose load
 - the diagnosis of GDM can be made if there is ≥ 1 abnormal value on the 75g OGIT

Table 32–4. Diagnostic criteria for gestational diabetes mellitus.

	100-g GTT Plasma/Serum Level (mg/dL) (Carpenter/Coustan)	100-g GTT Plasma Level (mg/dL) (National Diabetes Data Group)	75-g GTT Plasma Level (mg/dL) (IADPSG)
Fasting	95	105	92
1-hour	180	190	180
2-hour	155	165	153
3-hour	140	145	—

Screening is generally performed at 24 to 28 weeks' gestation, when the "diabetogenic state" of pregnancy has been established.

BOX 40-2 SCREENING STRATEGY FOR DETECTING GDM

GDM risk assessment should be ascertained at the first prenatal visit.

- **Low risk:** Blood glucose testing is not routinely required if *all* of the following characteristics are present:
 - Member of an ethnic group with a low prevalence of GDM
 - No known diabetes in first-degree relatives
 - Age <25 years
 - Weight normal before pregnancy
 - No history of abnormal glucose metabolism
 - No history of poor obstetric outcome
- **Average risk:** Perform blood glucose testing at 24 to 28 weeks by one of the following methods:
 - Two-step procedure: A 50-g GCT followed by a diagnostic oral GTT in those who meet the threshold value in the GCT
 - One-step procedure: Diagnostic oral GTT performed on all subjects
- **High risk:** Perform blood glucose testing as soon as feasible using the procedures described above if one or more of these are present:
 - Severe obesity
 - Strong family history of type 2 diabetes
 - Previous history of GDM, impaired glucose metabolism, or glucosuria

If GDM is not diagnosed, blood glucose testing should be repeated at 24 to 28 weeks or anytime a patient has symptoms or signs suggestive of hyperglycemia.

Complications of diabetes during pregnancy

Maternal	Fetal	Neonatal
▪ preeclampsia ▪ preterm delivery ▪ spontaneous abortion ▪ DKA (in type 1)	▪ Congenital malformations ▪ Macrosomia ▪ IUFD ▪ IUGR in those with vascular disease ▪ Polyhydramnios	▪ hypoglycemia, ▪ hyperbilirubinemia, ▪ hypocalcemia, ▪ hypomagnesemia, ▪ polycythemia, ▪ respiratory distress syndrome, ▪ cardiomyopathy

- In contrast to women with pregestational diabetes, women with true GDM typically do not have diabetes-related vasculopathy or an increased risk of infants with congenital malformations. It's because of
 - short duration of the disorder and
 - late pregnancy onset

Congenital Malformations

- Account for 30% to 50% of perinatal mortality.
- Cardiac anomalies are the most common malformations seen in IDMs
- The congenital defect thought to be most characteristic of diabetic embryopathy is sacral agenesis or caudal dysplasia, but it's not pathognomonic for diabetes

Table 32–1. Some congenital anomalies of infants of diabetic mothers.

Cardiac	Atrial septal defects Ventricular septal defects Transposition of the great vessels Coarctation of the aorta
	Tetralogy of Fallot Truncus arteriosus Dextrocardia Cardiomegaly
Central nervous system	Neural tube defects Anencephaly Holoprosencephaly
Renal	Hydronephrosis Renal agenesis Ureteral duplication
Gastrointestinal	Duodenal atresia Anorectal atresia Omphalocele
Spinal	Caudal regression syndrome, sacral agenesis

Proposed factors associated with teratogenesis in pregnancy complicated by diabetes mellitus include: -

- Hyperglycemia-Maternal hyperglycemia is the primary teratogenic factor
- Ketone body excess
- Somatomedin inhibition

- Arachidonic acid deficiency
- Free oxygen radical excess
- Genetic susceptibility to the teratogenic influence of diabetes

The profile of a woman most likely to produce an anomalous infant would include a patient with:

- poor periconceptional glucose control,
 - long-standing diabetes, and
 - vascular disease
-
- HbA_{1c} can predict the risk for malformations when measured in the first trimester

Initial Maternal Hemoglobin A1c Level	Major Congenital Malformations (%)
≤ 7.9	3.2
8.9–9.9	8.1
≥ 10	23.5

Fetal Macrosomia

- maternal hyperglycemia → fetal hyperglycemia (because while glucose crosses the placenta, insulin doesn't) & **hyperinsulinemia** (response to the hyperglycemia) → excessive fetal growth
- there is increase in subcutaneous fat and visceromegaly in macrosomic infants of diabetic mothers, which results in a relatively large abdominal circumference with normal head and skeletal growth.
 - ✓ the growth is disproportionate, with chest-to-head and shoulder to-head ratios larger than those of infants of women with normal glucose tolerance.
- 1. Leads to the higher rate of shoulder dystocia and birth trauma

Neonatal Hypoglycemia

- Neonatal hypoglycemia, a blood glucose level less than 35 to 40 mg/dL during the first 12 hours of life, results from a rapid drop in plasma glucose concentrations following clamping of the umbilical cord.
- It is particularly common in macrosomic newborns, in whom rates exceed 50%
- The degree of hypoglycemia influenced by maternal glucose control during the latter half of pregnancy during labor and delivery.
 - Prior poor maternal glucose control=>fetal β -cell hyperplasia→exaggerated insulin release following delivery
- The hyperinsulinemia and resultant hypoglycemia typically lasts for several days following birth.

Respiratory Distress Syndrome

- **Insulin excess** may interfere with the normal timing of glucocorticoid-induced pulmonary maturation in the fetus.
- **Cortisol** acts on pulmonary fibroblasts to induce synthesis of fibroblast-pneumocyte factor, which then acts on type II cells to stimulate phospholipid synthesis.
 - ✓ insulin blocks cortisol action at the level of the fibroblast by reducing the production of fibroblast pneumocyte factor
- Beside hormonal factors, **cesarean delivery** not preceded by labor and **prematurity** in diabetic pregnancies increases the likelihood of neonatal respiratory disease.

Hypocalcemia

- Neonatal hypocalcemia, with serum levels below 7 mg/dL or an ionized level less than 4 mg/dL, occurs at an increased rate in IDMs
 - Hypocalcemia in IDMs has been associated with a failure to increase parathyroid hormone synthesis following birth

Hyperbilirubinemia and polycythemia

- Neonatal jaundice occurs in 25% to 53% of pregnancies complicated by pregestational DM and 38% of pregnancies in women with GDM.
- Jaundice largely due to prematurity
- State of relative hypoxia in utero → increased EPO in the IDM. → increased RBC production → polycythemia and hyperbilirubinemia

Management of DM during pregnancy

Management of the patient with type 1 or type 2 DM

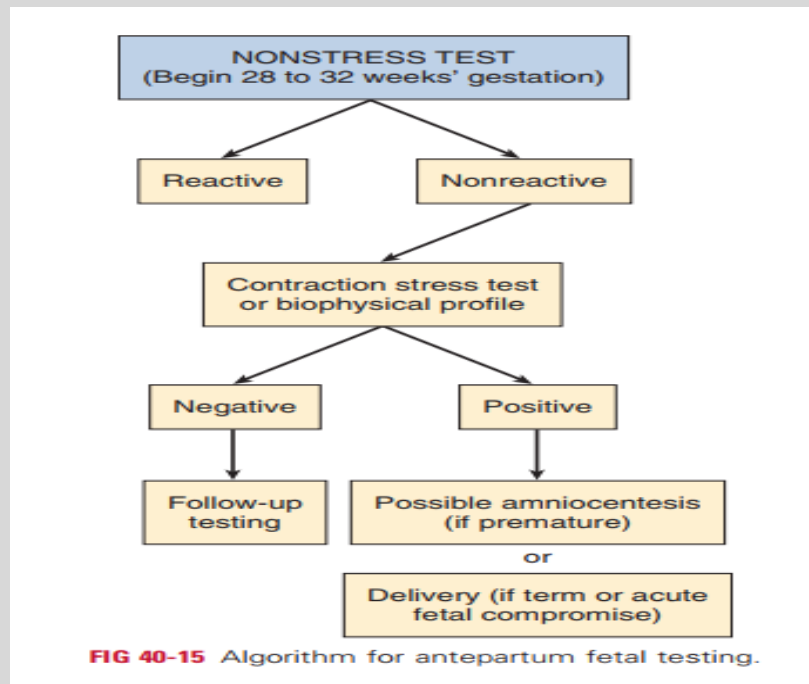
- Non-weight-bearing or low-impact exercise
 - 3 times per week for 30 – 60
 - Contraindications to exercise: PIH, PROM, preterm labor, incompetent cervix, persistent 2nd or 3rd trimester bleeding and IUGR.
- Diet-should be 40% carbohydrate, 40% fat, and 20% protein usually divided into 3 meals and 2 or 3 snacks per day.
 - A bedtime snack is important to prevent nocturnal hypoglycemia
- Self-monitoring of blood glucose: optimal glucose levels during pregnancy are:
 - fasting levels of 70-95 mg/dL and
 - 1-hour postprandial values ~ 140 mg/dL or 2-hour postprandial values ~ 120 mg/dL
- Insulin therapy
 - Due to its extensive safety record, insulin remains the first line treatment of diabetes in pregnancy

- While oral hypoglycemic agents have been used successfully for GDM, they are not currently recommended for overt diabetes
- Insulin requirement increases throughout pregnancy, most markedly in the period between 28–32 weeks of gestation.
- *Dosage:*
 - Starting insulin dose is 0.7-1.0 units/kg daily.
 - A combination of short & intermediate acting insulin is necessary to maintain glucose levels.
 - From total dose:
 - two-thirds before breakfast
 - From morning dose: $\frac{2}{3}$ is NPH and $\frac{1}{3}$ is regular insulin)
 - one-third before dinner
 - From evening dose: $\frac{1}{2}$ is NPH and $\frac{1}{2}$ is regular insulin)
 - Then each element of insulin is individually adjusted in order to keep blood glucose level between 70-130mg/dl.
 - Regular insulin should be given approximately 30 minutes before eating to reduce glucose elevations associated with eating.

Antepartum fetal evaluation

- **Maternal assessment of fetal activity**
 - During the third trimester, women are instructed to perform daily fetal movement counts.
- **Nonstress test (NST)**
 - preferred primary method to assess antepartum fetal well-being in the patient with DM.
 - if the NST is nonreactive, a biophysical profile (BPP) or contraction stress test (CST) is then performed

- In low-risk patients, do at least twice weekly after the patient reaches 32 weeks' gestation
- In patients with vascular disease, poor glucose control, or suspected FGR, testing is often initiated between 28- and 32-weeks' gestation.



▪ Doppler umbilical artery velocimetry

- in pregnancies at risk for placental vascular disease for women with nephropathy or hypertensive disease
- Elevated placental resistance, as evidenced by an increased systolic-to-diastolic ratio, is associated with FGR and preeclampsia in these high-risk patients

▪ Ultrasound

- Used for evaluating fetal growth, estimating fetal weight, and detecting hydramnios and malformations.
- MSAFP determination at 16 weeks' gestation is often used in association with a detailed ultrasound study during the midtrimester in an attempt to detect NTDs and other anomalies

- Levels of MSAFP, unconjugated estriol (uE₃), and inhibin A in diabetic women are lower than those in the nondiabetic population.
- A lower threshold for the upper limit of normal (ULN) for MSAFP,^{1.5} multiples of the median (MoM), is preferable
- A comprehensive ultrasound examination at 18 to 20 weeks, as well as fetal echocardiography performed at 20 to 22 weeks, is undertaken to diagnose fetal anomalies
- Ultrasound examinations should be performed during the third trimester to assess fetal growth.
 - sonographic measurements of the fetal abdominal circumference is most helpful in predicting fetal macrosomia.
 - abdomen is likely to be large because of increased subcutaneous fat deposition

TABLE 40-9 ANTEPARTUM FETAL SURVEILLANCE IN LOW-RISK INSULIN-DEPENDENT DIABETES MELLITUS*

STUDY	INDICATED
Ultrasonography at 4- to 6-wk intervals	Yes
Maternal assessment of fetal activity, daily at 28 wk	Yes
NST at 32 wk; BPP or CST if NST is nonreactive	Yes
Amniocentesis for lung profile	Yes, if elective delivery is planned before 39 wk

*Excellent control, no vasculopathy (classes B, C), no stillbirth.

BPP, biophysical profile; CST, contraction stress test; NST, nonstress test.

TABLE 40-10 ANTEPARTUM FETAL SURVEILLANCE IN HIGH-RISK INSULIN-DEPENDENT DIABETES MELLITUS*

STUDY	INDICATED
Ultrasonography at 4- to 6-wk intervals	Yes
Maternal assessment of fetal activity daily at 28 wk	Yes
NST; BPP or CST if NST is nonreactive	Initiate at 28-30 wk
Consider amniocentesis for lung profile prior to 38 wk	

*Poor control (macrosomia, hydramnios), vasculopathy (classes D, F, R), prior stillbirth.

BPP, biophysical profile; CST, contraction stress test; NST, nonstress test.

Delivery

Timing of delivery

- Delivery should be delayed until fetal maturation has taken place, provided diabetes is well controlled and antepartum surveillance remains normal.
- Delay delivery until 39 weeks in women without vascular disease whose diabetes is well controlled
- For diabetic women with vascular disease, it has been suggested that delivery may be undertaken at 37 to 39 *weeks.
 - Women with vascular disease are delivered prior to 37 weeks only if hypertension worsens, if significant FGR is present, or if biophysical testing mandates early delivery
- In women with poor glycemic control, use individualized approach
- late preterm or early term delivery or amniocentesis to assess lung maturity prior to delivery

Lung maturity test

- RDS may occur in the IDM with a mature lecithinto-sphingomyelin (L/S) ratio or fetal lung maturity index but absent PG.
- The presence of the acidic phospholipid phosphatidylglycerol (PG) is the final marker of fetal pulmonary maturation
 - onset of PG production is delayed in GDM
- lamellar body count can also be used to assess amniotic fluid for lung maturity

Indications for delivery before term

- When antepartum testing suggests fetal compromise
- Significant preeclampsia,
- Worsening renal function,
- Deteriorating vision secondary to proliferative retinopathy

Mode of delivery

- Cesarean section is reserved for obstetric indications.
- ACOG recommends consideration of cesarean delivery in diabetic women when estimated fetal weight exceeds 4500 g

Intrapartum management

- Strict control of maternal glycemia during labor and delivery is commonly recommended to optimize fetal outcomes.
- During labor, a variety of factors impact glycemic management and should be considered:
 - **Metabolic demands of labor** – During the active phase of labor, glucose uptake by the contracting uterus lowers blood glucose and reduces insulin requirements, analogous to exercise
 - **Food restriction** – Many people in labor are placed on a clear liquid diet.
 - ✓ Caloric restriction can reduce insulin requirements, but clear liquid diets that include sugar-sweetened beverages can cause hyperglycemia
 - **Dextrose-containing intravenous fluids** – Administration of dextrose-containing fluids may prevent dehydration, shorten the length of labor, and prevent ketosis, but may also lead to hyperglycemia if insulin is not present in sufficient quantities
- Glycemic targets
 - A reasonable target range for intrapartum glucose levels is **70 to 125 mg/dL** (uptodate)
 - ACOG; **70 to 110 mg/dL**
 - Current: **70-120**
- *Attention to intrapartum glycemic control is based on evidence that it reduces the risk of two major complications: fetal hypoxemia and neonatal hypoglycemia*

- fetal hyperglycemia and hyperinsulinemia (resulting from maternal hyperglycemia) may increase fetal oxygen consumption and can lead to fetal hypoxemia and acidemia
- Patient is kept NPO after midnight.
 - Usual dose of intermediate-acting insulin is given at bedtime.
 - Withhold morning (AM) insulin injection.
- Intravenous infusion of normal saline is begun
 - Once active labor begins or glucose levels fall to <70 mg/dL, the infusion is changed from saline to 5% dextrose
 - Begin and continue glucose infusion (5% dextrose in water) at 100 – 150 mL/hr.
- Add 10 U of regular insulin to 1000 mL of solution containing 5% dextrose.
 - Begin infusion of regular insulin if capillary glucose is greater than 80 mg/dl (if >100 mg/dL-William, if >140 mg/dl-Gabbe)
- Use fluid without dextrose if capillary glucose is greater than 180 mg/dL.
- Monitor maternal glucose levels hourly and adjust insulin infusion.
- Begin oxytocin as needed.

Insulin infusion protocol

Plasma/ Capillary Glucose (mg/dL)	Infusion Rate (U/hr)	Approximate Infusion Rate (Drops/min)	IV fluid
<80	Insulin off	40	5% dextrose
80-100	0.5	20	5% dextrose
101-140	1.0	30	5% dextrose
141-180	1.5	50	5% dextrose
181-220	2.0	60	Without dextrose
>220	2.5	80	Without dextrose

- When cesarean delivery is to be performed, it should be scheduled for early morning.
 - It simplifies intrapartum glucose control and
 - allows the neonatal team to prepare for the care of the newborn
- The patient is given nothing by mouth, and her usual morning insulin dose is withheld.
 - If her surgery is not performed early in the day, one third to one half of the patient's intermediate-acting dose of insulin may be administered
- Regional anesthesia allows detection of maternal hypoglycemia.
- Following surgery, glucose levels are monitored every 2 hours, and an IV solution of 5% dextrose is administered.

Postpartum

- Postpartum insulin requirements are usually significantly lower than during pregnancy
- The antepartum objective of tight glucose control is relaxed for the first 24 to 48 hours after delivery.
- Patients delivered vaginally who are able to eat a regular diet are given one third to one half of their end-of-pregnancy dose of NPH insulin and rapid-acting insulin the morning of the first postpartum day.

Management of the woman with GDM

- Mainstay of treatment is **nutritional counseling and dietary intervention**
- If a patient repetitively exceeds established thresholds, pharmacologic therapy is recommended
- Exercise
- Glycemic target thresholds
 - fasting glucose less than 95 mg/dL,
 - 1-hour postprandial glucose less than 140 mg/dL, and
 - 2-hour postprandial glucose less than 120 mg/dL

- ACOG--either insulin or oral agents may be appropriate first-line therapy for GDM.
 - The currently recommended oral agent is **only glyburide or metformin**
 - It appears that glyburide may be superior to metformin in achieving satisfactory glucose control in women with GDM

Antenatal care

- Women with GDM that is well controlled by diet alone usually do not require antenatal fetal testing
 - fetal surveillance with nonstress testing or biophysical profiles may be initiated at 40 weeks
- earlier initiation of fetal surveillance and ultrasound assessment of fetal growth are advised for women who
 - require medication for control of their blood sugars
 - are noncompliant, or
 - have GDM that is not well controlled

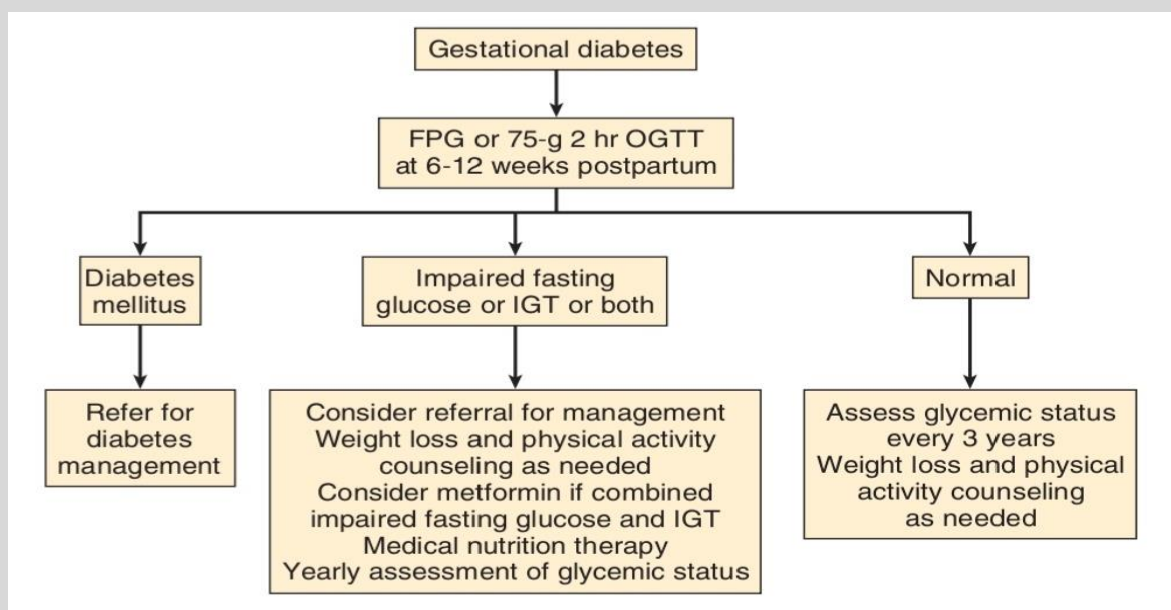
Time of delivery

- If GDM is well controlled on diet or medication, delivery should not be undertaken prior to 39 weeks' gestation.
- If glycemic control is suboptimum, ACOG recommends that the decision to schedule late-preterm or early-term birth in such cases should be individualized

Postpartum follow-up of women with GDM

- Women with GDM have a 7X increased risk for developing type 2 diabetes compared with women who do not have diabetes during pregnancy
- They also are at high risk for recurrence in future pregnancies

- Do postpartum glucose testing following a diagnosis of GDM
 - As many as one third of women with GDM will have overt diabetes, impaired fasting glucose, or IGT identified during postpartum testing conducted within 6 to 12 weeks of delivery
 - use either a fasting plasma glucose or a 75-g 2-hour oral GTT at 6 to 12 weeks' postpartum.
- repeat testing at least every 3 years for women with prior GDM and normal results of postpartum screening



Long-Term Effects of Glucose Intolerance on Mother and Fetus

Mother

- GDM is a significant risk factor for the later development of type 2 diabetes
 - 50% of patients with GDM will ultimately go on to develop type 2 diabetes later in life
- higher risk for developing the metabolic syndrome
- increased risk for cardiovascular dysfunction such as coronary artery disease and stroke

Fetus

- increased risk of diabetes, obesity and metabolic syndrome

Pre-pregnancy counseling of women with preexisting diabetes mellitus

- DM associated anomalies of the cardiac, renal, and central nervous systems arise during the first 7 weeks of gestation, to avoid these *management and counseling of women with diabetes should begin before conception.*

BOX 40-8 PREPREGNANCY CARE FOR DIABETIC WOMEN

- Multidisciplinary care team: Obstetrician, endocrinologist, diabetes educator, nutritionist
- Evaluation for vascular complications
 - Retinal examination
 - Assess renal function: Serum creatinine and evaluation for proteinuria
- Evaluation for cardiovascular status
 - Hypertension
 - Ischemic cardiac disease: Electrocardiogram if longstanding diabetes, hypertension, or symptoms
- Review medications
 - Angiotensin-converting enzyme inhibitors, angiotensin II receptor blockers, and statins are best avoided if planning conception.
 - Folate supplementation is recommended.
- Assessment of glycemic control
 - Measure HbA1c every 2 months.
 - Target HbA1c should be $\leq 7.0\%$.
 - Contraception is not advised until glucose is well controlled.
- Promote healthy lifestyle
 - Regular exercise
 - Nutrition counseling and weight loss for obese women
 - Smoking cessation

Chapter 11

Intrauterine Growth Restriction/Retardation

Prepared by Dr. Balkewkal Kebede (MI)

Introduction

Before going in to details on IUGR, what are differentials for small and large for date uterus?

- We say small/large for date uterus when there is a discrepancy of 4wks or more between the fundal height and GA

DDx for small for date uterus	DDx for large for date uterus
▪ IUGR ▪ IUFD ▪ Oligohydramnios ▪ PROM ▪ Transverse lie ▪ Engagement ▪ Anencephaly ▪ Wrong date (the LNMP date she told you is wrong)	▪ MG ▪ Polyhydramnios ▪ Hydrops fetalis ▪ Macrosomia ▪ Uterine/Adnexal mass (myoma) ▪ Distended bladder ▪ Wrong date

Definition and Pathogenesis

IUGR is a syndrome marked by failure of the fetus to reach its growth potential. Next to prematurity, IUGR is the second leading cause of perinatal mortality.

- Compared with appropriately grown counterparts, perinatal mortality rates in growth-restricted neonates are **6 to 10 times greater**;
 - perinatal mortality rates as high as 120 per 1000 for all cases of IUGR
 - As many as 53% of preterm stillbirths and 26% of term stillbirths are growth restricted.
 - In survivors, the incidence of intrapartum asphyxia may be as high as 50%

Placental and fetal growth

Placental and fetal growth across the three trimesters is characterized by *sequential cellular hyperplasia, hyperplasia plus hypertrophy, and lastly, by hypertrophy alone.*

Placental growth follows a sigmoid curve that plateaus in midgestation preceding exponential third-trimester growth of the fetus.

- During this exponential fetal growth phase of 1.5% per day, initial weight gain is due to longitudinal growth and muscle bulk and therefore correlates with glucose and amino acid transport.

Normal fetal growth also occurs in three phases

- The 1st phase; cellular hyperplasia
 - Occur in the 1st TM
 - Growth is a consequence of cell division
- The 2nd phase; cellular hyperplasia and hypertrophy
- The 3rd phase; cellular hypertrophy
 - Occur in the last 6 to 8wks of pregnancy

Disturbance of fetal growth dynamics can lead to a reduced cell number, cell size, or both, ultimately resulting in abnormal weight, body mass, or body proportion at birth

Eighty percent of fetal fat gain is accrued after 28 weeks' gestation, providing essential body stores in preparation for extra-uterine life.

- Rate of fetal growth (weight gain) increases from 5g/day at 14-15wks to 10g/day at 20wks, and to **30 to 35g/day at 32 to 34wks**
- From 32 weeks onward, fat stores increase from 3.2% of fetal body weight to 16%, which accounts for the significant reduction in body water content

- Mean weekly weight gain peaks at ~230-285g at 32-34wks of gestation, after which it decreases, progressively to zero even there might be weight loss after 41wks.

Incidence of IUGR

- Approximately $\frac{1}{4}^{\text{th}}$ to $\frac{1}{3}^{\text{rd}}$ of all low birthweight infants have sustained IUGR
- ~4-8% of all infants in developed and ~6-30% in developing countries have been classified as growth restricted
- Rate of **recurrence** is believed to be 20-25%

Classification of FGR

Two principal patterns of disturbed fetal growth have been described:

- **asymmetric** growth delay
 - is when growth of the trunk (abdomen and chest) and lower body shows a significant delay, whereas there is relative or absolute sparing of head growth
- **symmetric** growth delay
 - Both body and head growth are similarly affected.

The pattern of fetal growth depends on the cause of growth delay and on the timing & duration of the insult. For example: -

- Aneuploidy, non-aneuploidy syndromes, viral infections, and first trimester insults result in symmetric growth delay
- Disturbance in the fetal growth dynamics at earlier stages of growth (especially at hyperplasia) results in a reduced cell number, size, or both, resulting in abnormal body weight and body mass
- Uteroplacental insufficiency (from multiple causes) result in asymmetric growth delay, because of shifting of blood flow to the head (brain)

Summary of symmetric Vs. asymmetric growth delay

Criteria	Symmetric delay	Asymmetric delay
Body proportion	Uniformly small	Head larger than abdomen
HC/AC FL/AC	Normal	Elevated
Ponderal index (birthweight/CHL)	Normal	Low
Total cell number	Less	Normal
Cell size	Normal	Smaller
Neonatal course	Complicated with poor prognosis	Usually, uncomplicated having good prognosis
Etiology	Intrinsic to the fetus Aneuploidy Fetal infection	Extrinsic to the fetus Chronic uteroplacental insufficiencies

Classification of fetus/newborn based on Birthweight percentile

The terms SGA, simply identifies a fetus below a specific weight cut off without identifying the underlying cause.

IUGR/FGR imply that some pathological process has prohibited attainment of genetic growth potential

Classification based on birthweight percentiles (P)

- $<3^{\text{rd}}$ P= $\Rightarrow$ very small for GA
- $<10^{\text{th}}$ P= $\Rightarrow$ small for GA
- $10^{\text{th}}-90^{\text{th}}$ P= $\Rightarrow$ average for GA
- $>90^{\text{th}}$ P= $\Rightarrow$ large for GA

After confirming small fetal size, stratification into three patient groups is of particular importance.

- The first group consists of **constitutionally small** but otherwise normal fetuses.
 - Include ~70% of SGA newborns
 - They aren't at risk of adverse outcomes

- These will not usually require any intervention and therefore do not need antenatal surveillance.
- You have to reassure the parents
- The second group consists of **fetuses with aneuploidy, non-aneuploidy syndromes, or viral infection**.
 - In these conditions, prognosis is largely determined by the underlying disease, with little potential for impact by perinatal interventions.
 - Sensitive and knowledgeable counseling of the parents about the likely prognosis in these conditions is especially important in such cases.
- The third group consists of **fetuses with placental disease** in which progressive deterioration of the fetal condition worsens the prognosis.
 - This subset of patients is most likely to benefit from fetal surveillance and subsequent intervention

Etiologies of IUGR

- 40% of total birthweight variation is due to genetic contributions from mother and fetus (~half from each), the rest 60% is due to contribution from the fetal environment.

Conditions that result in FGR broadly consist of maternal, uterine, placental, and fetal disorders. These conditions result in growth delay by affecting:

- nutrient and oxygen delivery to the placenta (maternal causes),
- nutrient and oxygen transfer across the placenta (placental causes), and
- fetal nutrient uptake or regulation of growth processes (fetal causes)

Maternal causes

- **Vascular disease** such as: - ***hypertensive disorders of pregnancy***, pregestational DM (diabetic vasculopathy), collagen vascular disease, thrombophilia and chronic renal disease.
- Restrictive pulmonary disease

- Chronic renal insufficiencies
- Cyanotic cardiac disease
- High altitude (>10,000ft)
- Maternal malnutrition, (malabsorption or eating d/r)
- Smoking/substance abuse (cocaine, heroin)
- MG
 - The dx and prognosis of IUGR in twin pregnancies is critically determined by **chorionicity**
 - Incidence of FGR in twins is 15 to 25%
 - Causes
 - TTTS
 - High occurrence of congenital anomalies
 - Selective IUGR: is a specific term applied to FGR that occur in monochorionic twins
- Anemia from sickle-cell disease
- Hx of infertility

Placental cause

- Primary placental disease, chorioangioma
- Chronic placental abruption
- Placenta previa
- Placental mosaicism
- Single umbilical artery
- Abnormal cord insertion

*Maternal and placental causes can be summarized as **uteroplacental insufficiencies***

Fetal causes

- Teratogenic exposure, like
 - Cyclophosphamide
 - Valproic acid, phenytoin
 - Antithrombotic drugs, like Warfarin (Coumarin Embryopathy)
- Fetal infections
 - Contribute to <10% of FGR
 - HSV, CMV, Rubella, and T. gondii are well documented causes of symmetric IUGR
 - **Malaria** accounts for most of infection related FGR worldwide
- Chromosomal disorders and genetic syndromes
 - Contribute to <10% of FGR
 - Growth restriction has been observed in 53% of cases of trisomy 13 and in 64% of cases of trisomy 18 and may be manifest as early as the first trimester
- Structural anomalies

Check out the maternal and fetal manifestations of IUGR on Gabbe 7th ed.

Diagnosis of FGR

Hx and P/E

- Establish GA,
 - *Accurate determination of GA is the single most important part of evaluation for IUGR, -wrong GA=wrong Dx*
- Ask about weight gain, size of abdomen compared with previous pregnancy of same GA
- Risk factors/causes
- Past obstetric hx (recurrence)

Ultrasound

Fetal biometry

The primary measurements utilized to evaluate fetal growth include: -

- fetal head and abdominal circumference,
- measurement of long bones (Femoral length (FL))
- sonographically estimated fetal weight (SEFW)
 - is the most important variable of fetal growth evaluation

Accurate estimation of fetal growth from these fetal measurements requires knowledge of the gestational age as a reference point to calculate percentile ranks.

- An estimated date of confinement (EDC)/EDD should be based on the last menstrual period when the sonographic estimate of gestational age is within the predictive error (7 days in the first, 14 days in the second, and 21 days in the third trimester).
 - Once the EDC is set by this method or by a first-trimester ultrasound, it should not be changed because such practice interferes with the ability to diagnose IUGR

- **BPD and HC measurements**

- Measurement of the biparietal diameter (BPD) alone is a poor tool for the detection of IUGR
- Factors that interfere with a technically adequate measurement of the BPD include
 - ✓ alterations of the cranial shape by external forces (oligohydramnios, breech presentation) and
 - ✓ direct anteroposterior position of the fetal head
- HC is not subject to the same extrinsic variability as the BPD.
- Both HC and BPD detect asymmetric FGR late

▪ Abdominal circumference

- AC <10th percentile is the cut off for IUGR
- The most accurate AC is the smallest directly measured circumference obtained in a perpendicular plane of the upper abdomen at the level of hepatic vein between fetal respiration
- AC percentile has both the **higher sensitivity (98%)** and negative predictive value (lower positive predictive value (36%)) when compared with SEFW
- Its Sn can be further enhanced by serial measurements at least 14days apart
- *Reflects fetal nutrition, and it should be excluded from the calculation of composite GA after early 2nd TM*

▪ HC: AC ratio

- **To detect asymmetric FGR (is normal in symmetric GR)**
- Normally HC: AC is >1.0 before 32wks, =1.0 at 32-34wks and <1.0 after 34wks, therefore **ratio >1.0 after 32wks indicate disproportionately larger head**
- Detect 70-80% growth restricted fetus
- Has lower Sn and PPV as compared to AC alone or SEFW percentile

▪ FL (femoral length): AC ratio

- FL is relatively spared in asymmetric GR
- Can be used when HC measurement is difficult because of fetal position
- The ratio is 22 at all GA from 21wks on to term (this means it can be applied without GA)
- **Ratio >23.5 suggest IUGR**

▪ SEFW

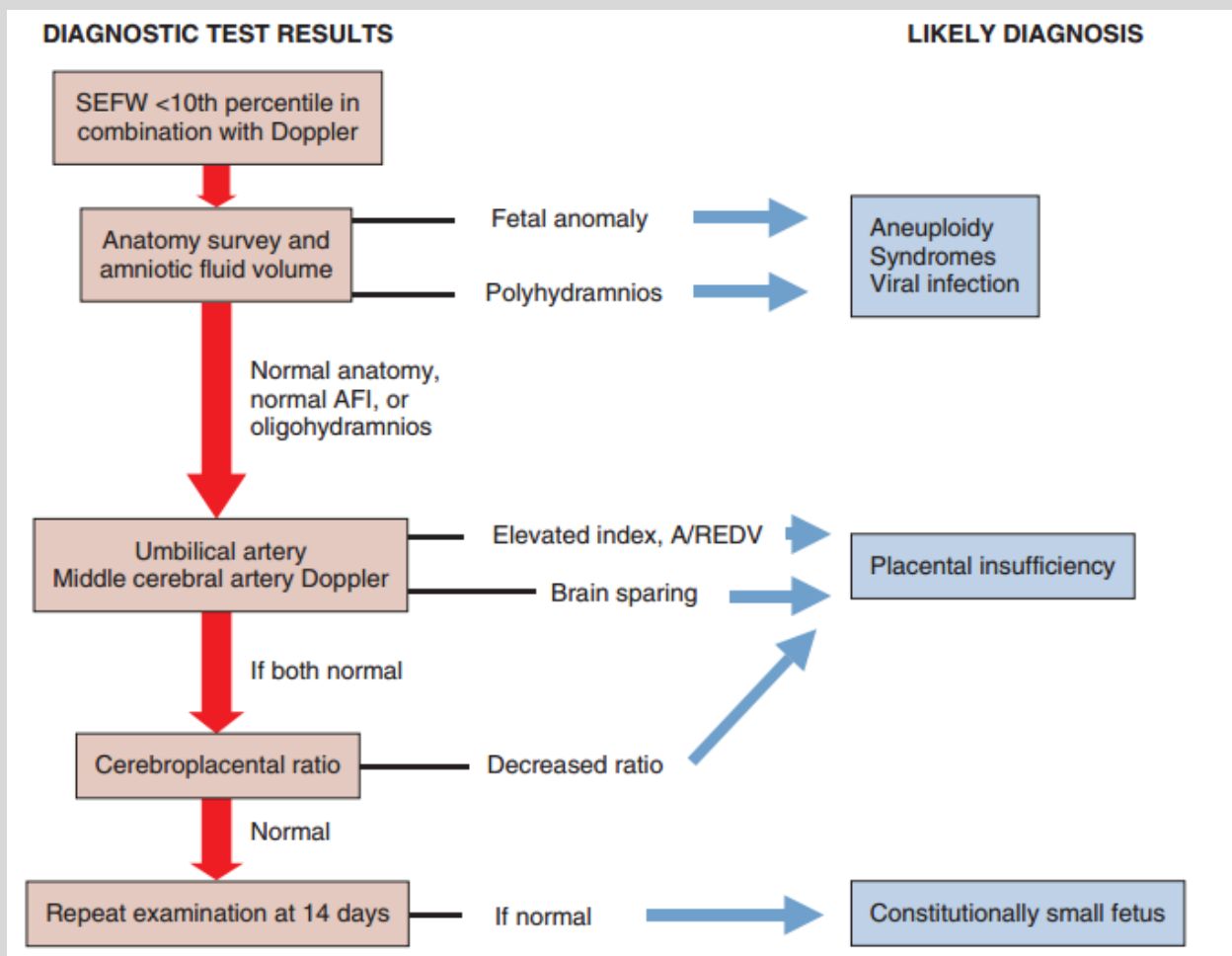
- Is calculated from the measurements above (especially the AC) using different formula. The accuracy of most formulas is $\pm 10\%$
- **<10th percentile suggests IUGR**
- Has lower Sn but higher PPV than AC percentile
- Is mostly used method for characterizing fetal size and growth

NB. ~70% of infants with BWt <10th percentile are normally grown (constitutionally small), the remaining 30% constitute those truly growth restricted, and are at risk for increased perinatal morbidity and mortality.

- When the cutoff is adjusted to 3rd percentile, the infants (truly GR) identified increases, but this might lead to missing some milder forms of IUGR

It is recommended to do U/S evaluation at 3wks interval, shorter interval increase likelihood of false positive dx.

The diagram below summarizes the diagnostic tests and the likely diagnosis



Other assessments to be done in IUGR

- Fetal anatomic survey
- Amniotic fluid assessment
- Doppler Velocimetry
 - It serves as both diagnostic and monitoring tool
 - Arterial Doppler waveforms provide information on downstream vascular resistance
 - Three Doppler indices are used to analyze arterial blood flow resistance, systolic/diastolic(S/D) ratio, the resistance index (RI), and the pulsatile index (PI)
 - ✓ An increase in blood flow resistance manifest with a relative decrease in end-diastolic velocity, that result in an increase in all three indices
 - ✓ Extreme resistance will cause absent end-diastolic velocity (AEDV) or reverse end-diastolic velocity (REDV)
 - ✓ Vascular damage affecting ~30% of the placenta produces elevation of Doppler index
 - Venous Doppler parameters complement evaluation of fetal cardiovascular status by providing an assessment of cardiac forward function
 - Umbilical artery (UA) and middle cerebral artery evaluation is important in detecting mild form of placental dysfunction
 - Ductus venosus (DV) Doppler has moderate predictive value for fetal acidemia and adverse outcome. And it is used to time delivery
 - Umbilical vein (UV) Doppler is important in detecting fetal cardiac compromise
- Invasive testing
 - Screen for TORCH
 - Karyotyping and microarray analysis to r/o chromosomal abnormalities, like trisomy 13, 18, or 21.

- ✓ Trisomy 18 (Edward's syndrome) is associated with unusual combination of *FGR and Polyhydramnios*

Management

▪ Antepartum fetal surveillance (AFS)

- UA Doppler should be used as a primary surveillance tool in IUGR fetuses, because it has shown to decrease perinatal mortality and morbidity
 - Normal indices=> repeat surveillance every 2wks
 - Abnormal indices, but no indication for delivery => twice weekly surveillance
- CTG or U/S assessment of AF shouldn't be used as an only form of surveillance
- *BPP shouldn't be used for fetal surveillance in preterm SGA fetuses*

▪ Delivery

- Depends mainly on the GA and Doppler Velocimetry results
- Delivery is recommended by 32wks, for those detected prior to 32wks with AEDV/REDV and abnormal DV Doppler or UV pulsation, after completion of steroid
- Deliver at 37wks for those detected after 32wks
- **In GA of 24-29wks**
 - Threshold for fetal indication should be high, i.e. don't rush to delivery, strong evidence of fetal compromise and risk of stillbirth is required
 - Manage maternal conditions like severe PE
 - Indications for delivery: -
 - FHR tracing with deceleration
 - BPP of ≤ 4
 - Reverse a-wave in DV, marker of acidemia
 - Despite maximum intervention PMR exceeds 50%

- **GA 29-34wks**
 - Firm evidence for termination, try to complete corticosteroid course whenever possible
 - Terminate immediately if the above indications are present
- **GA 34-37wks**
 - Lower delivery thresholds are acceptable, and may include: -
 - Absent fetal growth (particularly arrested head growth)
 - oligohydramnios
 - documented lung maturity on amniocentesis
 - Doppler evidence of accelerating disease
 - Maternal co-morbidity
- **GA >37wks**
 - Delivery can be delayed up to 38-39wks, provided normal AFS, AFV, and no maternal co-morbidity
- Mode of delivery
 - If UA Doppler show AEDV/REDV=>CD is recommended. Otherwise, induction of labor can be offered
 - Early admission and continuous FHR monitoring is recommended for those with spontaneous onset of labor

The diagram below shows an integrated fetal testing and management protocol

IUGR UNLIKELY		
Normal AC, AC growth rate and HC/AC ratio UA, MCA Doppler, BPS, and AFV normal	Asphyxia extremely rare low risk for intrapartum distress	Deliver for obstetric or maternal factors only, follow growth
IUGR		
AC <5th, low AC growth rate, high HC/AC ratio, abnormal UA and/or CPR, normal MCA and veins, BPS $\geq 8/10$, AFV normal	Asphyxia extremely rare Increased risk for intrapartum distress	Deliver for obstetric or maternal factors only, Every 2 weeks Doppler Weekly BPS
With blood flow redistribution		
IUGR diagnosed based on above criteria Low MCA, normal veins BPS $\geq 8/10$, AFV normal	Hypoxemia possible, asphyxia rare Increased risk for intrapartum distress	Deliver for obstetric or maternal factors only, weekly Doppler BPS 2 times/week
With significant blood flow redistribution		
UA A/REDV normal veins BPS $\geq 6/10$, oligohydramnios	Hypoxemia common, acidemia or asphyxia possible Onset of fetal compromise	>34 weeks: deliver <32 weeks: antenatal steroids Repeat all testing daily
With proven fetal compromise		
Significant redistribution present Increased DV pulsatility BPS $\geq 6/10$, oligohydramnios	Hypoxemia common, acidemia or asphyxia likely	>32 weeks: deliver <32 weeks: admit, Steroids, individualize testing daily vs. tid
With fetal decompensation		
Compromise by above criteria Absent or reversed DV a-wave, pulsatile UV BPS <6/10, oligohydramnios	Cardiovascular instability, metabolic compromise, stillbirth imminent, high perinatal mortality irrespective of intervention	Deliver at tertiary care center with the highest level of NICU care

Chapter 12

Amniotic fluid disorders

Prepared by Dr. Balkewkal Kebede (MI)

Introduction

AF formation and removal

- Early in pregnancy, the amniotic cavity is filled with fluid that is similar in composition to extracellular fluid.
- During the first half of pregnancy, transfer of water and other small molecules takes place
 - across the amnion—*transmembranous flow*;
 - across the fetal vessels on placental surface—*intramembranous flow*; and
 - across fetal skin— *transcutaneous flow*
- Fetal urine production begins between 8- and 11-weeks' gestation, but it does not become a major component of amniotic fluid until the second trimester, which explains why fetuses with lethal renal abnormalities may not manifest severe oligohydramnios until after 18 weeks.
- Water transport across the fetal skin continues until keratinization occurs at 22 to 25 weeks.
 - *This explains why extremely preterm neonates can experience significant fluid loss across their skin.*

Generally, AF is formed from **fetal urine (800-1200ml/day)**, & fetal lung fluid secretion (~340ml); and it is removed by fetal swallowing (500-1000ml/day) and intramembranous absorption (200-500/ml)

- AF volume increases from ~30ml at 10th wk., to 200ml by 16th wk., and reaches 800ml in the mid trimester; it can reach maximum volume of ~1000ml at around 31/32wks of gestation.

AF Function

- Creates a physical space for fetal movement, which is necessary for fetal normal musculoskeletal system development
- Permits fetal breathing, necessary for lung development
- Permits fetal swallowing, necessary for fetal GIT development
- Guards against umbilical cord compression
- Protect fetus from trauma
- Has bacteriostatic properties

Assessment of AFV

Clinically AF disorders should be suspected when: -

- the uterus measures too large or too small for the GA
- the fetal parts are either difficult to palpate or easily palpable

Ultrasound/Sonographic assessment

There are two ways of measuring AFV using U/S

- Measuring the **single Maximum Vertical Pocket (MVP)** of AF: - this measures the single deepest vertical pocket (**SDVP/SDP**) of AF visualized in a **single quadrant**.
 - *Normally it should be 2-8cm*, values below or above this shows oligohydramnios or polyhydramnios respectively
 - Is **preferred** over AFI for dx of oligohydramnios, but it has **low Sn** for Identifying the majority of pregnancy complications associated with oligohydramnios

- Measuring **Amniotic Fluid Index (AFI)**: - this is the sum of MVP in each (four) quadrants of the uterus.
 - **Normally it should be 5-25cm**, values below or above this shows oligohydramnios or polyhydramnios respectively
- Both MVP and AFI have high Sp but low Sn.

Dye dilution test

- Is the ***gold standard*** method of AF assessment, but it isn't practical

Oligohydramnios

- Refers to presence of inadequate amount of AF,
 - Anhydramnios refers to complete absence of AF.
- Is dx when the ***MVP/SDP is <2cm or AFI <5cm***, (although controversial, AFI 5-8cm is called borderline oligohydramnios)
- The incidence varies depending on the cutoff points used, the reported rate vary ***between 1 & 3%***

Etiologies

Fetal conditions

- Renal agenesis
 - Potter's syndrome
- Obstructive Uropathy
- PROM
- Abnormal placentation
- Severe IUGR
- Prolonged pregnancy
- TTTS (in the donor twin)

Maternal causes

- Uteroplacental insufficiencies
- Hypertensive disorders
- Medications (e.g., ACEIs, NSAIDs)
- Dehydration
- Antiphospholipid disorder

Fetal anomalies are common causes in 2nd TM oligohydramnios, while uteroplacental insufficiencies and PROM are common causes in 3rd TM

Evaluation and management**Hx and P/E**

Ask any hx consistent with ROM, any leakage of clear or bloody fluid, or wetness of underwear

- If there is possibility of ROM, do sterile speculum examination and other tests to rule in PROM, like: -
 - Pooling of fluid in posterior vaginal fornix
 - Ferning on microscopic examination
 - Neutral pH on Nitrazine paper test
 - Positive test for certain proteins from AF in the vagina

Ask hx of HTN, DM, and other possible causes of uteroplacental insufficiency

P/E,

- Look for signs of DHN,
- Do obstetric palpation and sterile speculum examination.

Targeted U/S

- Assess AFV
- Evaluate fetal anatomy (kidney & bladder)
 - If there is normally grown kidney & bladder, PROM is likely dx
- Look for placental abnormalities, like abruption (chronic PA)
- Determine fetal growth

Management

- Depends on the underlying etiology when feasible
- Maternal hydration with oral or IV fluid has shown to improve AFI, (≥L over 4hrs then reassess.)
- If PROM, manage accordingly
- Expectant management can be considered in those with isolated oligohydramnios in 3rd TM
 - Do NST or BPP once or twice weekly till delivery, depending upon fetal and maternal condition.
 - For idiopathic oligohydramnios, delivery is recommended at 36th to 38th wks of gestation, regardless of the Bishop score.
- All cases of oligohydramnios should be evaluated for IUGR and should be followed with antepartum fetal surveillance

Complications

It depends on the cause, severity, GA at onset and duration.

Major complications include: -

- Pulmonary hypoplasia
- Fetal deformation
 - Flexion contracture of the fetal extremity
- Increased perinatal mortality
 - In one study there is as high as **50x increase in PMR for px with MVP <1cm**
 - Especially in those dx in the 2nd TM, - because of pulmonary hypoplasia
 - PMR approaches 100% as the AFV decreased greatly
 - ✓ With renal agenesis, virtually 100% of newborns will die b/c of pulmonary hypoplasia
- In post term pregnancies it is associated with
 - MSAF
 - Fetal distress during labor (Variable deceleration in FHR)
 - Low 1st min APGAR score
- NRFHRP (bradycardia) during labor, especially if anhydramnios
 - Therefore constant/frequent follow up of the FHB is crucial.

Polyhydramnios/hydramnios

- Refers to excessive accumulation of AF
- Is dx when the MVP is >8cm or AFI is >25cm
- Its incidence is 1-2%

Classification based on severity

- **Mild**
 - AFI of 25-29.9cm or SDVP 8-9.9cm
 - Is the most common, accounts for 2/3rd of all cases
 - Is usually idiopathic and benign
 - Cases in 3rd TM are usually mild.
- **Moderate**
 - AFI of 30-34.9cm or SDVP 10-11.9cm
 - Account for around 20% of cases
- **Severe**
 - AFI ≥ 35 cm or SDVP ≥ 12 cm
 - Account for 15% of cases
 - Is likely to have an underlying etiology and consequences for the pregnancy

Etiologies

Fetal conditions

- Congenital anomalies
 - GI obstruction
 - CNS malformations
 - Cystic hygroma
 - Nonimmune hydrops
 - Sacrococccygeal teratomas
- TTTS, (in the receiving twin)
- Infections, like syphilis

- Aneuploidy
- Genetic disorders
 - Achondrogenesis type 1-B
 - Muscular dystrophies
 - Bartter syndrome
- Placental abnormalities

Maternal conditions

- Poorly controlled DM
 - maternal DM is present in **5% of cases**
- Fetomaternal hemorrhage
- Idiopathic

Evaluation and management

- **Hx and P/E**
 - Rapidly enlarging uterus in mid-pregnancy, (compare with previous pregnancy)
 - Maternal DM hx
- **U/S**
 - AFV evaluation
 - Fetal anomaly
 - ✓ Esophageal atresia, can present with early onset severe polyhydramnios
 - ✓ Duodenal atresia
 - ✓ Structural defects indicative of chromosomal abnormality
 - MG
- Screen for GDM, TORCH infection,
- Doppler of the MCA
 - Elevation >1.5 multiples of the median with moderate or severe fetal anemia
- **Kleihauer-Betke** test to look for fetomaternal hemorrhage
- Karyotype analysis, for those with severe polyhydramnios

Management

- Is dependent on underlying cause
- Antepartum surveillance
- The therapeutic options depend on the GA & degree of discomfort
- Treatment of polyhydramnios in singleton pregnancy is only indicated if it is severe and symptomatic (preterm labor, or significant maternal discomfort)
 - For severe symptomatic polyhydramnios at <32wks of gestation, **amnioreduction and Indomethacin is recommended.**
- **Amnioreduction**
 - Is done for polyhydramnios associated with development of maternal respiratory compromise
 - The **purpose is to prolong pregnancy**
 - ~1000 to 1500ml of fluid is slowly withdrawn during approximately 30min.
 - The procedure is terminated when the AFI is 15 to 20cm or intrauterine pressure is <20mmHg
 - Repeat amnioreduction may be required weekly
 - **Monitoring AFV every 1-3wks** is recommended
 - Risks of amnioreduction
 - PROM
 - Preterm labor
 - Placental abruption
- **Indomethacin**
 - Reduces fetal urine production
 - Has **onset** of effect within **5hrs**, and **it decreases AFV in 24hrs**
 - Is started 25mg p.o. QID, if there is no reduction in AFV, dose can be increased up to 2 to 3mg/kg per day.
 - **It is safe to use for short period of time, 72hrs.** prolonged administration or use after 32wks is associated with risk of
 - premature closure or narrowing of DA
 - renal abnormalities

Complications

- Maternal respiratory compromise
- Preterm labor
- PROM
- Fetal malformation
- UC prolapse, nuchal cord
- Placental abruption
- Uterine dysfunction and postpartum uterine atony (**PPH**)
- Pregnancy induced hypertension
- UTI
- Need of CD
- Low 5th min APGAR score
- Newborn NICU admission

Chapter 13

POSTPARTUM HEMORRHAGE (PPH)

Prepared by Dr. Daniel Delessa (MI)

Introduction

Definitions

There multiple definitions to PPH from different source/institutions. The different definitions include:-

- Excessive bleeding following delivery (>500 ml in vaginal delivery, >1000 ml in CD and twin vaginal deliveries, >1500 ml following cesarean hysterectomy)

(Classic definition)

- A drop in Hct > 10% from baseline
 - Bleeding that makes a patient symptomatic and/or results in signs of hypovolemia
- (best definition)*
- Blood loss of >15% of the total estimated blood volume

Limitation of the classic definition of PPH

- bleeding may not be visible externally,
- blood in collection devices may be mixed with amniotic fluid, and
- postpartum morbidity is relatively infrequent among patients with blood loss 500 to 999 mL

In acknowledgement of the limitations of the classic definition, ACOG revised their definition of PPH to the following:

- Cumulative blood loss ≥ 1000 mL, or Bleeding associated with signs/symptoms of hypovolemia within 24 hours of the birth process, and Regardless of route of birth

However, ACOG state that despite the new definition, a blood loss greater than 500 mL in a vaginal birth should be considered abnormal and should serve as an indication for the health care provider to investigate the increased blood deficit.

Normal Physiologic Mechanisms Preventing PPH

- **Mechanical hemostasis**, whereby contraction of the myometrium compresses the blood vessels supplying the placental bed
- **Local thrombosis**, whereby the presence or release of local decidual hemostatic factors (tissue factor and type-1 plasminogen activator inhibitor) and systemic coagulation factors lead to thrombosis of damaged blood vessels supplying the placental bed

Incidence

- PPH complicates *between 1 in 20 and 1 in 100 deliveries*. Generally reported to occur in 1 to 3% of births
- It is world's leading cause of maternal mortality

Classification

PPH can be classified temporally as:-

- Primary (early) PPH
- Secondary (late) PPH

Primary PPH

- Blood loss during the *first 24hrs after delivery*.
- Account for **98% PPH cases**

Etiology

- ✓ The etiologies for early PPH can easily remembered as the **fiveTs**: -
 - **Tone** (uterine atony)
 - **Trauma** (genital tract laceration and Uterine rupture)
 - **Tissue**(retained placenta and amniotic membranes]
 - **Thrombin**(coagulopathy) ,and
 - **Traction**(uterine inversion)

1. Uterine atony (80%)

- Is failure of the uterus to contract sufficiently

- **Risk factors:**

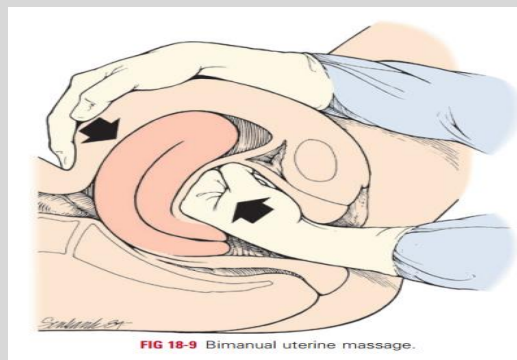
- | | |
|---|--|
| ✓ uterine over distention (MG, polyhydramnios, fetal macrosomia), | ✓ uterine inversion, |
| ✓ induction or augmentation of labor, | ✓ retained products of conception, |
| ✓ rapid or prolonged labor, | ✓ abnormal placentation, |
| ✓ grand multiparity, | ✓ use of uterine-relaxing agents (Tocolytics therapy, halogenated anesthetics, nitroglycerin), |
| ✓ uterine infection, | ✓ previous PPH |

- **Diagnosis:**

- ✓ Hypotonic (boggy) uterus with brisk bleeding and expression of clots when the uterus is massaged

- **Management**

- ✓ **Uterine massage**
- ✓ **Bimanual uterine compression:** the uterus should be compressed between the external, fundally placed hand and the internal, intra-vaginal hand



- ✓ **Uterotonic therapy**

- **Oxytocin**

- IV oxytocin is the recommended **first line uterotonic drug**
- **Dose:** 20 – 40 units in 1-liter of NS or RL solution infuse IV at fastest flow rate possible.
- Give oxytocin 10 units IM in women without IV access.
- **Maintenance:** infuse 20 units in 1 L IV fluids at 40 drops/min.

- When oxytocin fails to produce adequate uterine tone, second-line therapy must be initiated.
- The choice of a second-line agent depends on its side-effect profile as well as its contraindications.

TABLE 18-4 UTEROTONIC THERAPIES

AGENT	DOSE	ROUTE	DOSING INTERVAL	SIDE EFFECTS	CONTRAINDICATIONS
Oxytocin (Pitocin)	10 to 80 U in 500-1000 mL crystalloid solution	First line: IV Second line: IM or IU	Continuous	Nausea, emesis, water intoxication	None
Misoprostol (Cytotec)	600-1000 µg	First line: PR Second line: PO or SL	Single dose	Nausea, emesis, diarrhea, fever, chills	None
Methylergonovine (Methergine)	0.2 mg	First line: IM Second line: IU or PO	Every 2-4 hr	Hypertension, hypotension, nausea, emesis	Hypertension, migraines, scleroderma, Raynaud syndrome
Prostaglandin F _{2α} (Hemabate)	0.25 mg	First line: IM Second line: IU	Every 15-90 min (maximum of 8 doses)	Nausea, emesis, diarrhea, flushing, chills	Active cardiac, pulmonary, renal, or hepatic disease
Prostaglandin E ₂ (Dinoprostone)	20 mg	PR	Every 2 hr	Nausea, emesis, diarrhea, fever, chills, headache	Hypotension

✓ Uterine packing

- safe, simple, and effective way to control PPH
- The pack should be made of long, continuous gauze (e.g., Kerlix) rather than multiple small sponges.
- When packing the uterus, placement should begin at the fundus and progress downward in a side-to-side fashion to avoid dead space for blood accumulation.
- Insert urinary catheter and give prophylactic antibiotic to prevent urinary retention and infection
- Avoid prolonged packing (not more than 12 to 24 hours), and
- Follow vital signs and blood indices while the pack is in place to minimize unrecognized ongoing bleeding.
- This method is currently being replaced by balloon tamponade

✓ Intrauterine tamponade balloons

- The Bakri tamponade balloon
- the BT-Cath
- the Belfort-Dildy Obstetrical Tamponade System



18-12 The Belfort-Dildy obstetrical tamponade system. (Courtesy of Conveigh Medical.)

Figure: Types of Intrauterine Tamponade Balloons

- ✓ **Selective arterial embolization**
 - Has cumulative success rate of 90-97%
- ✓ **Surgical intervention**
 - **arterial ligation,**
 - The goal is to decrease uterine perfusion and subsequent bleeding.
 - Success rates have varied from 40% to 95% depending on which vessels are ligated.
 - Arterial ligation may be performed on
 - ✓ the ascending uterine arteries,
 - ✓ the uteroovarian arteries,
 - ✓ the infundibulopelvic ligament vessels, and
 - ✓ the internal iliac (hypogastric) arteries
 - **uterine compression sutures**
 - Techniques include:
 - ✓ B-Lynch suture,
 - ✓ **Hayman vertical sutures,**
 - ✓ Pereira transverse and vertical sutures, and
 - ✓ Cho's Multiple square sutures

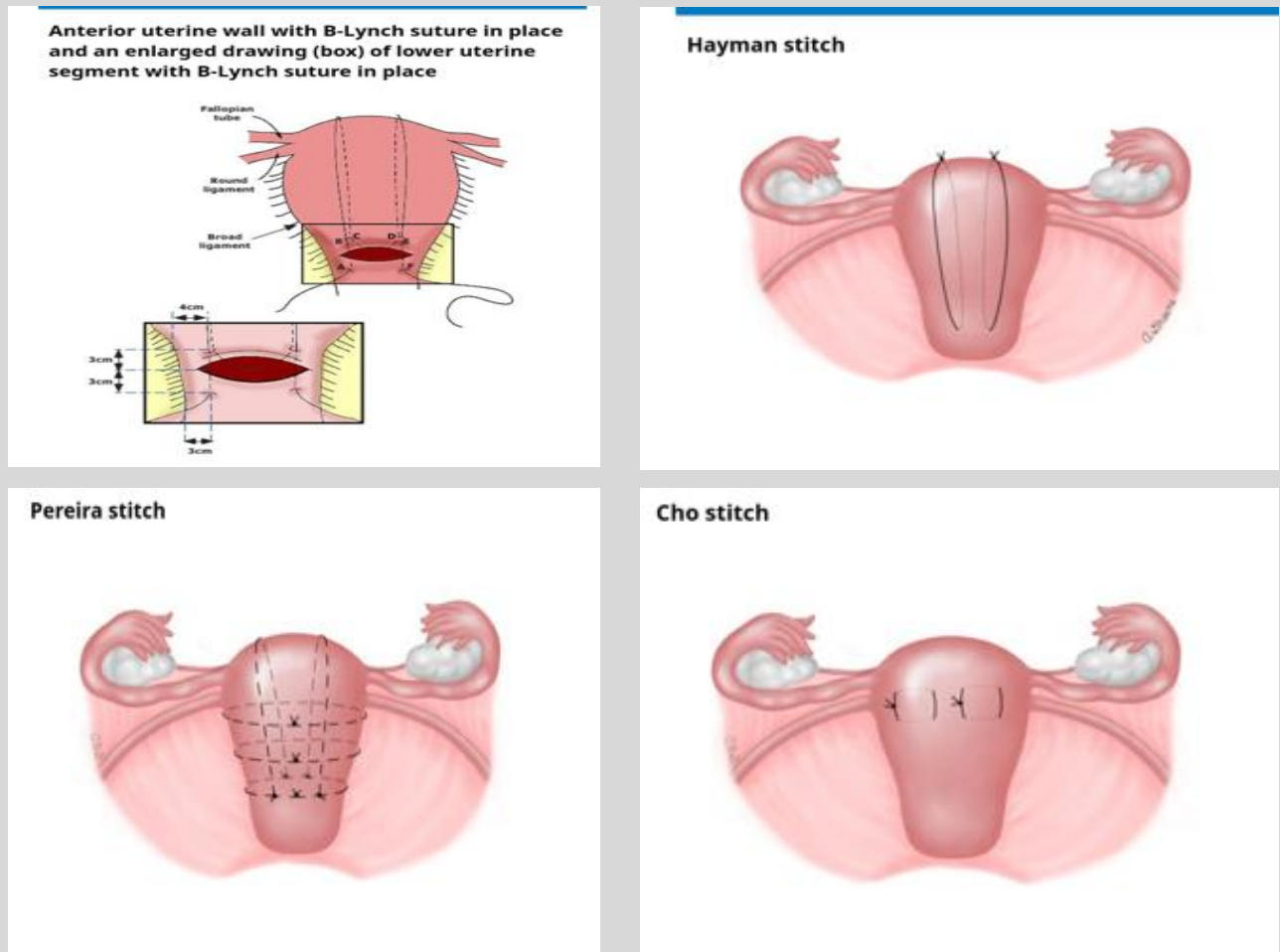


Figure: Types of Uterine compression sutures

- **Hysterectomy**

- The final surgical intervention for refractory bleeding due to atony is hysterectomy, which provides definitive therapy

2. Genital tract Laceration

- The second leading cause of postpartum hemorrhage
- The most common lower genital tract lacerations are perineal, vulvar, vaginal, and cervical.
- Upper genital tract lacerations are typically associated with broad ligament and retroperitoneal hematomas.

- **Risk factors:**

- ***Instrumental deliveries***
- Episiotomy,
- Precipitous delivery,
- Dührssen incisions,
- Fetal malpresentation or macrosomia
- Delivery through incompletely dilated cervix
- Prior cerclage placement
- Shoulder dystocia

- **Clinical manifestations**

- Suspect when bright red (arterial) bleeding occurs in the presence of a contracted uterus.
- Occasionally, the bleeding may be masked because of its location, such as with the broad ligament.
- Large amounts of blood loss may occur in an unrecognized hematoma.
- Pain and hemodynamic instability are often the primary presenting symptoms.

- **Classification of perineal lacerations**

- **First-degree lacerations** involve injury to the skin and subcutaneous tissue of the perineum and vaginal epithelium only.
- **Second-degree lacerations** extend into the fascia and musculature of the perineal body
- **Third-degree lacerations** extend through the fascia and musculature of the perineal body and involve some or all of the fibers of the external anal sphincter (EAS) and/or the internal anal sphincter (IAS).
 - ✓ **3a** – <50 percent of EAS thickness is torn
 - ✓ **3b** – >50 percent of EAS thickness is torn
 - ✓ **3c** – Both EAS and IAS are torn
- **Fourth-degree lacerations** – Injury to the perineum that involves both the anal sphincter complex (EAS and IAS) and anal mucosa

- Episiotomy is a type of second degree perineal tear

- Third- and fourth-degree perineal lacerations are called **Obstetric anal sphincter injuries (OASIS)**

- **Diagnosis**

- is made following exploration of the genital tract

- **Management**

- Repair the laceration with adequate exposure
 - Perineal repairs are the most common types of genital tract lacerations
 - on occasion, a blood vessel laceration may lead to the formation of a pelvic hematoma in the lower or upper genital tract
 - The three most common locations for a pelvic hematoma are: vulvar, vaginal, and retroperitoneal

Vulvar hematoma

- Primary treatment-Surgical drainage
- Typically, the bleeding is due to multiple small vessels; hence, vessel ligation is not possible
- Once the hematoma is evacuated, the dead space should be closed in layers with absorbable suture, and a sterile pressure dressing should be applied.
- A transurethral catheter should be placed until significant tissue edema subsides.

Vaginal hematoma

- Like vulvar hematomas, vaginal hematomas are due to multiple small vessel lacerations.
- Vaginal hematomas may or may not require surgical drainage.
 - Small, nonexpanding hematomas are often best managed expectantly.
 - Larger, expanding hematomas require surgical intervention.
- Unlike vulvar hematomas, the incision of a vaginal hematoma does not require closing; rather a vaginal pack or tamponade device should be placed on the raw edges.

- If bleeding persists, selective arterial embolization may be considered.

Retroperitoneal hematoma

- They are the most serious and life threatening.
- The early symptoms of a retroperitoneal hematoma are often subtle, with the hematoma being unrecognized until the patient is hemodynamically unstable from massive hemorrhage.
- They usually occur after a vessel laceration from the internal iliac (hypogastric) arterial tree
- Causes:
 - instrumented vaginal delivery,
 - inadequate hemostasis of the uterine arteries at the time of cesarean delivery, or
 - uterine rupture during TOLAC
- Treatment: laparotomy, hematoma evacuation, and arterial ligation.

3. Uterine rupture

- *Is defined as complete nonsurgical disruption of all uterine layers*
- **Uterine dehiscence** refers to an incomplete or occult uterine scar separation, in which the uterine serosa remains intact.
- Typically, no adverse obstetric outcomes are associated with a dehiscence

Incidence

- The overall incidence of uterine rupture (scarred and unscarred uteri) is **1 in 2000 deliveries.**

Risk factors

Uterine rupture: after previous cesarean birth

TABLE 18-5**RISK OF UTERINE RUPTURE BASED ON
INCISION OF PRIOR CESAREAN DELIVERY**

INCISION OF PRIOR CESAREAN DELIVERY	UTERINE RUPTURE RISK (%)
Classical	2-6
T or J shaped	2-6
Low vertical	2
Low transverse	0.5-1

- **Factors that increase the risk of rupture in those with scar**
 - The primary risk factors for uterine rupture after a previous cesarean birth are:
 - ✓ Previous uterine rupture
 - ✓ Previous fundal or high vertical hysterotomy
 - ✓ Patients with a previous low vertical hysterotomy
 - ✓ Induction
 - Possible risk factors — Factors inconsistently reported to be associated with an increased risk of rupture include:
 - ✓ Increasing maternal age
 - ✓ Gestational age >40 weeks
 - ✓ Birth weight >4000 gram
 - ✓ Inter-delivery interval less than approximately 18 months (and particularly <6 months)
 - ✓ Single-layer uterine closure, especially if locked
 - ✓ More than one previous cesarean birth
 - ✓ Previous second-trimester cesarean birth

None of these possible risk factors, alone or in combination, is sufficiently reliable to be clinically useful for prediction of rupture

Uterine rupture: unscarred uterus

- **Risk factors**
 - Exposure to uterotonic drugs (oxytocin, prostaglandins) – Exposure to uterotonic drugs is a consistent risk factor for rupture of the unscarred uterus
 - High parity
 - Uterine anomaly
 - Advanced maternal age
 - Dystocia

- Macrosomia.
- Multiple gestation
- Abnormal placentation (eg, placenta accreta spectrum).
- Short interpregnancy interval.
- Prior cerclage

Clinical manifestations

- Fetal
 - **Fetal bradycardia** with or without preceding variable or late decelerations-most common (33-70%)
 - **Loss of fetal station** in labor
- Maternal:
 - acute vaginal bleeding,
 - **cessation of contractions,**
 - **constant abdominal pain or uterine tenderness,**
 - change in uterine shape,
 - hematuria (if extension into the bladder has occurred), and
 - signs of hemodynamic instability

Diagnosis:

- Uterine rupture is suspected clinically but confirmed surgically.
 - Laparotomy-complete disruption of the uterine wall with hemoperitoneum and partial or complete extravasation of the fetus into the maternal abdomen.

Management

Management of patients in whom rupture is suspected before delivery

- Stabilize patients with hemodynamic instability
- Secure IV lines bilaterally with large bore cannulae.
- Resuscitation with IV fluids and blood products
- Prepare for operative interventions (e.g. determine HCT, BG and RH, avail cross-matched blood and organize the OR)

Management of patients with uterine rupture at laparotomy

- **Repair versus hysterectomy?**
 - **Hysterectomy:** based on a combination of factors, including
 - ✓ the patient's desire for future pregnancy,
 - ✓ the patient's intraoperative hemodynamic and anesthetic stability,
 - ✓ whether the uterine defect can be repaired,
 - ✓ whether hemostasis can be achieved, and
 - ✓ the skill of the surgeon for repairing a complicated rupture of the
 - surgeon for repairing a complicated rupture
 - **Uterine repair :** the goals of conservative surgery are
 - ✓ to repair the uterine defect,
 - ✓ control hemorrhage,
 - ✓ identify damage to other organs (eg, urinary tract),
 - ✓ minimize early postsurgical morbidity
 - ✓ reduce the risk of complications in future pregnancies

4. Retained products of conception (0.5-1%)

- Retained products of conception-->uterine atony-->PPH
- **Risk factors**
 - Mismanagement of third stage of labor,
 - Midtrimester delivery,
 - Abnormal placentation (morbidly adherent placenta, succenturiate lobe),
 - chorioamnionitis,
 - constriction of the cervix or lower uterine segment and
 - untimely use of uterotonics
- **Clinical manifestations**
 - It present with uterine bleeding and associated atony
 - Manual exploration is both diagnostic and therapeutic

- If manual access to the uterine cavity is difficult or limited owing to maternal body habitus or inadequate pain relief, transabdominal or transvaginal ultrasound may be used to determine whether retained placental fragments are present.
- **Diagnosis:**
 - when spontaneous expulsion of the tissue has not occurred within 30 to 60 minutes of delivery
- **Management**
 - **Manual extraction**
 - Nitroglycerin (50 to 200 µg IV) provides rapid uterine relaxation to assist with removal of the retained tissue.
 - **Uterine curettage.**
 - Uterine curettage may be performed in a delivery room; however, excessive bleeding mandates that an OR be used for the procedure.
 - Either a large blunt (Banjo or Hunter) curette or vacuum suction curette can be used.
 - Transabdominal ultrasound guidance is helpful in determining when tissue evacuation is complete.

5. Coagulopathy

- Can be Acquired or Hereditary
- **Acquired**
 - Idiopathic thrombocytopenic purpura
 - Thrombocytopenia with preeclampsia
 - HELLP syndrome
 - Disseminated intravascular coagulation
 - Fetal demise
 - Sepsis
 - Placental abruption
 - Amniotic fluid embolus
 - Severe hemorrhage

- **Hereditary**
 - Von Willebrand's disease

Clinical manifestations of consumptive coagulopathy include:

- Bleeding
 - Presence of bleeding from other sites in addition to the genital tract including (but not limited to) mucosal bleeding
- Hypotension out of proportion to blood loss
- Microangiopathic hemolytic anemia,
- Acute lung injury,
- Acute renal failure, and
- Ischemic end-organ tissue damage.

Diagnosis:

- Consumptive coagulopathy is a clinical diagnosis that is confirmed with laboratory data
- coagulopathy should be suspected in patients with one or more of the following:
 - low fibrinogen level (<300 mg/dL);
 - thrombocytopenia ($<100,000/\mu\text{L}$);
 - prolonged PT, (INR >1.5), and/or aPTT
- Bedside clotting time: draw 5 mL of maternal blood into an empty red-topped tube and watch for clot formation
 - If a clot is not visible within 6 minutes or forms and lyses within 30 minutes, the fibrinogen level is usually less than 150 mg/dL (Gabbe)
 - If the blood in the tube clots within 8 to 10 minutes and the clot remains intact, the patient likely has adequate fibrinogen stores. If the blood in the tube does not clot or an initial clot dissolves, it is likely that the patient is markedly deficient in key clotting factors (up-to-date)
 - *However, the test is highly subjective, insensitive, imprecise, and not validated in pregnancy.*

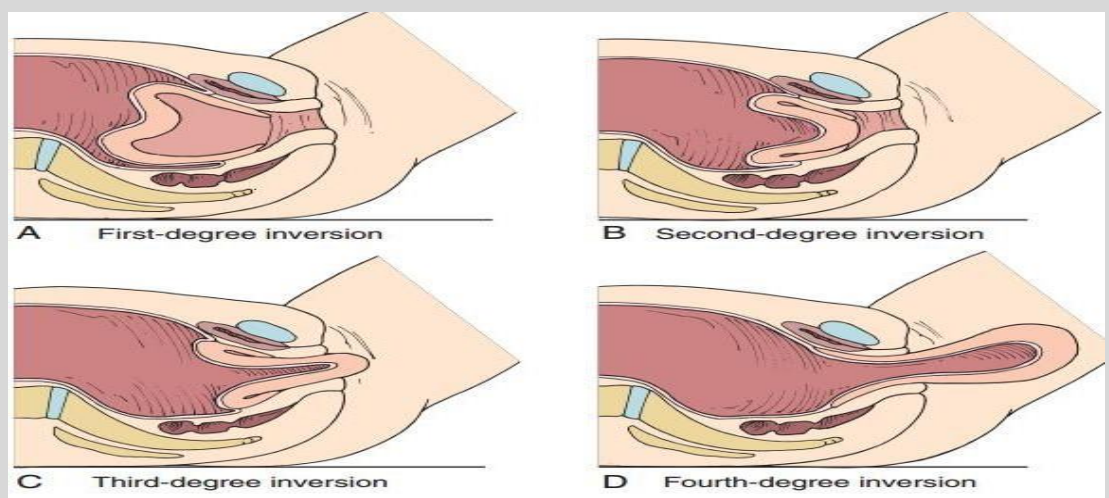
Management

- Identify and correct the underlying etiology- *the most important*.
- Rapid replacement of blood products and clotting factors.
- Maintain adequate oxygenation and normothermia
- For most obstetric causes, delivery of the fetus initiates resolution of the coagulopathy.
- adjuvant therapies: vitamin K, recombinant activated factor VII, fibrinogen concentrate, prothrombin complex concentrate, tranexamic acid, and hemostatic agents (topical agents for control of coagulopathic surface bleeding)
- Laboratory studies should be drawn serially every 4 hours until resolution of the coagulopathy is evident.
- Goal: you should attempt to achieve
 - hematocrit greater than 21%,
 - platelet count greater than 50,000/mm³,
 - a fibrinogen level greater than 100 mg/dL, and
 - PT and aPTT less than 1.5 times control.

6. Uterine inversion

- Collapse of the fundus in to t he uterine cavity
- It is classified it to:
 - **Three**, based on timing:-
 - ✓ Acute (within 24 hours of delivery)
 - ✓ Sub-acute (>24 hours postpartum but <4 weeks)
 - ✓ Chronic (>1 month postpartum)
 - **Four**, based on degree of inversion:-
 - ✓ **First degree** (incomplete): partial extrusion of the fundus into the uterine cavity

- ✓ **Second degree** (complete): the internal lining of the fundus crosses through the cervical os, forming a rounded mass in the vagina with **no palpable fundus abdominally**.
- ✓ **Third degree** (prolapsed): the entire uterus prolapsing through the cervix with the fundus passing out of the vaginal introitus.
- ✓ **Fourth degree** (total): both total uterine and vaginal prolapse through the vaginal introitus



- The two most commonly proposed etiologies are:
 - **Excessive umbilical cord traction** with a fundally attached placenta and
 - **Fundal pressure** in the setting of a relaxed uterus
- **Incidence:**
 - Rare event that complicates about **1 in 1200 to 1 in 57,000 deliveries**.

▪ Risk:- factors include

- | | |
|------------------------------------|----------------------------------|
| ✓ uterine over-distention | ✓ retained placenta |
| ✓ fetal macrosomia | ✓ short umbilical cord |
| ✓ rapid labor and delivery | ✓ use of uterine-relaxing agents |
| ✓ congenital uterine malformations | ✓ nulliparity |
| ✓ uterine fibroids | ✓ manual placental extraction |
| ✓ invasive placentation | ✓ Ehlers-Danlos syndrome |

▪ Clinical manifestation

- Depend on by degree and timing.
 - ✓ Incomplete-subtle in its clinical findings,
 - ✓ Complete-brisk vaginal bleeding- inability to palpate the fundus abdominally, and maternal hemodynamic instability
 - ✓ It may occur before or after placental detachment.

▪ Diagnosis

- Clinically with bimanual examination, during which the uterine fundus is palpated in the lower uterine segment or within the vagina.
- Sonography-to confirm the diagnosis if the clinical examination is unclear

▪ Management

- fluid resuscitation
- The uterus must be replaced to its proper orientation to resolve the hemorrhage, this best accomplished in an OR with the assistance of an anesthesiologist.

- **manual repositioning**

- ✓ the uterus and cervix should initially be relaxed with nitroglycerin (50 to 500 µg), a tocolytic agent (magnesium sulfate or β-mimetic), or an inhaled anesthetic
- ✓ gentle manual pressure is applied to the uterine fundus to return it to its proper abdominal location
- ✓ Uterotonic therapy should then be given to assist with uterine contraction and to prevent recurrence of the inversion

- **hydrostatic reduction**

- ✓ Warmed sterile saline is infused into the vagina.
- ✓ The physician's hand or a Silastic ventouse cup is used as a fluid retainer to generate intravaginal hydrostatic pressure and resultant correction of the inversion

- **surgical correction- includes:**

- ✓ ***The Huntington procedure:*** a laparotomy with serial clamping and upward traction of the round ligaments to restore the uterus to its proper position.
- ✓ ***The Haultain procedure:*** can be attempted, if the above technique fails
 - vertical incision within the inversion and subsequent repositioning of the fundus
- ✓ Laparoscopic assisted repositioning, and
- ✓ Cervical incisions with manual uterine repositioning.

Late PPH

- PPH occurring from 24 hours to 12 weeks after delivery
- Unlike primary PPH, bleeding usually is not catastrophic

Etiology

Common causes of secondary PPH are:

- Retained products of conception-most common cause secondary PPH
- Sub involution of the placental bed
- Infection

Rare causes include:

- Inherited or acquired bleeding diatheses
- Pseudoaneurysm of the uterine artery, internal pudendal artery, vaginal artery, or vulvar labial artery
- Arteriovenous malformations
- Choriocarcinoma
- Undiagnosed carcinoma of the cervix
- Adenomyosis
- Infected polyp or submucosal fibroid
- Uterine diverticulum
- Excessive bleeding with resumption of menses
- Hypoestrogenism
- Dehiscence of a cesarean scar

Sometimes the cause cannot be determined

Sub-involution of the uterus, signs of intrauterine infection and retained pieces of placental tissue are common in the first two weeks.

- If the mother has fever, uterine tenderness, and/or a foul- smelling discharge without massive bleeding, then endometritis should be suspected.

When bleeding occurs late in the postpartum period, (3rd to 6th wks) pregnancy test needs to be performed to rule out choriocarcinoma and the specimen of uterine evacuation needs to be sent for histological examination.

Management

- Treat anemia and shock
- Specific management
- **Sub-involution:** Oxytocin in drip or ergometrine (1 tablet PO twice a day for 2-3day). If bleeding is not controlled with these drugs give misoprostol 800 µg sublingually or rectally.

- **Infection:** Antibiotic
- **Retained placental tissue:** Evacuate the uterus using manual vacuum aspiration with large sized cannula (if there is active vaginal bleeding or medical management fails)

Complication

- **Death**
 - Maternal deaths are due to:
 - ✓ Underestimation of blood loss
 - ✓ Inadequate blood and fluid replacement
 - ✓ Delay in operative intervention
- **Short term morbidity**
 - Transfusion related complication: electrolyte abnormalities, allergic reactions, alloimmunization, and volume overload
 - Organ failure related to hemodynamic instability: renal failure, heart failure, respiratory failure, or hepatic failure.
 - Hysterectomy
 - Thromboembolism: DVT, pulmonary embolus, stroke, MI
 - Anemia
 - Fluid overload-pulmonary edema and dilutional coagulopathy
 - Abdominal compartment syndrome
- **Long term morbidity**
 - Sheehan syndrome
 - Asherman syndrome
- **Recurrence**
 - Patients with a prior PPH have as much as an
 - **18 percent risk of recurrence** in the subsequent pregnancy and
 - **27 percent** after two consecutive pregnancies with PPH

Prevention

- *AMTSL substantially reduces incidence of PPH due to atony*

NB, Risk factors do not predict PPH

- Only about 10% of women with any of the risk factors develop PPH and over two-thirds have no identifiable risk factors
- Hence *all pregnancies are at risk of PPH*

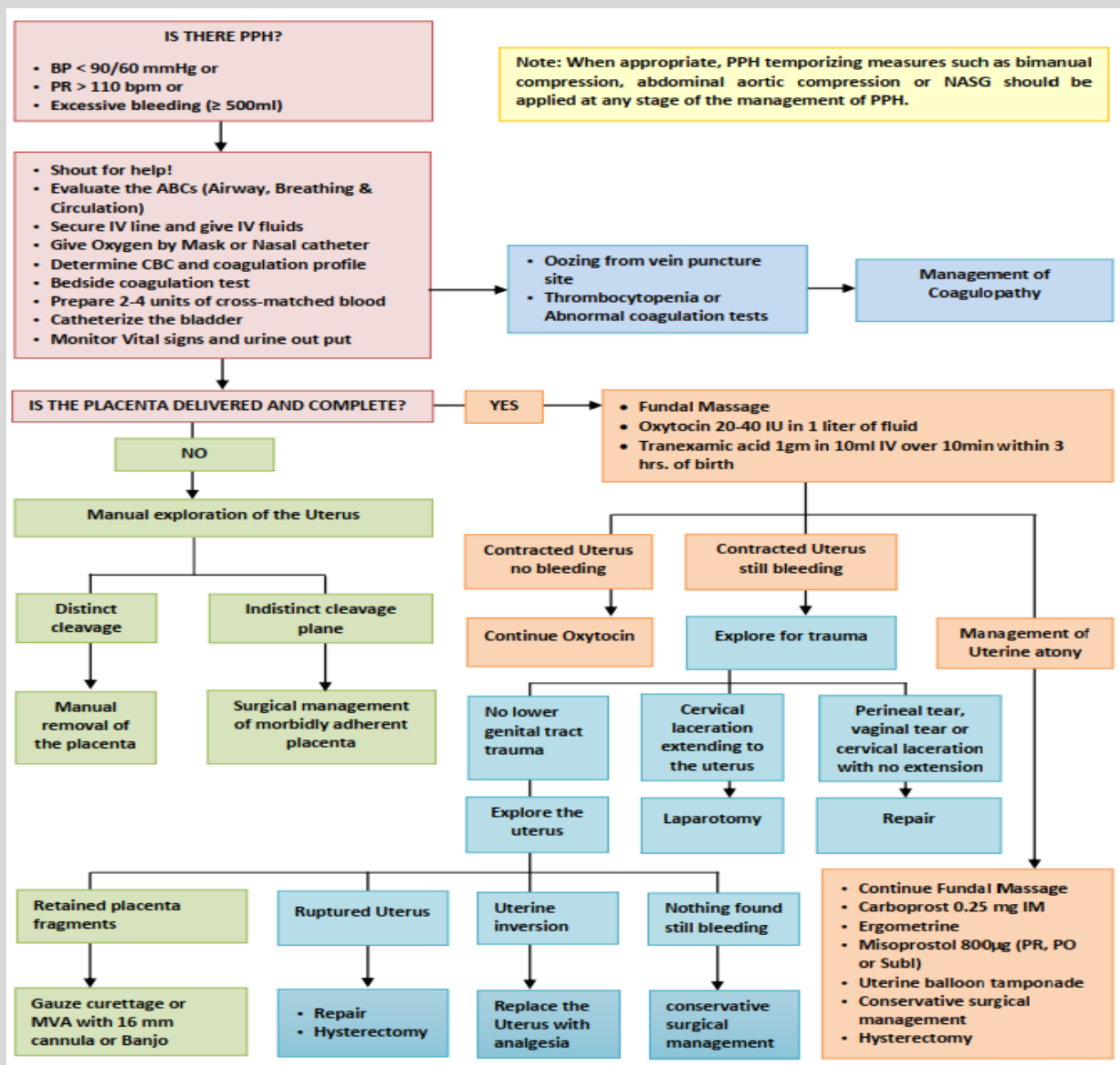


Figure 21. Postpartum hemorrhage (PPH) management algorithm.

The first step in management of PPH is call for help. The management is a race with time; every available practitioner should be involved

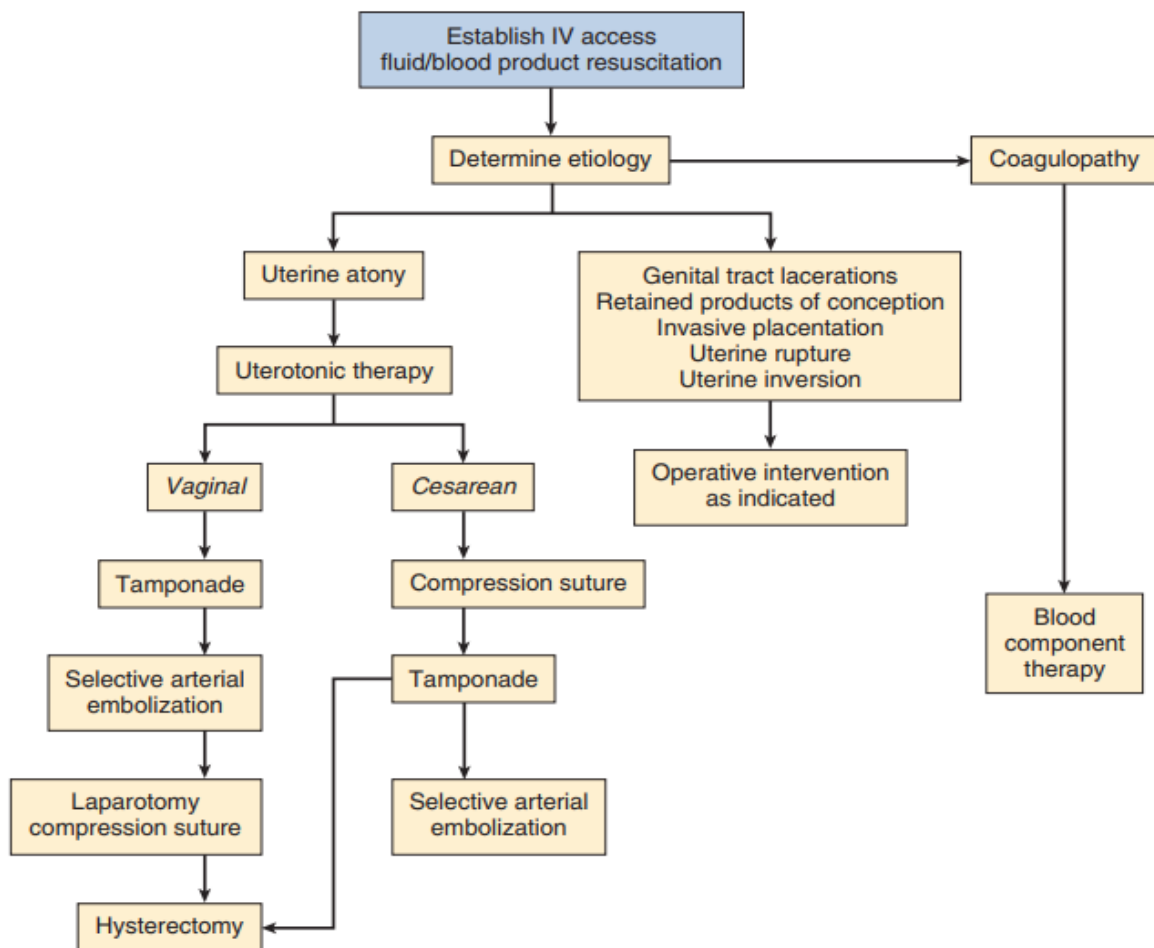
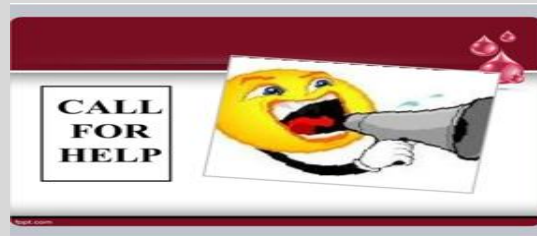


FIG 18-8 Management of postpartum hemorrhage. IV, intravenous.

Chapter 14

Intra-amniotic Infection (Chorioamnionitis)

Prepared by: Dr. Birhanu Olani (MI)

Introduction

Intraamniotic infection, also referred to as chorioamnionitis, is an infection with resultant inflammation of any combination of the amniotic fluid, placenta, fetus, fetal membranes, or decidua.

Intraamniotic infection can be associated with **acute** neonatal morbidity, including neonatal pneumonia, meningitis, sepsis, and death, as well as **long-term** infant complications such as bronchopulmonary dysplasia and cerebral palsy.

Pathogenesis

IAI can result from:

- **Migration of cervicovaginal flora** through the cervical canal is the most common pathway to IAI.
 - The principal pathogens are *Bacteroides* and *Prevotella* species, *E. coli*, anaerobic gram-positive cocci, GBS, and genital mycoplasmas.
- Uncommonly, **hematogenous spread** as a result of maternal bacteremia (eg, *Listeria monocytogenes*) infecting the intervillous space or
- **Invasive procedure** which leads to contamination of the amniotic cavity (eg, fetoscopy).

Infection from the peritoneum via the fallopian tubes has also been postulated but is likely rare.

*Subsequent activation of the maternal and fetal inflammatory response systems generally leads to **labor and/or rupture of membranes**.*

Epidemiology

Chorioamnionitis occurs **in 1% to 5%** of term pregnancies.

- In patients with preterm delivery, the frequency of clinical or subclinical infection may approach 25%.

Clinical risk factors

- *Extended duration of labor and ruptured membranes*
- *Multiple digital vaginal examinations (especially with ruptured membranes)*
- *Preexisting infections of the lower genital tract (e.g., **sexually transmitted infections, bacterial vaginosis and GBS infection**)*
- Internal fetal or uterine monitoring
- *Young age*
- *Low socioeconomic status*
- *Nulliparity*
- Cervical insufficiency
- Alcohol and tobacco use, and
- Previous history of chorioamnionitis

Antibiotic use in preterm PROM is associated with a statistically significant reduction in chorioamnionitis. In term PROM, consideration should also be given to:

- *minimizing use of vaginal digital examinations intrapartum once membrane rupture has occurred,*
- *antibiotic prophylaxis, especially when **latency greater than 12 hours is expected***

Microbiology

- IAI is typically **polymicrobial**, often involving vaginal or enteric flora.
 - Two-thirds of women with IAI have at least two isolates per specimen of amniotic fluid.
 - Regardless of gestational age, **genital mycoplasmas** (Ureaplasma and Mycoplasma species) are the most common isolates.
 - Anaerobes (including Gardnerella vaginalis), enteric gram-negative bacilli, and group B Streptococcus are other frequent pathogens.
 - Anaerobes appear to be more frequently involved in **preterm IAI than term IAI**.

Clinical presentation

IAI often occurs in women with premature rupture of membranes (PROM) but can occur with intact membranes, especially in laboring women.

The key clinical findings, which **are nonspecific** and their frequencies are as follows:

- Fever
- Maternal leukocytosis (variously defined as white blood cell count $>12,000/\text{mm}^3$ or $>15,000/\text{mm}^3$)
- Maternal tachycardia $>100/\text{min}$.
- Fetal tachycardia $>160/\text{min}$.
- Uterine tenderness.
- Bacteremia (most commonly associated with group B *Streptococcus* or *Escherichia coli* infection)
- Purulent or malodorous amniotic fluid.

IAI may be subclinical, which by definition does not present with the above clinical findings.

Subclinical chorioamnionitis criteria's: amniocentesis findings showing:-

- Culture: Growth of aerobic or anaerobic microorganism (gold standard, although results are not immediately available for clinical management).
- Presence of bacteria on gram stain
- Glucose level ≤ 10 to 15 mg/dl
- WBC > 30 /ml

Differential Diagnosis

- **Labor** – Labor can be associated with fever (if the patient has an epidural anesthetic), maternal tachycardia, leukocytosis, and uterine tenderness.
- **Abruptio-placenta** – Abruption can cause uterine tenderness and maternal tachycardia but is usually associated with vaginal bleeding and absence of fever.
- **Other infections** – Extrauterine infections associated with fever and abdominal pain (with or without labor) include pyelonephritis, influenza, appendicitis, and pneumonia.

Diagnosis

Isolated maternal fever – is when maternal oral temperature $\geq 39.0^{\circ}\text{C}$ (102.2°F) on any occasion or oral temperature 38.0°C (100.4°F) to 38.9°C (102.02°) on two occasions 30 minutes apart.

A presumptive diagnosis of IAI (suspected triple I) can be made in women with:

- **Fever** $\geq 39.0^{\circ}\text{C}$ [102.2°F] once or 38.0°C [100.4°F] to 38.9°C [102.02°F] on two or more measurements (ideally taken orally) 30 minutes apart, without another clear source PLUS one or more of the following:
 - **Baseline fetal heart rate** >160 beats/min for ≥ 10 minutes, excluding accelerations, decelerations, and periods of marked variability
 - **Maternal white cell count** $>15,000/\text{mm}^3$ in the absence of corticosteroids and ideally showing a left shift (bandemia)
 - **Purulent-appearing fluid** coming from the cervical os visualized by speculum examination

NB: Triple “T” refers to Intrauterine Inflammation or Infection or both.

A confirmed diagnosis of IAI can be made in women with:

- All of the above PLUS one or more of the following **objective laboratory findings:**
 - Positive amniotic fluid Gram stain for bacteria,
 - Low amniotic fluid glucose (≤ 14 mg/dl),
 - High amniotic fluid white cell count in the absence of a bloody tap (>30 cells/mm³), or
 - Positive amniotic fluid culture results, or
 - Histopathologic evidence of infection or inflammation or both in the placenta, fetal membranes, or the umbilical cord vessels (funisitis)

Laboratory studies should be performed on amniotic fluid obtained by amniocentesis. Histopathology is obtained **after delivery**.

Laboratory Tests:

- Basic lab tests like CBC, BG&RH, U/A, VDRL, HBsAg,
- Amniocentesis: Gram stain and culture of amniotic fluid, WBC count, Glucose concentration. Leukocyte esterase...etc.

Management

Delivery

Women with IAI (including suspected or confirmed) should be given antibiotics and delivered.

Antimicrobial therapy provides bactericidal concentrations of antibiotics to the fetus, membranes, and amniotic fluid **within one-half to one hour after infusion**, but IAI can only be cured by delivery of the infected products of conception.

Induction or augmentation of labor, as appropriate is suggested, with cesarean delivery reserved for standard obstetric indications.

Cesarean delivery in the presence of IAI increases the risk of

- Wound infection,
- Endomyometritis and
- Venous thrombosis

Antibiotic therapy

Broad-spectrum antibiotics should be given promptly following a diagnosis of IAI to initiate treatment of both the mother and fetus.

We administer antibiotics to women with a presumptive diagnosis of IAI even if epidural-related fever cannot be excluded. Early initiation of antibiotic therapy may reduce the frequency and severity of neonatal infection.

Primary Intrapartum regimens:

- **Recommended antibiotics**
 - Ampicillin 2g IV every 6 hours plus
 - Gentamicin 5 mg/Kg IV once daily
- **Recommended antibiotics (mild penicillin allergy)**
 - Cefazolin 2g IV every 8 hours plus
 - Gentamicin 5mg/Kg IV once daily
- **Recommended antibiotics (severe penicillin allergy)**
 - Clindamycin 900mg IV every 8 hours **or**
 - Vancomycin 1g IV every 12 hours **plus**
 - Gentamicin 5mg/Kg IV once daily
- ***NB! Vancomycin should be used if the woman is known to be GBS colonized unless Clindamycin inducible resistance testing is available & negative.***

Alternative Intrapartum regimens:

- Ampicillin 2g IV QID **plus** Gentamicin 1.5mg/Kg/dose IV TID
- Ampicillin-Sulbactam 3g IV QID
- Cefoxitin 2g IV TID
- Cefotetan 2g IV BID
- Ertapenem 1g IV daily

Cesarean Birth-anaerobic coverage should be added to prevent post-cesarean endometritis.

- Ampicillin 2g IV QID plus
- Gentamicin 5mg/kg IV daily plus
- Either metronidazole 500mg PO or IV TID **or** Clindamycin 900mg IV TID

Duration of treatment- The optimal duration of antibiotic treatment after delivery has not been determined conclusively.

- Continue IV antibiotic until the patient is afebrile and asymptomatic for at least 24-48 hours and then discontinue. **OR**
- Continue IV antibiotic until the patient is afebrile and asymptomatic for at least 24-48 hours and change to PO antibiotics for the next 7-10 days. But there is no evidence that oral antibiotics are beneficial of after discontinuation of parenteral therapy.
- **ACOG committee opinion:** Additional antibiotic doses **are not required after vaginal birth and at least one additional dose is indicated after cesarean birth.**

In our setup ceftriaxone 1gm IV BID is given till the women become afebrile for more than 48hrs together with Metronidazole 500mg P.O TID and then the ceftriaxone is changed to P.O cephalixin 500mg TID together with Metronidazole for 7ds

Fetal monitoring during labor

- Use of continuous electronic fetal monitoring is appropriate to detect dev't of fetal compromise due to sequel of IAI (villous edema, hyperthermic stress, fetal infection) or other factors.

Antipyretics - Acetaminophen is the **preferred** drug.

Postpartum care - Postpartum care is routine as IAI resolves after delivery in most women, particularly after vaginal delivery.

Women with persistent fever and/or pelvic pain should be evaluated for **postpartum endometritis**, wound infection after cesarean delivery, and, rarely, septic pelvic thrombophlebitis.

Complications

Maternal sequelae

- **Dysfunctional labor** - IAI is associated with an increased risk of labor abnormalities, which increase the risk for cesarean delivery, uterine atony, postpartum bleeding, and need for blood transfusion
- **Localized infection** - Patients with IAI who undergo cesarean delivery, which is common, are at increased risk for wound infection, endomyometritis, septic pelvic thrombophlebitis, and pelvic abscess.
- **Sepsis**-The risk of life-threatening sequelae, such as sepsis, coagulopathy, and adult respiratory distress syndrome related to IAI are low if treatment with broad-spectrum antibiotics is initiated.

Fetal and Neonatal outcome

- Perinatal death,
- Asphyxia,
- Early-onset neonatal sepsis,
- Septic shock,
- Pneumonia, meningitis,
- Intraventricular hemorrhage (IVH),
- Cerebral white matter damage, and
- Long-term disability including cerebral palsy, as well as morbidity related to preterm birth

Prevention

The main strategy to prevent IAI is ***administration of prophylactic antibiotics*** to women with preterm premature rupture of membranes (PPROM), which

- reduces the incidence of clinical chorioamnionitis,
- prolongs latency, and
- improves neonatal outcomes

For term PROM, we prefer delivery to expectant management with antibiotic prophylaxis.

Attention to modifiable risk factors may also reduce the incidence of IAI.

- Modifiable risk factors that pertain to the health care provider include conduct of labor (eg, minimizing the number of vaginal examinations) and use of prophylactic antibiotics in women with group B *Streptococcus* colonization.
- Modifiable risk factors that the patient can control include avoidance of tobacco and alcohol.

Chapter 15

Puerperal Fever

Prepared by Dr. Birhanu Olani

Definitions

Puerperal Fever- defined as temperature of 38.0°C or higher during the first 10 days postpartum, **exclusive of the first 24 hours.**

- Fever in the first 24 hours after delivery often resolves spontaneously and cannot be explained by an identifiable infection.

Any fever during 10 days postpartum should be aggressively investigated and timely managed.

- **Most persistent** fevers after childbirth are caused by genital tract infection.

Puerperal sepsis- is infection of the genital tract occurring at any time between the onset of ROM or labor and the 42nd day postpartum in which **two or more of the following are present:** (WHO technical working group)

- Pelvic pain
- Fever
- Abnormal vaginal discharge
- Abnormal smell/foul odor discharge
- Delay in uterine involution

SIRS (Systemic inflammatory response syndrome)

Criteria for the systemic inflammatory response syndrome

The systemic inflammatory response syndrome (SIRS) is clinically recognized by the presence of **two or more** of the following:

Temperature >38°C or <36°C

Heart rate >90 beats/min

Respiratory rate >20 breaths/min or PaCO₂ <32 mmHg

WBC >12,000 cells/mm³, <4000 cells/mm³, or >10 percent immature (band) forms

Sepsis = SIRS **plus** focus of infection

Severe sepsis- is sepsis plus sepsis induced organ dysfunction or tissue hypo-perfusion.

Septic shock-is defined as the persistence of hypo-perfusion despite adequate fluid replacement.

Differential Diagnosis for Puerperal fever

- Endomyometritis
- SSI,
- Episiotomy site infection
- Mastitis or Breast abscess
- Breast engorgement
- UTI/Pyelonephritis
- Aspiration pneumonia
- Pseudomembranous colitis

Postpartum endomyometritis

Postpartum endometritis refers to infection of the decidua (i.e, pregnancy endometrium).

Most infections are mild and resolve with antibiotic therapy; however, in a minority of patients, the infection extends into the peritoneal cavity potentially resulting in peritonitis, intraabdominal abscess, or sepsis.

Rare patients develop necrotizing myometritis, necrotizing fasciitis of the abdominal wall, septic pelvic thrombophlebitis, or toxic shock syndrome.

Epidemiology

Incidence varies depending on the route of delivery

- After vaginal delivery =1% to 3% (**0.2 -2% UTD 2022**).
- Following scheduled C/D before the onset of labor and ROM, the frequency of endometritis ranges from **5% to 10% without** antibiotic prophylaxis and is **<5% with** prophylaxis.
- When C/D is performed after an extended period of labor and ruptured membranes, the incidence of infection is **30% to 35% without antibiotic prophylaxis and about 60% less with prophylaxis.**
- **In indigent patient populations, the frequency of infection may be even higher**

Microbiology

Postpartum endometritis is typically a **polymicrobial infection**.

- Aerobes include groups A and B streptococci, Staphylococcus, Klebsiella, Proteus, Enterobacter, Enterococcus, and Escherichia coli.
- Anaerobes include Peptostreptococcus, Peptococcus, Bacteroides, Fusobacterium, Prevotella, and Clostridium.
- In HIV-infected patients, herpes simplex virus and cytomegalovirus.
- Rare, but potentially lethal, causes of endometritis include Clostridium sordellii, Clostridium perfringens and streptococcal or staphylococcal toxic shock syndrome

Risk factors

Route of delivery- the **single most important risk factor** for endometritis.

- **Cesarean birth-** the dominant risk factor for development of postpartum endometritis, esp. when performed after the onset of labor (**UTD 2022**)
 - With antibiotic prophylaxis, the frequency of postpartum endometritis is approximately 7.0 % for cesareans performed after the onset of labor and 1.5 % for those that are scheduled.

- In the absence of antibiotic prophylaxis are approximately 18 and 4 %, respectively.
- The frequency of postpartum endometritis after a vaginal birth is much lower than after a cesarean birth, ranging from 0.2 to 2.0 %.

Other risk factors are:-

- | | |
|--|--|
| ▪ Chorioamnionitis | ▪ Organ transplant recipients |
| ▪ Prolonged labor | ▪ Severe anemia |
| ▪ Prolonged rupture of membranes | ▪ Preterm or post-term birth |
| ▪ Multiple cervical examinations | ▪ Operative vaginal birth |
| ▪ Internal fetal or uterine monitoring | ▪ Obesity |
| ▪ Large amount of meconium in amniotic fluid | ▪ HIV infection |
| ▪ Manual removal of the placenta | ▪ Colonization with GBS |
| ▪ Low socioeconomic status | ▪ Nasal carriage of <i>S. aureus</i> |
| ▪ Maternal diabetes mellitus | ▪ Heavy vaginal colonization by <i>E. coli</i> |
| ▪ Chronic steroid therapy | |
| ▪ Cancer chemotherapy | |

Pathogenesis

During labor and birth, the endogenous cervicovaginal flora migrates into the uterine cavity, thereby contaminating its normally sterile contents.

The development of infection versus colonization is thought to be related to a complex interaction among:

- local factors (eg, presence of devitalized or otherwise damaged tissue, foreign bodies),
- host defense mechanisms and
- the size of the bacterial inoculum, and the virulence of the bacteria involved

The potential for infection is enhanced **10 to 30-fold** in cesarean deliveries compared with vaginal births because of:

- The presence of foreign bodies (eg, suture material),
- Myometrial injury and necrosis at the suture line, and
- Formation of hematomas and seromas

Clinical findings

Hx

- Fever
- Midline lower abdominal pain
- Malodorous purulent lochia/vaginal discharge
- Chills, rigor, headache, malaise, and/or anorexia
- Risk factors

P/E

- Fever
- Tachycardia that parallels the rise in temperature
- Uterine tenderness
- The uterus may be slightly soft and subinvoluted leading to excessive uterine bleeding.

Alarm findings (sepsis) -criteria's for suspecting severe infection/sepsis in febrile postpartum patients, **based on expert opinion:**

- Fever $\geq 103^{\circ}\text{F}$ (39.4°C) OR
- Fever $\geq 102^{\circ}\text{F}$ (38.9°C) **plus** one or more of the following
 - HR ≥ 110 beats/minute, sustained for at least 30 minutes
 - RR ≥ 20 respirations/minute, sustained for at least 30 minutes
 - WBC differential showing ≥ 10 percent bands
 - Blood pressure $\leq 90/60$ mmHg, sustained for at least 30 minutes (in the setting of infection, septic shock can be diagnosed if mean arterial pressure is < 65 mmHg after 30 mL/kg fluid load)
- An elevated lactic acid concentration (> 2 mmol/L) is also a sign of serious infection.

Diagnosis

Postpartum endometritis is **primarily a clinical diagnosis** based on signs and symptoms and presence of risk factors.

Diagnosis is made in a patient with at least two of the following (Up-to-date 2022)

- *Fever ($\geq 100.4^{\circ}\text{F}$ (38°C))*
- *Pain or tenderness (uterine or abdominal) with no other recognized cause*
- *Purulent drainage from the uterus*

The presence of tachycardia and/or leukocytosis supports the diagnosis, but are non-specific

Fever is a key sign because variable degrees of midline abdominal pain, uterine tenderness and leukocytosis are common after C/D, and to a lesser extent after V/D, in the absence of infection.

Laboratory

- **Complete blood count**
 - Leukocytosis with neutrophilia ($\geq 80\%$) or left shift /bandemia ($\geq 10\%$).
- **Imaging (Ultrasound scanning)**
 - **Not helpful for making the diagnosis**, but it can be helpful to exclude other diagnoses (eg, RPC, infected hematoma, uterine abscess)
- **Blood culture**
 - Only 5 to 20% have bacteremia.
 - Not routinely done as initial antibiotic therapy has to be made before the results are available, and the results usually do not lead to a change in the initial empiric antibiotic regimen.
 - **Indications for blood culture: To guide choice of antimicrobial treatment.**
 - ✓ Patients having alarm findings(above)
 - ✓ Immunocompromised patients

- ✓ Septic patients
- ✓ Failure to respond after 48hrs of antibiotic treatment
- **Endometrial culture**
 - Endometrial cultures are **not** performed because of the difficulty in obtaining an uncontaminated specimen through the cervix.

DDX

Surgical site infection

- include infection of abdominal wall incision, episiotomy incision, perineal lacerations
- is typically evident on physical examination of the surgical site (local erythema, edema, and/or tenderness)

Mastitis or breast abscess-

- usually evident on physical examination of the breast (local erythema, edema, and/or tenderness)
- typically occurs later in the postpartum course (the first three months of breastfeeding)

Breast engorgement-

- Breast fullness and firmness accompanied by pain and tenderness may also lead to a low-grade fever **24 to 72 hours postpartum**

Pyelonephritis

- is characterized by fever ($>100.4^{\circ}\text{F}$ [38°C]), chills, flank pain, costovertebral angle tenderness, and possibly lower urinary tract symptoms
- Pyuria and/or a positive urine culture supports the diagnosis

Aspiration pneumonia-

- presents with fever, dyspnea, and possibly hypoxemia
- Lung auscultation may reveal diffuse crackles, and a chest radiograph will show infiltrates.
- It primarily occurs in postpartum patients with compromise in the usual defenses that protect the lower airways, such as those with a recent history of a difficult or failed intubation.

Pseudomembranous colitis-

- is a rare, but potentially serious, cause of postpartum fever
- It should be considered in postpartum patients who have low-grade fever, abdominal and gastrointestinal symptoms, and recent antibiotic exposure

Septic pelvic thrombophlebitis

NB! It is also better to think of other acute febrile illnesses too! Esp, malaria.

*Unexplained fever with significant back pain after a neuraxial anesthetic, esp. when accompanied by neurologic symptoms may be due to **infection or inflammation of the spinal cord.***

MANAGEMENT

- Treatment is indicated for relief of symptoms and to **prevent sequelae, such as peritonitis, salpingitis, oophoritis, phlegmon or abscess, and septic pelvic thrombophlebitis.**
- Prompt administration of appropriate antibiotics is critical in septic patients. Treatment is the same, regardless of mode of birth

FMOH Obstetrics Management Protocol 2021**Supportive care**

- Institute supportive measures if necessary. This includes transfusion of blood, correction of fluid & electrolyte imbalance.

Antibiotics

- Give the woman a combination of antibiotics starting from presentation up to 24–48 hours after complete resolution of clinical signs and symptoms (fever, uterine tenderness, purulent lochia, and leukocytosis).
- Clindamycin phosphate 600 mg IV every eight hours PLUS Gentamicin 5 mg/kg body weight IV every 24 hours.
- If Clindamycin is not available administer Ampicillin 2 g IV every 6 hours PLUS gentamicin 5 mg/kg body weight IV every 24 hours.
- When available, Clindamycin (in combination with Gentamicin) is more effective than Ampicillin or a penicillin antibiotic for the treatment of postpartum endometritis.

Management of persistent fever

- If fever is still present 72 hours after starting antibiotics, re-evaluate and revise the diagnosis.
- Possible differential diagnosis for persistent fever includes peritonitis, pelvic abscess and septic thrombophlebitis.
- Do abdomino-pelvic ultrasound to assess for retained tissue and to check for other complications like abscess collection.
- If retained placental fragments are suspected, perform a digital exploration of the uterus to remove clots and large pieces. Use ovum forceps, aspiration with large cannula (12-14) or a wide curette if necessary (avoid sharp curette).
- If there are signs of general peritonitis (fever, rebound tenderness, general abdominal pain), perform laparotomy to drain the pus.

- If the uterus is necrotic and septic, perform subtotal hysterectomy

Williams 25th E.

TABLE 37-2. Antimicrobial Regimens for Pelvic Infections Following Cesarean Delivery

Regimen	Comments
Clindamycin + gentamicin	"Gold standard," 90–97% efficacy, once-daily gentamicin dosing acceptable PLUS
Clindamycin + aztreonam	Ampicillin added to regimen with sepsis syndrome or suspected enterococcal infection
Extended-spectrum penicillins	Gentamicin substitute for renal insufficiency
Cephalosporins	Piperacillin, piperacillin tazobactam, ampicillin/sulbactam, ticarcillin/clavulanate
Vancomycin	Cefotetan, ceftiofur, cefotaxime
Metronidazole + ampicillin + gentamicin	Added to other regimens for suspected <i>Staphylococcus aureus</i> infections
Carbapenems	Metronidazole has excellent anaerobic coverage
	Imipenem/cilastatin, meropenem, ertapenem reserved for special indications

Up-to-date 2022

Preferred initial regimen (no GBS colonization): results in resolution of infection in 90 to 97 percent of cases.

- Clindamycin 900 mg IV every 8 hours **plus**
- Gentamicin 5 mg/kg IV every 24 hours (**preferred**) **or** 1.5 mg/kg IV every 8 hours (without a loading dose).

Preferred initial regimen (GBS colonization) -Resistance to clindamycin in GBS isolates ranges from 13 to 43 percent.

- Clindamycin 900 mg every 8 hours **plus**
- Gentamicin 5 mg/kg every 24 hours (preferred) **or** 1.5 mg/kg every 8 hours (without a loading dose) **plus**
- Ampicillin 2 g IV every 6 hours **or**
- Ampicillin-sulbactam 3 g IV every 6 hours

In our setup

- Ceftriaxone 1 g IV BID plus
- Metronidazole 500 mg IV/PO TID is used because of **resistance** to Clindamycin & Gentamicin.

Duration of therapy

- A response to the initial antibiotic regimen should be evident within 24 to 48 hours.
- IV treatment is typically continued until the patient is clinically improved (no fundal tenderness) and afebrile **for 24 to 48 hours**.
- Oral antibiotic therapy after successful parenteral treatment is unnecessary as it did not improve outcome in randomized trials.
- **Note!** Here considering our working environment, oral antibiotics for 5-7 days after successful IV antibiotics are being practiced. **Options are:**
 - Cephalexin 500 mg Po TID **plus**
 - Metronidazole 500 mg PO TID

Options when IV therapy isn't possible, Up-to-date 2022

- Clindamycin 600 mg orally every 6 hours plus gentamicin 4.5 mg/kg intramuscularly every 24 hours **or**
- Amoxicillin-clavulanic acid 875 mg orally every 12 hours **or**
- Cefotetan 2 g intramuscularly every 8 hours **or**
- Meropenem or imipenem with cilastatin 500 mg intramuscularly every 8 hours **or**
- Amoxicillin 500 mg plus metronidazole 500 mg orally every 8 hours

They provide **>85** percent cure rates of early postpartum endometritis and were compatible with breastfeeding.

Duration of treatment:

- Oral regimen: 14 day course is suggested.
- Intramuscular regimen: continue 48 to 72 hours of intramuscular therapy and then switch to an oral antibiotic to complete a seven-day course

Complications

- PPH
- Sepsis >>septic shock
- Peritonitis
- Intraperitoneal abscess
- Necrotizing myometritis
- Necrotizing fasciitis of abdominal wall
- Septic pelvic thrombophlebitis
- Toxic shock syndrome.....etc

Prevention

- Standard clean surgery
- Proper intrapartum antibiotics prophylaxis-**indications:**
 - GBS colonization
 - Preterm PROM
 - Prolonged ROM (> 8hrs)
 - Manual removal of placenta
 - 3rd or 4th degree perineal tear
 - Elective or emergency Cesarean delivery
- Cleansing vagina with povidone iodine before C/D
- TAT administration, if not given TT vaccine previously
- Iron supplementation

SECTION TWO: GYNECOLOGY

Chapter 16

ABDOMINOPELVIC MASS (APM)

Prepared by Dr. Abdulhafiz Hussein (MI)

Definition

Abdominopelvic mass

- Is abnormal abdominal growth/mass/swelling that took its origin down from pelvic organs
- It's important to differentiate it from masses arising from abdominal wall, intra-abdominal and retroperitoneal structure

Approach

History

Take detail Hx and focus on: -

- **Age**- very important in narrowing DDx
 - In young premenopausal women u should focus on benign causes like: myoma, functional ovarian cyst, TOA,
 - In older patients malignant conditions like ovarian cancer should be top on DDx
- Elaboration of the c/c (swelling)
 - Duration, onset and location,
 - Progression- rapid growth is consistent with malignancy
- **Associated Sxs** - elaborate them accordingly,
 - Pelvic/abdominal pain, low back pain-eg. Inflammatory or invasive conditions
 - Vaginal bleeding- eg. Myoma, Ectopic pregnancy, Ovarian tumors
 - What are the three possible mechanisms of AUB in ovarian tumors?
 - Vaginal discharge- eg. FT cancer, TOA...

- Fever- eg. TOA, Pyometra...
- Urinary sx- urgency, incontinence, retention...
- GI sx- constipation, bloating, early satiety- eg. Ovarian masses
- Fatigue, significant weight loss (may indicate malignancy)
- Risk factors (RF)
- Family hx of any malignancy
 - Especially that of breast, ovarian, endometrial or colonic
- **Reproductive status**
 - Infertility/nulliparity can be RF (eg. low malignant potential ovarian tumors) or complication of some APM (myoma, endometrioma, hydrosalpinx).
 - Recurrent abortion, (can be from Submucous myoma)
- Menstrual and contraception hx
 - Age at menarche and menopause, COC use
 - For which gynecologic malignancy are COCs protective and for which are risks factors?
- Prior hx of PID
- Prior hx of any gynecological surgery, risk for ectopic Px, Teratoma

Physical Examination

- **General:-** CSL (?malignancy)
- **Vital Signs:-** including BMI (obesity is RF for some APM)
- **HEENT:-** Anemia (Pale conjunctiva)-of chronic disease in malignancies, from AUB
2° APM
- **LGS:-**
 - Is important in assessing for metastasis (LNs, Breast)
 - Hereditary ovarian cancer may occur synchronously with Breast cancer and/or the later may metastasize to ovary

- **R/S:**

- Pleural effusion 2° to Meig's syndrome (Ovarian fibroma) or pseudomeigs syndrome (eg. mature teratoma, stroma ovarii, Myoma, Benign FT)

Findings may include: - decreased tactile fremitus, Dullness, decreased air entry

- Metastasis to lung (eg. Choriocarcinoma, CRC, bladder cancer)

- **Abdomen:**

- **Inspection**-shape of abdomen, location of swelling, distended vein, any skin color change, any previous surgical scar.

- **Palpation**- superficial and deep

- ✓ Size- in weeks of pregnant uterus size. (...a X weeks gravid uterus sized APM...)

- ✓ 12 weeks at symphysis pubis

- ✓ 20 weeks at umbilicus

- ✓ 38 weeks at xiphisternum

- ✓ Origin-

- ✓ APM arising from the pelvis or an abdominal mass? It can be differentiated by identifying if one can go below the mass in to the pelvic cavity or not.

- ✓ Site-Location in the Abdomen

- ✓ Shape- globular, irregular

- ✓ Consistency- soft or cystic, firm or hard

- ✓ Surface contour-smooth, or nodular

- ✓ Border-regular or irregular

- ✓ Mobility- fixation may indicate adhesions or malignancy

A fixed mass that is mixed hard and cystic parts, nodular with irregular borders indicate malignancy

- ✓ Tenderness - indicate inflammatory process or torsion

- ✓ Surface Temperature -comparable or not with surrounding area.

E.g. there is a 14 weeks gravid uterus sized APM which occupied lower abdomen, mostly the left side. The mass is hard with cystic parts, smooth, immobile, non-tender and globular/irregular in shape with irregular border and comparable surface temperature with surrounding.

- **Percussion**

- ✓ Shifting dullness and fluid thrill to detect ascites
- ✓ Differentiation of a large ovarian tumor versus ascites

Large ovarian tumor has central dullness with tympanicity at the flanks as opposed to ascites with central tympanicity

- **Pelvic Examination**

- Complete (both abnormal and normal) finding should be reported for every Gynecologic case.

- 1. Examination of external genitalia- Inspection and palpation

- ✓ Discharge or bleeding from the introitus – should be noted.
- ✓ Hymen
 - Unruptured - septate, annular
 - Imperforate

All can result in hematometra/Pyometra

- 2. Speculum examination

- ✓ Look for Cervix- scarring from previous procedures or infection and/or cervical mass may interfere with menstrual outflow and end up with hematometra/Pyometra
- ✓ Submucous Myoma may deliver through cervical canal

- 3. Digital vaginal examination (PV)

- ✓ Appreciate Vaginal wall, esp. Bulging in posterior fornix (Cul-de-sac) (eg. In ovarian cancer metastasis)
- ✓ Transverse Vaginal septa- hemato/Pyometra
- ✓ Cervical Excitation (motion tenderness) – PID

4. Bimanual pelvic examination

- ✓ Help to differentiate whether origin of APM is uterus or adnexa (uterine origin mass move with cervical motion while that of adnexal will not)

5. Rectovaginal Examination

- ✓ Appreciate the rectal wall and pouch of Douglas (for nodularity of Cul-de-sac in case of metastasis and endometriosis)
- ✓ Inspect for blood on examining finger for possibility of Colorectal cancer (CRC)

▪ I/S & MSS:-

- Lower extremities edema may result as a complication of compressive APM (when it is large enough to compress large Pelvic Vessels)

Investigations

▪ Laboratory Ix's

- Pregnancy test - with Urine or Serum β -hCG should be performed in any reproductive age group women with pelvic mass
- CBC- Anemia, leukocytosis(TOA)
- Serum Biomarkers for malignancy- their utility is limited esp. in reproductive age group women because of low specificity
 - ✓ Mainly useful for postoperative follow-up of the patient (assessing the degree of response to therapies)
- Other baseline Ixs, especially if patient is going for surgery or chemotherapy
 - ✓ RFT, LFT,
- HBsAg, VDRL, PICT, U/A, basic Ixs for almost all gyne patients.
 - ✓ Any visit of a women to gyne opd is a good opportunity to screen for the aforementioned diseases.

▪ Imaging

- Ultrasonography(U/S)- TVS and/or Abdominal U/S is the best and 1st line imaging modality for evaluation of pelvic mass

- **MRI &/or CT-** for further evaluation (Anatomical details), staging malignancy and planning surgery
- Other Imaging modalities- depending on patient's presentation or primary image findings. E.g. Hysteroscopy (if bleeding), HSG(Infertility), Cystoscopy/urodynamic study (significant urinary symptoms)
- **Pathological study**
 - *definitive dx of type of pelvic mass can be made only with histopathologic evaluation of tissue sample , which is usually obtained after surgery*

Management

- *Will be discussed under each case.*

Differential diagnosis of APM

Causes of APM can be Gynecologic or non-gynecologic.

Gynecological masses:	Non-gynecologic masses:
▪ Ovarian origin: ○ Benign and malignant tumors ○ Functional cysts ○ Endometrium (Endometriosis to Ovary) ○ Metastasis to Ovary (Breast, GI...) ▪ Fallopian Tube: ○ TOA ○ Ectopic pregnancy (EP) ○ Hydrosalpinx ○ Para-ovarian/tubal cysts ○ FT neoplasm ▪ Uterus: ○ Fibroids/Myoma, ○ Sarcoma, ○ Hemato/Pyometra ○ Adenomyosis	▪ GI system: ○ Appendicular mass (Abscess, mucocele) ○ Diverticular abscess ○ Mesenteric cyst ○ Colorectal Cancer (esp. left side) ○ IBD (Crohn's disease) ○ Intestinal tuberculosis ▪ Urinary tract: ○ Pelvic kidney ○ Bladder tumors, ○ full bladder ▪ Miscellaneous: ○ LAP's (eg. mesenteric LAP) ○ Peritoneal carcinomatosis ○ Pelvic Musculoskeletal tumors ○ infections

Depending on Age and Reproductive status of the women different types of pelvic mass may be more likely than other

Infants and Adolescents:

- Ovarian tumor (germ cell tumors, mature cystic teratoma)
- Follicular cyst and other functional cysts
- Pregnancy and it's complications
- Hematometra- 2^o to imperforated hymen
- PID and its complications
- Endometrioma
- Appendiceal Abscess

Premenopausal (reproductive age):

- Functional cysts- follicular, corpus luteum
- Pregnancy related- Pregnancy, EP, theca lutein cyst
- Inflammatory-
 - PID related-TOA, Hydrosalpinx
 - Appendicular/diverticular Abscess
- Ovarian tumors- benign>>malignant
 - Endometrioma
- Leiomyoma (uterine fibroid)
- Adenomyosis
- Para ovarian/tubal cyst

Postmenopausal:

- Ovarian cancer (1^o and metastatic)
- Fallopian tube cancer
- Sarcoma of uterus

Discussion on some DDX of APM

*The most common causes of APM in our setup (ward admission) - **Ovarian Tumors and leiomyoma**- will be discussed in detail under separate chapter.*

Functional ovarian cysts

- **Follicular cyst**
 - Is the most common type of ovarian cyst
 - In menstruating women, a follicle containing the ovum (unfertilized egg) will rupture during ovulation. If this does not occur (failure in ovulation), most likely 2^o to disturbances in the release of the pituitary gonadotropins, the fluid of the incompletely developed follicle is not reabsorbed, producing an enlarged follicular cyst.
- **Corpus luteum cysts**
 - Appear after ovulation
 - Following normal ovulation, the granulosa cells lining the follicle become luteinized. In the stage of vascularization, blood accumulates in the central cavity, producing the corpus hemorrhagicum. Resorption of the blood then results in a corpus luteum, which is defined as a cyst when it grows **larger than 3 cm**.
- **Theca lutein cysts**
 - Occur within the thecal layer of cells surrounding developing oocytes
 - Under the influence of **excessive hCG** (in case of GTD, clomiphene therapy and rarely normal pregnancy), thecal cells may proliferate and become cystic.
 - Are usually bilateral

Presentation of functional cysts

- Dull aching pain within the abdomen or pelvis,
- AUB, menstrual irregularity and Dysmenorrhea.
- Fullness, heaviness, pressure, swelling, or bloating in the abdomen.
- pressure symptoms- urinary frequency, constipation
- Nausea or vomiting or weight gain

Complications

- Rupture of ovarian cysts
- Adnexal torsion

Ectopic pregnancy (EP)

It is an early complication of pregnancy in which the embryo attaches outside the uterus. Most EP (>95%) are tubal pregnancies. It can also occur in the cervix, ovaries, or within the abdomen.

Risk factors

- PID,
- Use of IUD (failed IUD),
- Previous exposure to DES,
- Intrauterine surgery (e.g. D&C),
- Smoking, previous EP,
- Endometriosis
- Tubal surgery and tubal ligation.

In up to one half cases, the RF cannot be identified

Presentation

- **Hx:** -
 - vaginal bleeding,
 - acute lower abdominal pain,
 - pelvic pain,
 - nausea, vomiting and diarrhea
 - ***Patient may be asymptomatic (10%)***
- **P/E:** -
 - Adnexal mass,
 - Tender cervix,
 - Adnexal tenderness
- **Ixs:-**
 - serum β -hcg (increased), TVS

Complications-

- Rupture of an EP leads to
 - abdominal distension,
 - tenderness,
 - peritonism and
 - hypovolemic shock

PID related TOA & Hydrosalpinx**Tubo-Ovarian Abscess (TOA)**

- It is an inflammatory mass involving the FT, ovary and, occasionally, other adjacent pelvic organs.
- Is one of the late complications of PID.
- Most commonly in reproductive age women

Presentation:

- **Hx:-**
 - Acute lower Abdominal/Pelvic pain,
 - Fever, Chills,
 - Purulent Vaginal discharge,
 - Abdominal swelling,
 - hx of prior STD
- **P/E:-**
 - Abdominal mass,
 - pelvic tenderness (lower abdomen)
 - Cervical motion tenderness (CMT)
- **Ix's:-**
 - CBC (Leukocytosis, elevated ESR),
 - U/S (complex mass that mimic malignancy)

Complications: -

- Sepsis if abscess ruptured- life threatening

Hydrosalpinx

- Is a distally blocked FT filled with serous or clear fluid.
 - Blockage is at fimbriated end and It is usually bilateral
- Cause can be PID (major cause), endometriosis, surgery, tubal TB, cancer of FT & Ovary

Presentation

- Commonly Infertility, recurrent lower Abdominal or pelvic pain, prior hx of PID, & Pelvic mass

Adenomyosis

Is characterized by uterine enlargement caused by ectopic rest of endometrial tissue, **both** glands and stroma, located **deep** within myometrium

Both= ectopic tissue should be both glands and stroma

Deep= superficial invasion of myometrium by endometrium is normal

Can be classified into **two**:

- **Diffuse type** -ectopic tissue rests scattered and the uterus becomes bulky and heavier.
- **Focal type** (Adenomyoma)-localized, nodular collection of ectopic tissues

Risk factors

- Parity-90% of cases occur in parous women
- Uterine surgery
- Age-nearly 80% occur in 40s and 50s
- Tamoxifen

Presentation:

- Mostly asymptomatic,
- if symptomatic-HMB, dysmenorrhea, dyspareunia, chronic pelvic pain and bladder irritation

Diagnosis

- TVS:-sonographic finding of diffuse Adenomyosis includes
 - a globular enlarged uterus
 - anterior/posterior wall thickness asymmetry
 - myometrial hypoechoic cysts
 - Myometrial texture heterogeneity
 - Diffuse myometrial echogenicity
 - Ill-defined endometrial echo
 - Striated projection extending from the endometrium to myometrium (sign of endometrial invasion to the myometria)
- MRI:-provides better diagnostic capability due to the increased soft tissue differentiation and when coexisting myoma distort anatomy

Hematometra/pyometra

- collection or retention of blood (or Pus-if blood is infected) in the uterus
- develops when the Uterus becomes distended with blood $\geq$ to menstrual outflow obstruction from blockage or atresia of the lower reproductive tract—the Ux, Cx or Vx
- Most commonly caused by congenital abnormalities, including imperforate hymen, transverse vaginal septum or vaginal hypoplasia.
- Other causes are acquired, such as cervical stenosis, intrauterine adhesions, endometrial Carcinoma, and Cervical Carcinoma.

Risk factors

- uterine surgery such as endometrial ablation
- cervical Surgery or procedures
- female genital mutilation(FGM)
- Abortion
- childbirth

Presentation

- cyclic, cramping pain in the midline of the pelvis or lower abdomen and Low back pain
- urinary frequency or retention and constipation
- dysmenorrhea (pain during menstruation)
- amenorrhea (lack of menstruation)
- Postmenopausal women are more likely to be asymptomatic.
- Fever and leukocytosis (Pyometra)
- May develop low blood pressure.
- firm and enlarged Ux which may become tender

Diagnosis

- often based purely on the pt's Hx of amenorrhea and cyclic abdominal pain + palpable pelvic mass
- confirmed by ultrasound

Para ovarian/tubal cysts

- Are epithelium-lined fluid-filled cysts in the adnexa adjacent to the FT and ovary.
- originate from the mesothelium and are presumed to be remnants of the Müllerian duct and Wolffian duct

- Eg. Hydatid cysts of Morgagni, or Morgagni's cysts, are common and appear as pedunculated, often tiny, frequently multiple cysts connected to the fimbriae of the FT. They thus appear to be a specific variant of paratubal cysts

Presentation

- Mostly small and asymptomatic
- Pressure sx
- Pain in the pelvis
- Infertility

SAMPLE HISTORY: for major cases (myoma and ovarian masses) is included in the respective portions of the topics

Chapter 17

Approach to Abnormal Uterine Bleeding

Prepared by Dr. Bethel Ayele (MI)

Definitions

Abnormal Uterine Bleeding (AUB)

- Is defined as any variation from the normal menstrual cycle, which is 28 ± 7 days, lasting ≤ 8 days (average 5 days), with 5-80 ml of blood loss per cycle. AUB can be acute or chronic.
 - **Acute AUB** is defined as bleeding sufficiently heavy to require immediate intervention to prevent ongoing loss.
 - **Chronic AUB** is defined as bleeding that has been present during most of the prior 6 months.

Menses is characterized by four qualities – **Volume, Duration, Frequency** and **Regularity**. The patterns of AUB present with variations in these categories as:

- **Heavy Menstrual Bleeding (HMB)** (formerly menorrhagia) – is menses with increased volume, that is objectively defined as >80 ml of blood loss per menstrual period.
- **Prolonged Bleeding** – is when the duration of menses exceeds 8 days per cycle.
 - In some cases, women may complain of both which is referred to as Heavy, prolonged menstrual bleeding.
- **Frequent bleeding** – describes the number of days between menses <24 days intervening.
- **Infrequent bleeding** (previously oligomenorrhea) – is defined by menses with >38 days intervening.
- **Amenorrhea** – absence of menstrual bleeding in a 90 days period.
- **Irregular cycle** – when the cycle length varies from cycle to cycle by ≥ 10 days.

Specific terms that describe nonmenstrual bleeding are:

- **Intermenstrual bleeding** – brief bleeding which occurs between fairly normal menses and it may further be defined by clinical context like in postcoital bleeding prompted by vaginal intercourse.
- **Breakthrough bleeding** and **Unscheduled bleeding** – are terms associated with hormone administration.
- **Withdrawal bleeding** – predictable bleeding which results from an abrupt decline in progesterone level.
- **Precocious menstruation** – is initiation of menses in childhood.
- **Postmenopausal bleeding** – menopause is marked by 12 months of amenorrhea after the last menstrual period, and any vaginal bleeding which occurs after more than 12 months is defined as postmenopausal bleeding.

How do you estimate the amount of Menstrual blood loss?

HMB evaluation can proceed based solely on patient's description of flow which suggests heavy flow, but there are objective methods however used for HMB diagnosis.

2. Hallberg and associates (1966): described a technique to **extract hemoglobin from sanitary napkins using sodium hydroxide**, which is then converted to hematin and measured spectrophotometrically. It is now revalidated and used with modern sanitary napkins which contain superabsorbent polymer granules, but it is rather used in research than routine clinical setting.
3. Hemoglobin and hematocrit evaluation: **hemoglobin <12g/dL** raise the chance of identifying HMB, but normal level does not exclude HMB.
4. Pictorial blood assessment chart (PBAC): there is a positive correlation between objective HMB and **passing clots more than 1 inch in diameter and changing pads more frequently than every 3 hours**.
 - This estimate of the number and type of pads or tampons used by a woman during menses is another method of assessing bleeding.

- In PBAC, using a scoring sheet, patients are asked to record daily the number of sanitary products that are lightly (1 point), moderately (5 points) or completely (10 points) soaked. Presence of small, 22-mm-diameter clots score 1 point, whereas 3-mm clots score 5. Points are then tallied for each day.
 - Totals >100 points per menstrual cycle correlate with >80 ml objective blood loss.

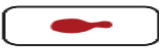
Pads	Points per each	Tampons	Points per each
	1		1
	5		5
	20		10
Large clots		5	
Small clots		1	

FIGURE E8-1 Scoring for the pictorial bleeding assessment chart. Patients are counseled to evaluate the degree of saturation for each sanitary product used during menstruation. The total number of points are tallied for each menses. Point totals greater than 100 indicate menorrhagia.

5. Menstrual cup – seated in the vagina like a diaphragm, these **cups allow trapping and collection of menstrual blood**. The woman can remove, empty, wash and replace the cup. Differing sizes and stiffness accommodates parous or nulliparous anatomy.
6. Menstrual calendars – can also be used to evaluate AUB and its pattern by **recording dates and blood flow quantity throughout the month**. This aids diagnosis and assessment of improvement during medical treatment.

Incidence and Etiology

- The true population rates of AUB can vary and data may also differ depending on the definitions used for AUB,
 - But the commonly cited rates range from **10 to 30%**.
- Age and reproductive status most greatly influence AUB incidence and help prioritize potential etiologies.
 - **Prior to menarche** – bleeding is investigated as an abnormal finding.
 - The vagina rather than the uterus is more frequently involved.
 - True uterine bleeding usually results from increased exposure to endogenous or exogenous estrogen
 - **In adolescence** – AUB results from ***anovulation (because of Immature HPO axis)*** and coagulation defects at disproportionately higher rates compared with older reproductive-aged women. Pregnancy, sexually transmitted diseases (STDs), and sexual abuse are also possibilities.
 - **Following adolescence** – the hypothalamic-pituitary-ovarian (HPO) axis matures, and anovulatory uterine bleeding is encountered less often.
 - Rates of bleeding related to pregnancy and STDs rise.
 - The incidence of bleeding from leiomyomas, adenomyosis, and endometrial polyps also increase with age.
 - **Perimenopause** – as with premenarchal girls, bleeding from ***HPO axis dysfunction*** is a more frequent finding. With aging, risks of benign and malignant neoplastic growth rise.
 - **After menopause** – bleeding typically can be traced to a benign origin such as endometrial or vaginal atrophy or polyp, even so, malignant neoplasms, especially endometrial carcinoma, are found more often in this age group. Less commonly, estrogen producing ovarian carcinoma may cause endometrial hyperplasia and bleeding. Similarly, ulcerative vulvar, vaginal or cervical neoplasms can be sources.

To organize the main causes of AUB, FIGO created a classification system using the acronym **PALM- COEIN**.

PALM (the structural causes) –

- **Polyps** (endometrial, endocervical),
- **Adenomyosis**,
- **Leiomyomas**,
- **Malignancy** (preinvasive like CIN, EIN and cancer of cervix, endometrium, sarcoma)

COEIN (non-structural causes) –

- **Coagulopathy** (inherited, liver failure, immunologic),
- **Ovulatory** (adolescent HPO immaturity, perimenopausal follicle aging, premature ovarian failure, hypothyroidism, obesity, hyperprolactinemia, excess androgen like PCOS, congenital adrenal hyperplasia, cushing syndrome/ disease),
- **Endometrial** (atrophy, intrinsic abnormality, endometrial hyperplasia, chronic endometritis),
- **Iatrogenic** (intrauterine device, medications like anticoagulants),
- **Not otherwise specified** (infections like PID, cervicitis, TB, Arterovenous malformation, partial sequestration of menses like in isthmocele and ascertain syndrome)

Common differential diagnoses include:

▪ Pregnancy (1st r/o pregnancy in women of reproductive age) ▪ Incomplete abortion ▪ Ectopic pregnancy ▪ GTD ▪ Myoma ▪ Adenomyosis ▪ Endometrial polyps ▪ Endometrial hyperplasia ▪ Endometrial cancer	▪ Cervical neoplasms ▪ Sex cord stromal tumors (ovarian tumor) ▪ Infections- chronic endometritis ▪ Systemic abnormality ▪ Thyroid disease (Hypothyroidism) ▪ Leech infestation
---	--

Ectopic Pregnancy

- Ectopic or extra uterine pregnancy is one in which the blastocyst implants anywhere other than the endometrial lining of the uterine cavity. Nearly 95% of ectopic pregnancies implant in the fallopian tube (ampullary 70%, isthmic 12%, fimbrial 11%, interstitial and cornual 2-3%). Other sites include ovarian 3%, abdominal 1%, cervical <1% and caesarean scar <1%.
- Heterotopic pregnancy is a uterine pregnancy in conjunction with an extrauterine pregnancy. The incidence is approximately 1 in 30,000 pregnancies.

Risk factors:

- Abnormal fallopian tube anatomy which alters normal embryo transport underlies most cases
- Prior tubal surgery (surgeries to restore fertility, to restore tubal potency or tubal sterilization)
- Prior ectopic pregnancy
- Smoking >20 cigarettes per day (threefold in women who smoke more than one pack of cigarette daily)
- PID confirmed by laparoscopy or positive test chlamydia trachomatis
- Greater than 3 prior miscarriages
- Age greater than 40 years
- Prior medical or surgical abortion
- Infertility >1yr
- Prior IUD use
- Assisted reproductive technology (ART)

Clinical manifestation:

- The classic symptom triad of ectopic pregnancy is amenorrhea followed by vaginal bleeding and ipsilateral abdominal pain. Other symptoms include, banal pregnancy discomforts such as breast tenderness and nausea.

- Vertigo and syncope may arise from haemorrhagic hypovolemia.
- **Physical examination:**
 - Speculum and bimanual examination may reveal an adnexal mass that is often tender, a uterus which is small for gestational age and bleeding from the cervix.
 - Patients with ruptured ectopic pregnancy may be hypotensive, tachycardia and unresponsive.

Diagnosis:

- Ectopic pregnancy is considered in any reproductive aged woman with pain, uterine bleeding, and/or anemia.
- Laboratory findings: serum beta-hCG to confirm pregnancy
- **Sonography:** In an attempt to unify the language used with sonographic evaluation of early pregnancies, a consensus statement was drafted with five categories:
 1. definitive ectopic pregnancy (extrauterine gestational sac with yolk sac and/or embryo),
 2. probable ectopic pregnancy (inhomogeneous adnexal mass or extrauterine halo-like structure),
 3. probable IUP (intrauterine echogenic sac)
 4. definite IUP (intrauterine gestational sac with yolk sac and/or embryo), and
 5. PUL (lacking signs of either ectopic pregnancy or IUP).

Adenomyosis

- *Is the extension of endometrial tissue (glands and stroma) into the uterine myometrium leading to abnormal bleeding and pain. These rests may be scattered throughout the myometrium, **diffuse adenomyosis** or maybe form a localized nodular collection, **focal adenomyosis**.*

- *Diagnosis is usually based on the histologic findings in surgical specimens.*
- *On gross examination the uterus is globally enlarged, but rarely exceeds a 12week pregnancy.* The surface contour is smooth and regular and generalized softening and reddish myometrial discoloration is common.
- *Adenomyosis, endometriosis, leiomyomas and endometrial hyperplasia frequently coexist.*

Risk factors:

- Parity: nearly 90 percent of cases are in parous women
- Age: nearly 80 percent develop in women in their 40s and 50s (it regresses after menopause)
- Adenomyosis is associated with other pathologies that are affected by cytochrome P450 aromatase expression and higher tissue estrogen level like leiomyomas, endometriosis, and endometrial cancer.
- Use of selective estrogen-receptor modulator like tamoxifen.

Clinical manifestation:

- Approximately one third of women have symptoms. Their severity correlates with increasing number of ectopic foci and extent of invasion
- Symptomatic adenomyosis occurs most often in parous women between age 35 and 50.
- Most common symptoms are heavy menstrual bleeding (HMB) (results from increased and abnormal vascularization of the endometrial lining) and dysmenorrhea (caused by increased prostaglandin production found in adenomyotic tissues). There may also be complaint of dyspareunia (10%).
- **Physical examination:**
 - Pelvic examination may reveal a diffusely enlarged globular uterus. The uterus is usually less than 12cm.

- The consistency of the uterus is typically softer and boggier than the firm, rubbery uterus containing fibroids. It should have normal motility and no associated adnexal pathology.

Diagnosis:

- CA125 levels are typically elevated in women with adenomyosis, but are not diagnostic.
- Transabdominal sonography does not consistently identify the changes of adenomyosis, thus imaging with **TVS is preferred**, and MR imaging may be complementary.
- Focal adenomyosis appears as discrete hypoechoic nodules that may sometimes be differentiated from leiomyomas by their poorly defined margins, elliptical rather than globular shape, minimal mass effect on surrounding tissues, lack of calcifications, and presence of anechoic cysts of varying diameter

Management:**▪ Medical Treatment:**

- The main objective of treatment is relief from pain and bleeding. First, cyclic NSAIDs are often given.
- Combination oral contraceptives and progestin-only regimens can be used to induce endometrial atrophy and decrease endometrial prostaglandin production to improve symptoms.

▪ Interventional Treatment:

- Hysterectomy is the definitive treatment and as with other conditions, the type of surgical procedure depends on uterine size and associated uterine or abdominopelvic pathology.

Endometrial polyp

- A polyp is a mass of tissue that develops on the inside wall of a hollow organ. Endometrial polyps are soft, fleshy intrauterine growths are composed of endometrial glands, fibrous stroma, and surface epithelium.

- Intact polyps may be single or multiple, measure from a few millimeters to several centimeters, and be sessile or pedunculated.

Risk factors:

- increasing age (commonly 40-50 years old),
- obesity
- tamoxifen use
- Use of oral contraceptive pills appear to be protective

Clinical manifestation:

- When symptomatic, polyps commonly present with abnormal vaginal bleeding, heavy cyclic or intermenstrual bleeding.
- Bleeding may stem from surface epithelium breaks associated with chronic inflammation and vascular fragility or from apical ischemic tissue necrosis.

Diagnosis:

- The main diagnostic tools include TVS with applied color Doppler, SIS (saline infusion sonohysterography), and hysteroscopy. In premenopausal women, TVS is best performed prior to day 10 of the cycle to lower the risk of false-positive finding from a normally thick secretory endometrium.

Management:

- It is directed by symptoms and malignancy risk. Most polyps are benign, and premalignant or malignant transformation develops in only 5%.
- Operative hysteroscopic polypectomy may be most effective for symptomatic women or those with risk factors for malignant transformation.

Endometrial hyperplasia

- It is defined as endometrial thickening with abnormal proliferation of both the glandular and stromal elements of the endometrium. If left untreated, it can

progress to endometrial carcinoma and can also coexist alongside endometrial carcinoma.

- Types include simple hyperplasia, complex hyperplasia, atypical simple hyperplasia and atypical complex hyperplasia.
- Patients at risk for endometrial hyperplasia, like those at risk for endometrial carcinoma, are at risk due to unopposed estrogen exposure.

Risk factors:

- Can be memorized with the mnemonic 'ENDOMETRIUM'
 - Excess exogenous estrogen use without progesterone
 - Nulliparity
 - Diabetes mellitus
 - Obesity
 - Menstrual irregularity
 - Elevated blood pressure
 - Tamoxifen use
 - Rectal cancer (personal history of HNPCC)
 - Infertility history
 - Unopposed estrogen
 - Menopause late (>age 55)

Clinical manifestations:

- Typically present with long periods of oligomenorrhea or amenorrhea followed by irregular or excessive uterine bleeding.
- Two thirds of women present with postmenopausal bleeding and should raise suspicion of endometrial hyperplasia or carcinoma until proven otherwise.
- **Physical examination:**
 - Occasionally the uterus will be enlarged from endometrial hyperplasia. This is attributed to both the increase in the mass of the endometrium and to the growth of the myometrium in response to continuous estrogen stimulation. *More commonly, the pelvic examination is unremarkable.*

- Patients may also have stigmata associated with chronic anovulation such as abdominal obesity, acanthosis, acne or hirsutism

Diagnosis:

- Hyperplasia is a histologic diagnosis, and so a Pipelle office endometrial biopsy (EMB) or outpatient dilation and curettage (D&C) can be done for sampling.

Management:

- Management depends mainly on a patient's age, comorbid risks for surgery, desire for fertility, and specific histologic features such as cytologic atypia. Hysterectomy is the most definitive treatment. Hormonal therapy is another option and includes oral or injectable progestin or the progestin (levonorgestrel releasing) IUD.

Endometrial cancer

- Is the fourth most common cancer in American women, exceeded only by cancer of the breast, bowel and lung. Patients usually seek medical attention early due to vaginal bleeding and endometrial biopsy leads quickly to diagnosis.
- Endometrial adenocarcinomas are categorized based on histology as *type I* (80%) which occurs in women with a history of chronic estrogen exposure unopposed by progestin (also known as estrogen dependent neoplasm) and *type II* (20%) which is an estrogen-independent neoplasm

Risk factors:

- Obesity
- Polycystic ovarian syndrome
- Long term, high dose unopposed menopausal estrogens
- Early age of menarche
- Late age of natural menopause
- Menstrual irregularities
- White race
- Infertility Nulliparity
- Older age
- Tamoxifen use, high cumulative doses
- DM, chronic hypertension, or gallbladder disease
- Cancer of the breast, ovary or colon
- Family history of endometrial cancer
- Endometrial hyperplasia

- Smoking cigarette and long-term COC use are protective of endometrial cancer.

Clinical manifestations:

- In premenopausal women history of prolonged, heavy menstruation or intermenstrual spotting
- Postmenopausal bleeding is particularly worrisome and carries 5-10% likelihood.
- In more advanced disease, pelvic pain, pelvic mass and weight loss are seen
- **Physical examination:**
 - May reveal obesity, acanthosis, nigricans, hypertension or stigmata of diabetes.
 - Look for signs of metastatic disease, including pleural effusion, ascites, hepatosplenomegaly, general lymphadenopathy, and abdominal masses.
 - Typically have a normal pelvic examination.
 - In more advanced stages of the disease, the cervical os may be patulous, and the cervix may be firm and expanded. The uterus may be of normal size or enlarged. The adnexa should be examined for evidence of extrauterine metastasis and/or coexistent ovarian carcinoma.

Diagnosis:

- **Lab investigation:** the only clinically useful tumor marker in the management of endometrial cancer is a serum CA 125 level. Preoperatively, an elevated level indicates the possibility of more advanced disease.
- **Imaging:** CT and MRI are usually not necessary. However, CT scanning can be obtained preoperatively in cases with higher-grade lesions to assess for lymph node involvement or metastatic disease.

Management:

The primary treatment is hysterectomy with bilateral salpingo-oophorectomy (BSO) and staging lymphadenectomy for most women. Most are diagnosed with stage I disease.

- **Surgical management:** most patients should undergo hysterectomy and BSO. Few contraindications for primary surgery include a desire to preserve fertility, massive obesity, high operative risk and clinically unresectable disease. Most surgically treated patients can simply be followed by pelvic examination every 3-6 months for the first 2 years then every 6-12 months for 3 years.
- **Chemotherapy:** is one of the adjuvant treatment options for advanced cancer following surgery. Only three classes of cytotoxic drugs are identified- paclitaxel (Iaxol), doxorubicin (Adriamycin), and cisplatin (f AP).
- **Radiation:** primary radiation is mainly for exceptionally poor surgical candidates, while adjuvant radiation is offered for women at risk for endometrial cancer recurrence

Prognosis:

- Many clinical and pathologic factors influence the likelihood of endometrial cancer recurrence and survival. Poor prognostic factors include:
 - Advanced surgical stage
 - Older age
 - Histologic type: UPSC or clear cell adenocarcinoma or carcinosarcoma

- Advanced tumor grade
- Presence of myometrial invasion
- Presence of lymphovascular space invasion peritoneal cytology positive for cancer cell increased tumor size

Patient Evaluation

The main goal of evaluation of a patient that presented with AUB is:

- Exclusion of pregnancy and malignancy and
- Identification of the underlying pathology

First confirm that the bleeding is through the vagina and not from the urethra or the rectum.

History

- Thorough menstrual history
 - Age at menarche
 - Date of last menstrual period
 - Duration, frequency, amount and regularity of menses
 - Contraception history (IUD can be associated with AUB)
- History of current bleeding
 - Timing of bleeding (if during sexual intercourse, intermenstrual...)
 - Quantity of bleeding (amount) (excessive bleeding is assessed by number of pads used, passage of clots (size and number) and duration of bleeding)
 - Tissue passage (r/o abortion)
 - Any passage of vesicles (r/o GTD)
 - Association with sexual intercourse (may indicate cervical or vaginal etiology)
- Associated symptoms
 - Fever (r/o infection)
 - Fatigue, dizziness and vertigo (anemia due to increased blood loss),

- Pain (dysmenorrhea often accompanies AUB caused by infection, structural abnormality, pregnancy complication),
- Bulk symptoms (due to abdominal mass when considering pregnancy and tumor)
- Assess risk factors
- History of use of medication
 - NSAIDs
 - Anticoagulants
 - Oral contraception
 - HRT
 - Corticosteroids
 - Antidepressants and antipsychotics

Physical examination

- **G/A:** well, looking, acute sick looking (in cases like EP), or chronically sick looking (in cases like ovarian neoplasms)
- **V/S:** tachycardia & tachypnea (? Severe anemia), febrile (?Infection) and hypotension (?Massive blood loss)
- **HEENT:** pale conjunctiva (signs of anemia)
- **LGS:** Thyroid enlargement
- **Abdomen:** abdominal swelling, tenderness with palpation
- **Pelvic examination:** thorough pelvic examination is important, including taking sample for Pap smear.
- **I/S:** skin pigmentation, ecchymosis, purpura, dry/sweaty and cold/hot skin (thyroid disorders), palmar pallor (sign of anemia), hair distribution (hirsutism)

Investigation

- **Pregnancy tests** (beta-hCG)
- **Complete blood count (CBC)** (assess the presence of anemia, infections and coagulopathy)
- **Thyroid stimulating hormone levels (TSH)**
- **Ultrasonography**
 - **Transvaginal sonography (TVS)**
 - ✓ Is chosen by many as a 1st line tool to assess AUB over endometrial biopsy (EMB), because it:
 - Allows assessment of both the endometrium and myometrium
 - Offers greater patient comfort
 - Allows suitable detection of post-menopausal endometrial hyperplasia
 - **Saline infusion sonography (SIS)** aka Sonohysterography
 - ✓ In which saline is instilled into the uterine cavity during TVS
 - ✓ It permits superior detection of intracavitary lesions
 - ✓ It can be used for sonography-guided Pipelle EMB.
- **Endometrial biopsy**
 - To rule out malignancies
 - Indications for EMB in a woman with AUB
 - ✓ Age >45 years (ACOG, 2012), but the JUSH guideline says age >35yrs.
 - ✓ ACOG suggest EMB in those younger than 45 years with a history of
 - ✓ unopposed estrogen exposure such as seen in obesity or polycystic ovarian syndrome (PCOS)
 - ✓ failed medical management
 - ✓ persistent AUB
- **Hysteroscopy** – involves direct visualization of the endometrial cavity. Usually performed with Dilation and curettage (D&C).
 - Particularly useful when polyp or submucosal fibroid is suspected, because these lesions can be confirmed and removed under direct visualization.
- **Cervical cytology**
- **Coagulation profile**- if indicated

Chapter 18

LEIOMYOMAS

Prepared by Dr. Abubeker Nuredin (MI)

Definition and Epidemiology

Leiomyomas, (also referred to as fibroids or simply myomas) are benign **monoclonal** tumors arising from the smooth muscle cells of the myometrium.

They are the most common pelvic tumor in women and they are more commonly multiple.

Epidemiology

- The incidence of myoma ranges from 20-25%, but is as high as 70-80% in studies using histologic or sonographic examination
- It is responsible for 27% of inpatient gynecologic admission and 1/3rd of all hysterectomies performed are for uterine fibroids.
- Typically occur in women of childbearing age.
 - The prevalence is highest between 35-40 years.
 - Leiomyomas have not been described in prepubertal girls, but they are occasionally noted in adolescents.
 - Most, but not all, women have shrinkage of leiomyomas after menopause.
- African American women are more likely to:
 - be younger at the time of diagnosis
 - have larger fibroids and a greater number of fibroids
 - have heavier bleeding, and more severe anemia

Factors associated with leiomyoma

Factors that increase incidences

- **Race:** the incidence is 2-3x greater in black women than in white women.
 - Differences in genetic factors, diet, lifestyle, psychosocial stress, and environmental exposures contribute this disparity.
- **Age**
 - Incidence of myoma increase with age until menopause.
- **Early menarche:** (<10 years old)
 - is associated with increase of estradiol to post-pubertal levels which can plausibly led to increased fibroid growth
- Reproductive and endocrine factors:
 - Leiomyoma parallels the ontogeny and life cycle changes of the reproductive hormones' estrogen and progesterone.
- Prenatal exposure to diethylstilbestrol
- **Obesity:**
 - production of more estrogens from increased conversion of androgens to estrogen in adipose tissue by aromatase
- Diet:
 - Significant consumption of beef and other reds meats or ham is associated with an increased relative risk of fibroids.
- Alcohol:
 - Consumption of alcohol, especially beer.
- Genetics:
 - Studies imply a familial predisposition to leiomyoma in some women.

Factors that decrease incidences

- **Parity**
- **Hormonal contraception:**
 - Long-acting progestin-only contraceptives (eg, depot medroxyprogesterone) appear to protect against development of leiomyomas
- **Smoking:**
 - Decreases the risk of fibroid developments possibly through the inhibition of aromatase
- **Diet:**
 - Consumption of green vegetables and fruit (especially citrus fruit), dietary vitamin A from animal sources may also be associated with decreased fibroid risk.

Pathophysiology

- The cause of uterine leiomyomas is unclear.
- Fibroids are benign monoclonal tumors, with ***each tumor resulting from proliferation of a single smooth muscle cell.***
- Genetic predisposition, steroid hormone factors, growth factors, and angiogenesis may all play a role in the formation and growth of uterine fibroids.
- Fibroids are hormonally responsive to both estrogen and progesterone but the relationship is complex. Evidence for the relationship include:
 - Myomas contain estrogen & progesterone receptors in higher concentration than surrounding myometrium
 - Myomas may increase in size with estrogen therapy & in pregnancy
 - They are not detectable before puberty
 - During menopause, the tumors usually stop growing and may atrophy in response to naturally lower endogenous estrogen levels.
 - Progesterone increases mitotic activity & reduce apoptosis
- There may be genetic predisposition
 - mutation in chromosome 6,7,12, and 14 are associated with the rate and direction of tumors growth

Classification

Uterine fibroids can be classified based on:

- **Simply their location in the uterus**
 - Submucous leiomyoma
 - Intramural or interstitial
 - Sub serous or sub peritoneal
 - Parasitic
 - Intra-ligamentary
 - Cervical
- The International Federation of Gynecology and Obstetrics (**FIGO**), as follows
 - Submucosal myomas (FIGO type 0, 1, 2)
 - protrude into the uterine cavity
 - Intramural myomas (FIGO type 3, 4, 5)
 - are located within the uterine wall
 - they may enlarge sufficiently to distort the uterine cavity or serosal surface
 - Subserosal myomas (FIGO type 6, 7)
 - They may have a broad or pedunculated base and may be intraligamentary (i.e., extending between the folds of the broad ligament).
 - Cervical myomas (FIGO type 8) – are located in the cervix rather than the uterine corpus.

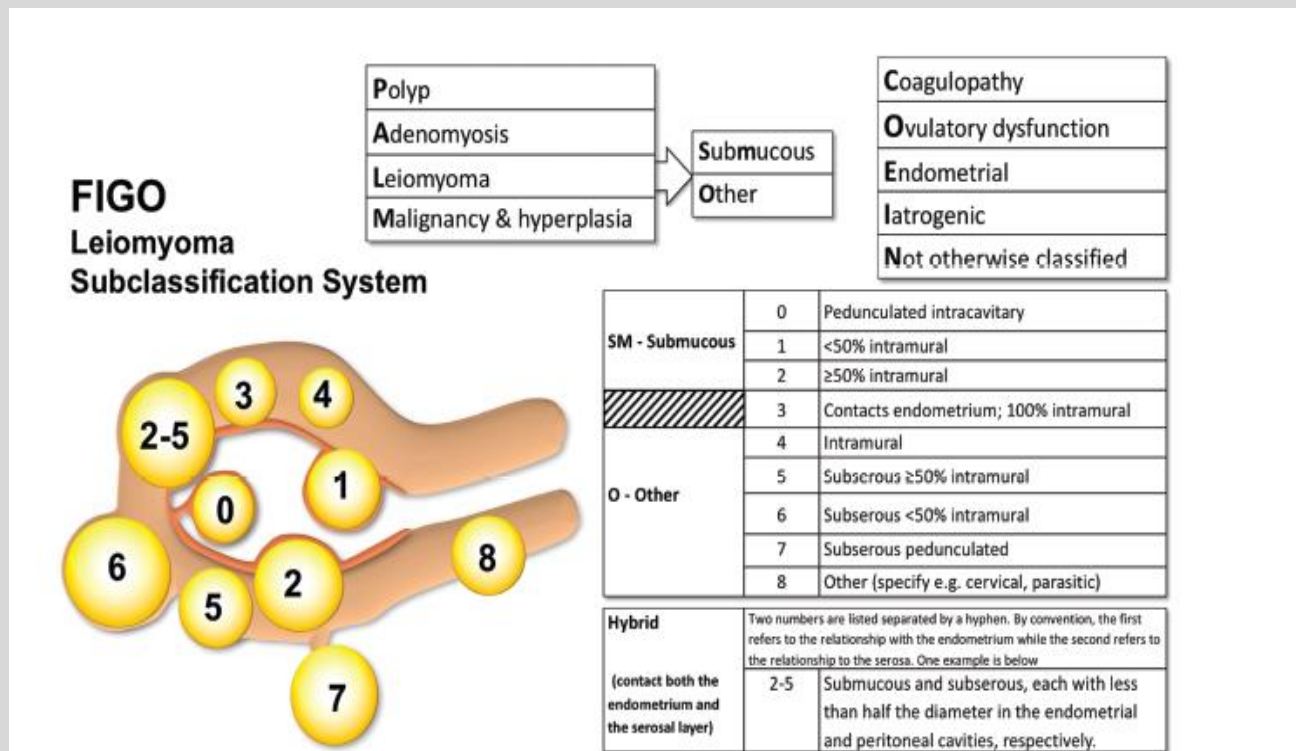


Figure 1: FIGO Classification of myoma

Secondary changes in myoma

Benign Degeneration

- **Atrophic:**
 - Signs and symptoms regress or disappear as tumor size decreases at menopause or after pregnancy.
- **Hyaline:**
 - Yellow, soft, and often gelatinous areas. These tumors are usually asymptomatic.
- **Cystic:**
 - Liquefaction follows extreme hyalinization, and physical stress may cause sudden evacuation of fluid contents into the uterus, the peritoneal cavity, or the retroperitoneal space.

- **Calcific** (Calcareous):
 - Sub serous leiomyoma are most commonly affected by circulatory deprivation, which causes precipitation of calcium carbonate and phosphate within the tumor.
- **Septic:**
 - Circulatory inadequacy may cause necrosis of the central portion of the tumor followed by infection and result **acute pain, tenderness, and Fever.**
- **Carneous** (Red)
 - Venous thrombosis and congestion with interstitial hemorrhage are responsible for the color.
 - Discrepancy between growth of myoma & blood supply results in aseptic degeneration & infarction
 - The process is usually accompanied by **pain** but is self-limited
 - *Most common during pregnancy,*
- **Myxomatous** (Fatty):
 - Uncommon and asymptomatic, it follows hyaline and cystic degeneration.
- **Metastasizing Leiomyomata**
 - Rarely, myomas spread beyond the uterus to distant locations such as the Peritoneum (Leiomyomatosis peritonealis disseminate/LPD) distant vasculature, and lung.
 - They are often asymptomatic.
 - Most women with these tumors have undergone a prior dilatation and curettage (D&C), myomectomy, or hysterectomy, suggesting the possibility of surgically induced vascular spread of leiomyoma cells.

Carcinomatous changes

- Very rare; mostly leiomyosarcoma de novo.

Clinical features

- The majority of myomas are small and asymptomatic
- The symptoms are related to the number, size, and location of the tumors.
- Symptoms are classified into three categories:
 - Abnormal uterine bleeding (AUB)
 - Bulk-related symptoms, such as pelvic pressure and pain
 - Reproductive dysfunction (i.e., infertility or obstetric complications)

AUB

- In 30% of women
- Heavy or prolonged menstrual bleeding —the most common fibroid symptom.
- Intermenstrual bleeding and postmenopausal bleeding should prompt investigation to exclude endometrial pathology.
- The presence and degree of uterine bleeding is determined, largely by the location of the fibroid (size is of secondary importance.)
 - Submucosal myomas that protrude into the uterine cavity (e.g., types 0 & 1) are most frequently related to significant HMB than women with myomas in other locations (34 versus 25 percent).
- *The mechanism of bleeding is: -*
 - *Interruption of the blood supply to the endometrium*
 - *Distortion and congestion of the surrounding vessels, particularly the veins*
 - *Ulceration of the overlying endometrium*
 - **Increased surface area**
 - *Alteration of normal myometrial contractile function in the small artery and arteriolar blood supply underlying the endometrium.*
 - *Inability of the overlying endometrium to respond to the normal estrogen/progesterone menstrual phases, which contribute to efficient sloughing of the endometrium.*
 - *Pressure necrosis of the overlying endometrial bed, which exposes vascular surfaces that bleed in excess of that normally found with endometrial sloughing.*

Bulk-related symptoms

- Irregularly shaped and enlarged myomatous uterus exerts pressure on the surrounding structures, resulting in: -
- Pelvic pressure or pain: it is likely to be chronic, intermittent, dull pressure or pain
- Pain result from degeneration associated with
 - vascular occlusion
 - infection
 - torsion of a large pedunculated tumor
 - myometrial contractions to expel a subserous myoma from the uterine cavity
- Urinary tract or bowel symptom
 - Urinary symptoms –frequency, difficulty emptying the bladder, or, rarely, complete urinary obstruction. It may also result in hydronephrosis
 - Bowel symptoms –constipation.
- Venous compression — Very large uteri may compress the vena cava and lead to an increase in thromboembolic risk
- Painful menses —in many women appears to be correlated with heavy menstrual flow and/or passage of clots.
- Painful intercourse — anterior or fundal fibroids are the most likely to be associated with deep pain with intercourse.
- Fibroid degeneration or torsion — fibroids cause acute pain from breaking down of the fibroid tissue.

Reproductive dysfunction

- Leiomyomas account for 1 to 3 % of infertility
- Fibroids, particularly those that impinge upon the endometrium, may affect fertility by: -
 - interfering with implantation over the myoma site
 - rapidly distending the uterus in early pregnancy,
 - impairing uterine contractility

- The location of a fibroid, and not its size, is the key factor regarding fertility
 - Leiomyomata with a submucosal or intracavitary component were associated with lower pregnancy rates
 - the likely mechanism is inhibition to normal implantation
 - In contrast, subserosal fibroids do not impact fertility.
- The role of intramural fibroids in infertility is controversial
- Fibroid near a fallopian tube ostium or near the cervix may impede fertilization.
- Spontaneous Abortion; both uterine leiomyoma and spontaneous miscarriage are common, and an association between these has not been shown convincingly.

Prolapsed fibroid

- Infrequently, a sub mucosal leiomyoma will prolapse through the cervix and present with a mass, bleeding, and possible ulceration or infection.

Endocrine effects

- Rare symptoms of fibroid tumors where fibroids can secrete ectopic hormones include:
- Polycythemia: from autonomous production of **erythropoietin**
- Hypercalcemia: from autonomous production of parathyroid hormone-related protein
- Hyperprolactinemia

The Mnemonic: “FIBROIDS” can be used to recall clinical symptoms of uterine leiomyomas

- **F:** Frequency and retention of urine, hydronephrosis
- **I:** Iron deficiency anemia
- **B:** Bleeding abnormalities (menorrhagia, metrorrhagia, menometrorrhagia, postcoital spotting), bloating
- **R:** Reproductive difficulties (dysfunctional labor, premature labor/delivery, fetal malpresentation, increased need for cesarean delivery)
- **O:** Obstipation and rectal pressure
- **I:** Infertility (failed implantation, spontaneous abortion)
- **D:** Dysmenorrhea, dyspareunia
- **S:** Symptomless (most common)

Differential diagnosis

- As mentioned above Patient with myoma can present usually with AUB (more commonly) or abdominal mass.

For AUB

- Endometrial hyperplasia
- Endometrial or tubal Ca
- Uterine sarcoma
- Endometrial polyps
- Adenomyosis
- DUB
- Endometrial Ca....

For mass

- Pregnancy
- Ovarian Ca
- Tubo-ovarian abscess
- Myometrial hypertrophy
- Adenomyosis
- Endometriosis...

Complications

- Anemia
- Urinary or bowel obstruction from large or parasitic myomas
- Malignant transformation is rare
- Red degeneration
- Pregnancy associated complications (see below)
- Surgical complications especially ureteric injury or ligation in cervical myomas.

Myomas and Pregnancy

- Incidence: Around 0.1 – 3.9%
- Influence of pregnancy on myoma
 - 1/3 of the myoma increases in size,
 - When an increase has been reported, most of the growth takes place in the early part of pregnancy, after which they remain static or decrease.
 - 1/3 remain same and
 - 1/3 decreases in size
 - Red degeneration

- Influence of myoma on pregnancy:
 - Early pregnancy bleeding and abortion
 - Preterm labor
 - Placental abruption: placentation over a fibroid appears to be a consistently strong risk factor for abruption
 - IUGR
 - Fetal compression syndromes: Dolicocephally, head molding, asymmetric ventricles.
 - Incarceration of the uterus:
 - When trapped within the pelvis with further uterine enlargement occurring by sacculation.
 - Present with abdominal pain and urinary retention.
 - At delivery:
 - Malpresentation, obstructed labor,
 - postpartum hemorrhage
 - post-myomectomy uterine rupture

Management

Choice of treatment depends on:

- Symptoms
- Age of patient
- Parity
- Presence of pregnancy
- Plan for future fertility
- General health of the patient
- Location of the myoma

General treatment

- Treatment of anemia
- Analgesia
- Observation
 - Asymptomatic leiomyomas usually can be observed and surveyed during annual pelvic examination
 - Exceptions:
 - ✓ Fibroids growing rapidly
 - ✓ If they are pedunculated and prone to torsion
 - ✓ If they are likely to complicate future pregnancy
 - ✓ If there is suspicion of malignancy
- Medical management
- Surgical management

Medical management

Drug therapy

Advantage:

- Reduces / relieves symptoms
- Correct preoperative anemia
- Option for a woman at late menopausal transition

Disadvantage:

- Recurrence of fibroids after stopping treatment

Indications for drug therapy of uterine leiomyoma and preferred agents

Agent	NSAIDs	COCs	DMPA	LNG-IUS	GnRH Agonist
Symptom					
Dysmenorrhea	+	+	+	+	+
Menorrhagia	-	+	+	+	+
Dyspareunia	-	-	-	-	+
Pelvic pressure	-	-	-	-	+
Infertility	-	-	-	-	+

Gonadotropin Releasing Hormone (GnRH) agonist

- Is used for 3 – 6 months
- Control bleeding and improve preoperative HCT
- Is temporarily used till surgery is planned
- Is used if menopause is anticipated
- Shrinks myomas (40 – 60%) to allow vaginal hysterectomy and technically easier surgery with diminished blood loss.

Mechanism of action (MOA)

There are two suggested MOA of GnRH agonists

1. Effect on HPO axis

- GnRH agonists stimulate receptors on pituitary gonadotropes => supraphysiological release of LH and FSH (called **Flare** and typically last 1 week)
- However, with their **long-term action** GnRH agonists down regulate receptors in gonadotropes- creates desensitization to further GnRH stimulation => decreased gonadotropin secretion
 - This leads to suppression of estrogen and progesterone levels 1 to 2 weeks after initial GnRH agonist administration

2. Direct effect on leiomyoma

- leiomyomas themselves may contain GnRH receptors, and agonists may directly decrease leiomyoma size
- GnRH agonists suppress leiomyoma cell proliferation and induce cell apoptosis by the fourth week of GnRH agonist therapy

Side effects of GnRH agonists

- Occur in up to 95 percent of women
- They result from the profound drop in serum estrogen levels
- It includes vasomotor symptoms, libido changes, vaginal epithelium dryness, accompanying dyspareunia, and loss in trabecular bone (6 months use will result in 6% trabecular bone loss),
- <10% of patients terminate treatment secondary to side effects

▪ **To obviate the side effects: -**

- These agents alone are not recommended for longer than 6-months' use
- Or add-back therapy is added
 - **Add back therapy**
 - ✓ Typically begun 1 to 3 months following GnRH agonist initiation
 - ✓ Includes: estrogen combined with a progestin or selective estrogen-receptor modulator (SERMs)
 - ✓ Advantages of SERMs include the ability to begin them concurrently with GnRH agonist treatment without negating the agonist effects of leiomyoma shrinkage.

LNG-IUS:

Advantage:

- Reduces menorrhagia
- Avoids surgery
- Prevents pregnancy

Disadvantage:

- No improvement of bulk symptoms
- May lose string as uterus enlarges

Androgens:

▪ **Danazol: -**

- Reduces endogenous production of ovarian estrogen

Surgical Management

- It is the mainstay of treatment in myomas
- It should be timed closely to planned pregnancy, if possible, to limit the risk of tumor recurrence
- Indications for emergency surgery are presence of complications:
 - Infected myoma
 - Acute torsion
 - Intestinal obstruction

- It includes: -
 - Hysterectomy
 - Myomectomy
 - Myolysis

Hysterectomy

- Hysterectomy is the definitive and most common surgery.
 - It can be performed vaginally, abdominally, or laparoscopically depending on patient and uterine factors.
- Symptomatic cervical or broad ligament leiomyoma are best treated with hysterectomy.
- Following hysterectomy, postoperative satisfaction rates are high, and data show superior rates compared with conservative options.
- However, benefits are balanced against the risks of major surgery
- In symptomatic women who have completed child bearing
- Conserve ovaries unless there is indication to remove them

Advantage: -

- Eliminates symptoms & recurrence
- The only definitive treatment
- Malignancy can be ruled out

Disadvantage: -

- Preclude future fertility
- Surgical complications

Myomectomy

- This uterus-preserving surgery excises myomas and is considered for women who desire fertility preservation or who decline hysterectomy.
- Contraindications are
 - active infection,
 - pregnancy, and
 - Suspected malignancy.

- Compared with hysterectomy, myomectomy poses a ***higher risk of myoma recurrence and incomplete symptom relief*** but has lower rates of injury to adjacent organs.
- Blood loss, surgery length, febrile morbidity, and hospital stays are comparable for surgery performed on uteri up to 16-week size.
- Myomectomy can be performed hysteroscopically, laparoscopically, robotic or via laparotomy.
- In general, predominantly intracavitary myomas are resected hysteroscopically, whereas subserosal or intramural myomas require laparotomy or laparoscopy for surgical excision.
 - For intracavitary myomas, hysteroscopic resection affords quicker recovery and less surgical morbidity than hysterectomy or laparoscopic myomectomy.
- Resection is most effective with type 0 and type 1 tumors and with those measuring ~3 cm.
- Although suitable for skilled surgeons, larger or type 2 myomas carry higher risks for incomplete resection, difficult visualization, uterine perforation, or reaching media volume limits.
- For hysteroscopic myomectomy in general, 15 to 20 percent of treated women eventually require additional surgery

▪ Advantage:	Disadvantage:
○ Fertility is preserved (since uterus will not be removed) or enhanced	▪ Hemorrhage during surgery requiring blood transfusion ▪ Persistence of menstrual symptoms ▪ Recurrence of fibroids: ○ 50–60% recurrence risk with 25 % requiring surgery ▪ Adhesion formation ○ Subsequent subfertility ○ Difficulty of subsequent myomectomy

When to conceive after myomectomy?

- After 3 – 6 months
- If pregnancy occurs following myomectomy, recurrence rates are diminished as pregnancy has protective effect on myoma growth

Myomectomy during pregnancy

- Generally, avoid myomectomy in pregnancy
- Indications for myomectomy during pregnancy
 - Intractable pain
 - to gain access to fetus
 - facilitate uterine repair at cesarean delivery

Myolysis

- Myolysis describes procedures using tools that directly contact the myoma to incite myoma necrosis and subsequent shrinkage.
- With the various procedures, radiofrequency energy, laser vaporization, or cryotherapy ablates the myoma.

Other managements**Endometrial Ablation**

- Several endometrium-destructive modalities are often elected for abnormal uterine bleeding due to endometrial dysfunction.
- These can be also used as a sole technique for myoma-related bleeding.
- most of these modalities have limitations regarding cavity length and degree of cavity distortion.
- some data show efficacy if treating submucous myomas measuring ~3 cm and some have used ablation as an adjunct following hysteroscopic leiomyoma resection in women with HMB

Uterine Artery Embolization (UAE)

- Uterine artery embolization is an angiographic procedure that delivers synthetic particulate emboli into both uterine arteries.
- Uterine blood flow is thereby obstructed, producing ischemia and necrosis.
- Because vessels serving leiomyoma have a larger caliber, these microspheres are preferentially directed to the tumors, sparing the surrounding myometrium.
- During UAE, an angiographic catheter is placed in one femoral artery and advanced under fluoroscopic guidance to sequentially catheterize both uterine arteries.
- Failure to embolize both uterine arteries allows collateral circulation and sustain leiomyoma blood flow and is associated with a significantly lower success rate.
- UAE is a management option for women who might otherwise be considered candidates for hysterectomy or myomectomy.
- Based on current evidence, women who have not completed childbearing may be better served by myomectomy.

TABLE 9-2. UAE Absolute and Relative Contraindications

Absolute

Pregnancy
Active uterine or adnexal infection
Suspected reproductive tract malignancy^a

Relative

Coagulopathy
Renal impairment
Severe contrast allergy
Desire for future fertility
Uterine size >20–24 weeks
Prior salpingectomy or SO
Prior pelvic radiation
Large hydrosalpinx
GnRH agonist use

Reason

Bleeding complications
Renal effects of contrast
Allergic reaction
See text
Difficult to embolize
Altered arterial anatomy
Altered arterial anatomy
Increased infection risk
Narrows vascularity

Magnetic Resonance Imaging-Guided Focused Ultrasound (MRgFUS)

- With MR-guided focused ultrasound (MRgFUS), ultrasound energy is focused to heat and incite coagulative necrosis in selected myomas.
- Concurrent MR imaging enables this energy to be precisely targeted and also provides real-time tissue temperature feedback to limit surrounding tissue injury.
- Early prospective studies show that MRgFUS improves quality of-life scores and is well tolerated.
- Potential minor complications include vaginal discharge, fever, hematuria, and abdominal wall edema.
- More serious conditions are venous thromboembolism, necrotic myoma tissue retained in the uterine cavity, heavy uterine bleeding, endometritis, and burns.
- Comparing UAE and MRgFUS, showed a 30-percent intervention rate within 3 years after MRgFUS and the rate was only 13 percent for UAE.

Sample History

Chief complaint: Abdominal swelling of 6 months duration.

HPI:

This is a 32years, Para 4(all alive no abortion, VD) mother, whose LMP was two weeks ago. She was relatively healthy 6 months back at w/c time she started to experience gradual onset of lower abdominal swelling w/c was initially small but later increased gradually to attain current size w/c is around mid-abdomen. Associated with the swelling she also developed mild lower abdominal pain of the same duration w/c is dull, intermittent, non-radiating with no known aggravating or relieving factors identified.

Her menses started at age of 14 years W/C comes every 28day, last for 3 to 5 days and she use 1to 2 pads per day. It was dark red, non-clotting type and has no associated pain. But since 2 months her menses become heavy and she started to use 4 to 5 pads per day and lasts for 9 to 11 days. It becomes bright red and clotting type.

Otherwise:

She has no history of previous self or family history of similar illness in the families

She has no history of use of any contraceptive or other medications

She has no history of cigarette smoking alcohol intake

She has no history of gynecologic surgeries and procedures

She has no history of constipation, diarrhea or blood in stool

She has no history of urgency, frequency or pain during urination

She has no history of pain during sexual intercourse or any bulging of swelling through vagina

She has no history of easy fatigability, weight loss or loss of appetite

She has no history of cough or shortness of breath

Prepared by Graduating Class of Jimma University 2015 E.C (2022/23)

She has no history of blurring of vision, vertigo or light-headedness

She has no history of fever, malaise or rigor

She has no history of itching, burning sensation around genitalia or foul-smelling vaginal discharge

She has no history of nausea, vomiting and absence of menstruation

She has no history of Orthopnea, PND or lower extremity swelling She has no history of RUQ pain or yellowish discoloration of eye She has no history of flank pain or decreased urine output

She has no history of personal or family history of DM, HTN, asthma, TB or renal disease

She is seronegative for RVI

Her usual diet is injera made of teff with shiro-wot made of Atter & she gets 3 times per day sometimes she gets meat, fruits & vegetables.

She was 18 years old during her marriage and first coitus; she is sexually active

Physical examination

G/A: well-looking

V/S: (all normal), look for any distressing signs (in case of complications)

HEENT: Pink connective and non-icteric sclera

Abdomen

- Inspection:
 - Abdomen is asymmetrically distended, more on the left side of lower quadrant.

- Flanks are not full.
- No striae gravidarum, linea nigra, scar
- Inverted Umbilicus
- no distended veins, no visible peristaltic movement, or pulsations
- hernia sites are free
- Superficial Palpation:
 - there is no superficial tenderness or mass
- Deep palpation:
 - There is 24 weeks sized mobile mass, that has irregular surface, non-tender, firm in consistency whose upper border is defined but lower border can't be reached (hand cannot pass below mass into pelvic inlet).
- Percussion: No shifting dullness or fluid thrill
- Auscultation: Bowel sounds well heard no bruits over the mass.

Pelvic Examination

- Bimanual pelvic examination:
 - Irregular enlarged with well-defined border firm in consistency mobile and tender mass on upper part of uterus.

Investigations

- CBC
 - r/o anemia/polycythemia, infection, thrombocytopenia (important for surgical patient)
- Blood group & Rh (for all surgical patients)
- Pregnancy test
- Ultrasound:
 - it is used to determine size, location, consistency, number of mass and hydro ureter & hydro nephrosis
- RFT (BUN & Serum creatinine)
- TFT (TSH, for surgical patient)

- Hysteroscopy
- KUB or IVU (if urinary symptoms are there)
- Endometrial biopsy or D&C
 - is essential in the evaluation of abnormal bleeding to exclude endometrial Ca

Chapter 19

OVARIAN TUMORS

Prepared by Dr. Abraham Gebeyehu (MI)

Epidemiology and Risk factors

- Worldwide each year, more than **295,000 women are diagnosed, and 185,000 women die from this disease**. Of these, epithelial ovarian carcinomas make up 90% of all cases, including the more indolent low malignant-potential (borderline).
- In the United States, 1 in 78 women (1.3%) will develop ovarian cancer during her lifetime
- Although it is not the commonest gynecologic malignancy, due to its high rate of mortality, **ovarian cancer remains the fifth leading cause of cancer-related death**.
- The average age at diagnosis is in **the early 60s**.
- Due to the absence of effectively screening tools for ovarian cancer, and early symptoms are few, **two thirds of patients have advanced disease when diagnosed**.
 - Debulking surgery sequenced with platinum-based chemotherapy usually results in clinical remission. However, up to 90 percent of these women will develop a relapse that eventually leads to disease progression and death.

Only about one quarter of patients will have stage I disease at presentation and have an excellent long-term survival rate.

Risk factors

- **Inherited genetic predisposition**
 - In up to **25% of patients**
 - For the other 75 percent with no identifiable genetic link for their ovarian cancer, risks have traditionally been attributed to a ***pattern of uninterrupted ovulatory cycles*** during the reproductive years
 - Repeated stimulation of the ovarian surface epithelium is hypothesized to lead to malignant transformation.
- **Family history of breast or ovarian cancer**
 - ***Is the most important identifiable risk factor***
 - Ovarian cancer in first-degree relative (mother, daughter, or sister) ***triples*** a woman's life time risk, the risk further increases if the number of affected family members increase
 - The risk increases if the diagnosed premenopausal
- Self or family **history of Lynch syndrome** (hereditary nonpolyposis colorectal cancer, (HNPCC)),

See detail on Hereditary Ovarian CA below
- **Nulliparity**
 - is associated with long periods of repetitive ovulation, and patients without children have double the risk of developing ovarian cancer
- **Infertility** (One theory suggests that pregnancy may induce premalignant ovarian cell shedding)
- **Early menarche and Late menopause**
 - Is associated with long duration of active ovulatory period
- **Increasing age** (allows accumulation of random genetic alterations within the ovarian surface epithelium)
 - NB, increasing age is risk only up to mid-70s, and the risk decreases beyond 80yrs.

- **White race**
 - Compared with that of black and Hispanic women, the risk is elevated by 30 to 40%.
 - Racial discrepancies in parity and rates of gynecologic surgery are believed to account for some of the differences.
 - NB, ovarian tumors are one of the few gynecologic abnormalities that are common in the white population (POP too)
- Residence in North America and Northern Europe (developing countries and Japan have the lowest rates, regional dietary habits may be partly responsible)
 - developing countries have ***approximately half the rate***
- Postmenopausal hormone therapy
- Pelvic inflammatory disease
- Use of coffee, tobacco, alcohol & dietary fat
- Use of perineal talc

Protective factors

- **Parity** (having at least one child is protective of the disease, with a risk reduction of 0.3 to 0.4%)
 - For all women, risks decrease with each live birth, eventually plateauing in women delivering five times.
- **Breastfeeding** (perhaps by prolonging amenorrhea)
- **Long-term (5 year) COC use**
 - Reduces the risk ***by 50%***.
 - The duration of protection lasts up to 30 years after the last use
- **Prophylactic salpingo-oophorectomy** (reduces, but *does not eliminate the risk* because the entire peritoneum is still going to be at risk)
- **Tubal ligation and hysterectomy**, (theoretically, any gynecologic procedure that precludes irritants from reaching the ovaries via ascension from the lower genital tract might plausibly exert a similar protective effect)
- Consumption of foods **low in fat but high in fiber**, carotene, and vitamins appears protective

Hereditary Ovarian Cancer

Approximately 75 percent of inherited ovarian cancers result from germline mutations in the *BRCA1*, *BRCA2*, or other homologous recombination deficiency genes.

- *Mutations of BRCA1 and BRCA2 genes lead to BRCA1 and BRCA2 protein dysfunction, which results in genetic instability and subject cells to a higher risk of malignant transformation.*
- Thus, any patient with a personal history of epithelial ovarian cancer or breast cancer in certain circumstances or those from a family carrying a known deleterious mutation should undergo testing.

If family history is mainly composed of colon cancer, clinicians may consider Lynch syndrome, also known as *hereditary nonpolyposis colorectal cancer (HNPCC)*.

- Patients with this syndrome have a high lifetime risk of colon cancer (85 percent) and ovarian cancer (10 to 12 percent).
- Endometrial cancer is the predominant gynecologic malignancy, and women with the syndrome have a 40 to 60 percent lifetime risk for this cancer.

Prevention of Ovarian Cancer

1. Screening

- Despite enormous effort, no evidence suggests that ***routine screening*** with serum markers, sonography, or pelvic examinations lowers mortality rates.
- Currently screening is recommended only to those *high-risk women*
 - *BRCA.1* or *BRCA2* carriers and
 - Women with a strong family history of breast and ovarian cancer
- In women at high-risk for short-term surveillance starting at age 30 to 35 years and continuing until risk-reducing surgery should be done.

- Screening modalities
 - Genetic testing
 - Tumor marker
 - ✓ CA 125: Alone, CA₁₂₅ is a poor marker for ovarian cancer detection. However, a more sensitive Risk of Ovarian Cancer Algorithm (ROCA) is based on the slope of serial CA₁₂₅ measurements drawn at regular intervals.
 - If a ROCA score exceeds 1% risk of having ovarian cancer, patients then undergo transvaginal sonography to determine whether additional intervention is warranted.
 - Transvaginal sonography

2. Chemoprevention

- COC
 - By how much percent does COC decrease risk?
 - NB, the slightly higher risk of **breast cancer** during current COC use should be presented during patient counseling.

3. Prophylactic surgery

- The only proven way to directly prevent ovarian cancer is surgical removal.
- *Currently, in these high-risk women, prophylactic bilateral salpingo-oophorectomy (BSO) still is recommended.*
- Prophylactic BSO should be offered to *BRCA 1* carriers at age 35 to 40 years. For *BRCA.2* carriers, BSO is recommended between 40 and 45, and at age 45 to 50 years for other HR deficiency (*BRIP1*, *RAD51C*) carriers.
 - In these patients, the procedure is approximately **90-percent effective in preventing epithelial ovarian cancer.**
 - Prophylactic BSO also reduces the risk of developing **breast cancer by 50 percent.**
 - Predictably, the protective effect is strongest among premenopausal women.

- In women with HNPCC, the ovarian cancer risk reduction approaches **100%**.
- The term *prophylactic* implies that the tubes and ovaries are normal at the time of removal. However, approximately 5 percent of *BRCA* mutation carriers undergoing prophylactic BSO will have an otherwise undetected, often microscopic, fallopian tube cancer at the time of surgery.
 - To account for this possibility,
 - ✓ Cytologic washings should be collected during surgery.
 - ✓ Peritoneal surfaces are thoroughly inspected, and abnormal areas are biopsied.
 - ✓ Last, the pathology requisition should clearly state that the BSO was performed for a prophylactic indication.

Typically, the BSO, washings collection, and peritoneal biopsy can be completed laparoscopically.

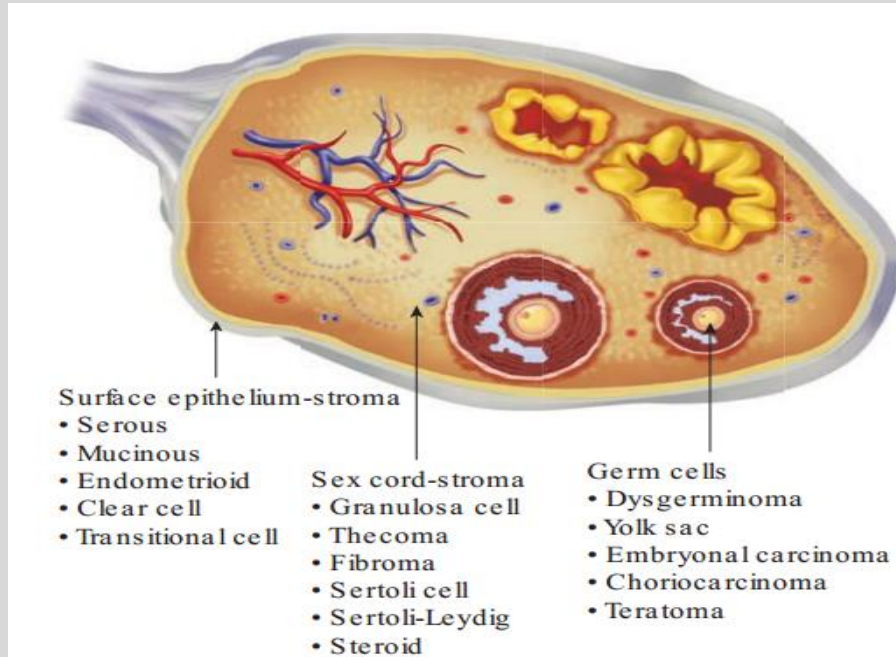
- Prophylactic BSO in young women will induce **premature menopause** and its associated
 - vasomotor and urogenital symptoms,
 - decline in sexual interest, and
 - osteoporosis

Estrogen therapy (ET) commonly is used to alleviate these symptoms in women without prior breast cancer

- In women with HNPCC syndrome, **hysterectomy is mandatory** when performing prophylactic BSO because of coexisting endometrial cancer risks.
- In low-risk patients' risk-reducing salpingectomy also is now considered in those undergoing hysterectomy or permanent sterilization.
 - Ovarian cancer risk is reduced up to 65%

Classification of Ovarian Tumors

- **Based on clinical behaviors:**
 - Benign
 - Borderline (low malignant potential)
 - Malignant
- **Based on morphologic characteristics:**
 - Cystic
 - Solid
 - Mixed
- **Based on histological features (cell origin)**
 - Epithelial ovarian tumors
 - Germ cell tumors of the ovary
 - Sex-cord stromal tumors



Benign Ovarian Masses

▪ Functional Cysts

- Follicular Cyst
- Corpus Luteum (Granulosa Lutein) Cysts
- Theca Lutein Cysts
- Hyperthecosis
- Polycystic Ovarian Syndrome (Stein-Leventhal Syndrome)
- Luteoma of Pregnancy

Endometriomas, (is not a functional cyst, but it is one of the most common ovarian cysts)

▪ Epithelial Tumors

- Serous Tumors
- Mucinous Tumors
- Endometrioid Lesions
- Clear Cell (Mesonephroid) Tumor
- Transitional Cell (Brenner) Tumors

▪ Sex Cord-Stromal Tumors

- Thecoma
- Fibroma
- Hilus Cell Tumors

▪ Germ Cell Tumors

- MATURE Teratomas (dermoid cysts)

Clinical feature of benign tumors

- Commonly seen in late child bearing age groups and most are asymptomatic
- Dermoid cyst & mucinous cystadenomas are common in reproductive age group
- Dermoid cyst & serious cystadenomas are common during pregnancy
- If symptomatic-
 - Swelling (distention),
 - Heaviness
 - Dull aching lower abdominal pain, are seen

- Signs are usually unaffected, but if there is any, it includes: -
 - Cachexia (in mucinous cystadenoma)
 - Pitting edema (from compression of draining lymphatic vessels and veins from the lower limb)
 - Mass (cystic/ solid, mobile (from side to side but not from above to downwards), usually smooth surface, non-tender)
 - Positive fluid thrill

Presence of an abdominopelvic mass (Ovarian fibroma), ascites (fluid thrill), right pleural effusion (reduce tactile fremitus & chest expansion dullness in percussion) is indicative of Meig's syndrome.

Differential diagnosis for benign ovarian masses

- Fibroid
- Functional cyst
- Full bladder
- Encysted peritonitis
- Pregnancy with or without fibroid
- Ascites
- Mesenteric cyst
- Para-ovarian cyst
- Textiloma (can be due to a forgotten gauze)

Complications of Benign ovarian tumors

- Torsion of the pedicle
 - NB. Benign ovarian cysts have higher risk for torsion than malignant ones.
- Intracystic hemorrhage
- Infection
- Rupture
- Pseudomyxoma peritonei
- Malignancy

Borderline Ovarian Masses

Ten to fifteen percent of epithelial ovarian cancers have histologic and biologic features that are **intermediate** between clearly benign cysts and frankly invasive carcinomas.

- They are termed as low malignant-potential (LMP) tumors or borderline tumors.

Although LMP tumors may develop at any age, on average patients are in their **mid-40s**, which is 15 years younger than women with invasive ovarian carcinoma.

Histologically, LMP tumors are distinguished from benign cysts by having at least two of the following: -

- nuclear atypia
- epithelial stratification
- microscopic papillary projections
- cellular pleomorphism or
- mitotic activity

Unlike invasive carcinomas, LMP tumors lack stromal invasion, LMP tumors lack stromal invasion. However, up to *10 percent of LMP tumors will exhibit areas of microinvasion*, defined as foci, measuring < 3 mm in diameter and forming < 5 percent of the tumor.

Clinical manifestation

- Ovarian LMP tumors present similar to other adnexal masses (see manifestation of benign masses above)

Malignant Ovarian Masses

Include: -

- Epithelial ovarian cancer
- Germ cell tumors
- Sex-cord stromal tumors

Epithelial Ovarian Cancer

- *Approximately 90% of ovarian cancers are derived from tissues that come from the coelomic epithelium or mesothelium.*

Pathogenesis

Most epithelial ovarian cancers are now believed to actually originate in the fallopian tube fimbria.

There are three distinct tumorigenic pathways: -

- Most carcinomas appear to originate *de novo* from ovarian surface epithelial cells that are sequestered in cortical inclusion cysts (CICs) within the ovarian stroma.
 - The replicative stress and DNA damage transforms the entrapped surface epithelial cells within CICs into any of the histologic ovarian cancer variants
- At least 10 percent of epithelial ovarian carcinomas, invariably high-grade serous tumors, result from an inherited predisposition.
 - Women born with a BRCA gene mutation require only one “hit” to the other normal copy (allele) to “knock out” the BRCA tumor-suppressor gene product. BRCA-related serous cancers appear to have a unique molecular pathogenesis, requiring p53 inactivation to progress.

- Accumulation of genetic alterations that leads to malignant transformation.
 - As such, mutation the ras family of oncogenes (K-ras, H-ras, and N-ras) are implicated in carcinogenesis by their inhibition of cellular apoptosis and promotion of cellular proliferation

Histopathology and pattern of spread

Histologic Type	Cellular Type
I. Serous	Endosalpingeal
II. Mucinous	Endocervical
III. Endometrioid	Endometrial
IV. Clear-cell “mesonephroid”	Mullerian
V. Brenner	Transitional
VI. Mixed epithelial	Mixed
VII. Undifferentiated	Anaplastic
VIII. Unclassified	Mesothelioma

Grading

Ovarian tumors are classified as:

- grade 1(well-differentiated)
- grade 2 (moderately differentiated)
- grade 3 (poorly differentiated) lesions

Patterns of Spread

- **Transcoelomic:** The most common and earliest mode of dissemination of ovarian epithelial cancer is by exfoliation of cells that implant along the surfaces of the peritoneal cavity. The cells tend to follow the circulatory path of the peritoneal fluid, to dependent areas
- **Lymphatic:** to the pelvic and para-aortic lymph nodes is common, particularly in advanced-stage disease
- **Hematogenous:** at the time of diagnosis is uncommon

Clinical manifestations of EOC

- More than 80% of epithelial ovarian cancers are found in postmenopausal women. The peak incidence of invasive epithelial ovarian cancer is at 56 to 60 years of age
- The age-specific incidence of ovarian epithelial cancer rises precipitously from 20 to 80 years of age and subsequently declines. Less than 1% of epithelial ovarian cancers occur before the age of 21 years, two thirds of ovarian malignancies in such patients being germ cell tumors
- About 30% of ovarian neoplasms in postmenopausal women are malignant, whereas only about 7% of ovarian epithelial tumors in premenopausal patients are frankly malignant

Symptoms:

The majority of women with epithelial ovarian cancer have vague and nonspecific symptoms:

In early-stage disease,

- Irregular menses if she is premenopausal.
- If a pelvic mass is compressing the bladder or rectum, she may report, urinary frequency or constipation
- Occasionally, she may perceive
 - lower abdominal distention,
 - pressure, or pain, such as dyspareunia.
- Acute symptoms, such as pain secondary to rupture or torsion, are unusual

In advanced-stage disease,

- Patients most often have symptoms related to the presence of ascites, omental metastases, or bowel metastases, include:-
 - abdominal distention
 - bloating, constipation, nausea, anorexia, or early satiety

- Premenopausal women may report irregular or heavy menses, whereas vaginal bleeding may occur in postmenopausal women

Signs:

The most important sign of epithelial ovarian cancer is the presence of a pelvic mass on physical examination.

- A solid, irregular, fixed pelvic mass is highly suggestive of an ovarian malignancy. If an upper abdominal mass or ascites is also present, the diagnosis of ovarian cancer is almost certain.

In patients who are at least 1 year past menopause, the ovaries should have become atrophic and not palpable.

- It has been proposed that any palpable pelvic mass in these patients should be considered potentially malignant, a situation that has been referred to as the postmenopausal palpable ovary syndrome.

Diagnosis

Hx and P/E

Investigations

- **Laboratory Tests**
 - **CBC:** Of affected women, 20 to 25 percent will present with thrombocytosis (platelet count $> 400 \times 10^9/L$). Malignant ovarian cells releasing cytokines are believed to increase platelet production rates
 - **serum electrolytes:** Hyponatremia, typically ranging between 125 and 130 mEq/L

- **serum tumor marker:**
 - ✓ **CA125** levels are elevated, an elevated value (false- positive) may be associated with various common benign indications such as PID, endometriosis, leiomyomas, pregnancy, and even menstruation
 - ✓ the human epididymal protein 4 (**HE4**) tumor marker
 - ✓ **OVA1** is another biomarker blood test panel: women with an identified ovarian mass when surgery is planned Ware Scores ≥ 5.0 in premenopausal and scores ≥ 4.4 in postmenopausal women suggest a need or gynecologic oncologist consultation
 - ✓ cancer antigen 19-9 (**CA19-9**) and carcinoembryonic antigen (**CEA**): when a mucinous ovarian tumor is identified
 - ✓ This markers are more important in post-surgery/chemo follow up than Dx of disease
- **Imaging**
 - **Transvaginal sonography:** In general, malignant tumors are multiloculated, solid or echogenic, and large (> 5 cm), and they have thick septa with areas of nodularity. Other features may include papillary projections or neovascularization demonstrated by adding color Doppler. Ascites, if present is easily detected,
 - **Radiographic** tests (CXR): detect pulmonary effusions or infrequently, pulmonary metastases.
 - **Barium enema:** helpful in excluding diverticular disease or colon cancer or in identifying ovarian cancer involvement of the rectosigmoid.
 - **Computed tomography (CT):** scanning has a primary role in treatment planning or women with advanced ovarian cancer. Preoperatively, implants in the liver, retroperitoneum, omentum, or other intraabdominal site are detected to thereby guide surgical cytoreduction or demonstrate obviously unresectable disease
 - **MRI**
 - **Bone scan**
 - **Positron emission tomography (PET)**
- **Paracentesis**
 - This procedure is typically avoided diagnostically as cytologic results are usually nonspecific and abdominal wall metastases may form at the needle entry site

Differential diagnosis for EOC

- Benign ovarian neoplasm
- Functional cysts of the ovaries.
- Pelvic inflammatory disease (TOA?)
- Endometriosis
- Uterine leiomyomata
- Nongynecological causes of a pelvic tumor, such as an
 - Inflammatory or neoplastic colonic masses
 - A pelvic kidney

Differentiating between malignant and Benign ovarian masses

One of the top differential diagnoses for EOC is benign neoplasms so in approaching patients with suspected ovarian malignancy students should know how to differentiate it from the benign mass using Hx, P/E, imaging evidences of u/s and different risk assessment models (RMI, IOTA) and if possible, appreciate the intra op findings.

Characteristics	Benign	Malignant
<i>Age</i>	<i>Young</i>	<i>Old age</i>
<i>Site</i>	<i>Unilateral</i>	<i>Bilateral</i>
<i>Mobility</i>	<i>Mobile</i>	<i>Fixed (due to adhesion)</i>
<i>Lymphadenopathy</i>	<i>No</i>	<i>Yes</i>
<i>Consistency</i>	<i>Cystic</i>	<i>Solid</i>
<i>Cul-de-sac</i>	<i>Smooth</i>	<i>Nodular</i>
<i>Ultrasound findings</i>	<i>Simple cyst, < 10 cm</i>	<i>Solid, or mixed cystic and solid</i>
	<i>Septation < 3cm</i>	<i>Septation= thick and >3cm</i>
	<i>No ascites</i>	<i>+ VE ascites</i>

There are several ways that help to differentiate before surgery between benign and malignant ovarian tumors, these includes:

1. *Risk of Malignancy Index (RMI)*
2. *The International Ovarian Tumor Assessment Group guidelines (IOTA)*

RMI is calculated as:

- $RMI = U \times M \times Ca-125$
 - Where:
 - U = ultrasound score (0, 1 or 3),
 - M = menopausal status (0 = premenopausal, 3 = postmenopausal) and
 - $Ca-125$ = absolute lab value of Ca-125 level (U/l).
- *Ultrasound scoring system (U): 1 point for each of the following features on ultrasound*
 - *Multilocular cystadenoma*
 - *Evidence of solid area*
 - *Evidence of metastasis*
 - *Ascites*
 - *Bilateral lesions*
- **Final U score:**
 - 0 = for none of the features above
 - 1 = if one feature from above is present
 - 3 = if two or more features from above are present

A RMI >200 is considered as elevated and suspicious for malignancy. Sensitivity and specificity for ovarian cancer versus benign pelvic mass for RMI ≥ 200 was 92% and 82%, respectively.

The IOTA

- It is based on ultrasound features
- It has “**B**” and “**M**” rules

B-rules	M-rules
▪ Unilocular cysts ▪ Presence of solid components where the largest solid component <7 mm ▪ Presence of acoustic shadowing ▪ Smooth multilocular tumor with a largest diameter <100 mm ▪ No blood flow in duplex Doppler	▪ Irregular solid tumor ▪ Ascites ▪ At least four papillary structures ▪ Irregular multilocular solid tumor with largest diameter ≥ 100 mm ▪ Very strong blood flow in duplex Doppler

- If one or more M-rules apply in the absence of a B-rule, the mass is classified as malignant.
- If one or more B-rules apply in the absence of an M-rule, the mass is classified as benign.
- If both M-rules and B-rules apply, the mass cannot be classified.
- If no rule applies, the mass cannot be classified.

So, women presenting with an ovarian mass with any M-rules on ultrasound need further investigation for suspicion of malignancy with good specificity (95%) and sensitivity (91%)

Staging of EOC

Staging of Ovarian ca is surgical, compared with clinical staging of cervical ca

Steps in surgical staging of EOC

- A midline or paramedian abdominal incision is recommended to allow adequate access to the upper abdomen
- Any free fluid, especially in the pelvic cul-de-sac, should be submitted for cytologic evaluation

- If no free fluid is present, peritoneal washings should be performed by instilling and recovering 50 to 100 mL of saline from the pelvic cul-de-sac, each paracolic gutter, and beneath each hemidiaphragm.
- A systematic exploration of all the intra-abdominal surfaces and viscera is performed, proceeding in a clockwise fashion from the cecum cephalad, the small intestine and its mesentery from the Treitz ligament to the cecum should be inspected
- Any suspicious areas or adhesions on the peritoneal surfaces should be biopsied. If there is no evidence of disease, multiple intraperitoneal biopsies should be performed.
- The diaphragm should be sampled either by biopsy or by scraping with a tongue depressor and obtaining a sample for cytologic assessment
- The omentum should be resected from the transverse colon, a procedure called an infracolic omentectomy
- The retroperitoneal spaces should be explored to evaluate the pelvic and paraaortic lymph nodes.

Stages

FIGO Staging of Carcinoma of the Ovary, Fallopian Tube, and Primary Peritoneal Carcinoma Stage Characteristics

I= Tumor confined to ovaries (or to fallopian tubes)

- IA, Tumor limited to 1 ovary (or 1 tube); capsule intact, no tumor on surface, negative washings
- IB, Tumor involves both ovaries (or both tubes), otherwise like IA
- IC₁, Tumor limited to 1 or both ovaries (or tubes), with surgical spill
- IC₂, Tumor limited to 1 or both ovaries (or tubes), with capsule rupture before surgery or tumor on ovarian surface
- IC₃, Tumor limited to 1 or both ovaries (or tubes), with malignant cells in ascites or peritoneal washings

II= Tumor involves 1 or both ovaries (or 1 or both tubes) a with pelvic extension (below the pelvic brim) or primary peritoneal cancer

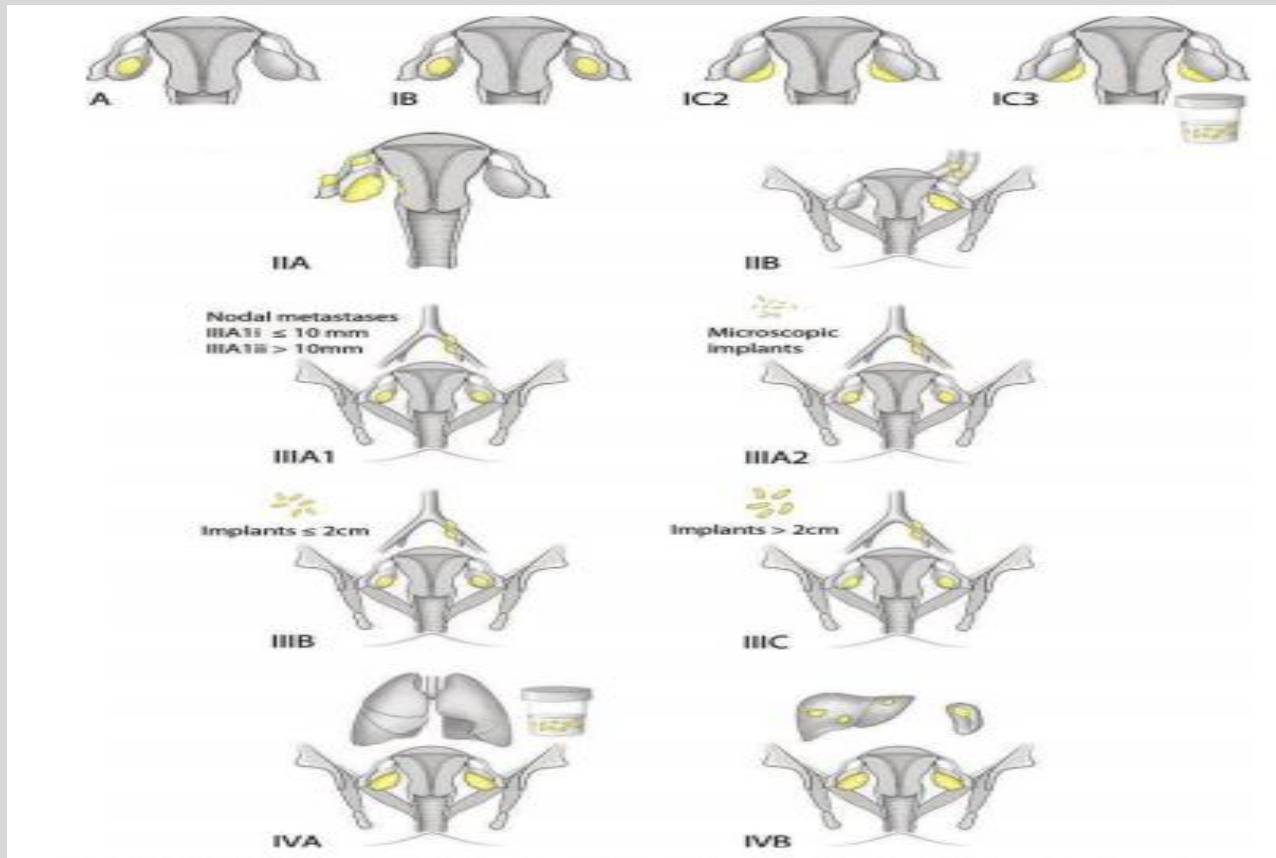
- IIA Extension and/or implants on uterus and/or fallopian tubes (and/or ovaries)
- IIB Extension to other pelvic intraperitoneal tissues

III= Tumor involves 1 or both ovaries (or 1 or both tubes) a with cytologically or histologically confirmed spread to the peritoneum outside the pelvis and/or metastasis to retroperitoneal lymph nodes

- IIIA₁ Positive retroperitoneal lymph nodes only (i) Metastasis ≤ 10 mm (ii) Metastasis > 10 mm
- IIIA₂ Microscopic, extrapelvic (above the brim) peritoneal involvement $\pm$ positive retroperitoneal nodes
- IIIB Macroscopic, extrapelvic, peritoneal metastasis ≤ 2 cm $\pm$ positive retroperitoneal nodes. Includes extension to capsule of liver/spleen
- IIIC Macroscopic, extrapelvic, peritoneal metastasis > 2 cm $\pm$ positive retroperitoneal nodes. Includes extension to capsule of liver/spleen

IV= Distant metastasis excluding peritoneal metastasis

- IVA Pleural effusion with positive cytology
- IVB Hepatic and/or splenic parenchymal metastasis, metastasis to extra-abdominal organs (including inguinal lymph nodes and lymph nodes outside of the abdominal cavity)



Management of EOC and Prognostic factors

Early stage (stage I-II)

- Low risk (Stage IA or IB, grade 1 and 2)
 - Surgery: Total abdominal hysterectomy (TAH) & bilateral salpingo-oophorectomy (BSO)
 - Adjuvant chemotherapy not recommended.
- High-risk
 - Includes: stage IA or IB, grade 3; Stage IC (tumor on external surface, ruptured capsule, ascites or positive peritoneal washing); clear cell histology and all stage II
 - Surgery: TAH, BSO & surgical staging
 - Chemotherapy: adjuvant chemo is recommended.
 - ✓ Regimens: carboplatin (AUC=7.5) and paclitaxel (175/m²) chemotherapy for 3-6 cycles.

Advanced-stage Epithelial Ovarian Cancer (stage III-IV)

- **Surgery:**
 - Primary cytoreductive surgery (debulking)
 - Includes: - TAH, BSO, complete omentectomy, and resection of any metastases from peritoneal surface or intestine.
 - The goal is residual lesion < 1.5 cm
 - Secondary cytoreductive surgery
 - Considered after completion of first line chemotherapy,
- **Chemotherapy**
 - Neoadjuvant chemotherapy (debulking chemotherapy)
 - May be considered for sub-optimally staged (resected) stage III & IV.
 - 2-3 cycles may be helpful in patients with massive ascites and pleural effusion

Prognostic Factors

- **Pathologic Factors**
 - High grade tumors and those with extensive invasion have poor prognosis
 - Clear cell carcinomas are associated with a prognosis worse than that of other histologic types
- **Biologic Factors: FIGO stage and ploidy**
 - Low-stage cancers tend to be diploid and high-stage tumors tend to be aneuploid.
 - Patients with diploid tumors have a significantly longer median survival than those with aneuploid tumors.
 - 30% of epithelial ovarian tumors expressed HER-2/neu oncogene and that this group had a poorer prognosis. The most commonly expressed tumor suppressor gene in ovarian cancer is p53
- **Clinical Factors:**
 - In addition to stage, the extent of residual disease after primary surgery, the volume of ascites, patient age, and performance status are all independent prognostic variables.

Ovarian cancers that undergo intraoperative rupture or spillage do not worsen prognosis, whereas tumors found to have already ruptured preoperatively do have a poorer prognosis.

For early stage disease, poor prognostic variables were tumor grade, capsular penetration, surface excrescences, and malignant ascites, but not iatrogenic rupture

Eastern Cooperative Oncology Group (ECOG) Performance Status Scale

- *It describes a patient's level of functioning in terms of their ability to care for themselves, daily activity, and physical ability (walking, working, etc.).*

GRADE	ECOG PERFORMANCE STATUS
0	Fully active, able to carry on all pre-disease performance without restriction
1	Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature, e.g., light house work, office work
2	Ambulatory and capable of all self-care but unable to carry out any work activities; up and about more than 50% of waking hours
3	Capable of only limited self-care; confined to bed or chair more than 50% of waking hours
4	Completely disabled; cannot carry on any self-care; totally confined to bed or chair
5	Dead

Malignant Ovarian Germ Cell Tumors

- Germ cell tumors arise from the ovary's germinal elements and make up ***one third of all ovarian neoplasms.***

Three features distinguish malignant germ cell tumors from epithelial ovarian cancers.

- typically present at a younger age, usually in their teens or early 20s
- most have stage I disease at diagnosis
- prognosis is excellent—even for those with advanced disease—due to ***exquisite tumor chemosensitivity***

Histopathology

Modified WHO Classification of Ovarian Germ Cell Tumors Germ cell tumors

Germ cell tumors

- Dysgerminoma
- Yolk sac tumor (endodermal sinus tumor)
- Embryonal carcinoma
- Nongestational choriocarcinoma
- Mature teratoma
 - Solid
 - Cystic (dermoid cyst)
- Immature teratoma
- Mixed germ cell tumor

Monodermal teratoma and highly specialized types arising from a mature cystic teratoma

- Thyroid tumors (struma ovarii: benign or malignant)
- Carcinoids
- Neuroectodermal tumors
- Carcinomas (squamous cell or adeno-)
- Sebaceous tumors

Diagnosis

Hx:

- Subacute abdominal pain in 85% of patients and reflects rapid growth of a large, unilateral tumor undergoing capsular distention, hemorrhage, or necrosis
- Severe acute abdominal pain. In 10 percent of cases, secondary to cyst rupture, torsion, or intraperitoneal hemorrhage
- Abdominal distention In more advanced disease, ascites may develop and cause it
- Heavy or irregular menses Because of the hormonal changes that frequently accompany these tumors

P/E:

- A palpable mass on pelvic examination is the most common, premenarchal patients may require examination under anesthesia (EUA) to adequately assess a suspected adnexal tumor.
- The remainder of the physical examination searches for signs of ascites, pleural effusion, and organomegaly

Investigations:

Laboratory Testing

- Serum Tumor Markers in Malignant Ovarian Germ Cell Tumors

Histology	AFP	hCG
Dysgerminoma	—	±
Yolk sac tumor	+	—
Immature teratoma	±	—
Choriocarcinoma	—	+
Embryonal carcinoma	+	+
Mixed germ cell tumor	±	±
Polyembryoma	±	±

AFP = alpha-fetoprotein; hCG = human chorionic gonadotropin.

- Placental alkaline phosphatase (PLAP), lactate dehydrogenase (LDH) α 1-Antitrypsin (AAT) can, are rarely be detected in association with germ cell tumors.

Imaging

- same as for EOC

Management

- Surgery:
 - Fertility preserving surgery is generally possible
 - The minimal surgery is unilateral oophorectomy and surgical staging
- Chemotherapy:
 - Is regarded as treatment of choice to preserve fertility as it occurs in young ladies.
- Adjuvant chemotherapy
 - Completely resected Stage I-III: Stage I, II, or III (endodermal sinus tumors, mixed cell tumors, embryonal carcinomas, choriocarcinomas, and immature teratoma) need adjuvant chemotherapy.
- Radiotherapy
- Sensitive but rarely used as primary therapy of germ cell cancers

Ovarian Sex Cord-Stromal Tumors (SCST)

- SCST are neoplasms that originate from the ovarian matrix.
- Cells within this matrix have the potential or hormone production, As a result, individuals with these tumors typically present with signs and symptoms of estrogen or androgen excess.

The overall prognosis of ovarian SCSTs is excellent—primarily due to ***early-stage disease at diagnosis and curative surgery.***

Epidemiology

- SCSTs account *for 3% to 5% of ovarian malignancies*.
- These tumors are more than twice as likely to develop in black women. In contrast with (EOC) or (GCT), ovarian SCSTs typically affect women of all ages.
- There are no proven risk factors for SCSTs.
 - However, obesity as a hyperestrogenic state was independently associated, whereas parity, smoking, and oral contraceptive use were protective.
- The etiology of SCSTs is largely unknown. However, a single, recurrent FOXL2 gene mutation (402C> G) is present in virtually all adult-type granulosa cell tumors.

Classification

Modified WHO Classification of Ovarian Sex Cord-Stromal Tumors

Pure stromal tumors	Pure sex cord tumors
▪ Fibroma/fibrosarcoma ▪ Thecoma ▪ Sclerosing stromal tumor ▪ Leydig cell tumor ▪ Steroid cell tumor	▪ Granulosa cell tumor ▪ Adult type ▪ Juvenile type ▪ Sertoli cell tumor ▪ Sex cord tumor with annular tubules
	Mixed sex cord stromal tumors ▪ Sertoli-Leydig cell tumors ▪ Sex cord-stromal tumors, NOS

Diagnosis

Hx

- Isosexual precocious puberty is the presenting sign in more than 80 percent of prepubertal girls ultimately diagnosed with an ovarian SCST.
- Adolescents often report secondary amenorrhea. Abdominal pain and distention are other common complaints in this age group
- In adult women, heavy, irregular bleeding and postmenopausal bleeding are the most frequent symptoms.
- mild hirsutism that rapidly progresses to frank virilization should prompt evaluation to exclude these tumors.
- The classic presentation is a postmenopausal woman with rapidly evolving stigmata of androgen excess and a complex adnexal mass.

P/E:

- Abdominal tenderness or a palpable mass

Investigations

Laboratory Testing:

- Elevated circulating levels of testosterone or androstenedione or both are strongly suggestive of an ovarian SCST in a woman with signs and symptoms of virilization.

Clinical hyperandrogenism is more likely to be idiopathic or related to polycystic ovarian syndrome, but serum testosterone levels > 150 g/dL or dehydroepiandrosterone sulfate (DHEAS) levels > 8000 g/L strongly suggest the possibility of an androgen-secreting tumor.

Imaging

- CT, MRI, U/S

Management

- Surgery:
 - Fertility preserving surgery is possible
 - Unilateral salpingo-oophorectomy and surgical staging for young ladies, however total abdominal hysterectomy and bilateral salpingo-oophorectomy for older ones
- Chemotherapy:
 - No recommendation of chemotherapy
- Radiotherapy
 - Usually for recurrence only

Ovarian cancer during pregnancy

- Adnexal masses occurs in 0.2-2% during pregnancy and 5% are malignant
- Found in 1% of 20,000 deliveries, 75% are early stage. Most are unilateral and virilization sometimes occur

Relative frequencies by histologic types

Histologic types	Frequency (%)
EOC	40
Germ cell tumors	30
SCST	17
Others	13

Diagnosis

- First trimester is the best time
- Based on clinical and ultrasound

Management

- Surgery
 - The optimum time for surgery is 16-18 weeks of gestational age
 - Symptomatic, complex and solid ovarian tumors should be operated immediately
 - Never perform TAH & BSO on the basis of frozen section diagnosis, discuss options with the patient
 - If Malignant or metastatic treat as non-pregnant or one can omit hysterectomy Progesterone replacement if before 10 weeks of gestation and pregnancy to continue
- Adjuvant chemotherapy

Prognosis

- Pregnancy doesn't worsen ovarian cancer, 5 year survival rate is 72-90%

Sample history

Chief complaint: Abdominal swelling of 8 month duration

HPI

This is a 59 years old nulliparous woman who has been amenorrheic for the past 7 years. She was relatively healthy till 8 month back at which time she started to experience abdominal swelling, w/c was initial in lower abdomen, but increase in size gradually to attain its current size within 5months. She also has lower abdominal pain for past 5mo w/c is non-radiating, intermittent and aching type, w/c has no any alleviating or exacerbating factors, the pain increase in severity recently and has been preventing her from doing her routine activities. Associated to this she has history of bloating, nausea, anorexia, early satiety, constipation, urinary frequency and urgency (11 times per day) and unquantified but significant weight loss of the same duration.

She started seeing her menses at age of 12, it used to come every 4wk and lasted 5 days, it did not show difference in amount month to month, and it was not associated with any lower abdominal discomfort. She stops seeing menses at age of 52. Her mother died of ovarian cancer 7 years back. She usually eat injera made of teff and shiro made ater with meat, she takes coffee 3x/day, and around 5 bottles of beer every weekend.

Otherwise she has no Hx of:

- Oral contraceptive use and Postmenopausal hormone therapy
- Gynecologic surgery and radiation to pelvic
- Talc exposure
- Breast mass
- Vaginal bleeding
- Yellowish discoloration of eyes and contact with yelloished discolored eye(chronic liver disease)
- Blood in stool(colonic cancer)
- Cough and SOB(plural effusion)

- Contact with chronicle coughing person(pelvic TB)
- Multiple sexual partner, vaginal discharge or itching sensation (PID, TOA)
- Gynecologic history and Past obstetric history: add to HPI if any pertinent Hx.

Physical examination:

- Do full examination
- Refer the Approach to APM for characterization of findings in abdominal and pelvic examination

Assessment: APM secondary to r/o ovarian tumor**Investigation:**

- CBC
 - r/o anemia(of chronic disease), infection, thrombocytopenia (important for surgical patient)
- Blood group & Rh (for all surgical patients)
- Ultrasound:
 - it is used to determine size, location, consistency, number of mass and hydro ureter & hydro nephrosis
- RFT (BUN & Serum creatinine) and TFT(TSH, for surgical patient)
- CXR
 - for all surgical patients above 6oyrs old
- ECG
 - for all surgical patients above 50

Chapter 20

CERVICAL NEOPLASIA

Prepared by Dr. Amanuel Tefera (MI)

Cervical Intraepithelial Neoplasm (CIN)

CIN, formerly called dysplasia, means disordered growth and development of the epithelial lining of the cervix.

The term intraepithelial neoplasia refers to epithelial lesions that are potential precursors of invasive cancer.

The severity of squamous intraepithelial lesions is graded by the proportion of epithelium with abnormal cells.

- These alterations ***begin at the basement membrane and continue upward*** toward the epithelium surface.

Classification of CIN

- Based on degree of dysplasia CIN is classified in to four:
 - ***CIN I (Mild dysplasia)***, is defined as disordered growth of the lower third of the epithelial lining.
 - ***CIN II (moderate dysplasia)***, abnormal maturation of the lower two-thirds of the epithelial lining
 - ***CIN III (Severe dysplasia)***, encompasses more than two thirds of the epithelial thickness.
 - ***Carcinoma in situ (CIS)*** representing full-thickness dysmaturity.

- According to the Bethesda system which based on cytological smear
 - Atypical Squamous cells of Undetermined Significance (ASCUS)
 - Atypical Squamous cells where high-grade lesions must be excluded (ASC-H).
 - Low-grade squamous intraepithelial lesion (LSIL) encompasses cytologic changes consistent with koilocytic atypia or CIN I
 - High-grade squamous intraepithelial lesion (HSIL) denotes the cytologic findings corresponding to CIN II and CIN III
- Because cytologic and histologic changes of **HPV infection and CIN I** are not distinguished reliably and because their natural histories are similar, they are categorized together as **low-grade squamous intraepithelial lesions (LSIL)**.
- Similarly, CIN 2, CIN 3, and CIS are difficult to distinguish, are truer cancer precursors, and are designated together as **high-grade squamous intraepithelial lesion (HSIL)**.

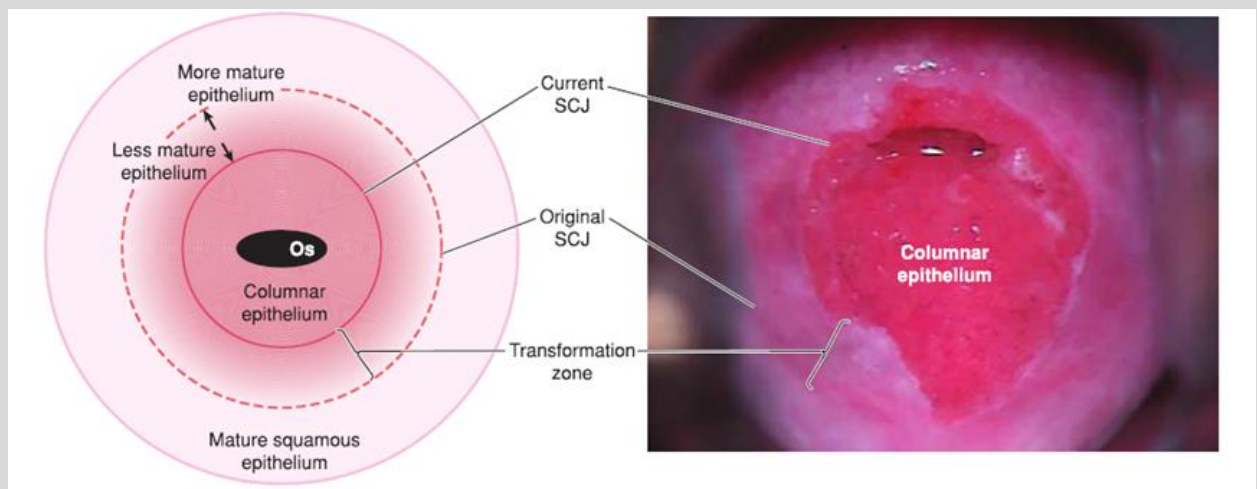
Pathogenesis and Risk factors

Squamo-columnar Junction

- During embryogenesis, upward migration of stratified squamous epithelium from the urogenital sinus is thought to replace müllerian epithelium; this process usually terminates near the external cervical Os, forming the original (congenital) squamocolumnar junction (SCJ).
- The location of the SCJ varies with age and hormonal status.
 - During the reproductive years, it everts outward onto the ectocervix in response to **estrogen** (especially during adolescence, pregnancy, and combination hormonal contraceptive use).
 - It regresses into the endocervical canal with the process of squamous metaplasia and low estrogen states like menopause, lactation, and progestin-only contraceptive use.

Squamous Metaplasia

- At puberty, the rise in estrogen levels leads to greater glycogen formation within non keratinized cervical and vaginal squamous epithelia.
- In providing a carbohydrate source, glycogen allows vaginal flora to be dominated by lactobacilli, which produce lactic acid.
- The resultant acidic vaginal pH is the suspected stimulus for squamous metaplasia, which is the normal replacement of columnar by squamous epithelium on the cervix.
- This creates a progressively widening band of newer metaplastic and maturing squamous epithelium that lies between the original squamous epithelium and the columnar epithelium and is termed the **transformation zone (TZ)**
- Nearly all cervical neoplasia, both squamous and columnar, develops within the TZ, usually adjacent to the new or current SCJ.



Figure, Relevant ectocervical landmarks depicted in a schematic drawing and corresponding photograph.

Risk factors of CIN

- Similar with risk factor of cervical Ca (see below)
 - **HPV infection** and *Immunosuppression* are the most important RFs

Natural History of CIN

	CIN I	CIN II	CIN III
Regression to normal (%)	57	43	32
Persistence (%)	32	35	<56
Progression to CIS (%)	11	22	-
Progression to invasion (%)	1	5	>12

CIN and Human papilloma Virus (HPV)

HPV

- Is a double stranded DNA virus with a protein capsid unique to each viral type
- More than 150 genetically distinct HPV types have been identified and approximately **40 types infect the LAGT**
- *HPV causes*
 - **nearly all cervical cancers** and
 - *approximately 90 percent of anal cancers,*
 - *70 percent of vaginal, and*
 - *40 percent of vulvar cancers.*
 - NB. Other than LAGT, where else in the body does HPV associated neoplasms occur?
- Analyses of cervical neoplasia lesions show the presence of HPV in more than 80% of all CIN lesions and in **99.7% of all invasive cervical cancers.**

HPV types

- Based on their malignant potential, HPV subtypes are categorized into
 - **Low-risk**
 - Include; HPV types 6, 11, 42, 43, and 44
 - Are associated with condylomata accuminata and low-grade lesions (CIN I)
 - **High-risk**
 - Include; HPV types **16, 18**, 31, 33, 35, 39, 45, 51, 52, 56, 58, 59, and 68
 - Are associated with invasive cancer in addition to CIN II and CIN III
 - HPV 16 is the most oncogenic and accounts for the largest percentage of CIN 3 lesions (45%) and cervical cancers (55%).
 - HPV 18 is found in 13% of cervical squamous cell carcinomas, and in 40% of cervical adenocarcinomas and adenosquamous Carcinomas.
 - Together, HPVs 16 and 18 account for approximately 70 percent of cervical cancers worldwide, 68 percent of squamous cell carcinomas, and 85 percent of adenocarcinomas.

The most important risk factors for genital HPV acquisition are number of lifetime and recent sexual partners and early age of first sexual intercourse

Once the epithelium is acutely infected with HPV, 1 of 3 clinical scenarios ensues:

1. Asymptomatic latent infection
2. Active infection in which HPV undergoes vegetative replication but not integration into the genome (eg, leading to condyloma or CIN I).
3. Neoplastic transformation following integration of oncogenic HPV DNA into the human genome

Outcomes of HPV infection:

- Resolution in more than 90% of *immunocompetent* women spontaneously over a 2-year period.
- Only approximately 5% will have cytologically detectable CIN
- This suggests that infection with *HPV alone is insufficient for the development of CIN or cervical cancer* and underscores the importance of other cofactors, such as cigarette smoking or immunosuppression

Infection Diagnosis

- HPV infection is clinically suspected based on lesions or results of cytology, histology, and colposcopy, all of which are subjective and often inaccurate
- HPV testing more accurately confirms HPV infection.
- Clinical HPV testing involves the direct detection of HPV nucleic acids by in situ hybridization, nucleic acid amplification testing (NAA T), polymerase chain reaction (PCR)

Infection Treatment

- The indications to treat HPV-related LAGT disease are
 - symptomatic warts,
 - high-grade neoplasia, or invasive cancer
- options of Tx
 - Mechanical removal or destruction,
 - topical immune modulators, and
 - chemical or thermal coagulation

HPV prevention

- **Behavioral**
 - Sexual abstinence,
 - Delaying coitarche, and
 - Limiting the number of sexual partners are some f behavioral modification

Condoms do not cover all potentially HPV-infected anogenital skin. Therefore, **condoms may not be completely protective**

▪ **HPV Vaccination**

- The Advisory Committee on Immunization Practices (ACIP) currently recommends that HPV vaccine be administered routinely to **all girls and boys aged 11 to 12 years (as early as age 9 years)**
- **Catch-up** vaccination is also recommended for **all 13- to 26-year-old persons**.
 - ✓ The ACIP does not recommend routine catch-up vaccination for persons aged 27 to 45 years due to limited benefit.
- The vaccines are:
 - ✓ **Cervarix** (HPV₂) is a bivalent vaccine against HPVs 16 and 18.
 - ✓ **Gardasil** (HPV₄) is a quadrivalent vaccine against HPV types 6, 11, 16, and 18.
 - ✓ **Gardasil 9** (HPV₉), a nonavalent vaccine, which protects against all HPV types in HPV₄ plus types 31, 33, 45, 52, and 58.
- They are administered in three intramuscular doses during a 6-month period; the second dose is given 1 to 2 months after the first dose, and the third 6 months after the first dose.
- The recent CDC guideline recommend as only two doses (first dose-11-12 years, second dose after 6months of the first dose) is enough for those 9-15 years.
 - ✓ If age is 15 and greater 3rd doses within six months is recommended.
- Vaccination can be given during lactation but is ***avoided during pregnancy***.

Cervical Cancer Prevention and Screening

- Generally consists of
 - Vaccination against human papillomavirus,
 - Screening and treatment of pre-cancer lesions,
 - Early detection and prompt treatment of early invasive cancers and palliative care have proven to be effective strategies to address cervical cancer across the continuum of care

- It is then organized around three key pillars: primary prevention, secondary prevention, and tertiary management of cervical cancer.

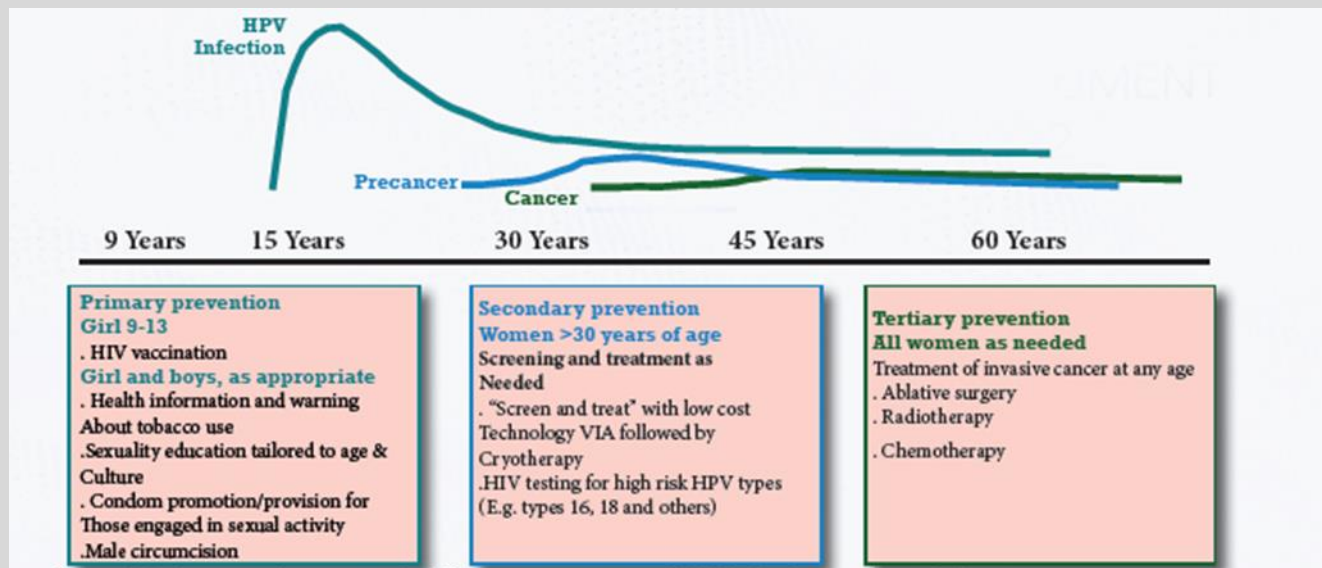


Fig...Intervention across the life course to prevent HPV infection and cervical cancer (taken from cervical screening Ethiopian guidelines 2021)

Screening Programs (Secondary Prevention)

- Cervical cancer has a long precancerous period, usually taking more **than 10 years** to progress from precancerous lesions to invasive cancer.
 - As a result, it is rare for cervical cancer to develop in a woman less than 30 years of age
 - This long precancerous stage provides an excellent opportunity for effective intervention measures
- WHO recommends three types of screening tests:
 - Cytology (Conventional Pap smear and liquid-based cytology (LBC),)
 - HPV testing for high-risk HPV Types,
 - Visual tests; with acetic acid or lugol iodine
- The objective of cervical screening is to prevent invasive cervical cancer by detecting and treating women with CIN II/III lesions (high-grade cervical cancer precursor lesions)

Recommendation from ACOG

- Cervical cancer screening should begin at age 21 years, with the exception of women who are infected with HIV or who are otherwise immune compromised,
- Women younger than 21 years should not be screened regardless of the age of sexual initiation or the presence of other behavior-related risk factors.
- Women aged 21–29 years should be tested with cervical cytology alone, and screening should be performed every 3 years.
- Co-testing and annual screening should not be performed in women younger than 30 years.
- For women aged 30–65 years, co-testing with cytology and HPV testing every 5 years is preferred; screening with cytology alone every 3 years is also acceptable.
- Screening by any modality should be discontinued after age 65 years in women with evidence of adequate negative prior screening test results and no history of CIN 2 or higher.
 - Adequate negative prior screening test results are defined as three consecutive negative cytology results or two consecutive negative co-test results within the previous 10 years, with the most recent test performed within the past 5 years.
- In women who have had a hysterectomy with removal of the cervix (total hysterectomy) and have never had CIN 2 or higher, routine cytology screening and HPV testing should be discontinued and not restarted for any reason.
- Women with any of the following risk factors may require more frequent cervical cancer screening than recommended in the routine screening guidelines, which were intended for average-risk women:
 - Women who are infected with HIV
 - Women who are immunocompromised (such as those who have received solid organ transplants)
 - Women who were exposed to diethylstilbestrol in utero
 - Women previously treated for CIN 2, CIN 3, or cancer

Table... screening method for cervical cancer for the general population

Population	Recommended Screening Method	Comment
Women younger than 21 years	No screening	
Women aged 21–29 years	Cytology alone every 3 years	
Women aged 30–65 years	HPV and cytology co-testing (preferred) every 5 years Cytology alone (acceptable) every 3 years	Screening by HPV testing alone is not recommended
Women older than 65 years	No screening is necessary after negative prior screening results adequate	Women with a history of CIN 2, CIN 3, or adenocarcinoma in situ should continue routine age-based screening for a total of 20 years after spontaneous regression or appropriate management of CIN 2, CIN 3, or adenocarcinoma in situ
Women who underwent total hysterectomy	No screening is necessary	Applies to women without a cervix and without a history of CIN 2, CIN 3, adenocarcinoma in situ, or cancer in the past 20 years
Women vaccinated against HPV	Follow age-specific recommendations (same as unvaccinated women)	

Cervical cytology screening

- Liquid-based and conventional methods of cervical cytology specimen collection are acceptable for screening.
- Exfoliated cells are collected from the transformation zone of the cervix and transferred to a vial of liquid preservative that is processed in the laboratory (liquid based technique) or
- Transferred directly to a slide and fixed (conventional technique)

HPV testing

- Primary screening modality in combination with cervical smear testing or alone
- HPV testing as primary screening modality should be used only for women >30 years
- Two types of HPV tests are available for the screening
 - Hybrid capture 2
 - Cervista HPV HR test, also referred as “cervista HPV 16 & 18” or HPV genotyping **identifies 14 high risk serotype**
- Positive test implies significant high risk of development of CIN 3 or cervical cancer.

Visual Inspection with Acetic Acid (VIA)

- Also called cervicoscopy, consists of naked-eye visualization of the uterine cervix after the application of 3% to 5% acetic acid then cervix is illuminated with a light source.
- Acetic acid dehydrates cells so that squamous cells with relatively large or dense nuclei (eg. metaplastic cells, dysplastic cells, cells infected with HPV) reflect light and therefore **appear white**.
- Blood vessels and columnar cells are not affected by acetic acid, but become easier to visualize against the white background

Visual Inspection with Lugol Iodine, *Schiller test*

- Lugol's solution is an aqueous iodine preparation.
- Normal mature squamous epithelium of the cervix contains glycogen, which combines with iodine to produce a deep mahogany-brown color.
- Non-staining, therefore, indicates abnormal squamous (or columnar) epithelium or immature metaplastic epithelium, and constitutes a positive Schiller test.

*Visual tests **cannot be relied on** in postmenopausal women, because the transformation zone of these Women is often inside the cervical canal.*

- The FMOH recommends screening every five years following normal results except for women living with HIV that should be rescreened every two years.
 - Following abnormal results and/or treatment, repeat screening in one year.
 - If follow-up screening is normal, return to screening every five years.

Colposcopic examination

- Colposcope is an instrument that uses illuminated low-power magnification (5–15×) to inspect the cervix.
- After the application of 3–5% aqueous acetic acid solution abnormalities in the appearance of the epithelium and its capillary blood supply can be identified by colposcopy
- **Indications for colposcopy are:**
 1. Abnormal cervical cytology smear or HPV testing;
 2. Clinically abnormal or suspicious-looking cervix;
 3. Unexplained intermenstrual or post coital bleeding;
 4. Vulvar or vaginal neoplasia; or
 5. History of in utero DES exposure
- **Normal finding from colposcopy**
 - The original squamous epithelium, which extends from the mucocutaneous vulvovaginal junction to the original squamocolumnar junction.
 - The columnar epithelium of the endocervical canal
 - The transformation zone
 - ✓ If the new SCJ is visualized in its entirety, the colposcopic examination is called satisfactory; if it cannot be fully visualized, the examination is called unsatisfactory.
- **Abnormal findings indicative of dysplasia and CIN are those of:**
 - Leukoplakia or hyperkeratosis
 - ✓ An area of white, thickened epithelium that is appreciated prior to the application of acetic acid and may indicate underlying neoplasia.
 - Acetowhite epithelium,

- ✓ Which is epithelium that stains white after the application of acetic acid
- Mosaicism or punctuation
 - ✓ Reflecting abnormal vascular patterns of the surface capillaries.
 - ✓ As general rule, capillary thickness and intercapillary distances correlate with the severity of the lesion and thus tend to be larger and coarser in higher -grade lesions.
- Atypical vessels with bizarre capillaries
 - ✓ With so-called **corkscrew**, **comma** shaped, or **spaghetti**-like configurations suggest early stromal invasion.

Diagnostic conization

- Indications:-
 - If colposcopy is unsatisfactory.
 - If the lesion extends into the cervical canal beyond the view afforded by the colposcope
 - If there is a significant discrepancy between the histologic and the cytologic examination,
 - If adenocarcinoma in situ is suspected,
 - If micro invasive carcinoma is suspected

Table...Management of Cervical Cancer Screening Results

Screening Method	Result	Management
Cytology screening alone	Cytology negative	Screen again in 3 years
	ASC-US cytology and reflex HPV negative	Co-test in 3 years
	All others	Refer to ASCCP guidelines
Co-testing	Cytology negative, HPV negative	Screen again in 5 years
	ASC-US cytology, HPV negative	Screen again in 3 years
	Cytology negative, HPV positive	Option 1: 12-month follow-up with co-testing Option 2: Test for HPV-16 or HPV-18 genotypes If positive results from test for HPV-16 or HPV-18, referral for colposcopy If negative results from test for HPV-16 and HPV-18, 12-month follow-up with cotesting
	All others	Refer to ASCCP guidelines

*Abbreviations: ASCCP, American Society for Colposcopy and Cervical Pathology; ASC-US, atypical squamous cells of undetermined significance (refer from Bethesda system for reporting cervical cytology 2014)

Cervical cancer

Epidemiology

- Worldwide, cervical cancer is common, and it ***ranks fourth*** among all malignancies for women after breast, colorectal, and lung cancer (WHO 2012). From this developing countries contribute 85 percent.
- It is the ***second most common cause of cancer-related morbidity and mortality*** among women in developing countries
- The age at which cervical cancer develops is in general earlier than that of other gynecologic malignancies, and the median age at diagnosis is 50 years

Risk factors

- HPV infection
- Cigarette smoking:
 - Both active and passive smoking increases cervical cancer.
 - Among HPV-infected women, current and former smokers have a two- to threefold increased incidence of high grade squamous intraepithelial lesion (HSIL) or invasive cancer.
 - Plausible explanations about how smoking might cause cervical cancer
 - ✓ One is that smoking inhibits the immune response to HPV.
 - ✓ second is that carcinogenic HPV-infected cells are exposed to smoking carcinogens that cause DNA damage while HPV oncoproteins block apoptosis and cell cycle arrest.
- Immunosuppression.
 - Cervical cancer is one of AIDS defining illness
- High parity:
 - Those with one or two full term pregnancy have a twofold risk when compared with nulliparous.
 - If she had seven pregnancies fourfold risk.

- Long-term combination oral contraceptive.
 - Increase risk by four fold if she is HPV positive.
- Young age at first intercourse (<20years)
- Multiple sexual partner
 - Having more than six lifetime sexual partners
- Race: black and Hispanic has highest incidence.

Tumor genesis

- In general, progression from dysplasia to invasive cancer requires several years, although times can vary widely.
- HPV is the primary etiologic infectious agent associated with cervical cancer.
 - HPV 16 is more commonly associated with squamous cell carcinoma of the cervix, whereas HPV 18 is a risk actor or cervical adenocarcinoma
- The E6 and E7 oncoproteins produced as a result of persistent HPV infection, interfere with and accelerate degradation of p53 and pRb which are key tumor suppressor proteins produced by the host.
 - This is central to host cell immortalization and transformation

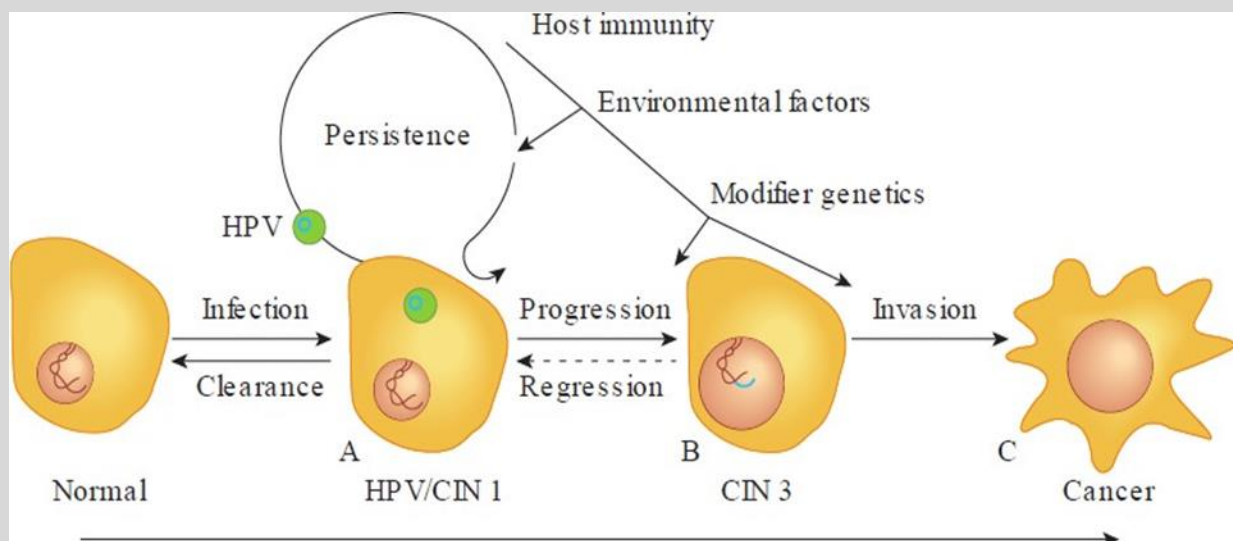


FIG...Molecular pathways of cervical CA

Patterns of spread

- **Direct invasion** into the cervical stroma and local invasion
 - To urethra then blockage frequently develops, resulting in hydronephrosis,
 - The bladder may be invaded by direct tumor extension through the vesicouterine ligaments
 - The rectum is invaded less often because it is separated from cervix by the posterior cul-de-sac
- **Lymphatic metastasis:**
 - The cervix has a rich network of lymphatic's, which follow the course of the uterine artery
 - These channels drain principally into the paracervical and parametrial lymph nodes → obturator lymph nodes→ the internal, external, common iliac lymph nodes, and ultimately the paraaortic lymph nodes.
 - Lymphovascular space involvement is when the tumor invades deeper into the stroma and enters blood capillaries and lymphatic channels.
 - Distant metastasis results from hematogenous dissemination, and the lungs, ovaries, liver, and bone are the most frequently affected.
 - Drainage from the posterior cervix course drains to the rectal lymph nodes.

Histologic types

- The two most common histologic subtypes of cervical cancer are ***squamous cell (70%) and adenocarcinoma (25%)***.
 - Squamous cell subtypes: Keratinizing, Non keratinizing and Papillar
 - Adenocarcinoma subtypes: Mucinous, Endometrioid, Serous, Clear cell and Mesonephric.
 - Mixed cervical carcinoma
 - Neuroendocrine cervical tumor
 - Others: Sarcoma, Lymphoma, Melanoma

Clinical manifestation

Symptoms

- Abnormal vaginal bleeding is the most common symptom and may take the form of a blood-stained leukorrheal discharge, scant spotting, or frank bleeding.
- A history of postcoital or post douching bleeding may be elicited on specific questioning
- Pelvic pain, radiating to the hip or thigh, is a manifestation of advanced disease.
- Incontinence sign of fistula formation.
- Acute blood loss and anemia may occur in a bulky or ulcerating stage I lesion.

Physical Examination

- Most women with this cancer have normal general physical examination findings.
- With speculum examination there may be
 - an exophytic or endophytic growth;
 - a polypoid mass, papillary tissue, or barrel-shaped cervix;
 - a cervical ulceration or granular mass; or necrotic tissue
 - A watery, purulent, or bloody discharge can also be seen
- During bimanual examination,
 - An enlarged uterus resulting from tumor invasion and growth
 - ✓ Alternatively, hematometra or pyometra may expand the endometrial cavity following obstruction.
- The size of the cervix and the tumor measured in cm (a normal cervix is 4×5 cm)
- Extension to adjacent structures
 - Extend into the vagina, and disease extent can be appreciated during anterior vaginal wall palpation or during rectovaginal examination.
 - During digital rectal examination, parametrial, uterosacral, and pelvic sidewall involvement can be appreciated.
 - Involved tissues feel thick irregular, firm, and less mobile
 - A fixed mass indicates that tumor has probably extended to the pelvic sidewalls.

- With advancing disease, enlarged supraclavicular or inguinal lymphadenopathy suggest lymphatic tumor spread
- Lower extremity edema and low back pain, which radiating down the posterior leg, may reflect compression the sciatic nerve root, lymphatics, or veins.

Investigations

- When obvious tumor growth is present, a cervical biopsy is usually sufficient for diagnosis.
- If gross disease is not present, a colposcopic examination with cervical biopsies and endocervical curettage is warranted
- If the diagnosis cannot be established conclusively with colposcopy and directed biopsies, cervical conization may be necessary

Table...Investigations/procedures used during Cervical Cancer Evaluation

Investigations		To Identify:
Laboratory	CBC	Anemia
	Urinalysis	Hematuria
	Chemistry profile	Electrolyte abnormality
	Liver function	Liver metastasis
	Creatinine/BUN	Renal impairment or obstruction
Radiologic	Chest radiograph	Lung metastasis
	Intravenous pyelogram	Hydronephrosis
	Nodal or CT Scan (abdominopelvic)	distant organ metastasis
	MR imaging	Local parametrial invasion
Procedural	Cystoscopy	Bladder tumor invasion
	Proctoscopy	Rectal tumor invasion
	EUA	Extent of pelvic tumor spread

Staging

Cervical cancers are staged clinically.

Allowable components of staging include

- Cold-knife conization,
- Pelvic examination under anesthesia,
- Cystoscopy,
- Proctoscopy,
- Chest radiograph, and
- Intravenous pyelogram (or this portion of the computed tomography [ct]scan can be used).

Table. Clinical Stages of Cervical Cancer (FIGO, Revised 2018)

Stage		Characteristics
0		Carcinoma in situ, cervical intraepithelial lesion (CIN) 3
I		Carcinoma is strictly confined to cervix (extension to corpus should be disregarded)
	IA	Microscopic lesion, invasion is limited to measured stromal invasion with a maximum depth <5 mm
	IA1	Measured invasion of stroma <3 mm in depth
	IA2	Measured invasion of stroma ≥3 mm and <5 mm in depth
	IB	Clinical lesions confined to the cervix or preclinical lesions greater than IA
	IB1	Invasive carcinoma <2 cm in greatest dimension
	IB2	Invasive carcinoma ≥2 cm and <4 cm in greatest dimension
	IB3	Invasive ≥4 cm in greatest dimension
II		Carcinoma extends beyond uterus but has not extended to pelvic wall; It Involves vagina, but not as far as the lower third
	IIA	No obvious parametrial invasion
	IIA1	Invasive carcinoma <4 cm in size

	IIA2	Invasive carcinoma ≥ 4 cm in size
	IIB	Obvious parametrial involvement
III		Carcinoma has extended to the pelvic wall; on rectal examination there is no cancer-free space between tumor and pelvic wall; tumor involves lower third of vagina; all cases with hydronephrosis or nonfunctioning kidney should be Included, unless they are known to be due to another cause; involves pelvic and/or para-aortic lymph nodes
	IIIA	No extension to pelvic wall, but involvement of lower third of vagina
	IIB	Extension to pelvic wall, or hydronephrosis or nonfunctioning kidney due to tumor
	IIIC1	Pelvic lymph node metastasis only
	IIIC2	Para-aortic lymph node metastasis
IV		Carcinoma has extended beyond true pelvis or has clinically involved mucosa of bladder or rectum
	IVA	Spread of growth to adjacent pelvic organs
	IVB	Spread to distant organs

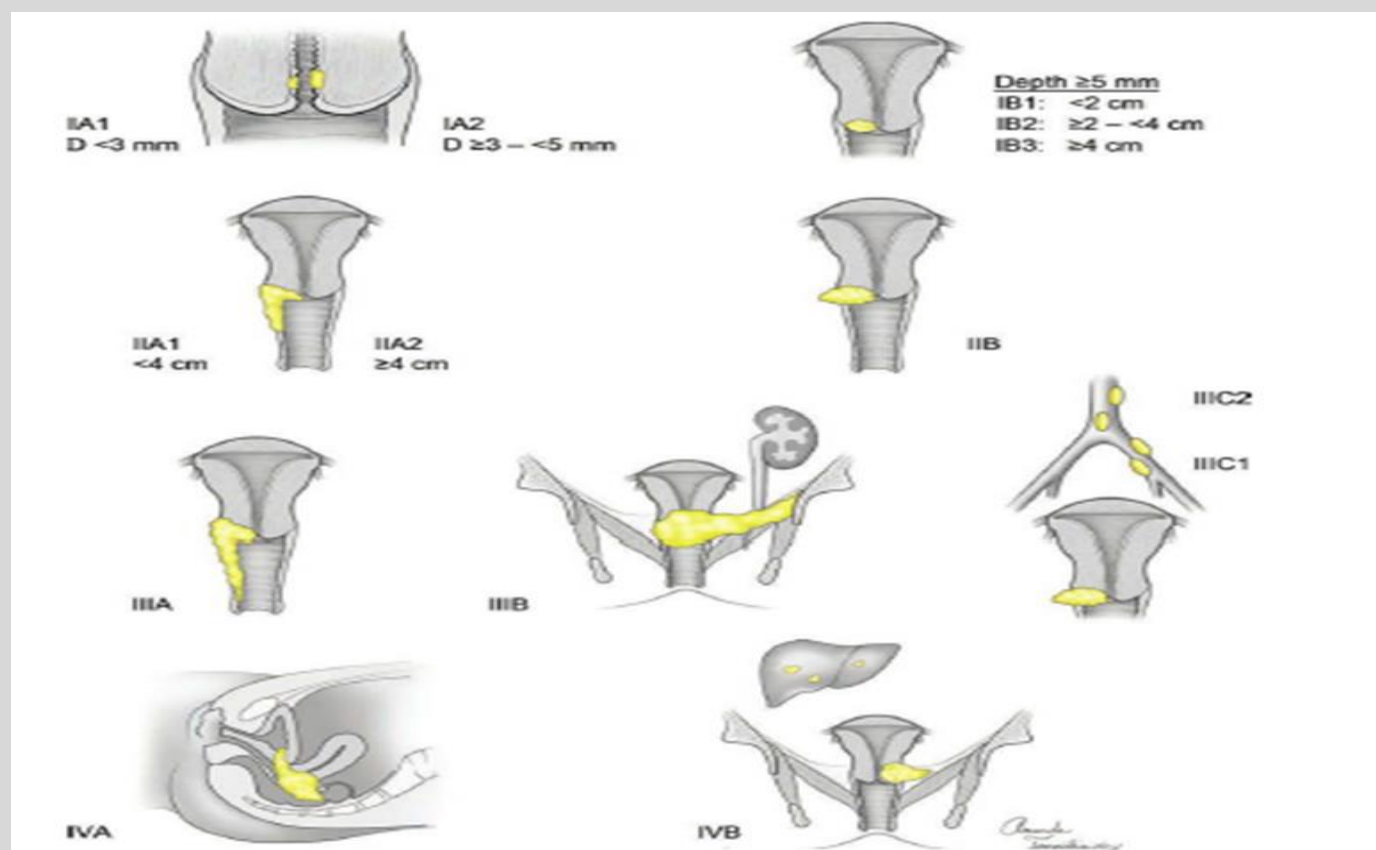


FIG...FIGO staging of cervical cancer

TABLE General Treatment for Primary Invasive Cervical Carcinoma

Cancer Stage	Treatment
IA ₁	Simple hysterectomy preferred if childbearing completed Or Cervical conization
IA ₁ (withLVSI)	Modified radical hysterectomy and pelvic lymphadenectomy Or Radical trachelectomy and pelvic lymphadenectomy for selected patients desiring fertility
IA ₂	Radical hysterectomy and pelvic lymphadenectomy Or Radical trachelectomy and pelvic lymphadenectomy for selected patients desiring fertility
IB _{1c} to IIA ₁	Radical hysterectomy & pelvic lymphadenectomy (consider SNLB if tumor <2 cm) Or Radical trachelectomy and pelvic lymphadenectomy for selected patients desiring fertility Or Chemoradiation
Bulky IB ₃ to IIA ₂	Chemoradiation
IIB to IVA	Chemoradiation Or Rarely, pelvic exenteration
IVB	Palliative chemotherapy and/or Palliative radiotherapy Or Supportive care (hospice)

Cervical Cancer Survival Rates According to Stage

Stage	5-Year Survival
IA	100%
IB	88%
IIA	68%
IIB	44%
III	18-39%
IVA	18-34%

Sample history

Chief complaint: vaginal bleeding of three month duration.

HPI:

This is a 40years old Para 7(all lived) o abortion mother whose LMNP was two week back. She was relatively healthy three month back at which time she start to experience vaginal bleeding of which was initially bright red scanty bleeding and has relationship with her menses. The bleeding was exacerbated after sexual intercourse and post douching and has no known relieving factor.

She also experience watery discharge which increase in amount and sometimes become blood stained. She also has intermittent moderate dull aching lower abdominal pain for the last two month which radiate to her thigh and has no known relieving and exacerbating factor. She also has loss of appetite significant but unqualified weight loss vertigo and tinnitus since the last one month.

Her menses started when she was 13 years old and come every 24 days. It stays for 3-4 days she uses 2 pads for it. The bleeding was black and non clotting.

She was married when she was 16years old and she claimed to be virgin before her marriage.

She used OCP for the last four year after delivery of her last baby. Other wise

She has no history of cigarettes smoking.

She has no history of foul smelling vaginal discharge, genital ulcer or treatment for STD

She has no history of multiple sexual partner.

She has no history of urinary incontinence, frequency, urgency or dysuria.

She has no history of cough shortness of breath chest pain or blood stained sputum.

She has no history of RUQ pain, yellowish discoloration of eye and bleeding from other site.

She has no history of bone pain or fracture. She has no history of pelvic trauma.

Physical examination

G/A: chronically sick looking, Look wasted and in pain

Vital sing: No vital sign derangement

HEENT: pale conjunctiva non icteric sclera.

LGS: no significant lymphadenopathy

Chest/CVS: No respiratory and cardiac finding

Abdomen:

- ✓ Flat abdomen that moves with respiration, No scar No distended vein.
- ✓ On palpation: there is no palpable mass and organomegaly
- ✓ On percussion there is no sign of fluid collection

MSS: no lower extremity edema.

GUS:

- ✓ No CVAT, The kidneys are not palpable.
- ✓ Pelvic examination
 - External genitalia examination: Inverted triangle type of pubic hair pattern. No discoloration of skin around vulva, no swelling, ulcer or discharge
 - Speculum examination: there is cervical ulceration and necrotic tissue are visible, Bloody discharge is also visible from cervix
 - Digital vaginal examination: there is irregular firm lesion on cervix which measures around 4 cm. cervix is mobile not attached to adjacent structure. No involvement of vagina
 - Bimanual examination: the uterus is normal size and position Digital rectal examination: no parametrial uteresacral and pelvic side wall involvement.

Chapter 21

GESTATIONAL TROPHOBLASTIC DISEASE

Prepared by Dr. Bulcha Lemma (MI)

Definitions

Gestational trophoblastic disease (GTD) refers to a spectrum of interrelated but histologically distinct tumors originating from abnormal proliferation of trophoblast of the placenta.

Gestational trophoblastic neoplasia (GTN) refers to the subset of GTD that develops malignant sequelae.

- These tumors require formal staging and typically *respond favorably to chemotherapy*.
- GTNs are among the rare human tumors that can be cured even in the presence of widespread dissemination.

Classification

Modified WHO Classification of GTD

Molar pregnancies

Hydatidiform mole

Complete

Partial

Invasive mole

Trophoblastic tumors

Choriocarcinoma

Placental site trophoblastic tumor

Epithelioid trophoblastic tumor

Epidemiology and Risk factors

Estimates of the incidence of GTD vary dramatically in different regions of the world. For example, the incidence of molar pregnancy in Japan (2 per 1,000 pregnancies) has been reported to be about threefold higher than the incidence in Europe or North America (about 0.6 to 1.1 per 1,000 pregnancies) .

- This variation result from differences between reporting population-based versus hospital based data.

In Ethiopia from various hospital based studies the magnitude is **1.8-3 per 1000 deliveries**.

Based on a thorough pathologic review, the incidence of *partial mole is higher* than complete mole which is:-

- Complete mole- 1 per 1,945 deliveries
- Partial mole- 1 per 695 deliveries

Risk factors

- The main risk factors for hydatidiform mole are extremes of maternal age and a history of pervious mole.
 - **Prior molar pregnancy:** - **10x** increased risk after 1= $>1\%$, 2= $>23\%$
 - ✓ There for women with prior history of molar pregnancy, 1st TM ultrasound examination is strongly recommended.
 - Extreme maternal age (<15 and >35years): - 2x for > 35, 7.5X for > 40 years, 45 years -1% , 50 years – 17%.
 - ✓ Adolescents have **a 7x increased** risk of developing a complete mole.
 - ✓ These findings suggest that ova from adolescent and older women may be more susceptible to abnormal fertilization, resulting in a complete hydatidiform mole.
 - ✓ This is much **greater for complete mole** than partial mole.

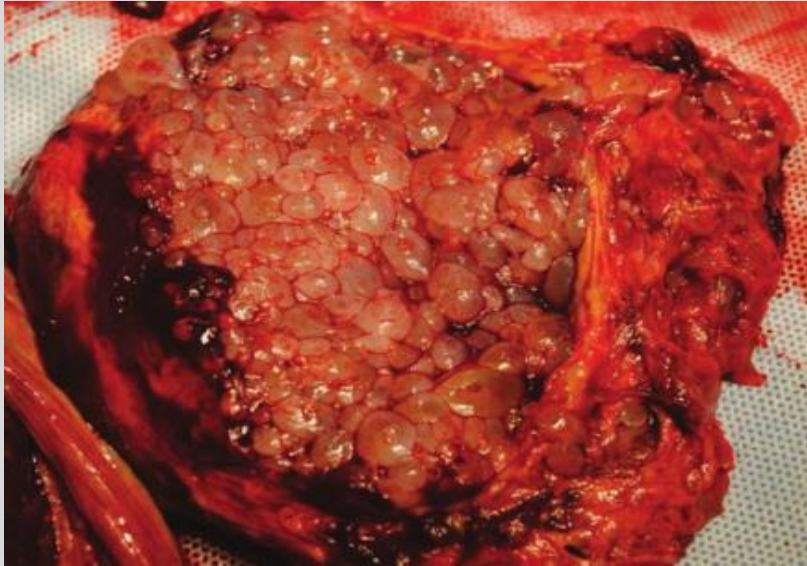
- Prior unsuccessful pregnancies: -
 - spontaneous abortion => 2x increased risk of GTD
- Combination oral contraceptive pills use: - doubles the risk
- Blood type AB, A, and B impregnated by blood group O....Those with blood group AB tend to have worse prognosis
- Vitamin A deficiency and low dietary intake of carotene are associated only with an increased risk of **complete moles**
- Higher educational levels, smoking, irregular menstrual cycles, and obstetric histories in which only male infants are among the prior live births are linked with **partial moles**

Pathogenesis

Two forms moles exist: Complete and partial moles, which are differentiated in **karyotype**, in **histologic** features, and **clinical** presentation.

Cytogenetic studies demonstrate that:

- A complete mole are usually euploid paternal in origin, and sex chromatin positive- 46 XX (90%) or 46 XY (10%).
 - It develops when an empty ovum with an absent or inactivated nucleus is fertilized by a haploid sperm that duplicates its own chromosomes or by two haploid sperm.
 - Although nuclear DNA is entirely paternal, mitochondrial DNA remains maternal in origin.



Picture of complete mole

A partial mole, on other hand, is triploid-69 XXX, 69 XXY or 69 XYY.

- It arises when an ovum with active nucleus is fertilized by a duplicated or two haploid sperm.

Comparison between Complete and Partial Mole

Feature	Complete Mole	Partial Mole
Karyotype	46,XX or 46,XY	69,XXX or 69,XXY
Pathology		
Fetus/embryo	Absent	Present
Villous edema	Diffuse	Focal
Trophoblastic proliferation	Can be marked	Focal and minimal
p57Kip2 immunostaining	Negative	Positive
Clinical presentation		
Typical diagnosis	Molar gestation	Missed abortion
Postmolar malignant sequelae	15%	4–6%

Clinical features

Patients with complete molar pregnancy are increasingly being diagnosed earlier in pregnancy and treated before they develop the classic clinical signs and symptoms.

- This may be because of changes in clinical practice, such as *the frequent use of serum hCG measurement and vaginal US in early pregnancy* in women with vaginal staining and even asymptomatic women.

Symptoms

- Vaginal bleeding:
 - The most common symptom causing patient to seek treatment in 97% previously, but currently 84 % of patients.
 - It results from separation of the molar villi from underlying decidua.
 - It usually is associated with passage of grape like vesicles.
- Pelvic pressure or pain: due to enlarged uterus in 28% or cystic ovaries
- Nausea and vomiting: which is excessive and requires medical attention (HEG due to excessively elevated β -hCG), currently 8%
- Chest pain and dyspnea: B/c of trophoblastic embolization in 2 % of patients.
- Heat and cold intolerance, palpitation, sweating: B/c of hyperthyroidism which was observed in 7% of patients with complete molar gestation as a result of thyrotrophic effect of hCG, currently rare
- Headache, blurring of vision, ABM, and epigastric pain: early onset pre-eclampsia previously 27%, currently 1 in 74
- Cramping lower abdominal pain: b/c of uterine contraction to evacuate its content
- Absent fetal kick:
 - NB. 100% feel fetal quickening by 24 weeks of gestational age, if not it needs to be evaluated.

Signs

- Acute sick looking: bleeding, in respiratory distress, in pain
- Tachycardia, tachypnea, elevated (pre-eclampsia) or decreased (heavy bleeding) blood pressure
- Pale conjunctiva, icteric sclera, dry tongue and buccal mucosa
- Thyroid swelling
- IC/SC retraction, crackles, decreased air entry
- Active precordium, S-3 gallop, systolic murmur- high output heart failure 2ndry to anemia
- Soft abdomen (Duffy abdomen), no palpable fetal parts, uterine size greater than gestational age (positive discrepancy by > 2 weeks), negative fetal heart sound.
- Adnexal mass,
- Warm moist skin, tremor

Patients with partial hydatidiform mole usually do not have the dramatic clinical features characteristic of complete molar pregnancy.

- In general, these patients have the signs and symptoms of incomplete or missed abortion, and partial mole can be diagnosed after histologic review of the tissue obtained by curettage.

Natural history

Complete mole

- Complete moles have a potential for local invasion and dissemination.
- After molar evacuation, local uterine invasion occurs in 15% of patients, and metastasis occurs in 4%.
- Risk factors associated with developing post molar tumor are:
 - hCG level >.100,000 mIU/mL
 - Excessive uterine enlargement > 14 weeks

- Theca lutein cysts 6 cm in diameter
- Older patients are also at increased risk of developing postmolar GTT

Partial mole

- Persistent tumor, usually nonmetastatic, develops in approximately **2% to 4%** of patients with a partial mole, and chemotherapy is required to achieve remission

Investigation

- **Pelvic Ultrasonography:-**
 - Ultrasonography is a reliable and sensitive technique for the diagnosis of complete molar pregnancy. ***Its sensitivity is 70 to 90 %***
 - ✓ Sonographic features suggestive of a complete mole include:
 - Absence of an embryo or fetus
 - Absence of amniotic fluid
 - Central heterogeneous mass with numerous discrete anechoic spaces- this has classically been described as a “**snowstorm or Swiss chess pattern**”
 - Ovarian theca lutein cysts
 - **Partial mole:** Based on ultrasound findings, a partial mole is diagnosed as ***missed or incomplete abortion in 15 to 60 % of cases***
 - ✓ Sonographics features suggestive of a partial molar pregnancy include
 - Placenta with one or more abnormal findings: enlarged, cystic spaces and increased echogenicity of chorionic villi
 - Increased transverse diameter of gestational sac- due to embryopathy of tripoidy
 - Amniotic fluid is present, but the volume is reduced
 - Theca lutein cysts are usually absent

- **CBC:** Anemia
- **Blood group and Rh**
- **hCG:** sometimes urine hCG is negative because when excessively elevated hCG level present, both the capture and tracer antibodies used in immune-radiometric assays become saturated, preventing the binding of the two to create sandwich, resulting in false negative result.
 - So after we have collected urine specimen we have to dilute it with normal saline in 1 to 9 ratios. This false negative test is called hook effect.
 - Minimum hCG level for a positive test
 - ✓ Urine pregnancy test
 - Qualitative test: 20 to 50milli-IU/mL, depending on test
 - ✓ Serum pregnancy test
 - Qualitative test: 5 to 10milli-IU/mL, depending on test
 - Quantitative test: 1 to 2milli-IU/mL for an ultrasensitive test
 - Causes of a false-negative test
 - ✓ Performed too soon after conception; hCG concentration is below threshold for a positive test
 - ✓ The hCG isoform measured is different from the hCG isoform in the sample (pertains mostly to urine tests)
 - ✓ **Hook effect** due to extremely high hCG concentration (>500,000 milli-IU/mL, these levels are most commonly seen in GTN)
 - Causes of a false-positive test
 - ✓ Recent first-trimester pregnancy loss (induced or spontaneous) in which hCG levels, though declining, are still elevated
 - ✓ Pregnancy loss very soon after implantation ("biochemical pregnancy")
 - ✓ hCG secretion from a tumor
 - ✓ Pituitary hCG secretion

- ✓ Interference from human antibodies against animal antibodies or heterophilic antibodies (serum test positive but urine hCG will be negative)
- ✓ Patient has received a medication containing hCG or certain antibodies
- Chest x-ray for those with pulmonary symptoms
- Histopathologic examination:
 - Two specimens
 - ✓ One specimen from the content of the uterus, to confirm GTD and determine whether it is complete or partial mole
 - ✓ Another specimen from the uterine wall to know whether it is invasive or not.
- Metastatic workup: RFT, LFT, TSH, serum electrolyte, chest X-ray, CT/MRI

Complications

- Hyper emesis gravidarum (HEG): in 8% of patients
- Pre-eclampsia before 20 weeks of GA: it was observed in 27% of patients with a complete hydatidiform mole.
 - Preeclampsia develops almost exclusively in patients with excessive uterine size and markedly elevated hCG levels.
- Hyperthyroidism: in 7% of patients with a **complete molar** gestation.
 - Anesthesia or surgery may precipitate thyroid storm.
 - ✓ Thus, if hyperthyroidism is suspected before the induction of anesthesia for molar evacuation, β -adrenergic blocking agents should be administered.
 - ✓ Thyroid storm may be manifested by hyperthermia, delirium, convulsions, tachyarrhythmia, high-output heart failure, or cardiovascular collapse.
 - Hyperthyroidism develops almost exclusively in patients with very high hCG levels.

✓ Some investigators suggest that hCG is the thyroid stimulator in women with molar pregnancy because positive correlations between serum hCG levels and total T₄ or T₃ concentrations were observed

- **Theca lutein cyst:**
 - Prominent theca lutein ovarian cysts (6 cm in diameter) develop in about one half of patients with a complete mole.
 - After molar evacuation, theca lutein cysts normally regress spontaneously within 2 to 4 months.
 - If it is bilateral it is sign of poor prognosis.
- **Anemia:** Because of heavy bleeding. Currently in 5%
- **Trophoblastic embolization:** in 2% of patients
 - These patients may have chest pain, dyspnea, tachypnea, and tachycardia and may experience severe respiratory distress during and after molar evacuation.
 - Auscultation of the chest usually reveals diffuse rales, and chest radiographic evaluation may show bilateral pulmonary infiltrates. Respiratory distress usually resolves within 72 hours with cardiopulmonary support.
- Respiratory insufficiency may result from trophoblastic embolization and the cardiopulmonary complications of thyroid storm, preeclampsia, and massive fluid replacement.
- Local invasion and dissemination

Treatment of Benign GTD

After the patient's condition has been stabilized, a decision must be made concerning the most appropriate method.

- **Suction Curettage**
 - Preferred method for who desire fertility & involves the following steps:
 - ✓ Oxytocin infusion must be setup/made ready before induction of anesthesia.

- ✓ Cervical dilatation

Prostanoids to ripen the cervix: are not recommended as these drugs can induce uterine contractions, which might increase the risk of trophoblastic embolization to the pulmonary vasculature

- ✓ Suction curettage/electrical is preferred.

For uterine size < 14 weeks better to put one hand on fundus and massage the uterus to stimulate contraction and reduce risk of perforation.

- **Sharp Curettage**

- To remove any residual molar tissue
- Send it separately for histopathology.

- **Anti-D immune globulin**

- for Rh negative give at the time of evacuation

- **Prophylactic Chemotherapy**

- For high-risk is recommended (one of the following required):
 - hCG level >100,000 mIU/mL
 - Excessive uterine enlargement (> 14-16 weeks)
 - Theca lutein cysts $\geq$ 6 cm in diameter
 - Poor compliance for follow up.

- **Hysterectomy**

- For patients who doesn't need their fertility.
- Hysterectomy does not prevent metastasis; patients still require follow-up with assessment of hCG levels. But it markedly reduces risk of subsequent GTN development.

Follow-up

- Patient needs *follow up by β -subunit hCG weekly until levels are normal for three consecutive weeks, then monthly until normal for 6 months* because there is risk of postmolar malignancy.

- The initial β -hCG level is obtained within **48 hours after evacuation**.
 - This serves as the baseline, which is compared with β -hCG quantification done thereafter every 1 to 2 weeks until levels progressively decline to become undetectable.
- The median time for such resolution is 7 weeks for partial moles and 9 weeks for complete moles.
- After achieving non-detectable serum hCG levels, the risk of developing GTN **approaches zero**
- Effective contraception during the entire follow-up is mandatory.
 - Because of the potential risk of uterine perforation, intrauterine devices should not be inserted until the patient achieves a normal hCG level.
 - Estrogen-progestin contraceptives are preferred because of their lower failure rate and relatively low incidence of irregular bleeding, since this symptom may raise concern about recurrence.

When follow-up is difficult as per protocol:

- Patient can be appointed at 3 months (average time when hCG expected to be normal).
- Counsel patient on symptoms of persistence

GESTATIONAL TROPHOBLASTIC NEOPLASIA

This term primarily encompasses pathologic entities that are characterized by aggressive invasion of the endometrium and myometrium by trophoblast cells.

Histologic categories include

- Common tumors: the invasive mole and gestational choriocarcinoma
- Rare tumors placental-site trophoblastic tumor and epithelioid trophoblastic tumor

Most cases follow a hydatidiform mole. *Rarely, GTN can develop after a term a live birth, miscarriage, or termination.*

Non metastatic Disease

- Locally invasive GTT develops in about 15% of patients after evacuation of a complete mole and infrequently after other gestations.
- These patients usually present with the following symptoms:
 - Irregular vaginal bleeding
 - Theca lutein cysts
 - Uterine sub-involution or asymmetric enlargement
 - Persistently elevated serum hCG levels.
- The trophoblastic tumor may perforate the myometrium, causing intraperitoneal bleeding, or erode into uterine vessels, causing vaginal hemorrhage.
- Bulky, necrotic tumor may involve the uterine wall and serve as a nidus for infection.
 - Patients with uterine sepsis may have a purulent vaginal discharge and acute pelvic pain.
- After molar evacuation, persistent GTT may exhibit the histologic features of either hydatidiform mole or Choriocarcinoma.
- After a non-molar pregnancy, however, persistent GTT always has the histologic pattern of choriocarcinoma.
 - Histologically, choriocarcinoma is characterized by sheets of anaplastic syncytiotrophoblast and cytotrophoblast without chorionic villi.

Placental-site Trophoblastic Tumor

- Placental-site trophoblastic tumor is an uncommon but important variant of choriocarcinoma that consists predominantly of intermediate trophoblast.

- Relative to their mass, these tumors ***produce small amounts of hCG*** and human placental lactogen (hPL), and they tend to remain confined to the uterus, metastasizing late in their course.
- In contrast to other trophoblastic tumors, placental-site tumors are relatively ***insensitive to chemotherapy***.

Metastatic disease

- Metastatic GTT occurs in about 4% of patients after evacuation of a complete mole, but it is seen more often when GTT develops after nonmolar pregnancies.
- Metastasis is ***usually associated with choriocarcinoma***, which has a tendency toward early vascular invasion with widespread dissemination.
- Because trophoblastic
- tumors often are perfused by fragile vessels, they are frequently hemorrhagic. Symptoms of metastases may result from spontaneous bleeding at metastatic foci
- The most common sites of metastases are lung (80%), vagina (30%), pelvis (20%), liver (10%), and brain (10%).
 - Pulmonary
 - ✓ At the time of diagnosis, lung involvement is visible by chest radiography in 80% of patients with metastatic GTT.
 - ✓ Patients with pulmonary metastasis may have chest pain, cough, hemoptysis, dyspnea, or an asymptomatic lesion visible by chest radiography.
 - ✓ GTT may produce four principal pulmonary patterns:
 - An alveolar or “snowstorm” pattern
 - Discrete rounded densities: cannon ball appearance
 - Pleural effusion
 - An embolic pattern caused by pulmonary arterial occlusion

- ✓ Because respiratory symptoms and radiographic findings may be dramatic, the patient may be thought to have a primary pulmonary disease.
- ✓ Pulmonary hypertension may develop in patients with GTT secondary to pulmonary arterial occlusion by trophoblastic emboli. T
- ✓ The development of early respiratory failure requiring intubation is associated with a poor clinical outcome.
- Vaginal
 - ✓ Vaginal metastases occur in 30% of the patients with metastatic tumor.
 - ✓ These lesions are usually highly vascular and may bleed vigorously when biopsied.
- Hepatic
 - ✓ Liver metastases occur in 10% of patients with disseminated trophoblastic tumor.
 - ✓ Hepatic involvement is encountered almost exclusively when there is a protracted delay in diagnosis and the patient has an extensive tumor burden.
 - ✓ Epigastric or right upper quadrant pain may develop if metastases stretch the hepatic capsule.
 - ✓ Hepatic lesions may be hemorrhagic, causing hepatic rupture and exsanguinating intraperitoneal bleeding.
- Central Nervous System
- Metastatic trophoblastic disease involves the brain in 10% of patients. Cerebral involvement is generally seen in patients with advanced disease; virtually all patients with brain metastasis have concurrent pulmonary or vaginal involvement or both.
- Because cerebral lesions may hemorrhage spontaneously, patients may develop acute focal neurologic deficits.

Diagnostic evaluation

All patients with persistent GTT should undergo a careful pretreatment evaluation, including the following:

- Measurement of the serum hCG value:

FIGO Criteria for Gestational Trophoblastic Neoplasia Diagnosis

β -hCG level plateau persists in four measurements during a period of 3 weeks or longer (days 1, 7, 14, and 21)

β -hCG level rise in 3 weekly consecutive measurements or longer, over a period of 2 weeks or more (days 1, 7, and 14)

β -hCG level remains elevated for 6 months or more

Histologic diagnosis of choriocarcinoma

- Hepatic, thyroid, and renal function tests
- Determination of baseline peripheral white blood cell and platelet count
- The metastatic workup should include the following:
 - Chest radiograph or computed tomography (CT) scans
 - Ultrasonography or CT scan of the abdomen and pelvic CT
 - Magnetic resonance imaging (MRI) scans of the head
 - In patients with choriocarcinoma or metastatic disease, hCG levels may be measured in the cerebrospinal fluid (CSF) to exclude cerebral involvement if the results of CT scanning of the brain are normal. The ratio of plasma-to-CSF hCG tends to be lower than 60 in the presence of cerebral metastases.

Staging

An anatomic staging system for GTT has been adopted by the International Federation of Gynecology and Obstetrics

Stage I	Patients have persistently elevated hCG levels and tumor confined to the uterine corpus.
Stage II	Patients has metastases to the vagina and pelvis or both
Stages III	Patients have pulmonary metastases with or without uterine, vaginal, or pelvic involvement.
Stage IV	Patients have advanced disease and involvement of the brain, liver, kidneys, or gastrointestinal tract. <i>These patients are in the highest risk category because they are most likely to be resistant to chemotherapy.</i>

Management

Depends on:

- WHO's PSI
- Disease stage
- Previous chemotherapy

Modified WHO Prognostic Scoring System as Adapted by FIGO

Scores	0	1	2	4
Age (yr)	< 40	≥ 40	—	—
Antecedent pregnancy	Mole	Abortion	Term	—
Interval months from index pregnancy	< 4	4–6	7–12	> 12
Pretreatment serum β -hCG (mIU/mL)	< 10^3	10^3 – $< 10^4$	10^4 – $< 10^5$	≥ 10^5
Largest tumor size (including uterus)	< 3 cm	3–4 cm	≥ 5 cm	—
Site of metastases	—	Spleen, kidney	GI	Liver, brain
Number of metastases	—	1–4	5–8	> 8
Previous failed chemotherapy drugs	—	—	1	≥ 2

- When the prognostic score is >7 , the patient is categorized as **high risk**
 - They require intensive combination chemotherapy to achieve remission.
- Patients with stage-I disease usually have a low-risk score, and those with stage-IV disease have a high-risk score.
- The distinction between low and high risk applies mainly to patients with stage-II or III disease.
- Malignant GTD is curable in 85 – 100% as they are Chemo sensitive.

Risk stratification: Approximately 80% are low risk group

Low risk	High risk
Stage I and low risk stage II & III	Stage IV & high risk stage II & III
Needs single agent chemoRx	Combination chemoRx

Low risk

- Hysterectomy plus single agent chemotherapy: If no need of fertility
- Chemotherapy alone
 - Commonly used agents → Methotrexate (M), Actinomycin-D (A).
 - Alternatives → Etoposide (E) & 5-Flourouracil (5-FU).
- **Refractory to single agent chemotherapy**
- When a patient's disease is resistant to single-agent chemotherapy and she desires to preserve fertility,
 - Combination chemotherapy should be administered.
 - MAC regimen:
 - MTX (methotrexate) 1mg/kg/d IM on day 1,3,5,7 & leucovorin 0.1mg/kg/d IM on day 2,4,6,8.

- Actinomycin 12mcg/kg/d IV for 5 ds. Cyclophosphamide 3mg/kg/d for 5 ds Cycles repeated Q 21ds. If resistant to this regimen EMA/CO used.
- Local uterine resection may be considered.

High risk: stage II, III & stage IV

- EMA/CO - preferred initial regimen.
 - Etoposide 100mg/m² iv/30' on day 1&2.
 - MTX 100mg/m² IV push followed by 200mg/m² iv/12hrs on day 1.
 - Actinomycin 0.5 mg IV bolus on day 1&2.
 - Leucovorin 15 mg Q 12hr for 4-doses Cyclophosphamide 600mg IV on day 8.
 - Oncovin 1mg/m² IV on day 8.

Repeated every 2wks until remission (normal serum hCG over 3 consecutive wks & negative radiographic evident disease). Complete remission rate 78%.

- Other regimens:
 - EMA as effective as EMA/CO
 - EMA/EP:
 - ✓ Given if relapse occur after EMA/CO and if refractory to MAC or EMA.
 - PVB/PEB:
 - ✓ if resistant to EMA/CO or EMA/EP and it is less toxic than EMA/EP

In patient with GTN Hysterectomy may be required for

- *To primarily treat PSTI, ETI, or other chemotherapy-resistant disease.*
- *To control uterine hemorrhage and sepsis*
- *To reduce the trophoblastic tumor burden and there by limit the need for multiple course s of chemotherapy*

However, the risk of GTN persistence after hysterectomy remains approximately 20 to 30 percent, and these patients should be monitored postoperatively.

Duration of therapy

- After normal hCG levels are attained, at least two additional courses of chemotherapy are administered to reduce the risk of relapse.

Monitoring during & after chemotherapy

- Stage I-III
 - Weekly serum hCG until 3- consecutive normal, monthly until 12 consecutive normal.
- Stage IV:
 - Weekly serum hCG until 3 consecutive normal, monthly until normal for consecutive 24 months.
- CBC, organ function tests based on clinical condition of the patient, U/S, chest X-ray, Serum electrolytes.
- Effective contraception should be used during serum hCG follow-up

Phantom HCG

- Occasionally, persistent mild elevations of serum β -hCG are detected and lead physicians to erroneously treat patients with cytotoxic chemotherapy or hysterectomy or both, when in reality no true β -hCG molecule or trophoblastic disease is present.
- This "phantom" β -hCG reading results from serum heterophilic antibodies that interfere with the β -hCG immunoassay and cause a false-positive result.

- Several steps can clarify the diagnosis
 - A urine pregnancy test can be performed.
 - ✓ With phantom β -hCG, the heterophilic antibodies are ***not filtered or excreted through kidney***. Thus, these test-altering antibodies will be absent from the urine, and urine testing will show true negative results for β -Hcg
 - Performing serial dilutions of the serum sample leads to a proportional decline in the β -hCG level if β -hCG is truly present.
 - ✓ However, phantom β -hCG measurements will be unchanged by dilution.

SAMPLE HISTORY

ID: 42 years old

C/C: Vaginal bleeding of 2 days duration

HPI:

This is a gravida 4 para III (all alive and males) mother whose LNMP was 25/04/11 E.C which makes her EDD on 29/01/11E.C, and gestational age 25 weeks. The cycle was regular coming every 4 weeks and she has never used contraception for 8 month before her LMNP. She suspected she was pregnant after she missed one menstrual cycle. Then diagnosis of pregnancy was confirmed by urine pregnancy test. She has one ANC contact at local health center. It was on 5th month of amenorrhea. Detailed history was taken and general physical examination and laboratory tests were done. She gave blood and urine sample. She was given TT vaccination and she was told everything was normal and that she should continue her follow up every 4 weeks.

She doesn't felt any fetal quickening still now.

Currently she is presented with vaginal bleeding of 2 days duration which is associated with passage of grape like vesicles. The bleeding is bright red and non-clotting. It soaked 2-3pads completely per day and her underwear is also soaked. But not associated with coitus or deep washing. In association to this she is also complaining cramp lower abdominal pain which does not radiate to another site and no aggravating or relieving factors noticed by the patient. The pain is severe enough to prevent her from her daily activities. She also complains rapid and excessive enlargement of her abdomen compared to her prior pregnancies. She has also history of excessive sweating which wets her night clothes and palpitation of the same duration. She complains tinnitus and vertigo of 1 day's duration. For all this complaint she was taken to local health center where they referred her directly to our hospital for better investigation and treatment.

Otherwise

She has no cough chest pain or shortness of breath

She has no excessive form of nausea and vomiting than her prior pregnancy

She has no history of headache, vision disturbances,

She has no history of epigastric pain or yellowish discoloration of skin.

She has no history of abnormal body movement or loss of consciousness

She has no history of smoking, menstrual irregularities.

She has no previous history of similar complaint

She has no history vaginal discharge or fever.

She has no history of trauma to abdomen and genital area

She has no history of bleeding from other sites

She has no history of river water contact

Her serostatus for RVI is negative

She has completed her family size

She eat 4 meals per day (injera, fruits, meat, vegetables and cereals)as the pre pregnancy time. The pregnancy was unplanned, but wanted and supported.

Ass't: R/o GTD + completed family size

NB: mentioning the completeness of the family size is important for the management purposes, for example, if she has completed her family size hysterectomy is the best treatment option.

DDX for GTD includes

- Abortion (incomplete, missed)
- Cervicitis
- Cervical cancer
- Ectopic pregnancy
- Vaginitis
- Trauma
- Ectropion

Chapter 22

Pelvic Pain

Prepared by Dr. Balkewkal Kebede (MI)

Classification

- Acute pelvic pain
- Cyclic pelvic pain
- Chronic pelvic pain

Acute Vs Chronic pelvic pain

	Acute pelvic pain	Chronic pelvic pain
Duration	Less than 7 days	Greater than 6 months
Onset	Rapid	Gradual
Type	Inflammatory	Neuropathic
Etiology	Infection, ischemia, or chemical irritation	Often it is obscure.
Association	Often associated with ▪ Profound autonomic reflex responses, such as ○ nausea, emesis, diaphoresis, & apprehension, ▪ unstable vital signs and obvious abnormalities on physical examination and laboratory assessment	Obvious autonomic reflex responses are not present ▪ is characterized by neurological, psychological, and behavioral alterations, like anxiety and depression ▪ vital sign derangements and obvious anatomic abnormalities may not be there
Treatment	Is directed against the underlying condition	If focused on treating the pain symptoms

Acute pelvic pain

- Acute pain is intense and characterized by sudden onset, sharp rise, and short course.
- It can be a somatic or visceral pain:
 - **Somatic pain** arises from the parietal peritoneum, muscle, subcutaneous tissue or the skin. it is typically sharp and localized. it is often unilateral
 - **Visceral pain** arises from the visceral peritoneum or the viscera.
 - The viscera are relatively insensitive to pain.
 - The first perception of visceral pain is a vague, deep, poorly localizable sensation associated with autonomic reflex responses; however, once the pain becomes localized, the pain is called referred pain.
 - The pain is often diffuse and dull ache, but it can be localized to the midline

DDx for Acute pelvic pain

Gynecologic Disease or Dysfunction ▪ Complication of pregnancy ○ Ectopic pregnancy ○ Abortion, threatened or incomplete ▪ Acute infections ○ Endometritis ○ Acute PID ○ Tubo-ovarian abscess ▪ Adnexal disorders ○ Hemorrhagic ovarian cyst ○ Torsion of adnexa ○ Rupture ovarian cyst ○ Prolapsing leiomyoma	Gastrointestinal ▪ Gastroenteritis ▪ Appendicitis ▪ Bowel obstruction ▪ Diverticulitis ▪ Irritable bowel syndrome Genitourinary ▪ Cystitis ▪ Pyelonephritis ▪ Ureteral lithiasis Musculoskeletal ▪ Abdominal wall hematoma ▪ Hernia

Ectopic pregnancy

- An ectopic pregnancy is defined as implantation of the fetus in a site other than the uterine cavity. The fallopian tube is the most common site of implantation (~95%)
- Implantation of the fetus in the fallopian tube produces pain only with acute dilation of the tube. If tubal rupture occurs, localized abdominal pain tends to be temporarily relieved and is replaced by generalized pelvic and abdominal pain as the hemoperitoneum develops.
- A mass in the cul-de-sac may produce an urge to defecate

Torsion of Adnexa

- Torsion (twisting) of the vascular pedicle of an ovary, fallopian tube, paratubal cyst, or rarely just a fallopian tube results in ischemia and rapid onset of acute pelvic pain.
- A benign cystic teratoma is the most common neoplasm to undergo torsion.
- Ovarian carcinoma and inflammatory masses are rarely affected by torsion (because of adhesions). It is also unusual for a normal tube and ovary to torque, although a polycystic ovary can undergo torsion
- The pain of torsion can be severe and constant or, if the torsion is partial and intermittently untwists, it could be intermittent.
- The onset of the torsion and the symptoms of abdominal pain frequently coincide with lifting, exercise, or intercourse.
- Autonomic reflex responses are usually present (e.g., nausea, emesis, apprehension).

Ruptured Ovarian Cyst

- Functional cysts (e.g., follicle, corpus luteum) are the most common ovarian cysts and rupture more readily than do benign or malignant neoplasms.

- A hemorrhagic corpus luteum cyst can develop in the luteal phase of the menstrual cycle. Rupture of this cyst can produce either a small amount of intraperitoneal bleeding or frank hemorrhage resulting in significant blood loss and hemoperitoneum.
- Nonmalignant neoplasms, most commonly cystic teratomas (dermoid cysts) or cystadenomas, as well as inflammatory ovarian masses, such as endometriomas, can also leak or rupture.
- A corpus luteum cyst is the most common cyst to rupture and lead to hemoperitoneum.
- Symptoms of a ruptured corpus luteum cyst are similar to those of a ruptured ectopic pregnancy.
- The onset of pain is usually sudden and is associated with increasing generalized abdominal pain and occasionally dizziness or syncope if a hemoperitoneum develops.

Uterine Leiomyomas

- Leiomyomas are uterine smooth muscle tumors (see the myoma section).
- Discomfort may be present when myomas are intramural, fundal, or encroaching on adjacent bladder, rectum, or supporting ligaments of the uterus.
- Although rare, fibroids are more likely to cause dyspareunia and noncyclic pelvic pain
- Acute pelvic pain can develop if the myoma undergoes degeneration or torsion. Degeneration of myomas occur secondary to loss of blood supply, usually attributable to rapid growth associated with pregnancy.
- A pedunculated subserosal leiomyoma can undergo torsion ischemic necrosis; when this situation occurs, it is associated with pain similar to that of adnexal torsion.

- When a submucous leiomyoma becomes pedunculated, the uterus contracts forcefully as if to expel a foreign body, and the resulting pain is similar to that of labor. The cramping pain is usually associated with hemorrhage.

Appendicitis

- Is most common intestinal source of acute pelvic pain in women
- The symptoms and signs of appendicitis can be similar to those of PID.
- The first symptom of appendicitis is typically diffuse abdominal pain, especially periumbilical pain, followed by anorexia, nausea, and vomiting. Within a matter of hours, the pain generally shifts to the right lower quadrant. Fever, chills, emesis, and obstipation may ensue. However, this classic symptom pattern is often absent.

Acute Diverticulitis

- Acute diverticulitis is a condition in which there is inflammation of a diverticulum or out pouching of the wall of the colon, usually involving the sigmoid colon.
- Diverticulitis typically affects postmenopausal women but can occur in women in their 30s and 40s.
- The severe, left lower quadrant pain of diverticulitis can occur following a long history of symptoms of irritable bowel (bloating, constipation, and diarrhea), although diverticulosis usually is asymptomatic. Diverticulitis is less likely to lead to perforation and peritonitis than is appendicitis
- Fever, chills, and constipation typically are present, but anorexia and vomiting are uncommon.

Intestinal Obstruction

- The most common causes of intestinal obstruction in women are postsurgical adhesions, hernia formation, inflammatory bowel disease, and carcinoma of the bowel or ovary.
- Intestinal obstruction is heralded by the onset of colicky abdominal pain followed by abdominal distention, vomiting, constipation, and obstipation.

- Higher and more acute obstruction results in early vomiting, whereas colonic obstruction presents with a greater degree of abdominal distention and obstipation.
- Vomiting first consists of gastric contents, followed by bile, then material with feculent odor, depending on the level of obstruction.

Ureteral colic

- Ureteral colic is a cramping lower abdominal pain that is due to ureteral lithiasis is caused by a sudden increase in intraluminal pressure and associated inflammation.
- The pain of lithiasis is typically severe and crampy; it can radiate from the costovertebral angle to the groin. Hematuria is often present

UTI

- Urinary tract infections producing acute pain include cystitis and pyelonephritis. The most common microbes causing urinary tract infections are *E. coli* followed by *Proteus*, *Klebsiella*, and *Pseudomonas*.
- Cystitis is associated with dull suprapubic pain, urinary frequency, urgency, dysuria, and occasionally hematuria. Because urethritis can occur secondary to chlamydia or gonorrhea and has similar symptoms, these infections must be ruled out if appropriate.
- Pyelonephritis is associated with flank and costovertebral angle pain, although lateralizing lower abdominal pain occasionally is present.
- On physical examination there is pain with firm pressure over the costovertebral angle in the case of lithiasis or pyelonephritis. Peritoneal signs are absent. Suprapubic tenderness may accompany cystitis.

Pelvic Inflammatory Disease (PID)

- Pelvic inflammatory disease (PID) comprises a spectrum of inflammatory disorders of the upper female genital tract (above the internal Os), including any

combination of endometritis, salpingitis, tubo-ovarian abscess, and pelvic peritonitis.

- Involvement the fallopian tube in PID is of special importance because of subsequent complication (infertility).

Etiology and Risk factors

Pelvic inflammatory disease is commonly caused by *N. gonorrhea*, *C. trachomatis*, and *T. vaginalis*, although the infection is usually polymicrobial.

Upper genital tract infections are usually caused by bacteria that ascend from the lower reproductive tract.

Risk Factors for PID

- *Douching*
- *Single status*
- *Substance abuse*
- *Multiple sexual partners*
- *Lower socioeconomic status*
- *Recent new sexual partner(s)*
- *Younger age (10 to 19 years)*
- *Other sexually transmitted infections*
- *Sexual partner with urethritis or gonorrhea*
- *Previous diagnosis of pelvic inflammatory disease*
- *Endocervical testing positive for *N. gonorrhea* or *C. trachomatis**
- *Not using mechanical and/or chemical contraceptive barriers*

Classification of PID

Pelvic inflammatory disease can be segregated into “silent” PID and PID. The latter can be further subdivided into acute and chronic.

Silent PID

- Silent PID is thought to follow multiple or continuous low- grade infection in asymptomatic women.
- Is not a clinical diagnosis. Rather, it is an ultimate diagnosis given to women with tubal-factor infertility who lack a history compatible with upper tract infection.
- Many of these patients have antibodies to C trachomatis and/or N gonorrhea.
- At laparoscopy or laparotomy,
 - affected women may have evidence of prior tubal infection such as adhesions, but for the most part, the fallopian tubes are grossly normal or they may have hydrosalpinx
 - fine adhesions between the liver capsule and anterior abdominal wall may also reflect prior silent disease

Acute PID

- Acute gonococcal PID is manifested by the acute onset of pelvic pain that increases with movement, fever, purulent vaginal discharge, and sometimes nausea and vomiting.
- The pain is often associated with a menstrual period.
- Chlamydial salpingo-oophoritis is associated with more insidious symptoms.
- On physical examination:
 - Direct and rebound abdominal tenderness with palpation are usually notable
 - Lack of a discrete mass or masses differentiates acute salpingo-oophoritis from TOA or torsion
- The most recent diagnostic criteria presented by the CDC (2015) are for sexually active women at risk for STDs who have pelvic or lower abdominal pain and in whom other etiologies are excluded or unlikely.

PID is diagnosed if uterine tenderness, adnexal tenderness, or cervical motion tenderness is present. One or more of the following enhances diagnostic specificity:

1. Oral temperature $> 38.3^{\circ}\text{C}$ (101.6°F)
2. Mucopurulent cervical discharge or cervical reliability
3. Abundant WBCs on saline microscopy of cervical secretions
4. Elevated erythrocyte sedimentation rate (ESR) or C-reactive protein (CRP)
5. Presence of cervical N gonorrhea or C trachomatis

Thus, a diagnosis of PID is one typically based on clinical findings.

Tubo-ovarian Abscess (TOA)

- TOA, a sequel of acute salpingitis, are usually bilateral, but unilateral abscess formation can occur.
- The symptoms and signs are similar to those of acute salpingitis, although pain and fever have often been present for longer than 1 week before presentation to the emergency room.
- A ruptured TOA is a life-threatening surgical emergency because gram-negative endotoxic shock can develop rapidly.
- Tubo-ovarian abscesses can be palpated on bimanual examination as very firm, exquisitely tender, bilateral fixed masses. The abscesses can be palpated in the pelvic cul-de-sac.

Based on the antecedent event, PID can also be classified as:

- Post-partum
- Post abortal
- Post-operative
- Post IUCD/Instrumental
- Post STD
- Post TB

Cyclic pelvic pain

- Cyclic pain refers to pain that occurs with a definite association to the menstrual cycle.

Dysmenorrhea or painful menstruation is the most common cyclic pain phenomenon, affecting as many as 50% of menstruating women. It is classified as primary and secondary dysmenorrhea.

	Primary Dysmenorrhea	Secondary dysmenorrhea
Underlying pelvic pathology	Absent	Present
Onset in relation to menarche	Within 1-2yrs	Years after menarche
Onset in relation to menses and duration	Begins a few hours before or just after the onset of a menstrual period and may last 48 to 72 hours	Begins 1-2wks before menstrual flow and persists until a few days after the cessation of bleeding
Mechanism	▪ Increased endometrial prostaglandin production ▪ Higher uterine tone with high- amplitude contractions causing dysmenorrhea.	▪ Diverse and not fully elucidated, ▪ excess PG production or ▪ hypertonic uterine contractions secondary to cervical obstruction, intrauterine mass, or the presence of a foreign body
Pain relief with treatment	NSAIDs are effective	NSAIDs and hormonal contraceptives are less likely to provide relief

Primary dysmenorrhea affects **younger women** but may persist into the 40s. The pain is similar to labor, with suprapubic cramping, and maybe accompanied by lumbosacral backache.

On examination, except suprapubic & uterine tenderness on bimanual palpation, there is no abnormal finding.

Chronic pelvic pain

- Chronic pelvic pain remains an inclusive, general diagnosis.
- Various forms of chronic pelvic pain affect 12% to 20% of women in the United States.
- Patients with chronic pelvic pain are frequently anxious and depressed. Their marital, social, and occupational lives are usually been disrupted.
- These patients have often had poor treatment outcomes and may have undergone multiple unsuccessful surgical procedures for pain.
- Approximately 60% to 80% of patients undergoing laparoscopy for chronic pelvic pain have no intraperitoneal pathology, nor do they have tissue distortion that correlates with the pain.

In Approach to chronic pelvic pain focus on:

History

- In addition to characterization the pain, history of any musculoskeletal trauma, surgical procedure (including cesarean delivery) and medical interventions done for similar or other complaint are very important.
- Current and past psychological history including psychosocial factors; history of past (or current) physical, sexual, or emotional abuse; history of psychiatric hospitalization; suicide attempts; and chemical dependency need to be included.

P/E

- Abdomen
 - Inspect for scars (esp. Surgical adhesion) and hernia
 - Demonstrate Carnett test, which is an evaluation of the abdomen with muscles tensed (head raised off the table or with straight leg raising) to differentiate abdominal wall and visceral sources of pain. Abdominal wall pain is augmented and visceral pain is diminished with these maneuvers
 - Perform superficial and deep palpation

- Increased bowel sound may indicate irritable or inflammatory bowel disease.

DDx Chronic Pelvic Pain

Gynecologic ▪ Non-cyclic ○ Adhesions ○ Endometriosis ○ Salpingo-oophoritis ○ Retained ovary syndrome ○ Pelvic congestion ○ Ovarian neoplasms ▪ Cyclic ○ Primary dysmenorrhea ○ Mittelschmerz ○ Secondary dysmenorrhea ○ Endometriosis ○ Asherman syndrome ○ Endometrial polyps ○ Uterine leiomyomata ○ Adenomyosis ○ Ovarian remnant syndrome ○ Chronic functional cysts Gastrointestinal ▪ Irritable bowel syndrome ▪ Ulcerative colitis ▪ Crohn's disease ▪ Carcinoma ▪ Infection	Genitourinary ▪ Recurrent/relapsing cystourethritis ▪ Urethral syndrome ▪ Interstitial cystitis ▪ Ureteral diverticuli or polyps ▪ Carcinoma of the bladder ▪ Ureteral obstruction Neurologic ▪ Nerve entrapment syndrome, ▪ neuroma, or other neuropathies Musculoskeletal ▪ Low-back-pain syndrome ▪ Scoliosis and kyphosis ▪ Spondylolysis ▪ Spondylolisthesis ▪ Spinal injuries ▪ Osteoporosis ▪ Degenerative changes ▪ Coccydynia ▪ Myofascial syndrome ▪ Hernia Systemic ▪ Fibromyalgia ▪ Acute intermittent porphyria ▪ Abdominal migraine ▪ Connective tissue disease including SLE ▪ Lymphoma
--	---

Endometriosis

Endometriosis is defined as the presence of endometrial glands and stroma outside the uterine cavity.

The annual incidence of surgically diagnosed endometriosis is **1.6 per 1000** women aged 15-49yrs.

- The prevalence is different in asymptomatic women (6-11%), in patient with infertility (20- 50%) and in patient with pelvic pain (40-50%, in Novak's gynecology 14th ed this figure is 15-40%).

Risk factors for Endometriosis

- Nulliparity
- Early menarche/late menopause
- Short menstrual cycles
- Prolonged menses
- Müllerian anomalies (increase ectopic implantation of endometrial tissue)

Endometriosis is an **estrogen dependent** disorder, any factor that increase estrogen exposure increases the risk. It is rarely seen in postmenopausal women and nonexistent in prepubertal girls

The definitive cause of endometriosis remains unknown, but different theories have been suggested:

- Retrograde menstruation through the fallopian tubes (the more favored one)
- The stem cell theory
- Aberrant lymphatic and vascular spread of endometrial tissue
- Coelomic metaplasia theory

Endometriosis **may develop anywhere** within the pelvis and on other extra-pelvic peritoneal surfaces. Most commonly, it develops in the dependent areas of the pelvis like

- the anterior and posterior cul-de-sacs,
- other parts of pelvic peritoneum,
- the ovary, and
- uterosacral ligaments are frequently involved

Additionally, the rectovaginal septum, ureter, and bladder and rarely, pericardium, surgical scars, and pleura may be affected. One pathologic review revealed that endometriosis has been identified on all organs except the spleen.

Implants may be: -

- **superficial** or
- **deep infiltrating endometriosis (DIE)**, that is, infiltrative forms that involve vital structures such as bowel, bladder, and ureters
 - Some definitions of DIE also quantify invasion as > 5 mm Ovarian endometriomas

Ovarian endometriomas

- are frequent manifestations of endometriosis
- these smooth-walled, dark-brown ovarian cysts are filled with a chocolate-appearing fluid and may be unilocular or, when larger, multilocular
- their pathogenesis is unclear, yet three theories include invagination of ovarian cortex implants, coelomic metaplasia, and secondary involvement of functional ovarian cysts by endometrial implants located on the ovarian surface

Endometriosis produces a low-grade inflammatory reaction; over time this results in adhesions between confluent pelvic organs.

However, the cause of the pain is not well established

Classification

The American Society for Reproductive Medicine allows endometriosis to be quantified and classified in to four stages.

- With this, endometriosis on the peritoneum, ovaries, fallopian tubes, and cul-de-sac is scored at surgery.
- At these sites, points are assigned for: -
 - disease surface area,
 - degree of invasion,
 - morphology, and extent of associated adhesions
- Also, endometriotic lesions are morphologically categorized as white, red, or black.
- In this system, endometriosis is classified as
 - stage I (minimal, score 1-5),
 - stage II (mild, score 6-15),
 - stage III (moderate, score 16-40), and
 - stage IV (severe, score >40)
- *NB, there is poor correlation with stage of disease and pain & infertility*

Clinical manifestation

Pain:

- Is the most common symptom.
- Endometriosis associated chronic pelvic pain (dysmenorrhea, dyspareunia and non-cyclic pain) is common.
- Less often dyschezia, dysuria and abdominal wall pain are seen.
 - (Dysmenorrhea, dyspareunia, dyschezia and dysuria are sometimes can be remembered as 4Ds of endometriosis)
- Vaginal, uterosacral ligament and rectovaginal septum endometrioses are associated with deep dyspareunia.
- Dyspareunia,

- may be caused by
 - ✓ pressure on inflamed tissues and neural invasion,
 - ✓ stretching and tearing during intercourse of pelvic structures bound by adhesions
- Endometriosis associated dyspareunia is suspected if pain develops after years of pain free intercourse.
- Dysuria and dyschezia are associated with DIE to the respective organs

Infertility:

- Endometriosis is seen in 20-30% of patients with subfertility.
- Adhesion, which is associated with impairment of normal oocyte pick-up and transport by the fallopian tube, is the main cause of infertility.
- Subtle defects like perturbations in follicle development, ovulation, sperm function, embryo quality and development, and implantation are also possible causes

Other symptoms depend on the system/site involved. Like: -

- Rectosigmoid lesions => dyschezia, constipation diarrhea and cyclic hematochezia
- Urinary tract lesion=> dysuria, hematuria, suprapubic pain and frequency
- Abdominal wall lesions=> abdominal pain
- Thoracic lesions=> cyclic chest pain, hemoptysis and pneumothorax associated with menses, which are sometimes refereed as **catamenial**.

Physical examination

- Often no abnormal findings are seen in physical examination. Most common finding is tenderness when palpating the posterior fornix.
- Other frequent findings:
 - Localized tenderness in the pouch of Douglas or uterosacral ligaments,
 - Palpable tender nodules in the posterior cul-de-sac, uterosacral ligament, or rectovaginal septum

- thickening and induration of uterosacral ligaments,
- pain with uterine movement,
- tender enlarged adnexal masses and
- fixation of adnexa or uterus in a retroverted position

Diagnosis

- Endometriosis is a surgical diagnosis based on identification of characteristic lesions.
- Laparoscopy is the gold standard investigation.
- Histologic and imaging studies are also required for definitive diagnosis. Although nonspecific CA125 is elevated in endometriosis.

Management

Management depends on age, symptoms, stage, and the need for fertility. It includes:

1. Expectant Management
2. Medical treatment
3. Hormonal treatment
4. Surgical treatment
5. Combination Medical and surgical treatment

Adhesion

Adhesions noted at the time of laparoscopy are often in the same general region of the abdomen as the source of the pelvic pain; however, neither the specific location nor density of the adhesions correlates consistently with the presence of pain symptoms.

Symptoms

- Noncyclic abdominal pain, sometimes increased with intercourse or activity, is a common source of pain in women with adhesions, but there is no symptom pattern specific for adhesions.
- Chronic pelvic pain developing from adhesions is thought to result from restriction of bowel mobility and distention. Furthermore, dense adhesions involving bowel can cause partial or complete bowel obstruction.

Signs

- The abdominal wall must be carefully evaluated for myofascial or neurological sources of pain.
- Most women with adhesions have had a prior surgical procedure with possible injury to abdominal wall structures that may be the cause of pain.
- Decreased mobility of pelvic organs or adnexal enlargement can often be detected in patients with adhesions.

Approach to Pelvic/lower abdominal pain

History

Detail elaboration of the pain. We can use the mnemonic “OLD CAARTS” to remember the contents we need to include.

- O=> onset
 - Sudden, indicate rupture of a cyst, an ectopic pregnancy, obstruction or perforation of a hollow viscus
 - Gradual, indicate a progressive or chronic condition
- L => location, (on history it is better to say lower abdominal pain)
 - Unilateral, usually indicate adnexal pathology in that side and a localized etiology
 - Bilateral,

- Midline (diffuse or generalized), indicates a visceral pain from distention, stretching or spasm of abdominal organs (which have their origin from the embryological hind gut, like the colon or intraperitoneal portions of the genitourinary tract)
- D => duration, (see the detail below)
 - Acute or Chronic
- C => character
 - Intermittent, usually associated with contraction of the bowel or the uterus
 - Continuous,
 - Sharp, kind of stabbing, indicate a localized cause,
 - Dull ache, diffuse pain from the viscera
 - Cramp, associated with contraction of a hollow viscus, like intestine or uterus in attempt to relive an obstruction
- A => alleviating or aggravating factors
 - Pain aggravated with motion may indicate peritonitis (from rupture of a cyst or appendix)
 - Pain relived with acquiring certain position may indicate obstruction of a vicious organ, as in renal colic. Is the pain relived by any medications?
- A => associated symptoms, (this should be included after completion of the pain description)
 - vaginal bleeding or discharge, amenorrhea, infertility...=> gynecologic pathology
 - dysuria, hematuria, frequency, or urgency.... => urinary pathology
 - diarrhea, constipation, or gastrointestinal bleeding.... => gastrointestinal (GI) disease.
 - Vomiting
 - ✓ In the acute surgical abdomen, if vomiting occurs, it usually follows as a response to pain and results from vagal stimulation. This vomiting is typically severe and develops without nausea. For

example, vomiting has been found in approximately 75 percent of adnexal torsion cases

- ✓ if vomiting is noted prior to the onset of pain, a surgical abdomen is less likely
 - Fever, chills ... indicate an acute inflammatory/infectious condition
- R => radiation, to
 - The shoulder, indicate hemoperitoneum, may be from ruptured ectopic pregnancy
 - The groin, from the costovertebral angle, indicate ureteral colic
- T => timing/ temporal
 - What time of day (relation to menstrual cycle and activities of daily living)?
 - Pelvic pain can occur associated with
 - ✓ Menses=> dysmenorrhea
 - ✓ Sexual intercourse=> dyspareunia
 - ✓ Urination=> dysuria
 - ✓ Defecation=> dyschezia
- S => severity
 - Has the patient stopped daily activity b/c of the symptom? Or lost normal sleep?
 - Does simple rest relieve the pain?
 - Is the pain associated any indicators systemic autonomic response like, diaphoresis, vomiting?

Physical Examination (especially for Acute pelvic pain)

G/A: Acutely sick looking, patient lying still (peritonitis) or restless (cramping)

V/S: tachycardia & tachypnea (severe pain), pyrexia (infection/inflammation) or hypotension (from hypovolemia)

Abdominal examination: -

- Inspection:
 - Distended abdomen: - bowel obstruction, torsion or hemoperitoneum
 - Abdomen doesn't move with respiration. -peritonitis
 - Scar, -bowel obstruction, ectopic pregnancy
 - Hernia sites
- Palpation
 - Generalized abdominal rigidity with significant tenderness and rebound tenderness => peritoneal irritation (in most of the cases)
 - Tenderness limited to the lower abdomen
- Percussion
 - Hyper tympanic=> bowel obstruction
 - Dullness=> blood or cystic fluid in the peritoneum
- Auscultation
 - Hyperactive bowel sound=>early stage of bowel obstruction
 - Absent bowel sound=> paralytic ileus

Pelvic examination

- Check the introitus and urethral orifice for discharge or bleeding
- Cervical motion tenderness, can be seen with PID, appendicitis, diverticulitis, and intraabdominal bleeding
- Uterine enlargement, in case of leiomyomas, pregnancy, and adenomyosis
- A tender adnexal mass may reflect ectopic pregnancy, tubo- ovarian abscess, or ovarian cyst with torsion, hemorrhage, or rupture

Rectal examination can add information regarding the source and size of pelvic masses and the possibility of colorectal pathologies.

Investigation

- Urine or serum β -human chorionic gonadotropin (hCG) testing is recommended in those of reproductive age without prior hysterectomy.
- Complete blood count (CBC) can identify hemorrhage, both uterine and intraabdominal, and can assess the possibility of infection.
- Urinalysis, to evaluate possible urolithiasis or UTI
- ESR & CRP, shows an inflammatory state
- Microscopic evaluation and culture of vaginal discharge
- Sonography (ultrasound), transabdominal or transvaginal
 - Is very important to assess most of pelvic pathologies and identify fluid in the abdomen.
- Computed tomography (CT scan), has superior performance in identifying GI and urinary tract causes of acute pelvic and lower abdominal pain
- Magnetic resonance imaging (MRI), if initial sonography is non-diagnostic
- Plain abdominal x-ray, (plain= no contrast) for bowel obstruction
- Intravenous pyelography (IVP), for ureteral colic
- Laparoscopy, for both diagnostic and management purpose
- Culdocentesis, in evaluation of suspected ruptured cyst

See special considerations in P/E and Ixs for chronic pelvic pain below

Chapter 23

ABORTION

Prepared by Dr. Balkewkal Kebede (MI)

Definition and Terminology

Abortion is defined as the spontaneous or induced termination of pregnancy before fetal viability.

- Viability lies between the lines that separate ***abortion from preterm birth***.
 - It is defined by pregnancy duration and fetal weight for statistical and legal purpose.
 - Both CDC and WHO define abortion as pregnancy termination before 20 weeks of gestation or with fetus born weighing <500g.
 - In Ethiopia and the UK, the age of viability is 28 weeks of gestation.

The term miscarriage is usually for spontaneous abortion.

The use of transvaginal sonography and precise measurement of serum hCG concentrations are used to identify extremely early pregnancies and makes it possible to distinguish between a chemical and a clinical pregnancy.

- Abortion of such early pregnancies is termed as *early pregnancy loss or early pregnancy failure or wastage*

Incidence

Although the true incidence of spontaneous abortion is unknown, approximately

- **15% of clinically** evident pregnancies and
- **up to 50% of chemically** evident pregnancies end in spontaneous abortion

Classification

Based on: -

- Mechanism/onset
 - Spontaneous (see further clinical classification below)
 - Induced
- Gestational age at onset
 - Early(<12wks) or first trimester
 - Late(>12wks) or mid-trimester
- Place of abortion
 - Safe (health facility)
 - Unsafe abortion,
- Presence or absence of infection
 - Septic
 - Aseptic

Spontaneous first trimester abortion

- More than 80% of spontaneous abortion occurs within the 1st 12 weeks of gestation.

Pathogenesis

In 1st trimester losses, death of the embryo or fetus nearly always precedes spontaneous expulsion which leads to hemorrhage in to the decidua basalis leading to necrosis of adjacent tissue that stimulate uterine contraction and expulsion.

- Thus, the key to determining the cause of early miscarriage is to ascertain the cause of fetal death.

In contradistinction, in later pregnancy losses, the fetus usually does not die before expulsion, and thus other explanations are sought.

Etiologies and Risk Factors

Fetal factors

Chromosomal abnormalities

- Approximately half of first-trimester miscarriages are anembryonic, that is, with no identifiable embryonic elements.
- The remaining (embryonic miscarriages), can be further grouped as either, those with normal chromosome (euploid abortions, in 45-55%) or those with chromosomal anomalies (aneuploid abortions)

Chromosomal abnormalities	Percentage
Aneuploidy	
▪ Autosomal trisomy	22-32
▪ Monosomy X (45, X)	5-20
▪ Triploidy	6-8
▪ Tetraploidy	2-4
▪ Double or triple trisomy	0.7-2
Structural anomaly	2

Almost 95 percent of chromosomal abnormalities in aneuploid fetuses are caused by **maternal gametogenesis errors**.

- Chromosomal abnormality decrease with advancing GA (55%, 35%, & 5% in 1st, 2nd and 3rd trimester abortuses/stillbirth respectively).

Aneuploid Abortion

- 50% of embryonic abortions are aneuploid
- Monosomy X or Turner's syndrome is the single most common aneuploidy, comprising approximately 18% of these gestations.

- As a group, the autosomal trisomies account for more than half of aneuploid losses, with trisomy 13, 16, 18, 21, and 22 being the most common. Autosomal trisomies have been noted for every chromosome except chromosome number 1.
- Most trisomic pregnancies end in miscarriage. The exceptions are trisomy 21, 18, and 13, which have survival to birth rates of 22%, 5%, and 3%, respectively.
- Polyploidy, usually in the form of triploidy, is found in approximately 20% of all miscarriages.
 - Polyploid conceptions typically result in an anembryonic gestation.
- About 75% of aneuploid fetuses aborted before 8wks.

Chromosomal structural abnormalities infrequently cause abortion.

Maternal factors

- **Age:** the rate of clinical miscarriages is almost doubled when either parent is older than 40yrs.
 - The incidence of euploid abortions increases dramatically after maternal age exceeds 35 yrs.
- **Infections:** systemic or local (GU) infections can cause pregnancy loss. Such infections include;
 - *Chlamydia trachomatis* (in 4% of abortuses),
 - Polymicrobial infections from periodontal disease,
 - HIV and Bacterial vaginosis (2nd trimester)
- **Medical disorders:** like DM, thyroid disorders (especially severe iodine deficiency), celiac disease, inflammatory bowel disease and SLE
- **Uterine and cervical factors:** Congenital uterine anomalies, such as unicornuate, bicornuate, or septate uterus; Asherman syndrome and cervical insufficiency.
- **Medications** and toxic factors: IUDs, antineoplastic drugs, anesthetic gases, alcohol, nicotine, or cocaine, lead, ethylene oxide, and formaldehyde.
- **Radiotherapy:** exposure to >5rads of radiation in abdominopelvic radiotherapy may later be at increased risk for miscarriage.

- **Surgical procedures:** like early removal of the corpus luteum or the ovaries,
- **Trauma:** direct abdominal trauma, amniocentesis or chorionic villus sampling. Generally, trauma seldom cause miscarriage
- **Nutrition:** both extremes of malnutrition (severe dietary deficiency and morbid obesity) are associated with miscarriage.
- **Social and behavioral factor:** substance abuse
- **Occupational factor:** exposure to toxic chemicals.

Paternal factors

- Chromosomal abnormalities in sperm and increasing paternal age had an increased abortion risk

Clinical classification of spontaneous abortion

Threatened abortion

- The clinical diagnosis of threatened abortion is presumed when bloody vaginal discharge or bleeding appears thru a closed cervical Os during the 1st 28weeks.
 - Bleeding is by far the most predictive risk factor for pregnancy loss.
- Approximately 25% of pregnancies experience 1st trimester bleeding. It must be differentiated from an implantation bleeding
 - With miscarriage bleeding usually begins 1st and crampy lower abdominal pain follows hours to days later.
 - The combination of bleeding and pain predicts a poor prognosis or pregnancy continuation.
 - About 43% of these cases will subsequently miscarry. Even if miscarriage does not occur, these pregnancies are associated with high risk of late adverse maternal and perinatal outcomes.

- The primary goal in evaluating a woman with early pregnancy bleeding is prompt diagnosis of an ectopic pregnancy.
 - Serial quantitative serum β -hCG levels, progesterone levels, and transvaginal sonography, alone or in combination, can help ascertain if the fetus is alive and if it is within the uterus.

Inevitable abortion

- Is characterized by bleeding with cervical dilation, often with back or abdominal pain
- Gross rupture of the membranes along with cervical dilatation is nearly always followed by either uterine contractions or infection, which is inevitably followed by abortion
- the products of conception have not yet passed from the uterine cavity

Incomplete abortion

- Incomplete abortion is defined as the passage of some but not all of the products of conception from the uterine cavity
 - Before **10wks** of gestation, the placenta and fetus are generally passed together, but after this time, they are passed separately.
 - Bleeding and cramping usually continue until all products of conception have been expelled
 - There is dilation of the cervical Os

Complete abortion

- All of the products of conception have passed from the uterine cavity and the cervix is closed.
- A history of heavy bleeding, cramping, and passage of tissue or a fetus is common

Missed abortion

- Is defined as a pregnancy that has been retained within the uterus after embryonic or fetal demise.
- Cramping or bleeding may be present, but often there are no symptoms.
- The cervix is closed, and the products of conception remain in situ.
- It may be complicated by bleeding disorders and infection

Recurrent abortion

- Is defined as ≥ 2 (for age <35 yrs) or ≥ 3 (for age ≥ 35 yrs) consecutive spontaneous abortions
- Approximately 1 percent of fertile couples have recurrent miscarriages; most of these are embryonic or early losses.
- The risk of spontaneous abortion is at least doubled in women experiencing recurrent pregnancy loss.
- Possible causes:
 - Parental chromosomal abnormalities
 - Immunologic factors (especially antiphospholipid antibody syndrome)
 - Anatomic factors
 - ✓ bicornuate uterus, is the most common uterine anomaly (39%) and has 40-70% pregnancy loss rate
 - ✓ septate or unicornuate uterus, seen in 14-24% of uterine anomalies and has 34-88% pregnancy loss rate
 - ✓ uterine leiomyoma
 - Endocrinologic abnormalities
 - ✓ Progesterone deficiency caused by luteal phase defect (LPD), which is associated with inadequate endometrial development at implantation.
 - ✓ Thyroid disorders

- Microbiologic (TORCH), and
 - Thrombophilia abnormalities. (See details on Current 11th ed).
 - ***NB. Approximately 50% have no definitive cause***
- The prognosis after repeated losses is good, with most couples having an approximately 60% chance of a viable pregnancy.

Septic abortion

- Usually occur in induced abortions that occur in an illegal setup, especially in developing countries.
- Infection usually begins as endometritis involving the endometrium and any retained products of conception.
- The 2 most common causes of septic abortion are retained products of conception and bacteria that have been introduced into the uterus via ascending infection.
- These patients present with fevers, chills, abdominal pain, vaginal bleeding, and malodorous vaginal discharge.
- Without treatment, endometritis may spread beyond the uterus, leading to peritonitis, bacteremia, and sepsis. Particularly worrisome are severe necrotizing infections and toxic shock syndrome caused by *S. pyogenes*.

Spontaneous midtrimester abortion

The timespan that defines a mid-trimester fetal loss extends from the end of the first trimester until the fetus weighs ≥ 500 g or gestational age reaches 20 weeks (28weeks).

Incidence and Etiology

- Abortion becomes much less common by the end of the first trimester, overall, spontaneous loss in the second trimester is estimated at 1.5 to 3 percent, and after 16 weeks, it is only 1 percent.
- First-trimester bleeding doubles the incidence of second- trimester loss

Risk factors for second-trimester abortion include race, ethnicity, prior poor obstetrical outcomes, and extremes of maternal age.

Some causes include;

- Uterine defects; Incompetent cervix, Congenital, or Leiomyoma
- Fetal anomalies; Chromosomal or Structural
- Placental causes; Abruption, Previa, Defective spiral artery transformation and Chorioamnionitis
- Maternal disorders; Autoimmune, Infections and Metabolic

Induced abortion

- Is surgical or medical termination of a live fetus that has not reached viability
- It can be classified in to two
 - indicated or therapeutic and
 - elective or voluntary.
- Indications:
 - Maternal health problems like: -
 - ✓ persistent cardiac decompensation
 - ✓ pulmonary arterial hypertension
 - ✓ advanced hypertensive vascular disease
 - ✓ diabetes with end-stage organ failure
 - ✓ malignancies
 - ✓ maternal mental illness
 - rape or incest
 - significant fetal anatomic, metabolic, or mental deformity
 - Age of woman less than 18 years
 - Extreme poverty

Complication of abortion

- Hemorrhage, shock, anemia
- Infection
- Intrauterine synechiae (Asherman syndrome)
- Perforation of the uterus
- Injury to bowel and bladder
- Fistula formation
- Rh isoimmunization
- Tetanus
- Ectopic pregnancy in later life
- Renal failure
- Death
- Psychological trauma - depression

Approach

History

- Age
- LNMP, hx of amenorrhea and symptoms of pregnancy like morning sickness (nausea and vomiting)
- Vaginal bleeding, assess onset, consistency (any embryonic/ fetal(solid) part
- Cramping (lower abdominal pain), assess onset in relation to vaginal bleeding
- Fever or chills, in case of septic abortion
- Risk factors
- Symptoms of anemia
- Sexual assault hx, in case of induce abortion

DDx

- ectopic pregnancy in intact tube
- cervical polyps
- vaginitis
- cervical carcinoma
- gestational trophoblastic disease
- trauma
- foreign body

Physical examination

- G/A: - well looking, distress or lethargic (from severe anemia or septic shock)
- V/S:
 - Tachycardia, tachypnea or hypotensive (from blood loss)
 - Febrile, in case of septic abortion
- HEENT: pale conjunctiva?
- **Abdominal examination:**
 - Inspect the skin changes associated with pregnancy
 - Severe unilateral tenderness in lower abdomen may indicate ectopic pregnancy
 - Determine uterine size: especially in case of mid-trimester it is important to determine size of the uterus, if it is less than the appropriately determined GA, it may indicate incomplete or complete loss of product of conception
 - Listen to FHT
- Pelvic examination
 - See if there is ongoing blood loss, or passage of amniotic fluid
 - Check the cervix (open or closed)

Investigations

- CBC: anemia? Or infection?
- Blood type and Rh
- Serial serum β -hCG:
 - Falling or abnormally rising serum levels of β -hCG are diagnostic of an abnormal pregnancy, either a failed intrauterine gestation or an ectopic pregnancy.

- **Ultrasound:**
 - Transvaginal ultrasound allows us to visualize gestational sac (GS) as early as 4–5 weeks of gestation.
 - At 5–6 weeks' gestation, a yolk sac will be present. In general, a GS with a mean sac diameter (MSD) of ≥ 8 mm should contain a yolk sac. Similarly, a GS with an MSD of >16 mm should also contain an embryo
 - Pregnancies with a large GS and no embryo are typically anembryonic gestations and are managed in a similar manner as a missed abortion
 - Fetal heart motion is expected in embryos with a crown to rump length of >5 mm or at 6–7 weeks' gestation.
 - If a repeat ultrasound in 1 week does not show embryonic cardiac activity, the diagnosis of embryonic demise is made.
- Urinalysis: to identify cause
- Endocervical cultures
- Blood cultures: in septic abortion

Management

Spontaneous abortion

- **Threatened abortion:**
 - There is no effective therapy for threatened intrauterine pregnancy other than pelvic rest.
- **Inevitable abortion:**
 - Awaiting spontaneous expulsion
 - Oxytocin use or Suction curettage or evacuation and curettage depending on the gestational age
 - Rh D Ig for Rh negative
- **Incomplete abortion:**

- Uterus should be evacuated either by suction curettage or E&C as quickly as possible depending on GA (suction being the recommended method for GA up to 12 weeks)
- Rh Ig for Rh negative
- Prophylactic antibiotics – doxycycline 200 mg stat dose or 100mg BID for three days
- **Missed abortion**
 - Immediate evacuation of the uterus is preferred treatment, but treatment choice between expectant management and immediate evacuation should depend on the woman's informed decision
 - Expectant management is possible for three weeks with assessment of coagulation profile every week
 - Risk of DIC is very low before 15 weeks of GA
 - Nearly in 80% of cases spontaneous expulsion during expectant management can occur in three weeks' time.
 - If immediate evacuation is decided, one has to use protocol for induced abortion
- **Septic abortion:**
 - Add broad spectrum antibiotics

Induced abortion

Medical or Surgical, (See technique and procedure in GYNEOLOGY MANAGEMENT GUIDELINE IN JUSH)

Components of Post abortion care (PAC)

- Emergency treatment of incomplete abortion and potentially life threatening complications
- Post-abortion family planning counseling and services
- Links between post-abortion emergency services and the reproductive health care system.
- Community service provider partnership
- Counseling

Chapter 24

Ectopic pregnancy

Prepared by Dr. Bemnet G/Michael (MI)

Definition and terminology

An **ectopic pregnancy** is one in which the blastocyst implants anywhere other than the endometrial lining of the uterine cavity.

The overall rate of ectopic pregnancy is **1%-2%** (Berek&Novak's Gynecology 14thed).

- Although ectopic pregnancy accounts for a small proportion of all pregnancies, it disparately accounts for **3%** of all pregnancy-related deaths (Williams Gynecology 4thed).

The most common site of implantation the fallopian tube in about 96%

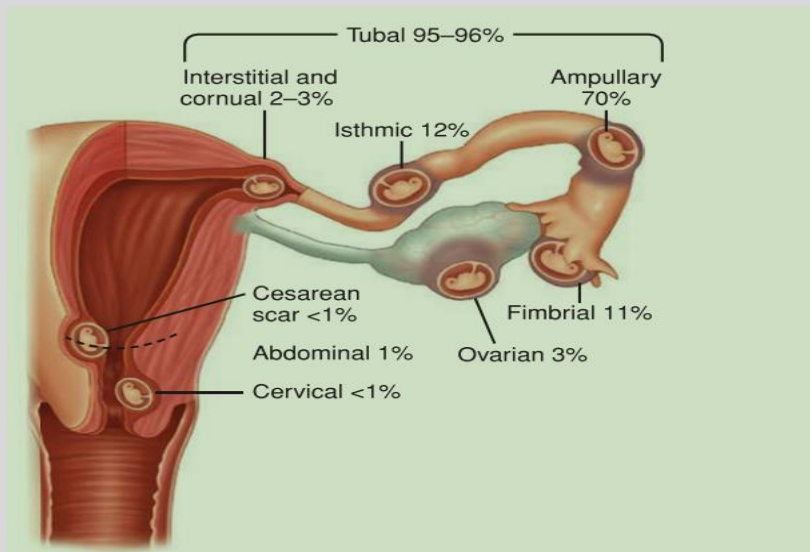


Fig. Various sites and frequency of ectopic pregnancy

Risk factors

- **Prior ectopic pregnancy**
 - Recurrence risk after one ectopic pregnancy is **10-15%** and **30%** after second ectopic pregnancy. (Berek&Novak'sGynecology¹⁴ed) (**12.5%** risk following prior ectopic pregnancy; Williams Gynecology 4thed).
- **Prior tubal surgery**; **4%** risk (Williams Gynecology 4thed)
- **PID** confirmed by laparoscopy or positive test for chlamydia trachomatis; **13%** risk after one episode of PID, **35%** after two and **75%** after three episodes. (Berek&Novak'sGynecology¹⁴ed)
- **Smoking >20 cigarettes per day**
 - Smoking elevates the ectopic pregnancy risk more than threefold in women who smoke more than one pack of cigarettes daily. (Williams Gynecology 4thed)
- **Infertility treatment**
 - Specific treatments provided for infertile women including reversal of sterilization, ovulation induction, and in vitro fertilization (IVF) have increased the odds of tubal pregnancy by at least four times (Berek&Novak'sGynecology¹⁴ed)
- **Increasing maternal age**;
 - raises risk of ectopic pregnancy especially in patients older than 40 years of age due to age-related hormonal changes that alter tubal function
- **Contraception**
 - In rare case that women conceive while using contraception methods like levonorgestrel-releasing intrauterine system (LNG-IUS) and copper intrauterine device (IUD), the pregnancy is more likely to be ectopic.
 - Progestin-only contraceptive pills also pose a slightly greater risk due to their effects to diminish tubal motility.

Pathophysiology

Ectopic pregnancy has three possible outcomes:

1. Tubal rupture

- The fallopian tube lacks a submucosal layer beneath its epithelium as a result the fertilized ovum can easily burrow through the epithelium and implant within the muscularis layer. As rapidly proliferating trophoblasts erode the muscularis layer, maternal blood pours into the spaces within the trophoblastic or the adjacent tissue and rupture usually occurs spontaneous or following trauma such as that associated with bimanual pelvic examination or coitus.

2. Tubal abortion

- The pregnancy may instead pass out the distal fallopian tube which usually occurs in distal implantations. Subsequently, hemorrhage may cease and symptoms eventually disappear. Bleeding may persist as long as products of conceptus remain in the tube.

3. Pregnancy failure with resolution

- In this case, the pregnancy dies and is reabsorbed

After tubal implantation, ectopic pregnancy development may follow an acute or chronic course.

- **Acute ectopic pregnancy;** is more common and is associated with higher serum beta-hCG levels at initial presentation and higher risk of tubal rupture.
- **Chronic ectopic pregnancy;** is much less common and minor repeated ruptures or tubal abortion incites an inflammatory response that leads to formation of pelvic mass which often prompts diagnostic surgery. Chronic ectopic pregnancies typically rupture late and beta-hCG testing may be negative or show very low, static levels.

Clinical manifestation

- The classic symptom triad of ectopic pregnancy is amenorrhea followed by vaginal bleeding and ipsilateral abdominal pain. This symptom group is present in about **50%** of patients and is more typical in patients with a ruptured ectopic pregnancy. (Berek&Novak'sGynecology14thed)
- Banal pregnancy discomforts such as breast tenderness and nausea.
- Shoulder pain worsened by inspiration, which is caused by phrenic nerve irritation from subdiaphragmatic blood.
- Vasomotor disturbances such as vertigo and syncope may reflect hemorrhage-related hypovolemia.

Physical Findings

- **Unruptured**
 - The findings before rupture and hemorrhage are nonspecific and vital signs are normal.
 - The abdomen may be non-tender or mildly tender, with or without rebound.
 - The uterus may be slightly enlarged with findings similar to a normal pregnancy.
- **Ruptured**
 - In ruptured ectopic pregnancy, patients may be tachycardic followed by hypotension.
 - In abdominopelvic examination, many will have marked tenderness on both abdominal and pelvic examination, and pain is aggravated with cervical manipulation.
 - A pelvic mass, including fullness postero-lateral to the uterus, can be gently palpated in some affected women.
 - An adnexal mass may be palpable in up to **50%** of cases, but the mass varies markedly in size, consistency, and tenderness. (Berek&Novak'sGynecology14thed)

Diagnosis

- Hx and P/E
- Laboratory findings
 - Serum beta-hCG measurement

A *single measurement* has limited usefulness in the evaluation of pregnancy of unknown location. But if levels are greater than the ultrasound discriminatory zone indicates possible extrauterine pregnancy.

A discriminatory zone is defined as a serum hCG level above which a gestational sac should be visualized by ultrasound examination if an intrauterine pregnancy is present. In most situations, this serum hCG level is 1500 or 2000 IU/L with transvaginal ultrasound (the level is higher [6500IU/L] with transabdominal ultrasound)

Serial measurement are usually required when the results of the initial ultrasonography examination are indeterminate (i.e., when there is no evidence of intrauterine gestation or extra-uterine findings consistent with an ectopic pregnancy).

- Serum progesterone levels
 - ✓ Serum progesterone concentration is used by some to aid ectopic pregnancy diagnosis when serum β -hCG levels and sonographic findings are inconclusive. Serum progesterone concentration varies minimally between 5 and 10 weeks' gestation, thus a single value is sufficient.
 - ✓ A single serum progesterone level $<6\text{ng/mL}$ ($< 20\text{nmol/L}$) has a specificity of 99% to predict a nonviable pregnancy in women without clear sonographic evidence of pregnancy location.

- **Sonography**

- With transvaginal sonography (TVS), an intrauterine gestational sac is usually visible between 4 ½ and 5 weeks, the yolk sac appears between 5 and 6 weeks, and a fetal pole with cardiac activity is first detected at 5 ½ to 6 weeks. With transabdominal sonography, these structures are visualized slightly later
- *The sonographic diagnosis of ectopic pregnancy rests on visualization of an adnexal mass separate from the ovary. Definitive criteria include either an extrauterine yolk sac or embryo.*

In an attempt to unify the language used with sonographic evaluation of early pregnancies, a consensus statement was drafted with five categories:

1. **Definitive ectopic pregnancy** (extrauterine gestational sac with yolk sac and/or embryo),
2. **Probable ectopic pregnancy** (inhomogeneous adnexal mass or extrauterine halo-like structure),
3. **Probable IUP** (intrauterine echogenic sac),
4. **Definite IUP** (intrauterine gestational sac with yolk sac and/or embryo), and
5. **PUL** (lacking signs of either ectopic pregnancy or IUP)

- **Culdocentesis**

With a 16- to 18-gauge spinal needle, the posterior cul-de-sac can be entered through the posterior vaginal fornix. The aspirate characteristics, in conjunction with clinical findings, may help clarify the diagnosis.

- **Dilation and curettage (D & C);**
 - Can help clarify the diagnosis in persistent or unclear pregnancy of unknown location (PUL) cases
 - Villi in the surgical sample support an intrauterine location
- **Laparoscopy;**
 - At the time of laparoscopy, the fallopian tubes are easily visualized and evaluated, but the diagnosis of ectopic pregnancy is missed in 3% to 4% of patients who have very small ectopic gestations.

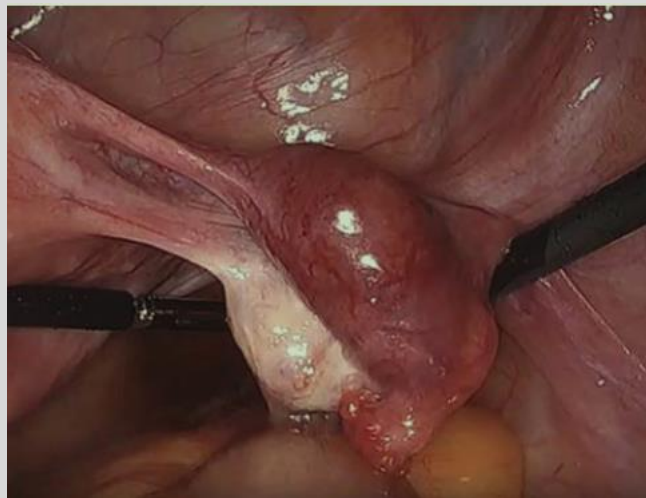


Fig Laparoscopic photograph of a right-sided ectopic pregnancy shows a distended tubal ampulla.

Management

Ectopic pregnancies can be managed **surgically**, **medically** with methotrexate (MTX) or **expectantly**.

Medical management

- For most cases, medical management is preferred, if feasible, to avoid surgical risks.

- For medical management to commence evidence for tubal rupture and hemodynamic instability must be absent and the patients must lack MTX contraindication.
- Other considerations include reasonably close access to an emergency department and a commitment to be compliant with surveillance laboratory testing.

Classic predictors of success with MTX therapy:

1. Initial serum beta-hCG levels (single best prognostic indicator); levels <5000mIU/mL are associated with a success rate of 92% (Williams Gynecology 4thed)
2. Ectopic pregnancy size; with ectopic mass ≤ 3.5 cm, a single-dose MTX success rate is 93%. (Williams Gynecology 4thed)
3. Fetal cardiac activity, seen sonographically is also a relative contraindication to medical therapy.

Table Medical treatment protocols for ectopic pregnancy

	Single Dose	Multidose
Dosing	One dose; repeat if necessary	Up to four doses of both drugs until serum β -hCG declines by 15%
Medication dosage		
Methotrexate	50 mg/m ² BSA (day 1)	1 mg/kg, days 1, 3, 5, and 7
Leucovorin	NA	0.1 mg/kg days 2, 4, 6, and 8
Serum β-hCG level	Days 1, 4, and 7	Days 1, 3, 5, and 7
Indication for additional dose	• If serum β-hCG level does not decline by 15% from day 4 to day 7 • <15% decline during weekly surveillance	If serum β -hCG declines <15% between days 1 & 3, give additional dose; repeat serum β -hCG in 48 hours and compare with previous value; maximum four doses
Surveillance	Once 15% decline achieved, draw weekly serum β -hCG level until undetectable	
Methotrexate Contraindications		
Sensitivity to MTX	Intrauterine pregnancy	Peptic ulcer disease
Tubal rupture	Hepatic, renal, or hematologic dysfunction	Immunodeficiency
Breastfeeding	Active pulmonary disease	
BSA = body surface area; β -hCG = β -human chorionic gonadotropin; MTX = methotrexate; NA = not applicable.		

With administration of MTX, women are counseled to avoid several aggravating agents until treatment is completed. These are:

- Folic acid-containing supplements, which can competitively reduce methotrexate binding to dihydrofolate reductase;
- Nonsteroidal anti-inflammatory drugs, which reduce renal blood flow and delay drug excretion;
- Alcohol, which can predispose to concurrent hepatic enzyme elevation;
- Sunlight, which can provoke MTX-related dermatitis; and
- Sexual activity, which can rupture the ectopic pregnancy (American College of Obstetricians and Gynecologists, 2018).

Surgical management

- There are two procedures *salpingectomies* or *salpingostomy*.
 - With *salpingectomy*, the entire fallopian tube is removed, and this is suitable for ruptured or unruptured tubes. With rare exception, the entire conceptus is removed with the fallopian tube. Thus, cases of persistent trophoblastic tissue with salpingectomy are uncommon.
 - In contrast, *salpingostomy* (done for hemodynamically stable patients and those who desire fertility preservation) is typically used to remove a small, unruptured pregnancy yet preserve the fallopian tube. For this, a 15-mm linear incision is made on the antimesenteric border of the fallopian tube over the pregnancy. The products usually will extrude from the incision and can be carefully removed. Main risks include postoperative bleeding from the tubal incision or trophoblast left within the tube that later causes tubal rupture.

Expectant management

- In anticipation that a tubal ectopic pregnancy will spontaneously resolve, close observation is reasonable in select women without pain or significant bleeding.

- A commitment to surveillance visits and relative proximity to an emergency department are other safeguards.

PERSISTENT ECTOPIC PREGNANCY

Incomplete eradication of trophoblastic tissue and its continued growth causes tubal rupture in 3 - 20% of women following conservative surgical or medical treatment of ectopic pregnancy. Rarely, surgically dislodged trophoblast can also secondarily implant on abdominal surfaces and bleed. Thus, abdominal pain following ectopic pregnancy treatment prompts immediate suspicion for persistent trophoblast proliferation.

Note that:

With the ending of any early pregnancy, Rh status is assessed. Alloimmunization is a possibility in D-negative women. Therefore, 300 ~of anti-D immune globulin is given to these women if the father of the baby has the o antigen or if his status is unknown.

Chapter 25

PELVIC INFLAMMATORY DISEASE

Prepared by Dr. Bulcha Lemma (MI)

Definitions

PID comprises a spectrum of inflammatory disorders of the upper female genital tract above internal cervical Os, including any combination of endometritis, salpingitis, tubo-ovarian abscess, and pelvic peritonitis.

Etiology

- Mostly **polymicrobial** infection caused by ascending infection
- Sexually transmitted organisms, especially *N. gonorrhoeae* and *C. trachomatis*, often are implicated.
 - Approximately 50% of women with diagnosis of acute PID tests positive for either of them.
 - *C. trachomatis* etiologic role is very different from *N. gonorrhea*.
 - ✓ *N. Gonorrhea* is Gram negative IC diplococcus with rapid cycle 20 to 40 minutes to divide. Is characterized by rapid and intense inflammatory response which makes patient symptomatic leads to early treatment. So there is less complication.
 - ✓ But *C.Trachomatis* is a slow-growing intracellular organism. It lacks of mitochondria and has growth cycle 48 to 72 hours, does not induce a rapid or violent inflammatory response and patients are mostly asymptomatic which results in delayed treatment. It causes destruction by rupture and associated with long term complications.
- Vaginal flora, such as strict and facultative anaerobes and *G. vaginalis*, *H. influenzae*, enteric gram-negative rods, and *Streptococcus agalactiae*
- Cytomegalovirus (CMV), *T. vaginalis*, *M. hominis*, and *U. urealyticum*

Pathogenesis

Physiologic barriers that keep the UFGT are

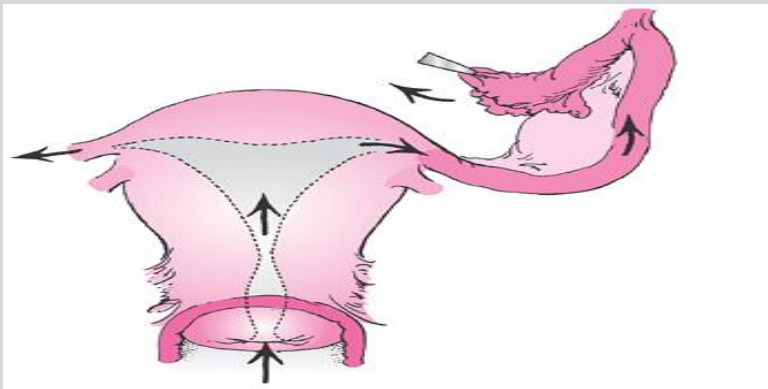
- Cervical canal that is closed & filled with mucus
- Presence of lysosome & secretion of IgA in cervical secretions
- Cyclic shedding of the endometrium
- Down ward beating tubal Cilia
- Systemic host defense

So for PID to occur there should be a breach in these barriers and the theories for this are

- Vector transport by spermatozoa and C. Trichomonas
- Pressure gradient formed during orgasm
 - Up to 75% of cases of PID occur within 7 days of menses due to loss of endocervical barriers & poor quality of the cervical mucus

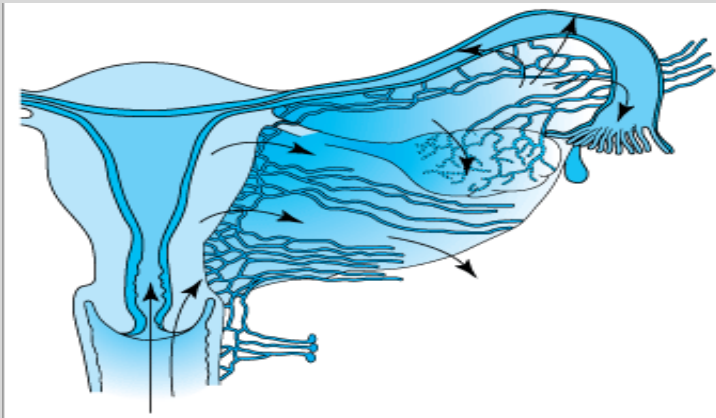
The 3 proposed pathways of spread of microorganisms in PID

1. **Ascending infection** endometrial-endo-salpingeal-peritoneal spread of microorganisms.
 - It is the most common forms of non-puerperal PID.
 - Pathogenic bacteria gain access to the lining of the uterine tubes → purulent inflammation & egress of pus through tubal ostia into the peritoneal cavity.
 - E.g. endometritis, adnexal infection, & peritonitis.



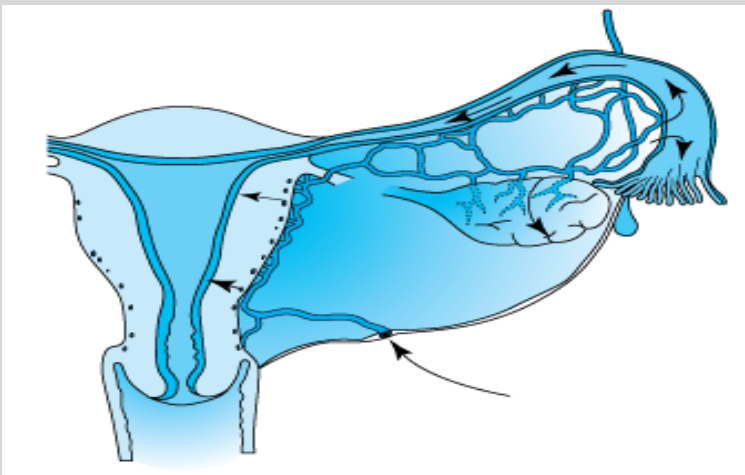
2. *Lymphatic dissemination*

- *Results in extraperitoneal parametrial cellulitis.*
- *Postpartum*
- *postabortal, &*
- *some IUD-related infections*



3. *Hematogenous routes*

- *Rarely, certain diseases (eg, tuberculosis)*
- *may gain access to pelvic structures*
- *by hematogenous routes*



Risk factors for PID

- STI
- Age
 - Adolescent 1:8 Vs 1:80 for a sexually active >24, b/c columnar epithelium
- Contraceptives
 - IUDs = threefold to fivefold
 - Barriers = ↓ 60%
 - OCP = ↓ risk, good Px fertility
 - previous tubal ligation = 1/450;
- Instrumentation ex. 1/200 induced abortion
- Previous acute PID = 25 %,
 - partner treatment

Classification of PID

- Based on duration
 - Acute,
 - Sub-acute,
 - Chronic
- Based on antecedent events
 - Post STI /menstrual
 - Post abortal
 - Postpartum
 - Post Instrumentation
 - IUD – Related
 - Secondary PID

Clinical manifestation

Acute PID is difficult to diagnose because of the considerable variation in symptoms and signs associated with this condition. Some women may present may develop PID without any symptoms.

Symptoms

- **Lower abdominal pain** is cardinal presenting symptoms.
 - It is usually bilateral and onset of pain during or shortly after menses is particularly suggestive.
- Abnormal uterine bleeding occurs in $>1/3$ of patients with PID.
- New vaginal discharge, urethritis, proctitis,
- Fever & chills can be associated signs
 - *Fever occurs in only 50% of cases.*

Signs

- *Abdominal examination reveals*
 - *Diffuse tenderness greatest in the lower quadrants, which may or may not be symmetrical.*
 - *Rebound tenderness & ↓ bowel sounds are common.*
 - *Marked tenderness in the RUQ does not exclude PID, since perihepatitis (Fitz-Hugh Curtis syndrome) occurs in about 10% of PID*
- *Bimanual examination*
 - *a purulent endocervical discharge &/or acute cervical motion & adnexal tenderness*
- *Recto-vaginal examination*
 - *Uterine & adnexal tenderness*
 - *Usually it has bilateral adnexal tenderness unlike unruptured ectopic pregnancy(unilateral adnexal tenderness*
 - *A palpable adnexal mass, a TOA complicating PID & other diseases processes*

DIAGNOSIS

Data indicate that a clinical diagnosis of symptomatic PID has a positive predictive value for salpingitis of 65%–90%, compared with laparoscopy.

Because of the difficulty of diagnosis and the potential for damage to the reproductive health of women, health care providers should maintain a low threshold for the clinical diagnosis of PID.

Diagnostic criteria

- **Major Criteria:-**
 - Cervical motion tenderness *or*
 - Lower abdominal / uterine tenderness *or*
 - Adnexal tenderness
- **Minor criteria**
 - Oral temperature $>38.3^{\circ}\text{C}$ ($>101^{\circ}\text{F}$)• Abnormal cervical mucopurulent discharge or cervical friability
 - Presence of abundant numbers of WBCs on saline microscopy of vaginal fluid
 - Elevated erythrocyte sedimentation rate
 - Elevated C-reactive protein
 - Laboratory documentation of cervical infection with *N. gonorrhoeae* or *C. trachomatis*

Presumptive treatment for PID should be initiated for sexually active young women and other women at risk for STIs if they are experiencing pelvic or lower abdominal pain,

- If no cause for the illness other than PID can be identified, or
- If one or more of the major criteria are present on pelvic examination.

Presence of one or more clinical criteria enhances the specificity of clinical diagnosis.

Investigation

More specific criteria

- Transvaginal sonography **90% accurate**
 - thickened, fluid-filled tubes with or without free pelvic fluid or tubo-ovarian complex
 - Doppler studies indicating pelvic infection (e.g., **tubal hyperemia**)
 - *Incomplete septation of the tubal wall ("cogwheel sign") acute PID*
 - *Thin wall ("beaded string") chronic PID*
- Laparoscopic abnormalities consistent with PID (eg, tubal erythema, edema, adhesions; purulent exudate or cul-de-sac fluid; abnormal fimbriae)
- Endometrial biopsy with histopathologic evidence of endometritis
- **Laboratory tests**
 - Always begins with pregnancy test
 - U/A (preferably on a catheterized specimen)
 - Stool for occult blood
 - Complete Blood counts(CBC) have limited value ,< 50% of acute PID patients exhibit leukocytosis.
 - If hematocrit of <30% è PID less likely
 - ESR
 - PIHTC
 - Gram stain & microscopic examination of vaginal & cervical discharge (↑probability of PID if positive for Gram negative intracellular diplococcic & little use if negative)
 - Nucleic acid amplification tests for chlamydia & gonococcus

- *Culdocentesis* is indicated whenever peritoneal material is needed for diagnosis.

Finding	Implications for Diagnosis
Blood	▪ Ruptured ectopic pregnancy. ▪ Haemorrhage from corpus luteum cyst. ▪ Retrograde menstruation. ▪ Rupture of spleen or liver. ▪ Gastrointestinal bleeding. ▪ Acute salpingitis.
Pus	○ Ruptured tubo-ovarian abscess. ○ Ruptured appendix or viscus. ○ Rupture of diverticular abscess. ○ Uterine abscess with myoma.
Cloudy	▪ Pelvic peritonitis (such as is seen with acute gonococcal salpingitis). ▪ Twisted adnexal cyst. ▪ Other causes of peritonitis: appendicitis, pancreatitis, cholecystitis, perforated ulcer, carcinomatosis, echinococcosis

Indications of laparoscopy

- A sick patient with high suspicion of a competing diagnosis (usually appendicitis, etc.)
- An acutely ill patient who has failed outpatient treatment for PID
- Any patient not clearly improving after approximately 72 hours of inpatient treatment for PID

Laparoscopic findings:

- Tubal serosal hyperaemia
- Tubal wall edema
- purulent exudate issuing from the fimbriated ends of the fallopian tubes and
- Pooling in the cul-de-sac confirm this diagnosis.

DDX. of PID

- Ectopic pregnancy,
- Infected or septic abortion,
- Torsion of an adnexal mass,
- Ruptured corpus luteum cyst with haemorrhage,
- Degeneration of a leiomyoma,
- Endometriosis ,
- Acute appendicitis
- Acute UTI,
- Regional enteritis,
- Ulcerative colitis.
- Diverticulitis ,

Complication**Early/ immediate**

- Peritonitis
- Pelvic abscess & TOA
- Sepsis → MOF → Death (**ruptured TOA = 10 %**)
- Fitz-Hugh-Curtis syndrome- 1%- 10%, RUQ- pain
- Develop from vascular or transperitoneal dissemination of either N. gonorrhea or C. trachomatis to produce the perihepatic inflammation.
 - DDX = acute cholecystitis or pneumonia

Late

- **Infertility** = 25%
- Ectopic Pregnancy = 6-10X higher; 12 %
- Chronic pelvic Pain = 20%
- Dyspareunia, dysmenorrhea
- Chronic PID
- Psychological consequences

Treatment

PID treatment regimens should provide empiric, broad spectrum coverage of likely pathogens.

- All regimens used to treat PID should also be effective against *N. gonorrhoeae* and *C. trachomatis* because negative endocervical screening for these organisms does not rule out upper genital tract infection.

Indication for admission

- Surgical emergencies (e.g., appendicitis) cannot be excluded.
- Tubo-ovarian abscess
- Pregnancy
- Severe illness, nausea and vomiting, or oral temperature $>38.5^{\circ}\text{C}$ (101°F)
- Unable to follow or tolerate an outpatient oral regimen
- No clinical response to oral antimicrobial therapy

N.B. Currently no data about improved clinical outcome after hospitalization of Adolescent females & HIV-infected women.

Parenteral Treatment

- Clinical experience should guide decisions regarding transition to oral therapy, which usually can be initiated within 24–48 hours of clinical improvement.
- For women with tubo-ovarian abscesses, >24 hours of inpatient observation is recommended.
- Because of the pain associated with IV infusion, doxycycline should be administered orally when possible.
 - Oral and IV administration of doxycycline and metronidazole provide similar bioavailability.

- After clinical improvement with parenteral therapy, transition to oral therapy with doxycycline 100 mg 2 times/day and metronidazole 500 mg 2 times/day is recommended to complete 14 days of antimicrobial therapy.

Recommended Parenteral Regimens for Pelvic Inflammatory Disease

Ceftriaxone 1 g by every 24 hours
plus
Doxycycline 100 mg orally or IV every 12 hours
plus
Metronidazole 500 mg orally or IV every 12 hours
or
Cefotetan 2 g IV every 12 hours
plus
Doxycycline 100 mg orally or IV every 12 hours
or
Cefoxitin 2 g IV every 6 hours
plus
Doxycycline 100 mg orally or IV every 12 hours

Oral or intramuscular treatment

- For mild to moderate PID
- Women who do not respond to IM or oral therapy within 72 hours should be reevaluated to confirm the diagnosis and be administered therapy IV

Recommended Intramuscular or Oral Regimens for Pelvic Inflammatory Disease

Ceftriaxone 500 mg* IM in a single dose
plus
Doxycycline 100 mg orally 2 times/day for 14 days with metronidazole 500 mg orally 2 times/day for 14 days
or
Cefoxitin 2 g IM in a single dose and probenecid 1 g orally administered concurrently in a single dose
plus
Doxycycline 100 mg orally 2 times/day for 14 days with metronidazole 500 mg orally 2 times/day for 14 days
or
Other parenteral third-generation cephalosporin (e.g., ceftizoxime or cefotaxime)
plus
Doxycycline 100 mg orally 2 times/day for 14 days with metronidazole 500 mg orally 2 times/day for 14 days

* For persons weighing ≥ 150 kg, 1 g of ceftriaxone should be administered.

From STG 2020

For outpatient management

For inpatient management

Ceftriaxone 250 mg IM stat/ Ciprofloxacin , 500mg po single dose/ Spectinomycin 2gm i.m stat PLUS Azithromycin 1gm po stat/ Doxycycline 100 mg po b.i.d for 14 days PLUS Metronidazole 500 mg po b.i.d for 14 days - Admit if there is no improvement within 72 hours Note : The preferred regimen is Ceftriaxone 250mg IM stat PLUS Azithromycin 1gm po stat PLUS Metronidazole 500 mg bid for 14 days	Ceftriaxone 250 mg i.m/i.v / Spectinomycin 2 gm i.m bid PLUS Azithromycin 1gm po daily / Doxycycline 100 mg po b.i.d for 14 days PLUS Metronidazole 500 mg po b.i.d for 14 days Note: For inpatient PID, ceftriaxone, spectinomycin or azithromycin should continue for 24hrs after the patient remain clinically improved, after which doxycycline and metronidazole should continue for a total of 14 days
---	--

Follow-Up

- Women should demonstrate clinical improvement (e.g., defervescence; reduction in direct or rebound abdominal tenderness; and reduction in uterine, adnexal, and cervical motion tenderness) <3 days after therapy initiation.
- If no clinical improvement has occurred <72 hours after outpatient IM or oral therapy, then hospitalization, assessment of the antimicrobial regimen, and additional diagnostics, including consideration of diagnostic laparoscopy for alternative diagnoses, are recommended.
- All women who have received a diagnosis of chlamydial or gonococcal PID should be retested 3 months after treatment, regardless of whether their sex partners have been treated.
 - If retesting at 3 months is not possible, these women should be retested whenever they next seek medical care <12 months after treatment.

Management of Sex Partners

- Persons who have had sexual contact with a partner with PID during the 60 days preceding symptom onset should be evaluated, tested, and presumptively treated for chlamydia and gonorrhea, regardless of the PID etiology or pathogens isolated.
- If the last sexual intercourse was >60 days before symptom onset or diagnosis, the most recent sex partner should be treated

Special Considerations

- **Pregnancy**
 - Pregnant women suspected of having PID are at high risk for maternal morbidity and preterm delivery.
 - These women should be ***hospitalized and treated with IV antimicrobials*** in consultation with an infectious disease specialist.
- **Intrauterine Devices**
 - The risk for PID associated with IUD use is primarily ***confined to the first 3 weeks after insertion***
 - If an IUD user receives a diagnosis of PID, the ***IUD does not need to be removed.***
 - But if no clinical improvement occurs within 48–72 hours of initiating treatment, providers should consider removing the IUD.
 - A systematic review of evidence demonstrated that treatment outcomes did not differ between women with PID

Chapter 26

HYPEREMESIS GRAVIDARUM (HEG)

Prepared by Dr. Amanuel Tefera (MI)

Definition and Incidence

Hyperemesis gravidarum [HEG] is defined as unexplained intractable nausea, retching, or vomiting beginning in the first trimester.

Severe unrelenting nausea and vomiting hyperemesis gravidarum is defined variably as being sufficiently severe to produce weight loss, dehydration, ketosis, alkalosis from loss of hydrochloric acid, and hypokalemia.

Incidence

- Mild to moderate nausea and vomiting are especially common in 50% to 90% pregnant women until approximately 16 weeks' gestation.
- The severe end of the continuum, hyperemesis gravidarum, may complicate up to 0.3–2% of pregnancies.

Pathogenesis

The pathogenesis of nausea and vomiting in pregnancy is unknown. The predominant theories are described below.

Hormonal changes

- No single hormonal profile can accurately predict the presence of hyperemesis gravidarum.
- **Elevated estrogen and progesterone**
 - Have long been implicated in the pathogenesis of this disorder.

- These hormones relax smooth muscle and thus slow gastrointestinal transit time and may alter gastric emptying.
- The fact that sex hormone levels peak in the third trimester, long after symptoms of hyperemesis gravidarum have typically resolved, is inconsistent with this theory
- **Human chorionic gonadotropin (hCG)**
 - Peak during the first trimester, the time when hyperemesis gravidarum is typically seen
 - The observations that serum hCG concentration is higher in women with hyperemesis than in other pregnant women and that nausea and vomiting are worse in women with multiple gestations and hydatidiform moles, conditions associated with high hCG levels, also support a possible etiologic role for this hormone.
- More recently implicated are other hormones that include; leptin, placental growth hormone, prolactin, thyroxine, and adrenocortical hormones ghrelin, leptin, nesfatin-1, and peptide

Psychological factors

- A feeling of ambivalence about the pregnancy has also been proposed as an etiologic or contributing factor.
- The woman's psychological response to persistent nausea and vomiting may exacerbate her symptoms as a result of conditioning.

Helicobacter pylori

- Most women with H. pylori do not develop severe nausea and vomiting in pregnancy, but the infection may play a role in pathogenesis of disease in some women. Case reports and small series have reported improvement in symptoms in women with severe disease after treatment of the infection.

Risk factors

- Hyperemesis in the first pregnancy
- Long interval between births
- Low maternal age <30
- Multiple gestation
- Multiparity
- Previous molar pregnancy
- Hyperthyroid disorders
- Pre-existing diabetes
- Gastrointestinal disorders Increased risk
- Asthma
- Non pregnant women who experience nausea and vomiting related to estrogen-based medication, motion, or migraine are at increased risk.
- Women who did not take multivitamins either prior to 6 weeks of gestation or during the peri conceptional period and women with heartburn and acid reflux.
- Daughters of women with hyperemesis were at significantly higher risk of developing hyperemesis in their own pregnancy.
- Interestingly, studies consistently report a preponderance of female fetuses among pregnancies complicated by hyperemesis
- Alcohol use and cigarette smoking (perhaps due to the effect of nicotine),

Protective factors

- Change in paternity.
- Pre-pregnancy underweight

Approach

History (Symptoms)

The mean onset of symptoms is at **5 to 6 weeks of gestation**, peaking at about **9 weeks**, and usually abating by **16 to 20 weeks** of gestation.

- However, symptoms may continue until the third trimester in 15 to 20 percent of gravida and until delivery in 5 percent.
- If vomiting persists beyond a few days postpartum other etiologies should be investigated.

Early manifestations	Late manifestations
✓ No evidence of dehydration and/or starvation ✓ Vomitus contains only food or bile stained ✓ Nutrition is not affected ✓ Blood biochemistry and urine analysis is normal	✓ Increased vomiting ✓ Oliguria ✓ epigastric pain ✓ Constipation

Physical examination

- **General examination**
 - Progressive emaciation with weight loss
 - Anxious look with sunken eyes
 - Icterus
 - Dry, coated tongue Teeth covered with sordes
 - Acetone smell in breath

- **Vital sign**
 - Tachycardia
 - Hypotension
- **Respiratory system**
 - Signs of acidosis, ketosis
- **Cardiovascular system**
 - Findings due to anemia
- **Abdominal examination**
 - Uterus may be less than/greater than weeks of gestation
 - Epigastric tenderness
- **Central nervous system**
 - Restlessness,
 - mental apathy
 - insomnia
 - convulsions and coma

Investigations

Laboratory evaluation is indicated in women with persistent nausea and vomiting to determine the severity of disease and to exclude other diagnoses that could account for the symptoms.

The standard initial evaluation of pregnant women with persistent nausea and vomiting includes: -

- Serum electrolytes.
 - Electrolyte and acid-base derangements, such as hypokalemia and hypochloremic metabolic alkalosis, from vomiting gastric secretions.
- Urine ketones and specific gravity.
 - Positive urine ketone and elevated urine specific gravity.
- An obstetrical ultrasound examination is performed to look for GTD and MG

Based on patient-specific factors, such as severity of disease and associated symptoms to assess the woman's volume/metabolic status or to exclude other diagnoses:

- **CBC**
 - An increase in hematocrit, indicating hemoconcentration due to plasma volume depletion.
- **RBS**
 - hypoglycemia
- **RFT**
 - an elevated blood urea nitrogen
- **LFT and liver enzymes**
 - Abnormal liver enzyme values occur in approximately 50 percent of patients who are hospitalized with hyperemesis. Alanine aminotransferase (ALT) is typically elevated to a greater degree than aspartate aminotransferase (AST).
 - Hyperbilirubinemia also can occur, but rarely exceeds 4mg/dL.
- **Serum amylase/lipase, and calcium**
 - Serum amylase and lipase may increase as much as 5-fold (as opposed to a 5- to 10-fold increase in acute pancreatitis) and are of salivary rather than pancreatic origin.
- **Thyroid function tests**
 - Mild hyperthyroidism, possibly due to high serum concentrations of human chorionic gonadotropin which has thyroid-stimulating activity
- **Serology for hepatitis A, B and C.**

Diagnosis

- *Common criteria for diagnosis of hyperemesis are **persistent vomiting** ($\geq 3x/day$) accompanied by **weight loss exceeding 5%** of pre-pregnancy body weight and **ketonuria** unrelated to other causes.*

- In contrast to women with mild disease like morning sickness, women with hyperemesis have: -
 - orthostatic hypotension, and physical signs of dehydration,
 - laboratory abnormalities such as electrolyte abnormalities (hypokalemia), and
 - often require hospitalization for stabilization
- In addition, many women with hyperemesis hypersalivate (ptyalism).

Differential Diagnosis

Acute appendicitis	Small bowel obstruction
Cholecystitis	Urinary tract infection
Cholelithiasis	Vestibular dysfunction
Diabetic ketoacidosis	Ovarian torsion
Esophagitis	Pancreatitis
Gastritis	Peptic ulcer disease
Gastroenteritis	Pre-eclampsia
Gastroesophageal reflux disease	Pseudotumor cerebri
Hepatitis	Pyelonephritis
Hydatidiform mole	Malaria
Hyperparathyroidism & Hyperthyroidism	Irritable bowel syndrome

Complications

- Acute kidney injury (may require dialysis)
- Depression (cause versus effect?)
- Diaphragmatic rupture
- Esophageal rupture (Boerhaave syndrome)
- Hypoprothrombinemia—vitamin K deficiency
- Hyperalimentation complications Mallory-Weiss tears, bleeding, pneumothorax, pneumomediastinum, pneumopericardium Rhabdomyolysis,

- Wernicke encephalopathy (thiamine deficiency)
 - The woman with hyperemesis gravidarum is prone to thiamine deficiency due to the increased demand for glucose metabolism, added to the inability to tolerate adequate food and vitamin/mineral supplements.
 - Thiamine deficiency can result in **beri-beri** symptoms that include fatigue, loss of appetite, emotional instability, sleep disturbances and abdominal discomfort.
 - Advanced neuropathic manifestations of beri-beri include paresthesias, weakness, tenderness and cramps of the lower extremities.
 - The cerebral progression of thiamine deficiency resulting in **Wernicke's encephalopathy**
 - Thiamine administration of 100mg intravenously (IV) or intramuscularly (IM) daily, or enterally if tolerated, has been suggested for any patient with more than 3–4 weeks of emesis.

NB. The initiation of *dextrose-containing intravenous fluids* or aggressive nutrition support, without the provision of thiamine, can precipitate Wernicke's encephalopathy. Why?

- Because the additional glucose will exhaust the thiamin left in the body for its metabolism, worsening the deficiency

Management

Principles

- Correction of Fluid and electrolyte deficits
- Identification and treatment of any co-morbidities
- Identification and management of complications

Inpatient management***Indications for admission***

- Weight loss > 5% from pre-pregnancy
- Ketonuria above +2
- Electrolyte imbalance
- Deranged renal and liver function tests
- Persistent vomiting / failed OPD management

1. Fluid management

- Oral feeding withheld for 24 to 48hrs
- Give 1 to 2 liters of isotonic saline or ringer lactate within 1 - 2hrs
- Continue fluid repletion at a rapid rate (1-2 L over the next 2-3hrs) until the clinical signs of hypovolemia improves.
- Avoid dextrose containing fluid until thiamine is supplemented with the initial rehydration fluid
- Give maintenance fluid after deficit is corrected
- In addition, replace ongoing loss

2. Vitamins

- Thiamine (vitamin B₁):
 - Give 100 mg IV with the initial rehydration fluids before administration of dextrose containing fluids and another 100 mg daily for the next two or three days
- Vitamin B₆:
 - Give 10-25 mg in every liter

3. Electrolyte management

- Depends on the electrolyte abnormality detected on lab tests

4. Antiemetics

- **First line**
 - Meclizine 25mg IV TID, or
 - **Metoclopramide- 5-10 mg IV TID**
 - Promethazine 5-10 mg IM every 6-8hrs
- **Second line**
 - Serotonin antagonists - ondansetron 4-8 mg IV or PO, TID
- **Third line**
 - Chlorpromazine 25mg IV or IM QID.

NB.

- Combinations of different drugs should be considered in women who do not respond to a single antiemetic.
- Shift from first to second line if emesis continues without improvement after 24hrs of therapy.

5. Diet

- PO diets that minimize nausea and vomiting can be resumed after a short period of gut rest.
- Advise the patient to avoid empty stomach
- Advise intake of small and frequent diet
- Counsel on restriction of coffee, and spicy, odorous, high fat, acidic and very sweet foods
- Counsel on preferably taking protein dominant, salty (e.g. nuts), low fat, tasteless and dry snacks/meals
- Encourage on fluid intake (better tolerated if cold, clear, and carbonated or sour)
- Advise on taking peppermint containing products (e.g. chewing gum) to reduce postprandial nausea

- Advice not to take drugs that may cause nausea and vomiting, e.g. iron supplement should be temporarily discontinued.
- Advise on taking ginger or ginger containing preparations.
- Counsel on avoiding of environmental triggers: - stuffy rooms, strong odors (eg, perfume, chemicals, food, smoke), heat, humidity, noise, and visual or physical motion (e.g., flickering lights, driving).

Outpatient management

- IV fluids:
 - Give 2Ls of IV fluid, infuse first liter over 1-2 hours and then 1000 ml over 4 hours (i.e., 2Ls over 5 to 6hrs),
- Followed by further assessment, including urine ketone testing.
- Discharge the patient from outpatient care with PO medications and dietary advice Or, transfer / admit for inpatient care.
- Medications
 - Vitamin B6 (pyridoxine): - 10-25mg PO BID-QID and
 - Meclizine 25 mg PO TID, or
 - Metoclopramide: - 5-10 mg PO TID, or
 - Promethazine: - 12.5-25 mg PO TID to QID, or
 - Chlorpromazine 12.5 mg IM BID

Chapter 27

PELVIC ORGAN PROLAPSE

Prepared by Dr. Amanuel Tefera (MI)

Definition and Epidemiology

- POP is a bulge or protrusion of pelvic organs and their associated vaginal segments into or through the vagina.
- In the United States, it is the **third** most common indication for hysterectomy
- A woman has an estimated lifetime risk of 12% to undergo surgery for prolapse or incontinence.
- 50% of women older than age 50 have some degree of pelvic organ prolapse, but fewer than 20% seek treatment.
 - This may be due to a number of reasons, including lack of symptoms, embarrassment, or misperceptions about available treatment options.

Pathophysiology

- Pelvic organ support is maintained by complex interactions among the:-
 - pelvic floor muscles
 - pelvic floor connective tissue, and
 - vaginal wall
- Pelvic organ prolapse results from attenuation of this supportive structures, whether by *actual tears* or “*breaks*” or by *neuromuscular dysfunction* or both.

Support systems that keep pelvic organs in place

- **Supports of the uterus,** are categorized in to three tier system
 - **Upper tier,** includes:
 - ✓ Endopelvic fascia covering the uterus
 - ✓ Round ligament
 - ✓ Broad ligament
 - ✓ *It is the weakest of the uterus and it maintain the uterus in anteverted position*
 - **Middle tier,** includes:
 - ✓ Condensations of endopelvic fascia
 - ✓ Cardinal / Transverse cervical/ **Mackenrodt's** ligament
 - ✓ Utero sacral ligament
 - ✓ Pubocervical ligament
 - ✓ Pubourethral ligament
 - ✓ *The strongest support of the uterus*
 - **Inferior/ Lower tier**
 - ✓ Pelvic diaphragm: Levator ani muscle
 - ✓ Perineal body: Fibro muscular condensation between the vagina & the rectum
- **Vagina has three levels of support, as described by DeLancey**
 - **Level I Support:**
 - ✓ consists of the cardinal and uterosacral ligaments which serve to maintain the vaginal length and axis
 - ✓ The uterosacral ligaments are posterior fibers that attach to the presacral region at the level of S2 through S4.
 - ✓ *Defects in this support complex may lead to apical prolapse.*

- **Level II Support:**

- ✓ These are the connective tissue attachments of the lateral vagina anteriorly to the arcus tendineus fascia pelvis and posteriorly to the arcus tendineus rectovaginalis.

- ✓ Detachment of this connective tissue from the arcus tendineus fascia pelvis leads to *lateral or paravaginal anterior vaginal wall prolapse*.

- **Level III Support:**

- ✓ The perineal body, superficial and deep perineal muscles.

- ✓ Damage to level III support contributes to *anterior and posterior vaginal wall prolapse*,

Factors believed to be involved in pelvic organ support failure:

- Genetic predisposition,
- Loss of pelvic floor striated muscle support,
 - The levator ani muscle muscles comprised of three regions iliococcygeal, pubococcygeus and puborectalis muscle
 - These muscles provide a supportive diaphragm through which the urethra, vagina, and rectum egress.
 - When the levator ani muscle has normal tone and the vagina has adequate depth, the upper vagina lies nearly horizontal in the standing female.
 - When the levator ani muscle loses tone, the vagina drops from a horizontal to a semi-vertical position
 - ✓ This widens or opens the genital hiatus and predisposes pelvic viscera to prolapse.
- Vaginal wall weakness, and loss of connective attachment

Risk factors**▪ Pregnancy**

- Specific obstetric risk factors remain controversial.
 - These include fetal macrosomia, prolonged second-stage labor, episiotomy, anal sphincter laceration, epidural analgesia, forceps use, and oxytocin stimulation of labor

▪ Vaginal childbirth

- Vaginal childbirth is the most frequently cited risk factor. The risk of POP increased 1.2 times with each vaginal delivery.

▪ Aging

- In women age 20 to 59 years, the incidence of POP roughly doubled with each decade.
- Aging is a complex process, and separating the effects of physiologic aging and estrogen deprivation is problematic.

▪ Menopause

- Hypoestrogenism
 - ✓ Reproductive hormones are essential to maintain pelvic organ support

▪ Chronically increased intraabdominal pressure

- COPD
- Constipation
- Obesity
 - ✓ Being overweight (BMI 25–30 kg/m²) was associated with an increased risk of 31 to 39%, and obesity (BMI >30 kg/m²) with an increased risk of 40 to 75%.

- **Race**

- POP is more common in White Women
- Black women more commonly have a narrow pubic arch and an android or anthropoid pelvis. These shapes are protective against POP compared with the gynecoid pelvis typical of most white women.

- Pelvic floor trauma
- Cigarette smoking
- Connective tissue disease
- Spinal bifida

Description and classification

Pelvic organ prolapse includes:-

- **Anterior compartment defects**

- Urethrocele
- Cystocele
 - ✓ Is herniation of the urinary bladder through the anterior vaginal wall

- **Apical defects**

- Uterine prolapse/Hysterocele
 - ✓ is generally the result of poor cardinal or uterosacral ligament apical support, which allows downward protrusion of the cervix and uterus toward the introitus.
- Vaginal vault prolapse(Post hysterectomy)

- **Posterior compartment defects**

- Enterocele
 - ✓ Is a herniation of the peritoneum and small bowel and is the only true hernia among the pelvic support disorders
- Rectocele
 - ✓ Is a protrusion of the rectum into the vaginal lumen

Procidentia, which involves prolapse of the uterus and vagina, and total vaginal vault prolapse, which can occur after hysterectomy, represent eversion of the entire vagina.

POP Quantification system

POP-Q system

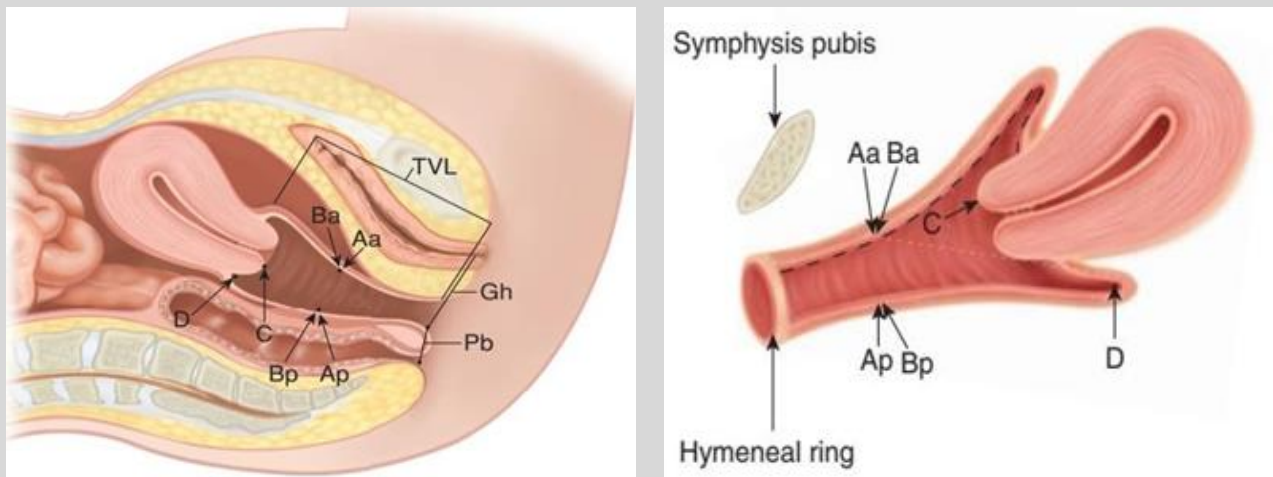
- This system contains a series of site specific measurements each measured relative to the hymen which is an anatomic landmark.
 - With the ***hymenal plane defined as zero***,
 - Points proximal to the hymen are described with a negative number and Points distal to the hymen are noted using a positive number
 - There are nine points; all except TVL are measured during patient Valsalva and should reflect maximum protrusion.

Possible Ranges of the Six Sites specific Pelvic Organ Prolapse.

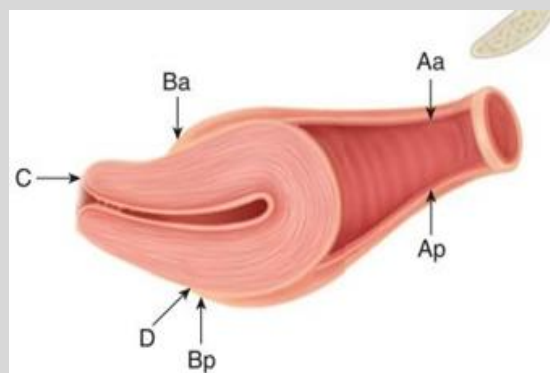
Points	Description	Range
Aa	Anterior wall 3 cm from hymen	-3 cm to +3 cm
Bb	Most dependent portion of rest of anterior wall	-3 cm to +TVL
C	Cervix or vaginal cuff	±TVL
D	Posterior fornix (if no prior hysterectomy)	±TVL or omitted
Ap	Posterior wall 3 cm from hymen	-3 cm to +3 cm
Bp	Bp Most dependent portion of rest of posterior wall	-3 cm to +TVL

- The total vaginal length (TVL) is the greatest depth of the vagina in centimeters when point C or D is reduced to its fullest position.

- Genital Hiatus and Perineal Body.
 - The genital hiatus is measured from the middle of the external urethral meatus to the midline of the posterior hymenal ring.
 - The perineal body is measured from the posterior margin of the genital hiatus to the mid anal opening.



Normal



When fully prolapsed

FIG. The anatomic landmarks used during pelvic organ prolapse quantification (POP-Q)

The POP-Q staging system of pelvic organ support

Stage 0	No prolapse is demonstrated. <i>Points Aa, Ap, Ba, and Bp are all at -3cm, and either point C or D is between -TVL cm and (TVL-2) cm</i>
Stage I	The criteria for stage 0 are not met, but the most distal portion of the prolapse is >1 cm above the level of the hymen(i.e., its quantitation value is < -1 cm)
Stage II	The most distal portion of the prolapse is ≤1 cm proximal to or distal to the plane of the hymen(i.e. its quantitation value is ≥ -1 cm but ≤ +1cm)
Stage III	The most distal portion of the prolapse is > 1 cm below the plane of the hymen but protrudes no further than 2 cm less than the total vaginal length in centimeters (i.e., its quantitation value is > +1 cm. But < [TVL-2] cm).
Stage IV	Essentially, complete eversion of the total length of the lower genital tract is demonstrated. The distal portion of the prolapse protrudes to at least (TVL-2) cm (i.e., its quantitation value is ≥ + [TVL-2] cm).

Baden-Walker Halfway system

- This descriptive tool is also used to classify prolapse during physical examination and is in widespread clinical use.

Grade 0	Normal position for each respective site
Grade 1	Descent halfway to the hymen
Grade 2	Descent to the hymen
Grade 3	Descent halfway past the hymen
Grade 4	Maximum possible descent for each site

Clinical manifestations of POP

Symptoms

- Bulge symptoms
 - Sensation of vaginal bulging or protrusion
 - Seeing or feeling a vaginal or perineal bulge
 - Pelvic or vaginal pressure
 - Heaviness in pelvis or vagina
- Urinary Symptoms
 - Stress urinary and urge incontinence (SUI), urinary frequency, urinary retention, recurrent urinary tract infection, or Manual reduction of prolapse to start or complete voiding.
- Bowel symptoms
 - Incontinence, feeling of incomplete emptying, constipation, urgency, digital evacuation to complete defecation.
- Sexual symptoms and pain
 - Dyspareunia, decreased lubrication, decreased sensation, decreased arousal or orgasm.
 - Pelvic and Back Pain

Physical examination

- Pelvic examination
 - The vulva and perineum are examined for signs of vulvar or vaginal atrophy, lesions, or other abnormalities.
 - Sacral reflexes is using a cotton swab.
 - ✓ The bulbocavernosus reflex: is elicited by tapping or stroking lateral to the clitoris and observing contraction of the bulbocavernosus muscle bilaterally.
 - ✓ Anal wink reflex: for evaluation of anal sphincter innervation is completed by stroking lateral to the anus and observing a reflexive contraction of the anus.
 - ✓ Intact reflexes suggest normal sacral pathways.

- Examination of prolapsed organ begins by asking a woman to attempt Valsalva maneuver prior to placing a speculum in the vagina.
- If unable to adequately complete a Valsalva maneuver; are asked to cough. If the full extent of prolapse cannot be demonstrated, a woman should be examined in a standing position.

During this assessment try to answer three questions:

- (1) Does the protrusion come beyond the hymen?
- (2) What is the presenting part of the prolapse (anterior, posterior, or apical)?
- (3) Does the genital hiatus significantly widen with increased intra-abdominal pressure?

- The use of a Graves speculum or Baden retractor can help to evaluate the apical compartment of the vagina.
- The anterior and posterior compartments are best examined with the use of a univalve or Sims' speculum.
- Rectovaginal examination may be useful in evaluating the posterior compartment to distinguish a posterior vaginal wall defect.
 - ✓ A clinician's index finger is placed in the rectum and thumb on the posterior vaginal wall.
 - ✓ Small bowel may be palpated between the rectum and vagina, confirming enterocele.
 - ✓ Enterocele can only definitively be diagnosed by observing small bowel peristalsis behind the vaginal wall.
- **Bimanual examination** is performed to identify other pelvic pathology and a stress test for associated stress incontinence.
- If the POP-Q examination is performed, the genital hiatus (Gh) and perineal body (Pb) are measured during Valsalva maneuver.
 - The total vaginal length (TVL) is then measured by placing a marked ring forceps, or a ruler, at the vaginal apex and noting the distance to the hymen.

- A bivalve speculum is then inserted to the vaginal apex. It displaces the anterior and posterior vaginal walls, and points C and D are then measured with Valsalva.
- A univalve speculum is then used to displace the posterior vaginal wall & allow for visualization of the anterior wall and measurement of points Aa and Ba;
- Then the speculum is rotated 180 degrees to displace the anterior wall and allow examination of the posterior wall and Points Ap and Bp are measured.

Management principle

Non-surgical measures

Considered in women with:-

- Mild to moderate prolapse (stage I & II)
- Those who desire preservation of future childbearing
- Those who do not desire surgical intervention.

The goals of a conservative therapy approach to the treatment of prolapse are as follows

- Prevent worsening prolapse
- Decrease the severity of symptoms
- Increase the strength, endurance, and support of the pelvic floor musculature
- Avoid or delay surgical intervention

Includes:-

1. Pelvic Floor Muscle trainings(PFMT)

- Referred to as Kegel exercises. Aimed to tighten and strengthen the pubococcygeus muscles.
- PFMT entails voluntary contraction of the levator ani muscles.
- A typical PFMT regimen begins with the contraction duration a patient can sustain (e.g., 3 seconds) and ask them to hold for this long and then relax for one to two times this duration (e.g., 6 seconds).
 - This squeeze and release is repeated 10 to 15 times.

- Three sets are performed throughout the day for a total of approximately 45 contractions.
- To help localize the correct muscle group, in an office setting, contraction of the levator ani muscle complex: can be palpated through the vagina by the practitioner to provide direct and immediate feedback to the patient

2. Mechanical Devices (Pessaries)

- Pessaries provide pelvic organ support within the vaginal vault.
 - Two categories of pessaries; support and space filling—exist for prolapse.
 - The ring pessary (with diaphragm) is a commonly used **support pessary**, the Gelhorn pessary is a commonly used **space-filling pessary**
 - Traditionally, the ring and other support pessaries are used for stage I and II prolapse, whereas the space-filling pessaries are used for stage III and IV prolapse
 - Possible complications associated with pessary use include vaginal discharge and odor.
 - More severe complications, including vesicovaginal or rectovaginal fistula, small bowel entrapment, hydronephrosis, and urosepsis,
3. **Estrogen**; in postmenopausal women, local estrogen therapy.
 4. Weight reduction in obese patients
 5. Modification of life style (avoid heavy exercise, other habits)

Surgical measures:

- The primary aims of surgery are
 - to relieve symptoms, which may be caused by prolapse, and
 - to restore vaginal anatomy so that sexual function may be maintained or improved
- Surgery should be offered to patients who have tried conservative therapy and were not satisfied with the results or who do not desire conservative therapy.

- Procedures for prolapse can be broadly categorized into three groups:
 - Reconstructive, which use the patient's endogenous support structures (native tissue repair);
 - Compensatory (augmentation), which attempt to replace deficient support with permanent graft material; and
 - Obliterative, which close or partially close the vagina.

Types of Pelvic Organ Prolapse Surgery (Penta surgery)

Surgical Technique	Aim	Indication
Abdominal sacral colpopexy	To correct upper vaginal prolapse	Most commonly used in women with recurrent cystocele, vault, or enterocele
Uterosacral ligament suspension	To correct upper vaginal prolapse	Performed at the time of hysterectomy or in patients with posthysterectomy vaginal vault prolapse
Sacrospinous fixation	To correct upper vaginal prolapse	Performed at the time of hysterectomy or in patients with posthysterectomy vaginal vault prolapse
Anterior vaginal repair (anterior colporrhaphy)	To correct anterior wall prolapse	May be used for the treatment of prolapse of the bladder or urethra (bladder, urethra, or both herniate downward into the vagina)
Posterior vaginal repair (posterior colporrhaphy) and perineorrhaphy	To correct posterior wall prolapse	May be used for the treatment of rectocele (rectum bulges or herniates forward into the vagina), defects of the perineum, or both

Compensatory (augmentation)

- Vaginal repair with synthetic mesh or the mesh to correct anterior wall prolapse, apical vaginal prolapse, or both

- Depending on the specific defect biologic graft augmentation augmentation can either be anterior, apical, or both
- This repair is not routinely recommended.

Obliterative Procedures

- Obliterative approaches for pronounced POP include LeFon colpocleisis and complete colpodeisis
- These procedures involve removing vaginal epithelium, suturing anterior and posterior vaginal walls together, obliterating the vaginal vault, and effectively closing the vagina.
- Obliterative procedures are appropriate only for elderly or medically compromised patients with no desire for future coital activity.

SAMPLE HISTORY

Chief complaint: Mass protruding per vagina of 2 years and 2 month duration.

HPI

This is a 55 years old para 8, patient who has been amenorrheic for the last 6 years; was relatively healthy until she presented with protruding mass of 2 years, which begins to get worse since 6 months back. The mass protrudes out up on standing, coughing and lifting heavy objects and returns on attaining supine position. The mass was soft and she was able to push back the mass manually. It is painless and not associated with abnormal bleeding or foul smelling discharge. She also complains of dragging type of lower abdominal pain which doesn't radiate and get exacerbated when she stands and strains while relieved upon rest. For this reason she went to a local health center and was referred to Jimma medical center.

She also has frequent urination, urgency, leakage of urine while coughing or sneezing and a burning sensation during urination. Due to urinary frequency, she was forced to reduce her water intake during work.

Her first delivery was 35years back and her last was 10 years back. All of her children were born through spontaneous vaginal delivery at her house attended by traditional birth attendants. Labor, lasted less than 6 hours in all cases. All her children are alive and healthy. None of her pregnancies had antepartum or post-partum complications.

She works in a factory as a daily laborer (animal food product factory). The job requires heavy lifting and lots of strenuous activities.

Otherwise:-

- She has no history of vaginal bleeding or discharge; She has no history of reddish discoloration of urine. She has no history of abdominal swelling or distention. She has no history of chronic cough,

- She has no previous history of constipation.
- She has no history of trauma to the pelvis or gynecological surgery She has no history of similar illness in the family.
- She has no personal and family history of chronic illness like DM, hypertension or asthma.
- She has no color change to the mass or ulceration.
- She has no history of urinary retention, change in bowel habits. She is not sexually active currently.
- She came to this hospital by walking.

Pertinent physical examination:

General appearance: Stable, comfortable, well-nourished and cooperative during physical examination.

Vital sign:

- BP=130/80 mm Hg from right arm, supine position.
- PR=88 beats/min from radial artery.
- RR=18 breaths/min
- To=36.00C from axilla Weight: 83kg height: 160cm BMI: 32.4kg/m²

Pelvic examination

- Examination of external genitalia: atrophic vulva, inverted triangle like hair distribution. Type I clitoridectomy, No ulcers, discolorations, pruritus or warts.
- Digital vaginal examination: a soft, non-tender and reducible mass felt bulging from the anterior vaginal wall.
- After straining a mass protruding per vagina is seen half way past the hymen. By Baden-Walker Half Way system = stage three pelvic organ prolapse.
- No cervical motion tenderness.
- Recto vaginal examination: no peristalsis felt behind the vaginal wall

Differential Diagnosis

- Tumors of the urethra and bladder
- Urethral diverticulum
- Skene's and Bartholin's glands cysts or abscesses.
- Soft tumors (lipoma, leiomyoma, sarcoma)
- Cervical/endometrial tumors (pedunculated myoma or endometrial polyps); if prolapsed through a dilated cervix.
- Elongated cervix

Investigations

- Hematocrit, Blood group & Rh, U/A
- RFT
- U/S for residual volume, kidneys, abdominal masses.
- Intravenous pyelogram (IVP)

Management

(See above)

Chapter 28

INFERTILITY

Prepared by Dr. Abubeker Nuredin (MI)

Definitions and Classification

Infertility is defined as the inability to conceive after 1 year of unprotected intercourse of reasonable frequency.

- Affects 10 to 15 percent of reproductive-aged couples.

Fecundability is the ability to conceive, a monthly probability of conceiving is 20-25%.

- In those attempting conception, more than 85 percent will be pregnant by 1 year.

Fecundity is clinically defined as the capacity to have a live birth.

Fertility is the ability to have clinical pregnancy.

Sterility is a permanent state of infertility.

Time to pregnancy refers to the length of time usually measured in months that it takes a couple to conceive.

Infertility can be subdivided into;

- Primary infertility, that is, no prior pregnancies, and
- Secondary infertility, referring to infertility following at least one prior conception.,

Most couples are more correctly considered to be *sub-fertile*, rather than infertile, as they will conceive if given enough time. Exceptions, such as

1. woman with bilaterally obstructed fallopian tubes
2. the azoospermic male

Etiologies

- In general, infertility can be attributed to:
 - the female partner one third of the time,
 - the male partner one third of the time, and
 - both partners in the remaining one third.
 - This approximation emphasizes the value of assessing both partners before instituting therapy

TABLE 20-2. Etiology of Infertility

Male	25%
Ovulatory	27%
Tubal/uterine	22%
Other	9%
Unexplained	17%

Approach to Infertility

History

Female History

- Gynecologic/Obstetrics
 - Prior history of conception
 - ✓ is important because it indicates ovulation and a patent fallopian tube at some point.
 - ✓ A prolonged time to conception may suggest borderline fertility and may raise the chance of determining an etiology.
 - Menstruation characteristics,

- Prior contraceptive use,
- A coital history, including frequency and timing of intercourse
- Possible menopause symptoms and infertility duration.
- Symptoms of endometriosis, recurrent ovarian cysts, leiomyoma's
- Prior cervical conization can diminish cervical mucus and cervical competence
- Symptoms of STIs, &/or PID is also pertinent.
- Pregnancy complications such as miscarriage, preterm delivery, retained placenta, postpartum dilation and curettage, chorioamnionitis, or fetal anomalies also are recorded.
- Medical and Surgical
 - seek symptoms of hyperprolactinemia or thyroid disease
 - Acne or hirsutism may point to polycystic ovarian syndrome (PCOS) or congenital adrenal hyperplasia.
 - Prior chemotherapy or pelvic irradiation may suggest ovarian failure.
 - Previous abdominal surgeries, especially if linked to endometriosis or adhesion formation, can lower fertility.
- Social history
 - Focuses on eating habits,
 - ✓ even modest weight reduction in overweight women is correlated with normalized menstrual cycles and subsequent pregnancies
 - Cigarette smoking lowers fertility rates & higher rates of miscarriage, abruption, fetal-growth restriction, and preterm labor.
- Environmental Factors
 - Agricultural pesticides and herbicides, lead, and bisphenol is implicated to underlie a broad range of women's reproductive disorders.

TABLE 20-3. Effects of Obesity and Environmental Factors on Fertility

Factor	Impact on Fertility
Obesity (BMI >35)	2-fold increase TTC
Underweight (BMI <19)	4-fold increase TTC
Smoking	1.6-fold increase RR
Alcohol (>2/day)	1.6-fold increase RR
Illicit drugs	1.7-fold increase RR
Toxins	1.4-fold increase RR
Caffeine (>250 mg/day)	45% decrease fecundability

BMI = body mass index; RR = relative risk of infertility;
TTC = time to conception.

- Ethnicity and Family History
 - A family history of infertility, recurrent miscarriage, or fetal anomalies may point to a genetic etiology that warrants genetic screening.eg, PCOS and endometriosis occur in familial clusters.

Male History

- Pubertal development and sexual function difficulties.
- Erectile dysfunction with diminished beard growth, may suggest lower testosterone levels
- search for developmental anomalies such as hypospadias, which could result in suboptimal semen deposition
- Sexually transmitted may lead to vas deferens inflammation and obstruction. mumps in an adult can create testicular inflammation and damage spermatogenic stem cells

- Prior cryptorchidism, testicular torsion, or testicular trauma may suggest abnormal spermatogenesis.
 - Compared with fertile males, male with unilateral or bilateral cryptorchidism have fertility rates of 80 %and 50 %, respectively.
 - Warm intrabdominal temperature may cause permanent stem cell damage.
- A history of varicocele is also important.
- Spermatogenesis, from stem cell to mature sperm, takes nearly 90 days any detrimental event in the prior 3 months can adversely affect semen characteristics
- Illness with high fevers or chronic hot tub use can temporarily impair sperm quality.
- Prior chemotherapy or local radiation treatment that may damage spermatogonial stem cells.
- Hypertension, diabetes mellitus, and neurologic disorders can be associated with erectile dysfunction or retrograde ejaculation.
 - Poor semen quality has been associated with higher rates of diabetes, ischemic heart disease, and mortality and an increased individual and familial cancer risk.
- Excessive weight has been associated with obesity-related secondary hypogonadism, erectile dysfunction, and infertility.
- Several medications are known to worsen semen parameters, including cimetidine, erythromycin, gentamicin, tetracycline, and spironolactone.
- cigarettes, alcohol, illicit drugs, and environmental toxins all adversely affect semen quality
- An exercise history also is obtained; bicycling has been associated with erectile dysfunction and decreased sperm concentration.

Physical examination

Examination of the Female Patient

A thorough physical examination may provide many clues to the cause of infertility.

- Vital signs, height, and weight should be recorded.
 - Short stature may indicate Turner syndrome.
- Hirsutism, alopecia, or acne indicates the need to measure androgen levels.
- Acanthosis nigricans is consistent with insulin resistance associated with PCOS or diabetes or, Cushing syndrome.
- The breast examination must be normal, and when indicated by age or family history, a mammogram is obtained prior to initiating hormonal treatment.
- Galactorrhea is often indicative of hyperprolactinemia.
- A pelvic examination may be particularly informative.
 - Inability to place a speculum through the introitus may raise doubts about coital frequency.
 - The vagina should be moist and rugated, and the cervix should have a reasonable amount of mucus.
 - ✓ Both indicate adequate estrogen production.
 - An enlarged or irregularly shaped uterus may reflect leiomyomas, whereas a fixed uterus suggests pelvic scarring due to endometriosis or prior pelvic infection.
 - Uterosacral nodularity or ovarian masses may implicate endometriosis or, malignancy.
 - All women should have cervical cancer screening that is up to date prior to treatment.

Examination of the Male Patient

- Normal secondary sexual characteristics that reflect androgen.
- Gynecomastia or eunuchoid habitus may suggest Klinefelter syndrome.

- The penile urethra should be at the glans tip for proper semen deposition in the vagina.
- Testicular length measures at least 4 cm & minimal testicular volume is 20mL.
 - Small testes are unlikely to produce normal sperm numbers.
- A testicular mass may indicate testicular cancer, which can present as infertility.
- Epididymal fullness may suggest vas deferens obstruction, the pampiniform plexus of veins is palpated for varicocele

Evaluation for Infertility

The infertility evaluation can be conceptually **simplified into confirmation of:**

- Ovulation,
- Normal female reproductive tract anatomy, and
- Normal semen characteristics

Evaluation of Female Infertility

Includes evaluation for:

- Anovulation
- Diminished Ovarian reserve
- Abnormalities of Female reproductive tract

TABLE 20-6. Infertility Testing

Etiology	Evaluation
Ovulatory function/ovarian reserve	Ovulation predictor kit Early follicular FSH $\pm$ estradiol level Antimüllerian hormone (AMH) Endocrine disorders (TSH, prolactin) Ovarian sonography (antral follicle count)
Tubal/pelvic disease	Hysterosalpingography Laparoscopy + chromotubation
Uterine factors	Hysterosalpingography Transvaginal sonography/saline-infusion sonography $\pm$ Magnetic resonance imaging Hysteroscopy $\pm$ laparoscopy
Male factor	Semen analysis

Evaluation for anovulation

- Ovulation may be perturbed by abnormalities within the:
 - hypothalamus,
 - anterior pituitary, or
 - Ovaries.
- **Clinical Evaluation**
 - A patient's menstrual history is an excellent predictor of regular ovulation. Signs suggestive of ovulation include:
 - ✓ cyclic menses at an interval of 25 to 35 days and duration of bleeding of 3 to 7 days
 - ✓ **mittelschmerz** (mid cycle pelvic pain associated with ovulation), or by minimal symptoms such as breast tenderness, acne, food cravings, and mood changes.

- ✓ dysmenorrhea, although severe dysmenorrhea may suggest endometriosis
- ✓ raised basal body temperature (BBT)
 - A postovulatory rise in progesterone levels increases basal temperature by approximately 0.4° to 0.8°F.
 - This biphasic temperature pattern is strongly predictive of ovulation.

▪ Ovulation Predictor Kits

- These kits measure urinary luteinizing hormone (LH) concentration by colorimetric assay.
 - ✓ In general, a woman begins testing 2 to 3 days prior to the predicted LH surge, and testing is continued daily.
 - ✓ Because the LH surge spans 48 to 50 hours, timing is probably not critical as long as the test is performed daily.
 - ✓ If equivocal results are obtained, the test can be repeated in 12 hours.
- *Urine LH surge assays were estimated to have 100% sensitivity and 96% accuracy, although this is undoubtedly an overestimate of typical-use results.*

▪ Serum Progesterone

- Adequate progesterone levels are required for endometrial preparation prior to implantation. This has led to the concept of luteal phase defect (LPD), defined as inadequate endometrial development due to suboptimal progesterone production.
- In a classic 28-day cycle, serum progesterone is obtained on
 - ✓ cycle **day number 21** following the first day of menstrual bleeding, or 7 days following ovulation.
- Levels during the follicular phase are generally **<2 ng/mL**.
 - ✓ Values above 4 to 6 ng/mL correlate with ovulation and progesterone production by the corpus luteum.

- The mid luteal progesterone level is best regarded as an acceptable test for ovulation but not an absolute indicator of adequate luteal function.
- **Endometrial Biopsy**
 - Luteal phase endometrial biopsy was hoped to reflect both corpus luteum function and endometrial response, and thereby provide more clinically relevant information than a serum progesterone level alone, but it has high intra-observer and inter-observer variability during histologic evaluation.
 - Currently endometrial biopsy is no longer considered a routine part of infertility evaluation.
- **Sonography**
 - Serial ovarian sonographic evaluations can demonstrate the development of a mature antral follicle and its subsequent collapse during ovulation.
 - Sonography is an excellent approach for supporting the diagnosis of PCOS.

Evaluation for diminished ovarian reserve

Ovulatory status does not provide a complete picture of ovarian function. A woman may have regular, ovulatory menses but have reduced follicular response to ovarian stimulation compared with other women of similar age due to a smaller ovarian follicle pool.

- Although most often the result of **advancing age**, a small ovarian reserve can be linked to:
 - smoking,
 - genetic conditions
 - prior ovarian surgery
 - chemotherapy
 - pelvic irradiation

Reproductive Aging

- Female age and fertility have a clear inverse relationship. This loss is primarily attributable to a decline in oocyte quality and quantity.
- Ongoing atresia of non-dominant follicles proceeds throughout a woman's reproductive life span.
 - the risks of genetic abnormalities and mitochondrial deletions in the remaining oocytes substantially rise as a woman ages.
- These factors result in lower pregnancy rates and higher miscarriage rates in both spontaneous and stimulated cycles.
- Fertility testing is ideally performed
 - starting at age 35 in all patients desiring conception
 - For any woman with:
 - ✓ an unexplained change in menstrual cyclicity,
 - ✓ family history of early menopause, or
 - ✓ risk factor for POI

Tests for assessment of ovarian reserve

- Currently, early follicular measurement of follicle-stimulating hormone (FSH) is probably the most cost-effective approach for the general practitioner.
- Serum anti mullerian hormone level is rapidly moving into standard practice.
- Measuring serum inhibin B levels and the clomiphene citrate challenge test have fallen out of favor.
- Abnormal test findings correlate with a ***poorer prognosis for achieving pregnancy***, and referral to an infertility specialist is advisable.
- A normal test result does not negate the effect of a woman's age on her fertility status.
- Borderline results in a younger woman with infertility may suggest a need for more intensive treatment.

- **Follicle-Stimulating Hormone and Estradiol**

- Elevated menstrual cycle **day 3 FSH** is a marker of decreased oocyte number and quality: > 10 to 12 mIU/ml
- An elevated estradiol level (> 80 pg/mL) on the third day can also represent a decrease in ovarian reserve, but it is not as predictive as the FSH level.
- An elevated estradiol level may suppress the FSH level through negative inhibition and falsely portray a normal FSH level.
- FSH level greater than 40 mIU/mL associated with amenorrhea is the definition of ovarian failure. If these findings occur in a woman under 40 years old, she has premature ovarian failure

- **Anti-mullerian Hormone (AMH)**

- Is expressed by the fetal testes during male differentiation to prevent development of the mullerian system.
- Also expressed by the granulosa cells of small preantral follicles & role in dominant follicle play recruitment.
- Measurement of AMH levels has advantages compared with FSH and inhibin testing. Because:
 - ✓ AMH expression is gonadotropin-independent and therefore is relatively independent of cycle stage and is consistent across cycles
 - ✓ AMH levels correlate with ovarian primordial follicle number more strongly than FSH or inhibin levels
 - ✓ AMH levels may drop prior to observable changes in FSH or estradiol levels, thereby providing an earlier marker of waning ovarian function
 - ✓ AMH levels are under consideration as a tool for diagnosis of PCOS.
 - Levels are raised two- to threefold in affected women compared with normal cycling women, this observation is consistent with the multiple early follicles found in patients with PCOS.

- **Antral Follicle Count (AFC)**

- Sonographic evaluation of the follicular phase antral follicle count (AFC) is commonly used as a reliable predictor for subsequent response to ovulation induction. The total AFC usually ranges between 10 and 20 in a reproductive-aged woman.
- An AFC with **< 3 to 6 total antral follicles** predicts poor response to gonadotropin stimulation during IVF cycles, but this measure is limited by low sensitivity.

Evaluation for female anatomic abnormalities

- **Tubal and Pelvic Factors**

- Chronic pelvic pain or dysmenorrhea may suggest tubal obstruction, pelvic adhesions, or both.
- Adhesions can prevent normal tubal movement, ovum pick-up, and transport of the fertilized egg into the uterus.
- Etiologies include tubal disease, especially pelvic infection; endometriosis; and prior pelvic surgery
- Approximately ***one third to one fourth of all infertile women*** are diagnosed with tubal disease in developed countries.
 - ✓ With PID, tubal infertility has been estimated to follow in **12%, 23%, and 54%** of women following **one, two, and three** cases of PID, respectively.
 - ✓ an absent PID history is not overly reassuring, ***as nearly one half of patients who have tubal damage have no clinical history*** of antecedent disease

- In developing countries, **genital tuberculosis** may account for 3 to 5% of infertility.
 - ✓ Tubal damage and endometrial adhesions are underlying causes
 - ✓ The likelihood of a return to fertility after Anti-TB treatment is low, and IVF with embryo transfer remains the most reliable approach.
- Endometriosis affect fertility by:
 - ✓ Chronic inflammation and intra peritoneal bleeding can lead to pelvic adhesions.
 - ✓ increase in peritoneal fluid inflammatory factors,
 - ✓ alterations in endometrial immunologic function,
 - ✓ poor oocyte or embryonic quality, or
 - ✓ Impaired implantation
- **Uterine Abnormalities**
 - Uterine abnormalities can be congenital or acquired.
 - ✓ Common congenital anomalies include uterine septum, bicornuate uterus, unicornuate uterus, and uterine didelphys, the fertility effects of these anomalies have been difficult to verify, although associated with pregnancy complications.
 - ✓ Acquired anomalies include **intrauterine leiomyoma's, adenomyosis, polyps, and Asherman syndrome.**
 - ✓ Adenomyosis is thought to impact fertility and obstetric outcomes via functional and structural defects in the endometrium and myometrium
 - ✓ Endometrial polyps are found in an estimated 3 to 5 percent of infertile women. The prevalence is higher in women with symptoms such as inter menstrual or postcoital bleeding.

✓ The presence of intrauterine adhesions (synechiae), is termed ***Asherman syndrome***.

- Asherman syndrome develops most frequently in women with prior uterine dilation and curettage, particularly in the context of infection and pregnancy.
- The clinical history will often include an acute postsurgical decline in menstrual bleeding or even amenorrhea.
- A woman with genital tuberculosis is also at high risk for intrauterine adhesions.

▪ **Anatomy Evaluation for tubal and uterine anomalies**

1. Hysterosalpingography (HSG)
2. transvaginal sonography (TVS) with or without saline instillation,
3. 3-dimensional (3-D) TVS
4. hysteroscopy,
5. laparoscopy, and
6. magnetic resonance (MR) imaging

TABLE 20-8. Advantages and Disadvantages of Various Methods for Evaluating Pelvic Anatomy

	Tubal Patency	Uterine Cavity	Developmental Defects	Endometriosis or PAD	Ovaries
HSG	+	+	-	+/-	-
TVS	-	+/-	+/-	-	+
3-D TVS	-	+	+	-	+
SIS	-	+	+	-	+
MR imaging	-	+	+	+	+
Hysteroscopy	-	+	+	-	-
Laparoscopy	+	-	+	+	+

HSG = hysterosalpingography; MR = magnetic resonance; PAD = pelvic adhesive disease; SIS = saline-infusion sonography; TVS = transvaginal sonography.

- HSG allows visualization of the internal contour of the uterine cavity.
 - ✓ It detects synechiae, congenital anatomic anomalies, and polyps and fibroids if they distort the uterine cavity.
- If an abnormality is detected on HSG, a **hysteroscopy** can be performed to confirm the abnormality.
 - ✓ Hysteroscopy: direct visualization of the uterine cavity.
- If a congenital uterine anomaly is suspected after HSG, a **pelvic magnetic resonance image (MRI)** may be helpful to noninvasively assess the external and internal contours of the uterus.
- **Laparoscopy** is a more invasive method of obtaining information about the external contour of the uterus.

- **Sonohysterography** is performed by placing a small balloon-tipped cannula into the uterus via the cervix and infusing the cavity with sterile saline.
 - ✓ A vaginal ultrasound is performed simultaneously to visualize the internal uterine contour.
 - ✓ Small polyps and fibroids impinging on the uterine cavity can be detected by this method.
 - ✓ However, this method does not allow assessment of the external surface of the uterus.
- **Cervical factor**
 - Post-coital test (Sims-Huhner test): assess quality of cervical mucus & ability of sperm to survive in it
 - Sperm cervical mucus contact test (SCMCT): invitro cross over test, indicated when three post coital tests are negative.

Evaluation of male infertility

Causes of male infertility can roughly be categorized as:

- ✓ abnormalities of sperm production,
- ✓ abnormalities of sperm function,
- ✓ Obstruction of the ductal outflow tract.

Semen Analysis

This is a core test in male fertility evaluation.

For this test,

- ✓ the male is asked to refrain from ejaculation for 2 to 3 days, and a specimen is collected by masturbation or a couple can use specially designed Silastic condoms without lubricants.
- ✓ the sample should arrive in the laboratory **within an hour** of ejaculation.
- ✓ Ideally, **two semen samples separated by at least a month are analyzed**. In practice, frequently only a single sample is analyzed if parameters are normal.

TABLE 20-9. Semen Analysis Terminology

Term ^a	Definition ^b
Oligospermia	Sperm concentration 10–15 million sperm/mL (mild-moderate) or <5–10 million/mL (severe)
Asthenospermia	Greater proportion of immotile sperm or sperm with decreased motility
Teratospermia	Increased proportion of morphologically abnormal sperm
Azoospermia	Semen present but lacks sperm
Aspermia	Sperm and seminal plasma lacking (anejaculation)
Leukocytospermia	Increased white blood cell count
Necrospermia	All sperm nonviable or immotile

^aAlternative terminology includes oligozoospermia, asthenozoospermia, and teratozoospermia.

^bAs compared with World Health Organization (1999) reference values.

In depth investigation is needed in cases of azoospermia, oligospermia, low ejaculate volume & problems of sexual potency.

TABLE 20-10. Semen Analysis Reference Limits

Volume	>1.5 mL ^a
Count	>15 million/mL ^a
Total Motility	>40% ^a
Morphology	>4% ^a
WBCs	<1 million/mL ^b
Round cells	<5 million/mL ^b

^aData from Cooper, 2010.

^bData from World Health Organization, 1999.

WBCs = white blood cells.

Other tests include:

- FSH, LH, Testosterone, prolactin, TSH
- Fructose content in semen
- Testicular biopsy
- Trans rectal ultrasound (TRUS)
- Immunological tests: sperm agglutinating & sperm immobilizing

Unexplained

- No obvious cause for infertility following all standard investigations i.e., semen analysis, ovulation detection, tubal & peritoneal factors, endocrinopathy & post-coital test
- *With expectant management, 60 % conceive in three years.*

Treatment**Couple instructions**

- Assurance: when faults found in both, treat both at a time
- Optimal body weight
- Avoid excessive alcohol ingestion & smoking
- Coital problems need to be carefully evaluated by intelligent interrogation

Treatment of male infertility

- Improve general health
- Avoid smoking, alcohol ingestion, tight underwear
- Clomiphene citrate: 25mg/day for 25 days, for three cycles
- hCG: 5000iu IM weekly
- hMG(human menopausal gonadotropins) & hCG: gonadotropin deficiency or failed CC cases
- Testosterone, GnRH therapy
- Surgery: vaso-vasotomy
- ART: intra-cytoplasmic sperm injection, IUI, IVF

Treatment of female infertility

- Drugs used for ovulation induction
 - **Clomiphene citrate** (start with **50mg** from 5th day of menstrual cycle for five days, doses can be escalated by 50mg every month if there is no documentation of ovulation, no need to go beyond 150mg)
 - Metformin in cases of insulin resistance
 - hMG (human menopausal gonadotropin)
 - FSH
 - hcG (human chorionic gonadotropin)
 - GnRH analogs
- Dexamethasone: reduction of level of androgen
- Bromocriptine: reduction of prolactin
- Substitution therapy for hypothyroidism
- Surgery: wedge resection, ovarian diathermy for polycystic ovarian syndrome, surgery for prolactinomas
- Tubal & peritoneal factors
 - **Tuboplasty**
 - Adhesiolysis
 - Fimbrioplasty
 - Tubal anastomosis
 - Tubo-cornual anastomosis
- Cervical factors
 - Treat proven infections
- Immunological factors
 - Dexamethasone
- ART: - IUI, IVF, GIFT, ZIFT

Chapter 29

AMENORRHEA

Prepared by Dr. Abubeker Nuredin (MI)

Introduction

Amenorrhea is defined as absence of menses

- It can be a transient, intermittent, or permanent condition resulting from dysfunction of the hypothalamus, pituitary, ovaries, uterus, or vagina.

It is classified as

- Primary (absence of menarche by age 15 years)
 - Primary amenorrhea should prompt a thorough evaluation to identify the cause
- Secondary (absence of menses for more than three cycle intervals or six months in women who were previously menstruating).

Primary amenorrhea

Is defined as the absence of menses at:

- **age 15 years** in the presence of normal growth and secondary sexual characteristics or
- **age 13 years**, if no menses have occurred and there is a complete absence of secondary sexual characteristics

Etiologies

Primary amenorrhea is usually the result of a genetic or anatomical abnormality. However, all causes of secondary amenorrhea can also present as primary amenorrhea.

The most common etiologies are

- Gonadal dysgenesis, including Turner syndrome – 43 %
- Müllerian agenesis (absence of vagina, sometimes with absence of uterus) – 15 %
- Physiologic delay of puberty (constitutional delay of puberty, chronic systemic disease, acute illness) – 14 %.
- Polycystic ovary syndrome (PCOS) – 7 %
- Isolated gonadotropin-releasing hormone (GnRH) deficiency – 5%
- Transverse vaginal septum – 3 %
- Weight loss/anorexia nervosa – 2 %
- Hypopituitarism – 2 %

Less common etiologies (≤ 1 percent each) included

- imperforate hymen,
- complete androgen insensitivity syndrome,
- hyperprolactinemia/prolactinoma, other pituitary tumors,
- congenital adrenal hyperplasia,
- hypothyroidism,
- central nervous system defects,
- craniopharyngioma, and Cushing's disease.

A logical approach to the woman with either primary or secondary amenorrhea is to consider disorders based upon the level of control of the menstrual cycle: hypothalamus and pituitary, ovary, and uterus and vagina

Hypothalamic and pituitary disease

- Hypothalamic causes of primary amenorrhea are
 - functional hypothalamic amenorrhea and isolated GnRH deficiency
 - tumors and infiltrative lesions of the hypothalamus or pituitary
- **Functional hypothalamic amenorrhea —**
 - It is characterized by abnormal hypothalamic GnRH secretion leading to
 - ✓ decreased gonadotropin pulsations,
 - ✓ low or normal serum LH concentrations,
 - ✓ absent LH surges,
 - ✓ absence of normal follicular development,
 - ✓ anovulation, and
 - ✓ low serum concentrations of estradiol
 - FSH concentrations are often in the normal range, with a high FSH-to-LH ratio similar to the pattern in prepubertal girls.
 - Cause of functional hypothalamic amenorrhea, are
 - ✓ eating disorders (such as anorexia nervosa)
 - ✓ exercise (like in athletes)
 - ✓ stress
- **Isolated GnRH deficiency-**
 - due to complete congenital gonadotropin-releasing hormone (GnRH) deficiency
 - This syndrome is called idiopathic hypogonadotropic hypogonadism or, if it is associated with anosmia, *Kallmann syndrome*.
- **Constitutional delay of puberty —**
 - is characterized by both delayed adrenarche and gonadarche, and
 - it is often difficult to distinguish clinically from congenital GnRH deficiency
 - is common in boys with a family history of delayed puberty, but it is 5x less common in girls
 - have completely normal pubertal development

- **Hyperprolactinemia**

- The presentation is similar to hypothalamic amenorrhea except for the additional finding of galactorrhea in some patients.
- Is a common cause of secondary amenorrhea but an uncommon cause of primary amenorrhea

- **Infiltrative diseases and tumors**

- Can result in diminished GnRH release or gonadotrope destruction and amenorrhea
- Include craniopharyngioma, germinoma, and Langerhans cell histiocytosis.
- Main indications for MRI are hypogonadotropic hypogonadism, visual field defects, headaches, other evidence of hypothalamic or pituitary dysfunction, or symptoms suggestive of other diseases (such as sarcoidosis.)

- **Systemic illness –**

- include celiac disease, type 1 diabetes mellitus, and inflammatory bowel disease

Gonadal dysgenesis/primary ovarian insufficiency

- The most common cause of primary amenorrhea is ***gonadal dysgenesis caused by chromosomal abnormalities.***
 - result in premature depletion of all ovarian oocytes and follicles
 - These women have **significantly elevated FSH levels** due to the absence of ovarian oocytes and follicles, leading to reduced negative feedback on FSH from estradiol and inhibin B.
- The largest number of patients have ***Turner syndrome*** (45, X as well as other karyotypes), followed by 46, XX gonadal dysgenesis and, rarely, 46, XY gonadal dysgenesis.
 - **Turner syndrome**
 - ✓ Amenorrhea occurs because the oocytes and follicles undergo accelerated apoptosis (in utero in most cases).

- ✓ The ovaries are replaced with fibrous tissue, and in the absence of follicles, there is no ovarian estrogen secretion.
- ✓ The external female genitalia, uterus, and fallopian tubes **develop normally** until puberty when estrogen-induced maturation fails to occur)
- **Primary ovarian insufficiency (POI)**
 - Primary ovarian insufficiency (POI) is defined as the development of clinical menopause before the age of 40 years in women who have a normal karyotype (46, XX)
 - Usually presents as secondary amenorrhea and can be due to chemotherapy or radiation, autoimmune oophoritis

Polycystic ovary syndrome —

- Most cases of PCOS present with oligomenorrhea, but occasionally secondary amenorrhea and rarely primary amenorrhea, may be present.
- The menstrual disturbances in women with PCOS classically have a peripubertal onset.
- The women with PCOS who present with primary amenorrhea typically have higher androgen levels and are more overweight.
- However, most have a slightly early menarche related to overweight, followed by irregular cycles (oligomenorrhea) or secondary amenorrhea.

Outflow tract disorders

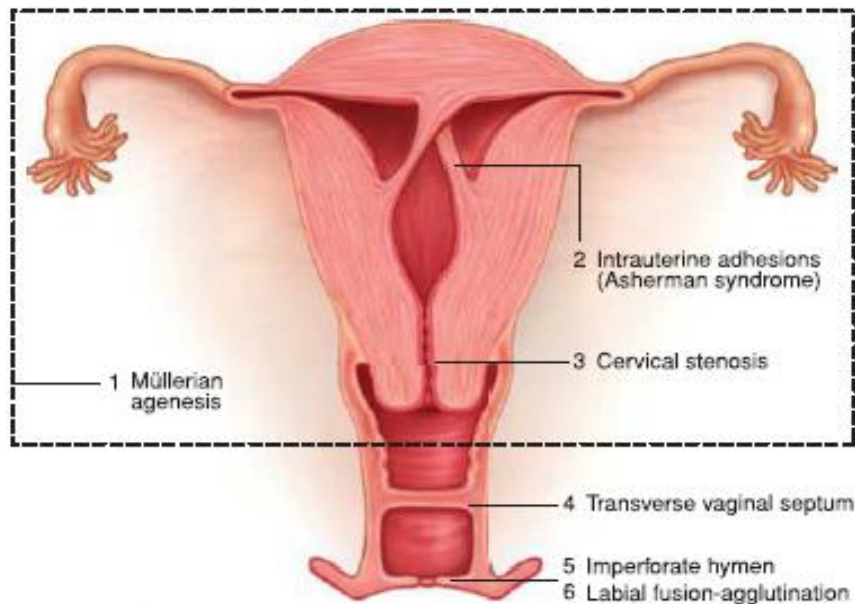


FIGURE 17-1 Anatomic defects that may lead to amenorrhea. Müllerian agenesis results in an absence of the upper reproductive tract structures contain within the dash-lined box.

- Congenital abnormalities of the female reproductive organs account for approximately 20 percent of cases of primary amenorrhea.
- Pelvic or lower abdominal pain is a common presenting symptom in girls with primary amenorrhea and an obstructed reproductive tract.
- **Imperforate hymen**
 - associated with cyclic pelvic pain and a perirectal mass from sequestration of blood in the vagina (hematocolpos)
 - An imperforate hymen is easily corrected with surgery.
- **Transverse vaginal septum**
 - Can occur at any level between the hymenal ring and the cervix.
 - After menarche, the major symptoms are similar to those associated with an imperforate hymen.

- **Müllerian agenesis**

- also known as vaginal agenesis or *Mayer-Rokitansky-Küster-Hauser* (MRKH) syndrome
- refers to congenital absence of the vagina with variable uterine development
- usually accompanied by cervical and uterine agenesis; however, 7 to 10 percent of women have a normal but obstructed or rudimentary uterus with functional endometrium
- The defect results from agenesis or hypoplasia of the müllerian duct system.

Complete androgen insensitivity syndrome —

- is an X-linked recessive disorder in which 46, XY subjects have a normal female phenotype.
- These patients are resistant to testosterone due to a defect in the androgen receptor and, therefore, fail to develop all of the male sexual characteristics that are dependent upon testosterone.
- The external genitalia are typically female in appearance, but testes may be palpable in the labia or inguinal area.
- The testes make müllerian-inhibiting substance, which is functional and causes regression of all müllerian structures: the fallopian tubes, uterus, and upper third of the vagina.
- At puberty, breast development occurs, but the areolae are pale and pubic and axillary hair is sparse.
- Carrier females (46, XX) develop normal internal and external genitalia.
- The diagnosis of this disorder is based upon
 - the absence of the upper vagina, uterus, and fallopian tubes on physical examination and pelvic ultrasonography;
 - high serum testosterone concentrations (in the range for normal men); **and**

- a male (46, XY) karyotype.
- The testes should be surgically excised after puberty because of the increased risk (2 to 5 percent) of developing testicular cancer after age 25 years.

TABLE 17-5. Comparison of Müllerian Agenesis and Androgen Insensitivity Syndrome

Presentation	Müllerian Agenesis	Androgen Insensitivity
Inheritance pattern	Sporadic	X-linked recessive
Karyotype	46,XX	46,XY
Breast development	Yes	Yes
Axillary and pubic hair	Yes	No
Uterus	No	No
Gonad	Ovary	Testis
Testosterone	Female levels	Male levels
Associated anomalies	Yes	No

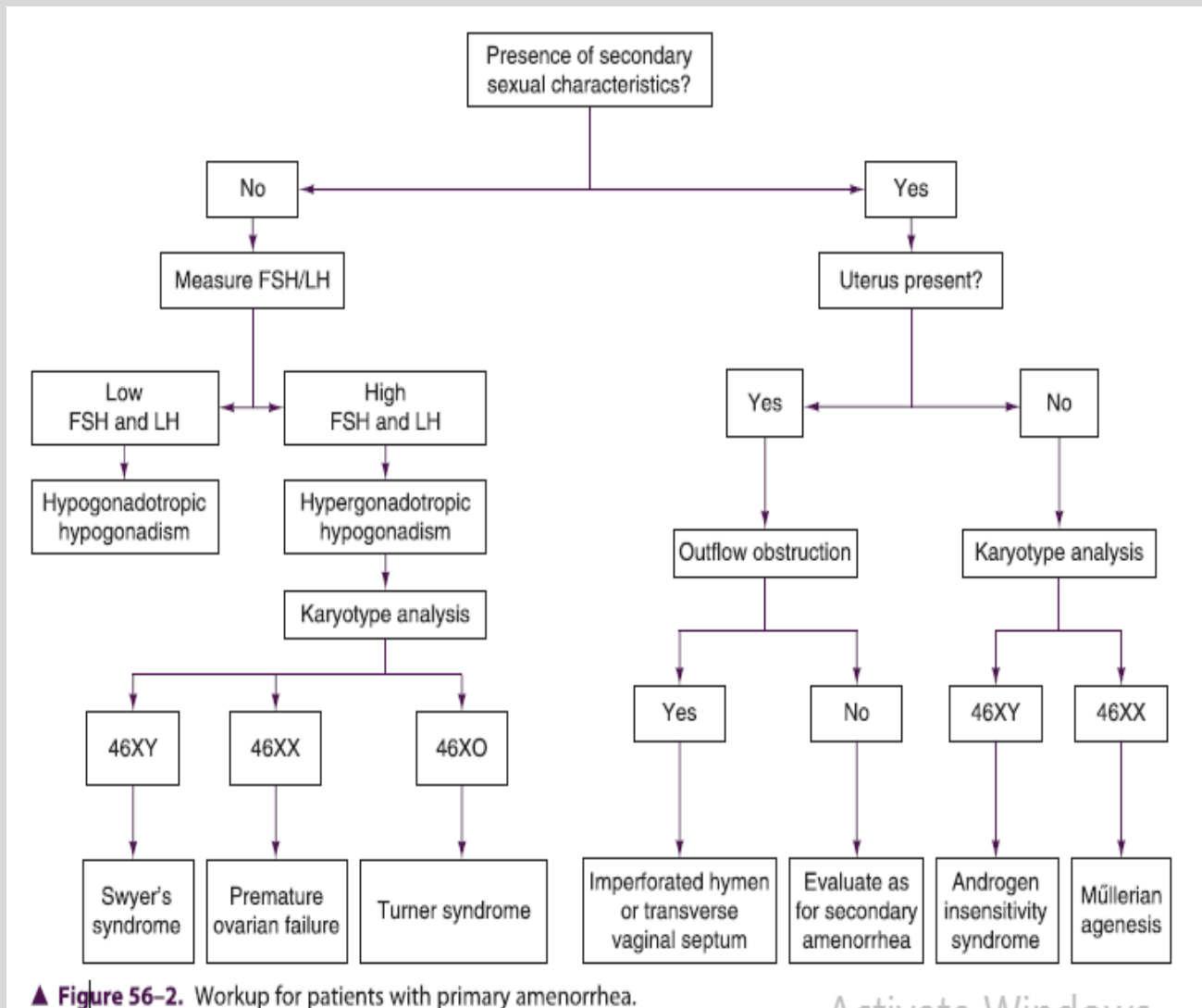
Evaluation of primary amenorrhea

Overview of approach

Primary amenorrhea is evaluated most efficiently by focusing on

- the presence or absence of breast development (a marker of estrogen action and therefore function of the ovary),
- the presence or absence of the uterus (as determined by ultrasound, or in more complex cases by magnetic resonance imaging [MRI]), and
- the follicle-stimulating hormone (FSH) level

- If the serum FSH concentration is elevated, the probable diagnosis is gonadal dysgenesis (including Turner syndrome) and a karyotype should be obtained.
- If FSH is normal and the ultrasound indicates that the uterus is absent, the probable diagnosis is müllerian agenesis or androgen insensitivity syndrome.
 - ✓ In the case of müllerian agenesis, the circulating testosterone is in the normal range for women, and in the case of **androgen insensitivity, the circulating testosterone is in the male range.**
- If the FSH is normal, breast development is present, and the ultrasound detects blood in the uterus (hematometra) or vagina (hematocolpos), an obstructed outflow tract is present.
- If the FSH is low or normal and the uterus is present, further evaluation is guided by the degree of pubertal development.



History

The following questions should be asked of a woman with primary amenorrhea:

- Has she completed other stages of puberty, including a growth spurt, development of axillary and pubic hair, apocrine sweat glands, and breast development?
 - Lack of pubertal development suggests deficient estradiol secretion, which could be due to a hypothalamic or pituitary disorder, ovarian failure, and/or a chromosomal abnormality.
- Is there a family history of delayed or absent puberty (suggesting a possible familial disorder)?

- What is the woman's height relative to family members?
 - Short stature may indicate Turner syndrome or growth hormone deficiency due to hypothalamic-pituitary disease.
- Were neonatal and childhood health normal?
 - Neonatal crisis suggests congenital adrenal hyperplasia. Alternatively, poor health may be a manifestation of hypothalamic-pituitary disease.
- Are there any symptoms of hyperandrogenism (acne, hirsutism) or virilization?
 - The presence of acne or hirsutism is consistent with a diagnosis of PCOS, while virilization suggests more severe androgen excess, due to an androgen-secreting ovarian or adrenal tumor, or 5-alpha-reductase deficiency.
- Has there been stress; change in weight, diet, or exercise habits; or illness that might result in hypothalamic amenorrhea?
- Is she taking any drugs that might cause or be associated with amenorrhea?
 - The medication may be taken for a systemic illness that itself can cause hypothalamic amenorrhea (eg, sarcoidosis).
 - Alternatively, drugs such as heroin and methadone can decrease GnRH and, therefore, gonadotropin secretion.
- Does she have galactorrhea, which would suggest excess prolactin?
 - This could be caused by hypothalamic or pituitary disease or by drugs, such as metoclopramide and antipsychotic drugs.
- Are there symptoms of other hypothalamic-pituitary disease, including headaches, visual field defects, fatigue, or polyuria and polydipsia?
- Any abdominal pain. Or distension?
 - Might suggest hematometra

Physical examination

- The single most important step in the evaluation is to determine by physical examination (or ultrasonography if needed) if a ***uterus is present***.
- Careful examination for features of secondary sexual characteristics development is also very important, (breast and genitalia)
 - A careful genital examination should be performed for clitoral size, pubic hair development, intactness of the hymen, vaginal length, and presence of a cervix, uterus, and ovaries.
 - In addition, the vagina and cervix should be examined for anatomic abnormalities.
- Growth, including height, weight, and arm span (normal arm span for adults is within 5 cm of height) and the growth chart.
- Skin findings such as hirsutism, acne, striae, increased pigmentation, and vitiligo.
- Physical features of Turner syndrome such as low hairline, webbed neck, shield chest, and widely spaced nipples.
 - The blood pressure should be measured in both arms if Turner syndrome is suspected because it is associated with an increased incidence of coarctation of the aorta.

Investigations

Pelvic ultrasound

- If a normal vagina or uterus is not obviously present on physical examination, pelvic ultrasonography should be performed to confirm the presence or absence of ovaries, uterus, and cervix.

Uterus absent

- Further evaluation should include a karyotype and measurement of serum total testosterone.
 - The history, physical exam, and results of these tests should distinguish between
 - ✓ abnormal müllerian development (a normal 46, XX karyotype, female phenotype, and normal female serum testosterone concentrations) and
 - ✓ complete androgen insensitivity syndrome (46, XY karyotype, normal female phenotype, sparse axillary and pubic hair, and normal male serum testosterone concentrations)

Initial laboratory testing

All women with primary amenorrhea should have serum

- human chorionic gonadotropin (hCG),
- FSH,
- thyroid-stimulating hormone (TSH), and
- prolactin (PRL) measured,

NB. Similar to the approach for women with secondary amenorrhea

Additional testing depends upon the results of the physical exam; in particular, whether müllerian structures are present or absent.

High FSH

- A high serum follicle-stimulating hormone (FSH) concentration is indicative of primary ovarian insufficiency.

- A karyotype is then required and may demonstrate complete or partial deletion of the X chromosome (Turner syndrome) and/or the presence of Y chromatin.
- The presence of a Y chromosome material (SRY) is associated with a higher risk of gonadal tumors and makes gonadectomy mandatory.

Low or normal FSH

- A low or normal serum follicle-stimulating hormone (FSH) concentration suggests a central hypothalamic-pituitary process, outflow tract disorder due to an anatomic abnormality, or an endocrine disorder.
- The presence or absence of breast development (an indicator of ovarian function and estrogen secretion) helps to further categorize these disorders.
 - Girls with low/normal FSH and **normal breast development** have either an anatomic abnormality (which is identified on ultrasound) or an endocrine disorder such as PCOS, hyperprolactinemia, or thyroid disease, disorders that more commonly cause secondary amenorrhea.
 - ✓ *Approximately 15 percent of girls with primary amenorrhea will have an anatomic abnormality* identified on ultrasound or exam such as imperforate hymen, transverse vaginal septum, or müllerian agenesis (congenital absence of the vagina with variable uterine development).
 - ✓ A high serum prolactin should be repeated once to confirm hyperprolactinemia prior to performing pituitary MRI.
 - ✓ Hypothyroidism as a cause of hyperprolactinemia should be ruled out by measuring serum TSH.
 - ✓ If there are signs or symptoms of hyperandrogenism, serum testosterone and dehydroepiandrosterone sulfate (DHEAS) should be measured

- Girls with **low or normal FSH and no evidence of breast development** most likely have a central hypothalamic-pituitary disorder; they should have a second serum sample obtained for both luteinizing hormone (LH) and FSH measurements
 - ✓ If **LH and FSH are both very low** (undetectable or near the lower limit of the assay), congenital GnRH deficiency, constitutional delay of puberty, or other disorders of the hypothalamic-pituitary axis should be considered.
 - ✓ If **LH is low and FSH is low or normal**, functional hypothalamic amenorrhea is likely if there is also a history of an eating disorder, excessive exercise, or stress.
- Contrast-enhanced MRI of the sella region is indicated in most cases of primary amenorrhea due to hypogonadotropic hypogonadism to evaluate for hypothalamic or pituitary disease.
- We recommend pituitary MRI in all women with hypogonadotropic hypogonadism, visual field defects, headaches, and/or any other signs of hypothalamic-pituitary dysfunction.

Management

Treatment of primary amenorrhea is directed at:

- correcting the underlying pathology (if possible),
- helping the woman to achieve fertility (if desired), and
- prevention of complications of the disease process (e.g., estrogen replacement to prevent osteoporosis)

All women with primary amenorrhea should be counseled regarding its cause, potential treatment, and their reproductive potential.

- Psychological counseling is particularly important in patients with absent müllerian structures and/or a Y chromosome.

Surgery may be required in patients with either congenital anatomic lesions or Y chromosome material.

- surgical correction of a vaginal outlet obstruction is necessary as soon as the diagnosis is made after menarche to allow passage of menstrual blood
- Creation of a neovagina for patients with müllerian failure is usually delayed until the women are emotionally mature and ready to participate in the postoperative care required to maintain vaginal patency.
- In those patients in whom Y chromosomal material is found, gonadectomy should be performed to prevent the development of gonadal neoplasia.

In women with polycystic ovary syndrome (PCOS), treatment of hyperandrogenism is directed toward

- achieving the woman's goal (e.g., relief of hirsutism, resumption of menses, fertility)
- preventing the long-term consequences of PCOS
 - e.g., endometrial hyperplasia/cancer, obesity, and metabolic defects)

Functional hypothalamic amenorrhea can usually be reversed by weight gain, reduction in the intensity of exercise, and/or resolution of illness or emotional stress.

- For women who want to continue to exercise or are unable to improve their nutritional health, estrogen-progestin replacement therapy should be given to those not seeking fertility as it has been demonstrated to improve bone mineral density.

For women who want to become pregnant,

- Either exogenous gonadotropins or pulsatile GnRH can be given.
- Advances in assisted reproductive technologies (ART) now make it possible for many women with primary amenorrhea to participate in reproduction.
- For women with gonadal dysgenesis, the use of donor oocytes and their partners' sperm with in vitro fertilization (IVF) allows the women to carry a pregnancy in their own uterus.
- For women with an absent uterus, use of their own oocytes in IVF and transfer of their embryos to a gestational carrier can allow these women to have genetically related children.

Secondary amenorrhea

Etiology

- The most common causes of pathologic secondary amenorrhea,
 - Hypothalamus – 35% (almost all functional hypothalamic amenorrhea)
 - Pituitary – 17%
 - Ovary – 40%
 - 30% PCOS, & 10% POI (aka premature ovarian failure)
 - Uterus – 7% (all due to intrauterine adhesions)
 - Other – 1% (congenital adrenal hyperplasia, ovarian and adrenal tumors, hypothyroidism)
- **Pregnancy**
 - Pregnancy ***is the most common cause of secondary amenorrhea.***
 - It is also important to note that apparent menstrual bleeding does not exclude pregnancy, since a substantial number of pregnancies are associated with some early first trimester bleeding.

- Thus, a pregnancy test (measurement of serum or urinary human chorionic gonadotropin [hCG]) is recommended as a first step in evaluating any woman with amenorrhea.
- **Hypothalamic dysfunction**
 - One of the most common types of secondary amenorrhea is functional hypothalamic amenorrhea, which, by definition, excludes pathologic disease.
 - Although isolated gonadotropin-releasing hormone (GnRH) deficiency most commonly presents as primary amenorrhea, it extremely rarely presents as secondary amenorrhea.
 - Benign and malignant tumors in the hypothalamus, such as craniopharyngiomas, radiation of sellar tumors, and infiltrative diseases of the hypothalamus, can cause secondary amenorrhea
 - Systemic illness may be associated with menstrual cycle disorders when it is severe enough to result in a decrease in hypothalamic GnRH secretion and/or when it is associated with nutritional deficiencies.
- **Functional hypothalamic amenorrhea**
 - Is a disorder that, by definition, *excludes pathologic disease*.
 - It is characterized by a presumed decrease in hypothalamic GnRH secretion
 - The term hypothalamic amenorrhea is often used interchangeably with functional hypothalamic amenorrhea.
 - **Risk factors**
 - ✓ Include eating disorders (such as anorexia nervosa), excessive exercise, and stress.
 - ✓ However, in a few women with functional hypothalamic amenorrhea, no obvious precipitating factor is evident.
 - Both weight loss below a certain target level (approximately 10 percent below ideal body weight) and excessive **exercise** are associated with amenorrhea.

- Most cases of amenorrhea associated with exercise are also associated with weight loss, with evidence that normal cycles are maintained when caloric intake is sufficient to match the energy expenditure
- The "**female athlete triad**" is defined as the presence of amenorrhea, disordered eating, and osteoporosis or osteopenia.
 - ✓ This syndrome is especially common in amenorrhea associated with activities that tend to be associated with low body weight (e.g., running, ballet dancing) and sports in which scoring are subjective (e.g., figure skating or gymnastics).
- Hypothalamic amenorrhea can be caused by **nutritional deficiencies** that are not associated with weight loss or strenuous exercise.
- **Emotional stress** and stress induced by illness (e.g., myocardial infarction, severe burns) are additional causes of hypothalamic amenorrhea.
 - ✓ With severe illness, the hypothalamic GnRH deficiency is transient; the hypothalamic-pituitary-ovarian axis recovers once the patient is well.
- **Recovery**
 - ✓ The natural history of hypothalamic amenorrhea has been studied.
 - ✓ Women with a history of hypothalamic amenorrhea with a clear precipitant (eating disorder, stress, and/or weight loss) had a better prognosis for recovery than those with no clear precipitant (71 and 29 percent, respectively).
- **Hypothalamic tumors and infiltrative lesions**
 - Hypothalamic tumors, (e.g., craniopharyngiomas, lymphomas) and infiltrative diseases (e.g., Langerhans cell histiocytosis, sarcoidosis) may result in decreased GnRH secretion, low or normal serum gonadotropin concentrations, and amenorrhea.
 - However, these lesions are uncommon compared with functional hypothalamic amenorrhea.

- Most women with infiltrative disease of the hypothalamus who have amenorrhea will have one or more neurologic symptoms, such as severe headache, change in personality, or marked mood changes.
- **Pituitary disease**
 - lactotroph adenomas (prolactin-secreting pituitary adenomas, prolactinomas) are the most common.
 - They are responsible for 13 percent of cases of secondary amenorrhea and 90 percent of the cases due to pituitary disease
 - Other types of pituitary adenomas and other sellar masses, and other types of pituitary disease, account for the majority of the remaining cases of pituitary origin.
- **Hyperprolactinemia**
 - Prolactin appears to cause amenorrhea by suppressing hypothalamic GnRH secretion, leading to low gonadotropin and estradiol concentration.
 - Unlike the other pituitary hormones, prolactin release is mostly controlled by inhibition, primarily by hypothalamic dopamine.
 - The negative regulation by dopamine is so potent that disruption of the pituitary stalk, by trauma or a large tumor, leads to hyperprolactinemia.
 - Several drugs, estrogen, and increased thyrotropin-releasing hormone (TRH) release due to hypothyroidism can also reversibly stimulate prolactin secretion.
- **Systemic illness**
 - include type 1 diabetes mellitus and celiac disease, which may also present with autoimmune ovarian insufficiency
 - **Type 1 diabetes mellitus**
 - ✓ Adolescent girls with type 1 diabetes mellitus have an increased prevalence of oligomenorrhea and amenorrhea.

- ✓ In one study, 39 of 56 (70 percent) adolescents with type 1 diabetes mellitus had amenorrhea or oligomenorrhea compared with 12 of 56 (22 percent) controls.
- ✓ The adolescents with type 1 diabetes mellitus and glycated hemoglobin (A₁C) concentrations >7.6 percent were more likely to have menstrual abnormalities.
- Type 2 diabetes in adolescent girls may be associated with polycystic ovary syndrome (PCOS) and menstrual cycle disorders.
- **Celiac disease**
 - ✓ It is estimated that approximately 40 percent of women with untreated celiac disease have menstrual cycle disorders
 - ✓ Other reproductive problems in these women include delayed menarche, infertility, miscarriage, and pregnancy complications.
- **Thyroid disease**
 - Menstrual cycle disorders are common in women with thyroid disease.
 - Women with severe hyperthyroidism, amenorrhea and hypomenorrhea occurred in 2.5 and 3.7 percent, respectively.
 - Although heavy bleeding is the typical bleeding pattern seen with hypothyroidism, secondary amenorrhea can also occur.
 - It is important to recognize hypothyroidism as a potential cause of a reversibly enlarged pituitary gland (due to thyrotrope hyperplasia, lactotroph hyperplasia, or both) and hyperprolactinemia and not to confuse this entity with a lactotroph adenoma
- **Polycystic ovary syndrome**
 - PCOS, the most common reproductive disorder in women, accounts for approximately 20 percent of cases of amenorrhea but may account for approximately 50 percent of cases of oligomenorrhea
 - The principal features of PCOS include androgen excess, ovulatory dysfunction, and/or polycystic ovaries. In addition,

- Many women with PCOS are overweight or obese and have insulin resistance. Women with PCOS may present with amenorrhea, but they more commonly have irregular menses (oligomenorrhea).
- The minimal criteria for the diagnosis of PCOS are two out of three of the following:
 - ✓ hyperandrogenism,
 - ✓ oligomenorrhea or amenorrhea, and
 - ✓ polycystic ovaries on ultrasound
- **Ovarian disorders**
 - **Primary ovarian insufficiency (premature ovarian failure)**
 - ✓ The depletion of oocytes before age 40 years is called primary ovarian insufficiency (POI, or premature ovarian failure).
 - ✓ Most women experience intermittent follicular development, estradiol production, LH surges, ovulation, and menstrual bleeding between months of hypoestrogenemia.
 - ✓ When POI is complete, lack of ovarian function leads to estrogen deficiency, endometrial atrophy, and cessation of menstruation.
 - ✓ Despite the intermittent ovarian function, conception is rare once a diagnosis of POI has been made.
 - ✓ POI may be due to complete or partial loss of an X chromosome (Turner syndrome), the fragile X premutation, autoimmune ovarian destruction, or, most commonly, unknown and rare causes.
 - ✓ Radiation therapy or chemotherapy with alkylating agents such as cyclophosphamide may also result in POI.
 - **Ovarian tumors**
 - ✓ Rare cases of ovarian tumors secreting inhibin may present with secondary amenorrhea.

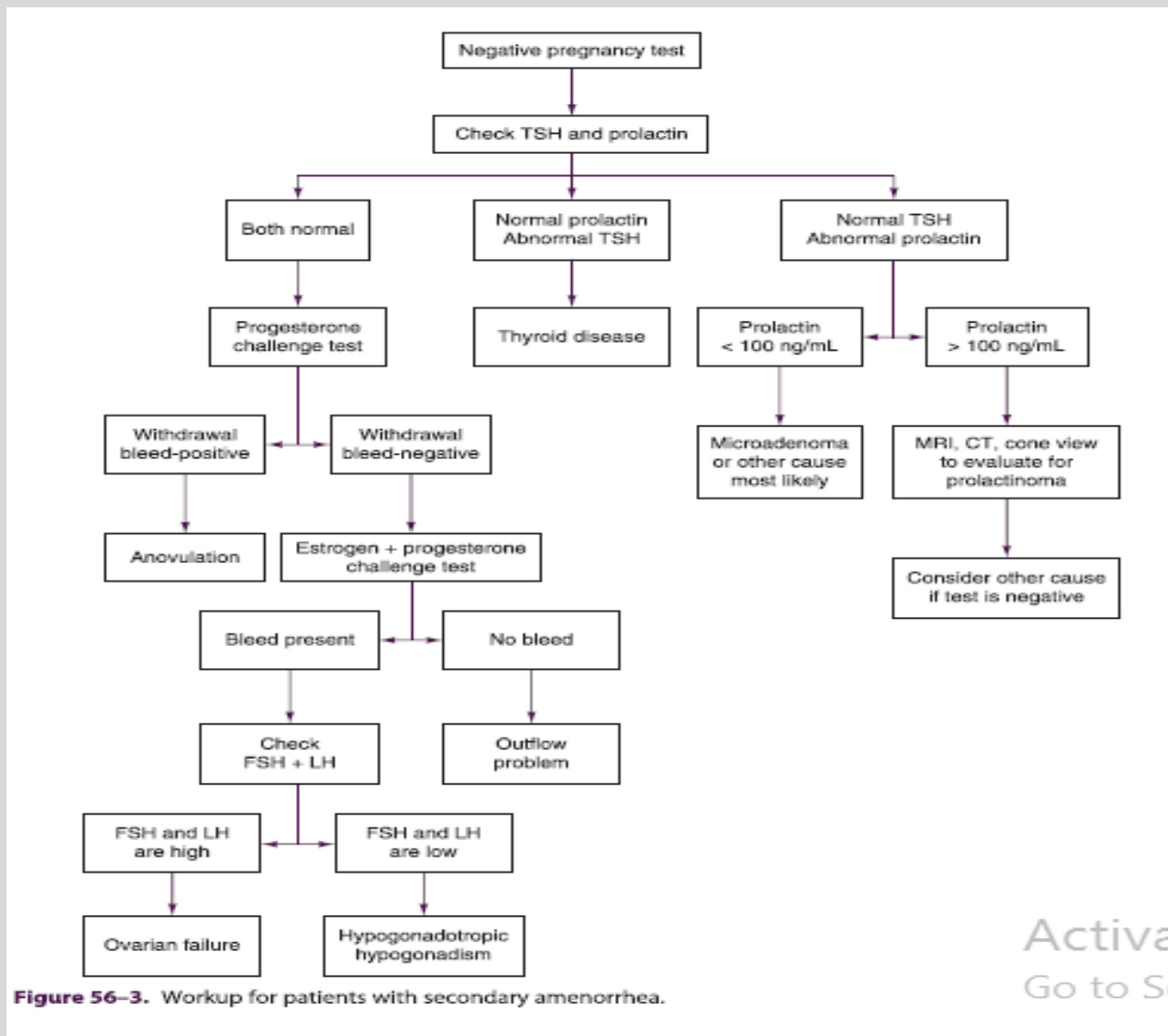
- **Uterine disorders**

- Intrauterine adhesions (Asherman syndrome) are the only uterine cause of secondary amenorrhea.
 - ✓ Results from acquired scarring of the endometrial lining, usually secondary to postpartum hemorrhage or endometrial infection followed by instrumentation such as a dilatation and curettage.
 - ✓ This abnormality prevents the normal build-up and shedding of endometrial cells, leading to very light or absent menses.

Evaluation of secondary amenorrhea

- **Rule out pregnancy**

- A pregnancy test is recommended as a first step in evaluating any woman with secondary amenorrhea.
- Measurement of serum beta subunit of human chorionic gonadotropin (hCG) is the most sensitive test.



History

- Has there been stress, change in weight, diet, or exercise habits,
- Is there an eating disorder or illness (that might result in functional hypothalamic amenorrhea)?
- Is the woman taking any drugs that might cause or be associated with amenorrhea? The drug may be taken for a systemic illness that itself can cause hypothalamic amenorrhea.

- Newly initiated or discontinued oral contraceptives can be associated with several months of amenorrhea, as can androgenic drugs like [danazol](#) or a high-dose progestin.
- Other drugs cause amenorrhea by increasing serum prolactin (PRL) concentrations, including metoclopramide and antipsychotic drugs.
- Is there hirsutism, acne, and a history of irregular menses (suggestive of hyperandrogenism)?
- Are there symptoms of hypothalamic-pituitary disease, including headaches, visual field defects, fatigue, or polyuria and polydipsia?
- Are there any symptoms of estrogen deficiency, including hot flashes, vaginal dryness, poor sleep, or decreased libido?
 - These symptoms may be prominent with primary ovarian insufficiency (POI).
 - In contrast, women with hypothalamic amenorrhea do **not** usually have these symptoms, despite the presence of similarly low serum estradiol (E₂) concentrations.
- Has the patient had galactorrhea, which suggests hyperprolactinemia?
- Is there a history of obstetrical catastrophe, severe bleeding, dilatation and curettage, or endometritis or other infection that might have caused scarring of the endometrial lining (Asherman syndrome)?

Physical exam

- Height and weight,
 - BMI > 30 kg/m² is observed in 50% or more of women with PCOS.
 - Women with a BMI less than 18.5 kg/m² may have functional hypothalamic amenorrhea due to an eating disorder, strenuous exercise, or a systemic illness associated with weight loss.

- The patient should also be examined for hirsutism, acne, striae, acanthosis nigricans, vitiligo, and easy bruisability.
- Breasts should be examined for evidence of galactorrhea, and
- A vulvovaginal exam should look for signs of estrogen deficiency.
- Parotid gland swelling and/or erosion of dental enamel would suggest an eating disorder (bulimia nervosa).

Work up

- see the above algorithm.

Management

Goals

- The overall goals of management in women with secondary amenorrhea include:
 - Correcting the underlying pathology, if possible
 - Helping the woman to achieve fertility, if desired
 - Preventing complications of the disease process (eg, estrogen replacement to prevent osteoporosis)

Management depends on the etiologies

- **Hypothalamic amenorrhea**
 - **Lifestyle modifications**
 - ✓ For many athletic women, explaining the need for adequate caloric intake to match energy expenditure sometimes results in increased caloric intake or reduced exercise, followed by resumption of menses.

- ✓ Nonathletic women who are underweight or who appear to have nutritional deficiencies should have nutritional counseling, and they can be referred to a multidisciplinary team specializing in the assessment and treatment of individuals with eating disorders.
- **Cognitive behavioral therapy –**
 - ✓ Cognitive behavioral therapy (CBT) may be effective for restoring ovulatory cycles in some women.
- **Management of low bone density –**
 - ✓ The effect of estrogen therapy on bone and the approach to women with exercise-associated amenorrhea are discussed separately.
- **Hyperprolactinemia –**treat based on the cause
- **Primary ovarian insufficiency (premature ovarian failure)**
 - should receive estrogen therapy for prevention of bone loss. This can be either an oral contraceptive or replacement doses of estrogen and progestin
- **Intrauterine adhesions**
 - hysteroscopic lysis of adhesions followed by a course of estrogen treatment to stimulate regrowth of endometrial tissue
- **Polycystic ovary syndrome**
 - is directed toward achieving the woman's goal (eg, relief of hirsutism, resumption of menses, fertility) and preventing the long-term consequences of polycystic ovary syndrome (PCOS), such as endometrial hyperplasia, obesity, and metabolic disorders.
- **Thyroid disease –**treat accordingly.

Chapter 30

GENITOURINARY FISTULA (GUF)

Prepared by Dr. Abraham Gebeyehu (MI)

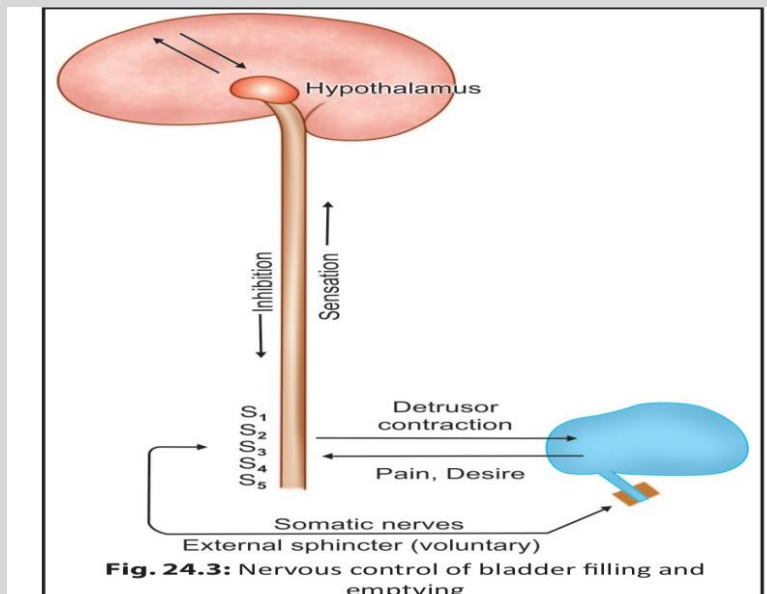
Physiology of Micturation

Storage phase:

- Filling rate: 0.5-5 ml/min
- As bladder fills, walls stretch to maintain a constant tone
- Intra-vesicle pressure is maintained to a steady level of 10cmH₂O even with a volume of 500ml.

Voiding Phase:

- *When volume of 250 ml reached, a sensation of bladder filling is perceived.*
- *Desire to void is not by increased intra-vesical pressure but by stimulation of stretch receptors in the bladder wall.*
- *Involves both spinal and higher centers*



Urinary Incontinence

Urinary incontinence (UI) is an involuntary loss of urine that is of social and hygienic problem which is demonstrable objectively.

Classification of incontinence with possible Differential diagnosis (causes)

1. Stress incontinence
 - Intrinsic sphincter dysfunction
 - Urethral hyper mobility
2. Urge incontinence
 - Idiopathic
 - Neurologic detrusor hyperreflexia
3. Mixed incontinence (stress and urge combined)
4. Overflow incontinence with urinary retention
 - Obstruction
 - Bladder hyporeflexia
5. Bypass incontinence
 - **Genitourinary fistulas**

Additional DDX

- Urethral diverticulum
- Congenital urethral abnormalities (eg, epispadias, bladder extrophy, ectopic ureter)
- Functional and transient incontinence can be due to: (DIAPPERS)
 - Delirium
 - Infection
 - Atrophic vaginitis
 - Pharmacologic agents
 - Psychological
 - Endocrine disorders
 - Restricted mobility
 - Stool impaction

Genitourinary fistula

A **genitourinary fistula** is defined as an abnormal communication between the urinary (ureters, bladder, urethra) and the genital (uterus, cervix, vagina) systems.

Incidence

- The true incidence of genitourinary fistula is unknown and varies according to whether the etiology is obstetric or gynecologic.
- In Asia and Africa, up to 100,000 new cases of obstetric genitourinary fistula are added each year to the estimated pool of 2 million women with unrepaired fistulas.
- For industrialized countries, most fistulas occur iatrogenically from pelvic surgery, and the generally accepted incidence derives from data on surgeries to correct these fistulas.
- Of genitourinary fistulas, the vesicovaginal fistula is most common.

Classification

Many classification systems exist for genitourinary fistula:

Classification of GUF Based on Anatomic Communication

- Communication between the urinary tract and vagina:-
 - Ureterovaginal
 - Vesicovaginal (VVF)
 - Urethrovaginal
- Communication between the cervix and the urinary tract
 - Ureterocervical
 - Vesicocervical
 - Urethrocervical
- Communication between the uterus and the urinary tract
 - Ureterouterine
 - Vesicouterine

Vesicovaginal fistulas can also be further classified by their **size and location** in the vagina.

- high vaginal, when found proximally in the vagina
- low vaginal, when noted distally;
- midvaginal, when identified centrally

Others classify vesicovaginal fistula **based on the complexity and extent of involvement**

- **Simple**
 - Size ≤ 3 cm
 - Located near the cuff (supratrigonal)
 - No prior radiation or malignancy
 - Normal vaginal length

- **Complicated**

- Size > 3 cm
- Located distant from cuff or has trigonal involvement
- Prior radiation therapy
- Pelvic malignancy present
- Vaginal length shortened

In one obstetric classification system, high-risk vesicovaginal fistulas are described by

- their size (> 4 to 5 cm in diameter);
- involvement of urethra, ureter(s), or rectum;
- juxtacervical location with an inability to visualize the superior edge; and
- reformation following a failed repair

A surgical classification to objectively evaluate obstetric urinary fistula repair has also been introduced:-

- **Type I** fistulas are those that do not involve the urethral closure mechanism,
- **Type II** fistulas, .type II fistulas are divided into: (A) without or (B) with subtotal or total urethra involvement. type IIB fistulas are further subdivided as: (a) without or (b) with a circumferential configuration around the urethra. And
- **Type III** fistulas involve the ureter and include other exceptional fistulas.

To aid objective comparison of surgical outcomes, a more comprehensive and standardized classification system has been developed

- It integrates fistula distance from the external urethral meatus, fistula size, degree of surrounding tissue fibrosis, and extent of vaginal length reduction
- This new classification divides genitourinary fistulas **into four main types**, depending *on the distance of the fistula's distal edge from the external urinary meatus.*

- Type 1: Distal edge of fistula > 3.5 cm from external urinary meatus
- Type 2: Distal edge of fistula 2.5–3.5 cm from external urinary meatus
- Type 3: Distal edge of fistula 1.5 to < 2.5 cm from external urinary meatus
- Type 4: Distal edge of fistula < 1.5 cm from external urinary meatus

These four types are further sub-classified by the size of the fistula, extent of associated scarring, vaginal length, or special considerations.

▪ ***By the size of the fistula***

- a) Size < 1.5 cm, in the largest diameter
- b) Size 1.5–3 cm, in the largest diameter
- c) Size > 3 cm, in the largest diameter

▪ ***By the extent of scarring/fibrosis***

- i. None or only mild fibrosis (around fistula and/or vagina) and/or vaginal length > 6 cm, normal capacity
- ii. Moderate or severe fibrosis (around fistula and/or vagina) and/or reduced vaginal length and/or capacity
- iii. Special consideration, e.g., postradiation, ureteric involvement, circumferential fistula, or previous repair

- *For example, a particular dx for a patient with a VVF that is 3cm from external urethral meatus measuring 2cm in diameter (subjectively admit 1 finger) which shows severe fibrosis will be **Type 2bii VVF***

Etiology and Risk factor

- Congenital genitourinary fistulas are rare, but if found, are commonly associated with other renal or urogenital abnormalities.
- Most GUF are acquired and typically result from either obstetric trauma or pelvic surgery
 - ***Obstetric Trauma:*** In developing countries, more than 70 percent of genitourinary fistulas arise from obstetric trauma, specifically from prolonged or obstructed labor or complicated cesarean delivery.

- *When OL is unrelieved, the presenting fetal part is impacted against soft tissue of the pelvis and a wide spread ischemic vascular injury develops, that results in **pressure tissue necrosis and subsequent fistula formation***

- Factors associated with fistula formation in **developing countries**

- Childbearing at a young age
- Female genital mutilation, may significantly narrow the vaginal introitus and obstruct labor
- Prolonged obstructed labor or
- Anatomic malpresentation of the presenting fetal part

These factors can cause pressure and ischemic necrosis of the anterior vaginal wall and bladder, subsequently resulting in fistula formation.

- Instrumental delivery used to deliver stillborn infants or perform abortion, may damage the vagina
- Malnutrition and limited health care in many of these countries can further diminish wound healing.

- Fistula in **developed countries**

- Obstetric procedures or deliveries are unusual causes of fistula in developed countries.
- **Iatrogenic injury** during pelvic surgery is responsible for 90 percent of vesicovaginal fistulas in these nations
 - The accepted incidence of fistula formation after pelvic surgery is 0.1 to 2 percent.
 - In industrialized countries, **hysterectomy** is the most common surgical precursor to vesicovaginal fistula, accounting for approximately 75 percent of fistula cases

- Other Causes include:-
 - ✓ Radiation therapy (6%)
 - ✓ Malignancy (1.8%),
 - ✓ Trauma (4%),
 - ✓ foreign bodies
 - ✓ infections, pelvic inflammation, and
 - ✓ inflammatory bowel disease
- Other rare causes of fistula formation include infections such as lymphogranuloma venereum, urinary tuberculosis, pelvic inflammation, and syphilis; inflammatory bowel disease; and autoimmune disease.
- Additionally, conditions that interfere with healing such as poorly controlled diabetes mellitus, smoking; local infection, peripheral vascular disease, and chronic corticosteroid use are potential risks.

Pathophysiology

The principles and phases of wound healing aid the understanding of genitourinary fistula pathogenesis. After injury, tissue damage and necrosis stimulate inflammation, and the process of cell regeneration begins.

- Angiogenesis phase: Initially new blood vessels form
- Fibrosis phase:
 - 3 to 5 days after injury, fibroblasts proliferate and subsequently synthesize and deposit extracellular matrix, in particular collagen.
- Remodeling phase: Subsequent scar maturation and organization

These phases are interdependent, and any disruption of this sequence eventually may create a fistula.

Most defects tend to present 1 to 3 weeks after tissue injury. This is a time during which tissues are most vulnerable to an altered healing environment that may include hypoxia, ischemia, malnutrition, radiation, and chemotherapy. Eventually, edges of the wound epithelialize, and a chronic fistulous tract is thus formed.

Clinical manifestation

Symptoms

- Classically presents with unexplained continuous urinary leakage from the vagina after a recent operation.
- Other less specific symptoms of genitourinary fistula include fever, pain, ileus, and bladder irritability

Diagnosis

- A thorough history and physical examination identifies most cases of vesicovaginal fistula.
- Visual assistance with an endoscopic lens and translucent vaginal speculum can sometimes help identify a vaginal-apex fistula, which can be more difficult to detect.

Investigations

- Measurement of the vaginal fluid's creatinine content can sometimes be used to confirm its origin.
 - Although creatinine levels in urine vary, with mean levels reaching 113.5 mg/dL, a value > 17 mg/dL is consistent with urine.
 - In contrast, fluid with a concentration < 5 mg/dL is highly unlikely to be human urine.
- Retrograde bladder instillation of visually distinct solutions such as sterile milk or dilute methylene blue or indigo carmine can often indicate a fistula and aid in its localization.
 - A **three-swab test**, commonly known as the “tampon test” is recommended.
 - ✓ Two to four pieces of gauze sequentially packed into the vaginal canal.
 - ✓ A diluted solution of methylene blue or indigo carmine is instilled into the bladder using a transurethral catheter.

- ✓ After 15 to 30 minutes of routine activity, the gauze is removed serially from the vagina, and each is inspected for dye.
 - ✓ The specific gauze colored with dye suggests the fistula location-a proximal or high location in the vagina or the innermost gauze and a low or distal fistula or the outermost.
- Cystourethroscopy is another valuable diagnostic tool and vaginoscopy.
 - CT-scanning
 - Intravenous pyelography (IVP)
 - can adequately confirm integrity of the upper collecting system and exclude ureteral involvement in a fistula
 - Voiding cystourethrography (VCUG)
 - can help confirm the presence, location, and number of fistulous tracts
 - Transabdominal sonography

Complications

The complication of GUF is beyond the hole between the bladder and vagina. GUF is one in the spectrum of the multifaceted nature of Obstructed Labor(OL) complications, termed *OL injury complex*. *These injuries include:*

- *Fetal loss*
- *Maternal complications*

<i>Physical injuries</i> ▪ <i>Total urethral loss</i> ▪ <i>Stress incontinence</i> ▪ <i>Hydroureteronephrosis</i> ▪ <i>Renal failure</i> ▪ <i>RVF</i> ▪ <i>Rectal atresia</i> ▪ <i>Anal sphincter incontinence</i> ▪ <i>Cervical destruction</i> ▪ <i>Amenorrhea</i> ▪ <i>Secondary infertility</i> ▪ <i>PID</i> ▪ <i>Vaginal stenosis</i> ▪ <i>Osteitis pubis</i> ▪ <i>Foot drop</i>	<i>Socioeconomic</i> ▪ <i>Divorce</i> ▪ <i>Exclusion from religious activities</i> ▪ <i>Separation from family</i> ▪ <i>Worsening poverty</i> ▪ <i>Malnutrition</i> <i>Psychological</i> ▪ <i>Depression</i>
---	--

Management

Supportive

- **The four “D” of supportive care**
 - Drink
 - Drain
 - Dry
 - Diet
- **Psychological support**

Conservative

- Approximately **12 percent** of women treated by sustained catheterization alone had fistulas that healed spontaneously.
- Fibrin sealant also colloquially called fibrin glue can be used

Surgical Treatment:

Fundamentals of genitourinary fistula repair include accurate fistula delineation; adequate assessment of surrounding tissue vascularity; timely repair; multilayer, tension- free, and watertight defect closure; and postoperative bladder drainage.

Timing of Repair

- One principle of fistula repair dictates that, a repair be performed in non-infected and non-inflamed tissues.
- Early surgical intervention of uncomplicated fistulas within the **first 24 to 48 hours** following the inciting surgery.
- In instances of extensive and severe inflammation, we recommend delaying operative repair or 6 weeks until the inflammation subsides.
- Subsequent repair after a failed prior repair is done after 3 months.

Route of Surgical Repair

- Vaginal: Latzko technique, classical technique
- Abdominal (Transperitoneal): This approach is used in situations in which the fistula:
 1. is located proximally in a narrow vagina,
 2. lies close to the ureteral orifices
 3. is complicated by a concomitant ureteric fistula
 4. persists after prior repair attempts,
 5. is large or complex in configuration, or
 6. requires an abdominal interposition graft, described in the next section

Prevention

Prevention of GUF in developing country focus on prevention of OL, including:

- Prevention of maternal malnutrition
 - In-utero
 - ✓ IUGR
 - Childhood
 - ✓ Appropriate sunlight exposure
 - ✓ Adequate balanced diet
- Avoid teenage pregnancy
 - contraception
- ANC follow up
- Avoid delay in getting appropriate management during delivery
 - Primary delay: delay in deciding to **seek** health care
 - Secondary delay: delay in **arriving** at suitable obstetric care facility
 - Tertiary delay; delay in **receiving** care at health facility

*There is also so called 4th delay, which is failure to **refer** to appropriate higher health facility*

Sample history

C/c: failure to control of urine of 5 months duration

HPI:

This is an 18yrs old P-I (Stillborn, by C/D) lady who is relatively healthy 3yrs back at which time she start to experience inability to control urine, which started 2wks after she deliver by C/D, after 36hrs of pushing down pain. For this compliant she visited Metu Karl hospital 2and half years back, and repair was done. Currently she presented with failure to control of urine of 5month duration, she is referred from Metu Karl hospital for further management. She is divorced and lives by herself. She doesn't attend any social or religious events because of her condition.

Otherwise she has on Hx of

- vaginal discharge and itching sensation
- fever, pain,
- radiation therapy,
- malignancy ,
- trauma ,
- foreign bodies

P/E:

- V/S; stable
- GUS
 - The vagina is soaked with urine
 - There is a 2cm anterior vaginal wall defect communicating with the bladder 3cm from external urethral meatus.

Ass't- Type 2biii VVF

References

- Gabbe obstetrics 7th edition
- Williams' Obstetrics 25th and 26th edition
- Williams' Gynecology 4th edition
- Current Obstetrics and Gynecology 12th edition
- Berek and Novak's Gynecology, 14th edition
- DC Dutta's Textbook Of Obstetrics 7th edition
- ACOG Practice Bulletin, Gestational hypertension and Preeclampsia 2019
- JUSH Obstetrics management guideline
- Obstetric Management Protocol for Hospitals by MOH, 2021
- Ayder's Obstetrics and Gynecology
- Up-to-date 2022
- Obstetrics: Normal and Problem Pregnancies, (Gabbe): Seventh edition. | Philadelphia, PA: Elsevier, [2017]
- NHS Guideline for use of anti-D for prevention of HDFN, London; 2019